
GENERAL THEORY OF RELATIVITY

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Lecture 1. The Equivalence Principle: Gravitation is Curvature

Einstein's general theory of relativity, or **the** theory of gravitation, is often said to be the greatest single achievement in theoretical physics. It was developed by Einstein (finally, in 1916) in a purely deductive manner and only later has been confirmed by (now numerous) astronomical observations. It is also said to be the most beautiful of all existing physical theories. The starting point for this deduction is the equivalence principle.

There were two fundamental interactions known by the beginning of the last century: electromagnetism and gravity. The similarity between electrostatic forces and gravity is compelling:

$$\boxed{F = k \frac{q_1 q_2}{r^2}} \quad \boxed{F = G \frac{m_1 m_2}{r^2}} \quad (1.1)$$

Both describe long-range forces with only a “small” difference that charges could be either positive or negative but masses are always positive.

One of the cornerstones of special relativity is that such an instantaneous action on distance is impossible. We know that the electrostatic Coulomb force appears in the SR as $\mathbf{F} = q_1 \mathbf{E}$ where the field $\mathbf{E}$ is the solution to one of the Lorentz-covariant Maxwell's equation, $\text{div } \mathbf{E} = 4\pi\rho$ (in Gaussian units where $k = 1$ in Eq. (1.1), with $\rho = q_2\delta(\mathbf{r} - \mathbf{r}_2)$ corresponding to a single charge) in a reference frame where both the charges are at rest. In general, the electromagnetic interaction is not instantaneous. The Lorentz force experienced by the charge q_1 , $f^\mu = (q_1/c)F^{\mu\nu}v_\nu$, is due to the electromagnetic field $F^{\mu\nu}$ created by the charge q_2 . The field propagates at speed of light, and the laws of electromagnetism are Lorentz-covariant that is essentially the same in all inertial frames.

It was tempting to construct a similar approach to gravitation forces, making them relativistically invariant by introducing a scalar gravitational field (not a tensor, as there is no gravitational analogue of magnetism). However, there exists a great difference between charges and masses:

While electric charges are invariant under coordinate transformations, masses are not.

We know from SR that inertial masses are temporal components of the energy-momentum tensor. What is most important is that the same masses govern the Newton law of universal gravity. This is the biggest “mystery” of gravitation: why masses in the law of gravity are the same as in the law of inertia (Newton's 2nd law, $F = ma$)? In the scalar theory of gravitation this would remain unexplained. That is why after a few years of attempts in this direction, this idea has been abandoned by Einstein. Starting from the equivalence of inertial and gravitational masses and from the relativity principle, he came to a conclusion that gravity is fundamentally different from electromagnetism (or from any other “fundamental” interaction, as we can add today), as it is intimately related to geometry of the space-time. Our first (and main) task is to understand this relation.

The starting point is the famous discovery by **Galileo**:

Every object falls in vacuum with the same acceleration.

From this immediately follows **equivalence of the inertial and gravitational masses**. The inertial mass enters the 2nd Newton law, $\mathbf{F} = m\mathbf{a}$, while the gravitational one is the coefficient of proportionality in $\mathbf{F} = -m_g \nabla \Phi$ where Φ is the gravitational potential. If we *assume* that m and m_g are the same (or proportional), then we *find* that $\mathbf{a} = -\nabla \Phi$ ($= GM/r^2$ in the field of a planet or star of mass M) is independent of m , *i.e.* the same for a feather and a cannon ball (obviously, in the absence of air resistance).

As we want to make no distinction between these inertial and gravitational mass, we are forced to introduce the coefficient of proportionality, G , into the law of gravity. Otherwise, we could simply write $F = m_g M_g / r^2$, introducing new gravitational masses, in a similar way as we introduced electric charges in Coulomb's law with $k = 1$ in Gaussian units. The fundamental difference between the two classical laws is in the fact that the gravitational mass is proportional to the inertial mass with a universal coefficient of proportionality while the electric charge has nothing to do with the mass.

Note that the equivalence of the gravitational and inertial masses holds with the accuracy of at least 10^{-11} . It is totally mysterious in Newton's theory which would work perfectly well without it.

Let us now consider a passenger in a falling lift (following Einstein; nowadays it could be a spacecraft on a stable orbit around the Earth). As both fall with the same acceleration, the passenger feels no gravitational force; classically, one can say that it is *exactly* compensated by the force of inertia. In other words, the passenger is weightless as in an inertial frame in the space where no gravity exists. Hence we arrive at the **Galileo equivalence principle**:

A freely falling framework is locally equivalent to an inertial one.

It is also called the weak equivalence principle (WEP) as it, strictly speaking, refers **only to mechanical forces**. More loosely it can be formulated as a sound-bite:

Forces of gravity are locally equivalent to forces of inertia.

"Locally" here means that the gravitational forces should be parallel; otherwise, their difference from the inertia reveals itself in tidal forces which, as we shall learn later, are crucial for formulating the relativistic gravity laws – the Einstein equations.

Einstein has generalised the WEP **postulating** that it should be valid for any form of motion. Such a generalisation is called the Einstein equivalence principle. To make it explicit, we choose the following formulation¹:

There is no way of distinguishing between effects on a local observer of a uniform gravitational field and of a constant acceleration.

The main difference between the EEP and WEP:

EEP generalises WEP from the laws of mechanics to all the laws of nature.

Let us consider, for illustration, four different observers: **A** is in a falling lift; **B** is in a spacecraft which is in a free fall far from all stars; **C** is in a lift hanging on a rope (gravity), and **D** is in an accelerated (with g) spacecraft. The equivalence principle states that **A** $\longleftrightarrow$ **B** and **C** $\longleftrightarrow$ **D**. The trajectory of a thrown ball or, more strikingly, a light beam is a straight line for **A** and **B**, while for **C** and **D** it is a parabola.

The fact that free motion in a non-inertial system is not along a straight line is obvious. The EP claims that the same is true in the presence of gravity – not a surprise by itself (the light trajectories will be bent in the Newtonian world if we assume that *the light is heavy*, i.e. massive, albeit we will learn later that this approach would not correctly describe the curvature of light trajectories in the gravitational field). Their full equivalence, however, has very deep and surprising consequences. We

¹There are many equivalent formulations of the Einstein equivalence principle and the so-called strong equivalence principle – I will not discuss here a subtle difference between the two. Let me only stress again the importance of *locality*, whose meaning is that the temporal and spatial scales of any experiment are negligible compared to the scale of variation of the gravitational field.

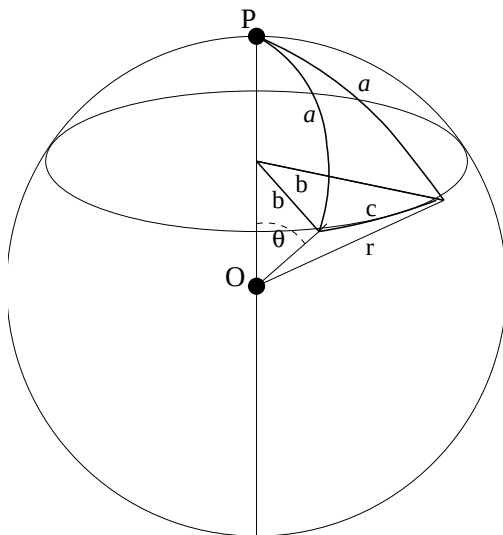
claim that an object, which is acted upon only by forces of inertia and gravity, moves along *geodesic* lines, which are the shortest paths connecting two close points.²

We need to understand why the shortest path in a non-inertial frame is no longer a straight line. But first we should ask ourselves what is actually an inertial frame? Let us recall the special relativity principle: **The laws of nature are covariant in all *inertial* frames.** We want to understand (following Einstein) whether Nature singles out such preferred frames. What are special about them? Do they exist at all? (We only know that if one such frame does exist, then all frames that move in it at a constant speed are also inertial). SR abolished the concept of Absolute Space. However, the existence of a privileged class of non-accelerated frames allows the Absolute Space through the back door (non-accelerated with respect to what?).

We argue that a special, distinctive property of inertial frames is Euclidean geometry. Hence, it is freely falling frames (with straight trajectories for light or a thrown ball) which will take place of inertial ones. On the contrary, geometry of non-inertial frames is non-Euclidean, as can be illustrated by the following simple argument.

Consider a frame (Σ') which is uniformly rotating with respect to an inertial frame (Σ) around their common z axis. A circle in the xy plane in Σ remains a circle of the same diameter in Σ' . Indeed, using an *instantaneous* inertial frame with the x axis tangential to the circumference of the circle in Σ' , we see that the yardstick at rest in Σ' suffers the Lorentz-Fitzgerald contraction in the tangential but not in radial direction. Hence the ratio of the circumference ℓ to the diameter d in Σ' is smaller than π which is their ratio in the Euclidean frame Σ .

On the other hand, geometry where $\ell/d \neq \pi$ is typical for any curved surface in a usual Euclidean space. Let us consider a spherical surface.



A circle at the colatitude θ on the surface has radius $r \sin \theta$, and the circumference $\ell = 2\pi b = 2\pi r \sin \theta$. For two-dimensional inhabitants of this sphere the existence of a third dimension is unknown. For them, radius of the circle is the distance from the pole to the circle, i.e. $a = r\theta$. Hence they would have the ratio of the circumference to the radius as $2\Pi = 2\pi \sin \theta / \theta$. For small θ , i.e. for $a \ll r$, they have $\Pi = \pi(1 - \frac{1}{6}\theta^2) = (1 - \frac{1}{6}(a/r)^2)$. When $a \ll r$, the difference between π and Π is negligible but still detectable if the inhabitants either can travel over distances comparable to r , or can measure the ratio of the circumference to the radius extremely accurately.

Even after the discovery of corrections to ℓ/ℓ_0 , they could not prove (or disprove) the existence of a third dimension: all they know is that their world has some *intrinsic* curvature, which measure is related to $1/r$. Note that in such a 2D world inertial frames (whose spatial part is Euclidean by definition) do not exist globally (small local frames still could be approximately inertial).

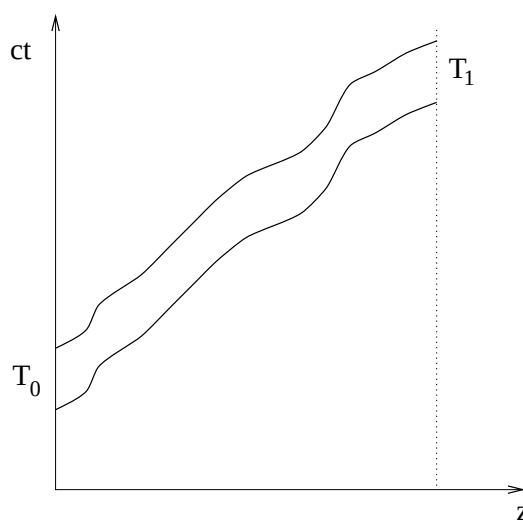
In a curved world, the fifth Euclidean postulate (stating that on a plane with a given straight line and a point outside of it, one can draw only a single straight line parallel to the given one going

²This is not an additional assumption; as we will learn later, this follows from a precise mathematical formulation of the equivalence principle.

through this point) is wrong. On a spherical surface straight (geodesic) lines are great circles so that “parallel” lines that never intersect each other do not exist. If we took a *saddle* as another example of curved surface, then there is an infinite number of non-intersecting straight lines (geodesics) passing through a given point parallel to a given straight line.

These two simple examples suggest two ways for the realisation of non-Euclidean geometry: the space could have intrinsic curvature; or it could be perfectly flat (Euclidean) but viewed from a non-inertial reference frame. On the face of it, these two ways seem to be fundamentally different. The equivalence principle states that in the physical world they are *locally* equivalent. In space-time, *any* reference frame is locally equivalent to some inertial frame and thus suitable for further considerations.

The main point here that in the presence of gravity we have no chances to maintain a special role of inertial frames, mandatory in SR. We argue that in the presence of gravitational forces global inertial frames do not exist, not speaking about their special role. To illustrate this, let's consider a beam of light travelling against an uniform gravitational field. Due to the famous $\mathcal{E} = mc^2$, each photon has some mass (although this is not a rest mass!). The equivalence principle requires then that this mass is subject to gravity (we could consider experiments where massive particles annihilate emitting only photons which then again transform to the particle-antiparticle pairs; the energy conservation law would require photons to feel gravity). Then, we have $\mathcal{E}_1 = \mathcal{E}_0 - mgH = \mathcal{E}_0(1 - gH/c^2)$, so that the frequency, $\omega = \mathcal{E}_1/\hbar$, at the height H is ‘red-shifted’ with respect to that, $\mathcal{E}_0/\hbar$, at the ground level.



Consider now the emission of a light-beam on the Minkowski space-time diagram. Two successive nodes of an electromagnetic wave must travel along identical paths. They are separated by the period $T = 2\pi/\omega$ which is different – due to the gravitational red shift – at the top and at the bottom ($T_1 > T_0$). An observer at the top therefore sees a clock at the bottom running slowly. However, as paths must be congruent in the Minkowski geometry, it would required $T_0 = T_1$. Thus, geometry is not Minkowski, and the frame is not a global Lorentz frame. As there is no empty space, global inertial frames seem to be impossible. This example illustrates that a temporal part of the space-time is also not ‘flat’ in the presence of gravity. **Locally**, however, gravitation is eliminated in freely falling frames; therefore, the red shift is absent in such frames so that flat Minkowski geometry can exist in some ‘close’ neighbourhood of an observer.

Since in the presence of gravity we do not have a special class of frames which are inertial, we must reformulate the relativity principle. The role of inertial frames is played now by **locally inertial** free falling frames (that are ‘flat’). We will show in the following lectures that *any non-singular frame of reference can be made flat in a neighbourhood of any point* – this is just a definitive property of the Riemannian space. Thus, all frames of reference got equal rights. Since all physical laws are mathematically formulated as differential equations which are local by construction, we can argue in favour of the **general relativity principle**:

The laws of nature are locally covariant in all frames, no matter what is the frame's state of motion .

The word *covariant* in this context means that differential equations describing physical laws can be written in the same form in any frame of reference. Of course, in this formulation GRP is not precise as we need to define exactly what is meant by the same form. This we will do in a great detail later in the course.

The equivalence principle and the general relativity principle make the foundation of the general theory of relativity. While the former postulate (in its weak form) is experimentally confirmed with a very high accuracy (at least 10^{-15}), the latter cannot be directly verified. But accepting these postulates leads to the plethora of experimentally verifiable (and verified!) consequences. All we need to get grasp of these consequences is to follow Einstein's way of thinking (to a degree available for us) in building mathematical dressing for these principles. This maths will lead to some fantastic consequences. If we accept the general principles, however, we have no choice but to accept all the consequences, unless they contradict to experimental evidence.

*Note: the main principle of **Special Relativity**, the constancy of the speed of light in all inertial frames, is not generalised: speed of light does not remain constant in noninertial frames!*

We have stressed several times that the equivalence principle has a local nature. As a gravitational field is usually nonuniform, globally it cannot be excluded by choosing a properly accelerated system. If we look at a very large body moving in a (necessarily) nonuniform gravitational field, we see that only its centre of mass falls freely. The body is kept as a whole by some forces of non-gravitational origin (elasticity, etc) which compensate the gravitational forces due to non-uniformities of the field (they are called **tidal forces**: tides bulge due to the difference of the Moon's gravitational field across the diameter of Earth). As tidal forces cannot be excluded, they are regarded as the fundamental manifestation of gravity in GR. Thus we again come to the conclusion that global inertial frames are impossible to construct.

We have accepted that geometry in an accelerated frame is non-Euclidean. The equivalence principle tells us immediately that geometry is non-Euclidean also in the presence of gravitation. As we saw that 2D geometry is non-Euclidean on curved surfaces, we should anticipate the appearance of the curved space (or spacetime). It is hard for us to imagine what is the curvature of space itself, without embedding it into something of a higher dimension (although a clarity of such an embedded higher-dimensional space for our imagination is also doubtful). However, remember our 2D beings on the curved surface: they are able to detect the curvature *intrinsically*, and so are we.

We conclude that our principal task is to understand that

GRAVITATION IS SPACETIME CURVATURE.

There are two major topics hidden in the above sound-bite which constitute the two principal aims of the course. We will use the relativistic covariance and the correspondence with Newtonian gravity in the nonrelativistic limit

- To describe free motion of particles and light in a curved spacetime.
- To understand how the presence of matter makes the spacetime curved.

We shall start constructing a necessary geometric technique by revising in the next lecture what we know about vector and metric spaces (e.g., from Special Relativity), and describing geodesic motion in metric spaces. This would allow us to cope with the first topic using nothing more than the rules of Lagrangian mechanics. The second topic, starting from Lecture 8, would require considerably more involved maths.

Lecture 2. Geodesic lines in metric spaces

The most important immediate consequence of the equivalence principle is that particles do not move along straight lines in general frames. We will use the least action principle demanding particles and light to move along geodesic lines – the shortest (and straightest) lines connecting two points in a curved space-time (postponing a formal proof to the second half of the course). Our aim in this lecture is to revise Lagrangian approach to geodesic lines curved spaces. But we start with a flat Euclidean space.

2.1 Metric tensor

The shortest distance between two points in Euclidean space is the length of the straight line segment connecting them. In a curved space, we need to start with defining the distance between two infinitesimally close points. A space where such a distance is defined is called *metric space*. We will be interested in a particular definition of metrics which is due to Riemann.

Let us denote coordinates of a point in frame Σ in a d -dimensional space as $x^\mu \equiv \{x^1, \dots, x^d\}$. In the context of Relativity, x^μ are called event coordinates in the 4-dimensional spacetime, $x^\mu \equiv \{x^0, x^1, x^2, x^3\} \equiv \{x^0, x^m\}$ with $x^0 \equiv ct$ being the time coordinate, and $x^m \equiv \{x^1, x^2, x^3\}$ being the space coordinates. Then a close point, $x^\mu + dx^\mu$ is given by infinitesimal shift of each coordinate, and the “distance” between these points (events) is given by

$$ds \equiv \sqrt{ds^2}$$

where ds^2 is the Riemannian metric defined as follows:¹

Definition 2.1. $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$, $g_{\mu\nu} = g_{\nu\mu}$

Here the symmetric metric tensor $g_{\mu\nu}$ is (in general) an arbitrary function of coordinates with the only restriction $\det g_{\mu\nu} \neq 0$. While $g_{\mu\nu}$ depends on the choice of coordinates, ds^2 is a **scalar** which tells about an intrinsic structure of the underlying space.

Example 2.1. On a 2D plane, we can use either Cartesian or plane polar coordinates to write

$$ds^2 = dx^2 + dy^2 = dr^2 + r^2 d\phi^2.$$

In Cartesian coordinates, $g_{\mu\nu}$ is a unit matrix, and the above equation is the Pythagoras theorem; the expression in polar coordinates follows from $x = r \cos \phi$ and $y = r \sin \phi$ so that $g_{\mu\nu} = \text{diag}(1, r^2)$.

In a generic d -dimensional space, $g_{\mu\nu}$ has up to $\frac{1}{2}d(d+1)$ independent elements which are varying from point to point. Taken locally at a given point, such a matrix can be diagonalised (as it is symmetric) and rescaled to the form $\text{diag}(\pm 1, \dots, \pm 1)$. What cannot be changed is the signature of the metric,² i.e. a difference between the number of positive and negative diagonal elements.³ Moreover, the signature stays the same in all points of a given space.

If $g_{\mu\nu}$ can be globally (i.e. simultaneously in all points) diagonalised to the unit matrix, the space is Euclidean, while such coordinates are called Cartesian. If $g_{\mu\nu}$ can be globally diagonalised to

¹We always employ Einstein's summation agreement that means summation over two repeated indices (one sub- and one super-index): $g_{\mu\nu} dx^\mu dx^\nu \equiv \sum_{\mu, \nu=1}^d g_{\mu\nu} dx^\mu dx^\nu$. The proper definition of tensor will be given in the 2nd part of the course; for now, it is enough to think about $g_{\mu\nu}$ as a real symmetric $d \times d$ matrix.

²This is true up to the simultaneous change of signs of all the elements, so that the Minkowski matrix $\text{diag}(1, -1, -1, -1)$ used in the present course is equivalent to $\text{diag}(-1, 1, 1, 1)$ used in Year 3.

³The appropriate mathematical theorem is called Sylvester's law of inertia.

$\text{diag}(\pm 1, \dots, \pm 1)$ with not all the elements of the same sign, it is called pseudo-Euclidean. In particular, the 4D spacetime is called Minkowski space where the metric tensor can be globally diagonalised to $\text{diag}(1, -1, -1, -1)$. The (pseudo)Euclidean space is called flat (we will see, after defining the curvature, that it is zero).

Any space where such a global diagonalisation is not possible is curved. Any curved space is locally either Euclidean or pseudo-Euclidean, with the same signature at all points. While ds^2 is always real, in the pseudo-Euclidean space it could be of any sign, so that the ‘distance’ ds could be imaginary. In the Minkowski space, ds is called the interval between events, and it could be real (time-like), zero (light-like), or imaginary (space-like). As ds is local, this classification remains valid in GR.

2.2 Euler–Lagrange equations for geodesic lines

Our first task, before addressing the space-time of GR, is geometric: we need to express intrinsic geometric properties of the curved space via its metric. For this task, the fact that a d -dimensional curved surface could be embedded into a $(d + 1)$ -dimensional linear space is totally irrelevant. However, it is helpful for our geometric imagination, which should be stretched beyond limits to imagine curved space of higher dimensions, not speaking about the curved space-time. Note that relations between geometry of the space and the metric tensor are not one-to-one mapping as there exist infinite number of different coordinates (i.e. different metric tensors) describing the same metrics of a flat or curved space.

Any given metric tensor contains full information about a **particular coordinate system**, in addition to describing geometric properties of the “space itself”. So we need to extract invariant characteristics of the curved space from the metric tensor. We postpone until the 2nd part of the course⁴ a detailed mathematical description of curved spaces. But certain essential features can be understood by considering geodesics – the optimal (i.e. the shortest or the longest) path between two points. This is the basis of our 1st part – particles and light motion in a given gravitational field characterised by its metric.

The starting point is the definition of a geodesic line:

Definition 2.2.

The line, which optimises the length, $S = \int_1^2 ds$, of a path s connecting two points in a metric space, is called geodesic.

I remind that $ds \equiv \sqrt{ds^2}$ where the metric is given by Eq. (2.1). Optimal could mean shortest or longest. You might remember that the optimal way connecting two points on a spherical surface is an arc of the great circle passing through these points – the shortest and the longest ways are two complementary arcs on this circle.

In classical Lagrangian mechanics one is usually interested in the shortest path that corresponds to stable trajectories. In the Minkowski space, the stable worldlines have **the longest interval**.⁵ Indeed, if a particle is at rest in an inertial frame, its worldline is obviously geodesic. The interval between any two events for this particle is nothing more than the time elapsed, $\Delta s_0^2 = c^2 \Delta t^2$, since

⁴Eventually, we will be trying to build a **differential geometry**. Indeed, the statement that local geometrical properties are Euclidean could be made precise only when we are dealing with infinitesimal quantities.

⁵This statement is valid for motion of material particles with $ds^2 \geq 0$, while for light, the interval is always zero.

its spatial coordinates do not change. Any non-inertial motion deviates from this geodesics, and the corresponding interval would be shorter than the geodesics, $\Delta s^2 = c^2 \Delta t^2 - \Delta r^2 < \Delta s_0^2$, so that the *stable* optimal worldline of material particle corresponds to the longest interval.

Let us parameterise any path as $x^\mu = f^\mu(\tau) \equiv x^\mu(\tau)$ where τ is some scalar parameter, x^μ are coordinates in the curved space, and $f^\mu(\tau)$ are some functions of τ that we simply call $x^\mu(\tau)$. This means that we have a set of d equations that define a line in a d -dimensional space,⁶

$$x^1 = x^1(\tau) \quad \dots \quad x^d = x^d(\tau).$$

where we call the functions $f^\mu(\tau)$

Writing $dx^\mu = \frac{dx^\mu}{d\tau} d\tau \equiv \dot{x}^\mu d\tau$, so that the metric becomes

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu = g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} d\tau^2 \equiv g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu d\tau^2 \quad (2.1)$$

Hence we rewrite the distance S introduced in Definition 2.2

$$ds = \underbrace{\sqrt{g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu}}_{\hat{\mathcal{L}}} d\tau \quad \Rightarrow \quad S = \int_{\tau_1}^{\tau_2} ds = \int_{\tau_1}^{\tau_2} \hat{\mathcal{L}} d\tau \quad (2.2)$$

It is fully analogous to the action, S , expressed in terms of the Lagrangian defined as⁷

$$\hat{\mathcal{L}} \equiv \sqrt{g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu}. \quad (2.3)$$

The value of S , Eq. (2.2), depends on a path connecting the end points. If the path is a geodesic, its length (i.e. action S) is optimal.

The standard procedure of optimisation is based on the fact that any small distortion of a geodesic path would make the resulting path suboptimal. Let $x_0^\mu(\tau)$ and $\hat{\mathcal{L}}_0$ denote the optimal geodesic path, and the functions $y^\mu(\tau)$ specify a distortion, $x^\mu(\tau) = x_0^\mu(\tau) + \epsilon y^\mu(\tau)$ where ϵ is small and $y(\tau_1) = y(\tau_2) = 0$ as the ends of the path remain the same for any distortion. Then we should have $dS/d\epsilon = 0$. First, we expand $\hat{\mathcal{L}} - \hat{\mathcal{L}}_0 = d\hat{\mathcal{L}}$ into Taylor's series to the lowest power in ϵ :

$$d\hat{\mathcal{L}}(x_0^\mu(\tau) + \epsilon y^\mu(\tau), \dot{x}_0^\mu(\tau) + \epsilon \dot{y}^\mu(\tau)) = \frac{\partial \hat{\mathcal{L}}}{\partial x^\mu} \epsilon y^\mu + \frac{\partial \hat{\mathcal{L}}}{\partial \dot{x}^\mu} \epsilon \dot{y}^\mu$$

where Einstein's summation agreement is implied. Then, as $\hat{\mathcal{L}}_0$ is ϵ -independent, we have

$$\frac{dS}{d\epsilon} = \int_{\tau_1}^{\tau_2} \left\{ \frac{\partial \hat{\mathcal{L}}}{\partial x^\mu} y^\mu + \frac{\partial \hat{\mathcal{L}}}{\partial \dot{x}^\mu} \frac{dy^\mu}{d\tau} \right\} d\tau = \int_{\tau_1}^{\tau_2} \left\{ \frac{\partial \hat{\mathcal{L}}}{\partial x^\mu} - \frac{d}{d\tau} \left(\frac{\partial \hat{\mathcal{L}}}{\partial \dot{x}^\mu} \right) \right\} y^\mu d\tau = 0. \quad (2.4)$$

In the last term, we have integrated by parts and used $y(\tau_1) = y(\tau_2) = 0$. As y^μ is an arbitrary function, this leads to the Euler–Lagrange equations for geodesics:

$$\frac{\partial \hat{\mathcal{L}}}{\partial x^\mu} - \frac{d}{d\tau} \left(\frac{\partial \hat{\mathcal{L}}}{\partial \dot{x}^\mu} \right) = 0, \quad \mu = 1, \dots, d.$$

⁶For example, $x^1 = a \cos \tau$, $x^2 = a \sin \tau$, and $x^3 = 0$ define a circle of radius a in the equatorial plane $x^3 = 0$ in a 3D space with Cartesian coordinates $x^{1,2,3}$.

⁷You do not need to know Lagrangian mechanics in advance, albeit it might be helpful. If you want to know more, look at my old lecture notes on Lagrangian and Hamiltonian mechanics available on the GR page on Canvas.

These equations could be considerably simplified if τ is chosen in such a way that $\mathcal{L}$ is constant along the curve (such τ is called an *affine parameter*). This is always possible; for example, one can choose⁸ $d\tau = ds$, which means $\mathcal{L} = 1$. Then, we introduce the Lagrangian without a hat, sometimes called the geometric Lagrangian:

$$\mathcal{L} \equiv \hat{\mathcal{L}}^2 = g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = 1 \quad (2.5)$$

The Euler – Lagrange equations do not change with $\hat{\mathcal{L}} \rightarrow \mathcal{L}$ since for $\hat{\mathcal{L}} = \sqrt{\mathcal{L}} = \text{const}$, one has

$$\left. \begin{aligned} \partial_\mu \hat{\mathcal{L}} &= \frac{1}{2} \mathcal{L}^{-1/2} \partial_\mu \mathcal{L} \\ \frac{d}{ds} \partial_\mu \hat{\mathcal{L}} &= \frac{1}{2} \frac{d}{ds} (\mathcal{L}^{-1/2} \partial_\mu \mathcal{L}) = \frac{1}{2} \mathcal{L}^{-1/2} \frac{d}{ds} (\partial_\mu \mathcal{L}) \end{aligned} \right\} \Rightarrow \frac{\partial \mathcal{L}}{\partial x^\mu} - \frac{d}{d\tau} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \right) = 0, \quad (2.6)$$

where $\partial_\mu \equiv \partial / \partial \dot{x}^\mu$, and we have canceled the overall constant factor of $\frac{1}{2} \mathcal{L}^{-1/2}$. Eqs. (2.6) and (2.5) completely define a geodesic line through any point and in any direction as a function of the metric tensor $g_{\mu\nu}$.

The geometrical meaning of $\dot{x}^\mu$: it is the tangent vector to a geodesic for a given s .

Intuitively, however, $\dot{x}^\mu \equiv dx^\mu / d\tau$ is a velocity of a ‘particle’ moving along the line parametrised with the ‘time’ τ (we can use this terminology even for purely geometric problems). Then, $\mathcal{L}$ is proportional to its square, *i.e.* it’s proportional to the kinetic energy of a particle. In the absence of any force (*i.e.*, in the absence of potential energy) that is what we expect for the Lagrangian ($= T - V$). Therefore, considering *geometry*, not *mechanics*, we have naturally arrived at the Lagrangian way of describing geodesic lines. Choosing the statement that free particles in the gravitational field move along geodesics lines as an axiom, one would arrive at the geometric description of gravity even *before* SR.

2.3 First integrals of the Euler – Lagrange equations.

The first integrals are conservation laws for a given $\mathcal{L}$. While the Euler-Lagrange equations in general are not analytically solvable (as differential equations of the 2nd order), the first integrals are differential equations of the 1st order which are always solvable. As we always use the affine parameter for any curve, the condition $\mathcal{L} = \text{const}$ is one of the conservation laws. If $g_{\mu\nu}$ is independent of one of the coordinates, say x^1 , then the Euler–Lagrange equation for $\mu = 1$ simplifies to⁹

$$\frac{d}{d\tau} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^1} \right) = 0, \quad \Rightarrow \quad p_1 \equiv \frac{\partial \mathcal{L}}{\partial \dot{x}^1} = \text{const along a geodesic.}$$

This derivative is called *generalised momentum* p_1 . Such “momenta” may appear as linear or angular momentum, energy, and so on, but this terminology is useful even in purely geometric problems not related to the particle motion.

Now let us consider a few examples of geodesics for different 2D metrics.

⁸Unless otherwise stated, in geometric examples we always use s as the affine parameter corresponding to $\mathcal{L} = 1$.

⁹I distinguish upper and lower indices without any comments, in the hope you remember this from *Radiation and Relativity*, postponing their proper consideration till Lecture 8. They are equivalent in Cartesian coordinates.

Example 2.2. The simplest case is that of a constant unit metric $g_{\mu\nu} = \text{diag}(1 \dots 1) \equiv \mathbb{1}$ corresponding to Cartesian coordinates in Euclidean space.

In this case $\mathcal{L} = \sum_{\mu=1}^d (\dot{x}^\mu)^2$ does not depend on any of coordinates. There are d first integrals of the Euler's equations, each giving $p_\mu = \dot{x}_\mu = \text{const}$. Solutions of these equations are just usual straight lines, as expected.

Example 2.3. A flat surface with a non-unit metric.

We choose the metric $ds^2 = dr^2 + r^2 d\phi^2$ (2D polar coordinates; therefore, we know that this is the metric of a flat space). The Lagrangian is $\mathcal{L} = \dot{r}^2 + r^2 \dot{\phi}^2$. The two first integrals are

$$\mathcal{L} = \dot{r}^2 + r^2 \dot{\phi}^2 = 1 \text{ and } \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = 2\dot{\phi}r^2 = \text{const} \equiv 2r_0.$$

The notation is dimensionally justified: $\mathcal{L}$ is dimensionless and $\dot{\phi} \equiv d\phi/ds = r_0/r^2$ has dimensionality of the inverse length. Substituting this into $\mathcal{L} = 1$, we find (choosing $ds \geq 0$)

$$\dot{r}^2 = 1 - \frac{r_0^2}{r^2}, \quad r \geq r_0 \Rightarrow \frac{r dr}{\sqrt{r^2 - r_0^2}} = ds \Rightarrow s = \frac{1}{2} \int_{r_0}^r \frac{d(r^2)}{\sqrt{r^2 - r_0^2}} = \sqrt{r^2 - r_0^2}.$$

so that $r^2 = r_0^2 + s^2$. Substituting this into $\dot{\phi}r^2 = r_0$, we find

$$\dot{\phi} = \frac{r_0}{r_0^2 + s^2} \Rightarrow \phi = r_0 \int \frac{ds}{r_0^2 + s^2} = \tan^{-1} \frac{s}{r_0} + \phi_0 \Rightarrow s = r_0 \tan(\phi - \phi_0),$$

with ϕ_0 being the integration constant.

It is not immediately obvious that these two equations describe straight lines – geodesics in a flat space. However, excluding s , we find $r^2 = r_0^2(1 + \tan^2(\phi - \phi_0)) = (r_0/\cos(\phi - \phi_0))^2$, so that $r(\cos \phi \cos \phi_0 + \sin \phi \sin \phi_0) = \pm r_0$ which leads to a standard $ax + by = \pm r_0$ (with $a \equiv \cos \phi_0$ and $b \equiv \sin \phi_0$). It is so easy, however, because we know the relations between the polar and Cartesian coordinates.

In general, we do not know whether such relations even exist. Moreover, Euler's equations for geodesics are not easy to solve, especially in the absence of symmetries (i.e. first integrals on top of $\mathcal{L} = 1$). Later in the course, we shall look for some fundamental *intrinsic* characteristic to judge by the metric whether or not the corresponding space is flat. But for now – a couple of solvable examples for positively and negatively curved 2D spaces.

Example 2.4. Geodesics on a spherical surface.

Choosing s as the affine parameter along geodesic, we have for a unit sphere ($r = 1$) in spherical polar coordinates:

$$ds^2 = d\theta^2 + \sin^2 \theta d\phi^2 \Rightarrow g_{\mu\nu} = \text{diag}(1, \sin^2 \theta) \Rightarrow \mathcal{L} = \dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2 = 1.$$

Since $\mathcal{L}$ does not depend on ϕ , we have the other (besides $\mathcal{L} = 1$) first integral of the Euler equations:

$$\frac{\partial \mathcal{L}}{\partial \dot{\phi}} = 2\dot{\phi} \sin^2 \theta = 2J \quad (\text{We call this constant } J \text{ because in Lagrangian mechanics a quantity conjugate to angular coordinate is angular momentum})$$

The second Euler equation is

$$\frac{d}{ds} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) = \frac{\partial \mathcal{L}}{\partial \theta} \Rightarrow \ddot{\theta} = \sin \theta \cos \theta \dot{\phi}^2.$$

There is no need to solve it, however. As we know the two first integrals, we can exclude ϕ from them and obtain a differential equation for θ of the first rather than the second order:

$$\mathcal{L} = \dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2 = \dot{\theta}^2 + \frac{J^2}{\sin^2 \theta} = 1 \implies \dot{\theta} = \pm \sqrt{1 - \frac{J^2}{\sin^2 \theta}}.$$

As the expression inside $\sqrt{}$ should be non-negative, we have $\sin \theta \geq |J|$ so that for any given J (< 1) the colatitude is bounded by θ_0 and $\pi - \theta_0$ where $\sin \theta_0 \equiv J$. Then the variables in the differential equation are separable:

$$\frac{\sin \theta d\theta}{\sqrt{\sin^2 \theta - \sin^2 \theta_0}} = ds \quad \begin{array}{l} \text{change of sign} \Leftrightarrow \text{change of direction of motion} \\ \text{along the curve} \implies \text{irrelevant} \end{array}$$

which is integrated to give

$$s - s_0 = \int \frac{d\cos \theta}{\sqrt{\cos^2 \theta_0 - \cos^2 \theta}} = \arcsin \left(\frac{\cos \theta}{\cos \theta_0} \right) \implies \boxed{\cos \theta = \cos \theta_0 \sin s} \quad (2.7)$$

where we have put a constant of integration, s_0 , to zero (it defines the choice of a starting point on the geodesic).

It remains to determine ϕ as a function of s :

$$\dot{\phi} = \frac{J}{\sin^2 \theta} = \frac{J}{1 - \cos^2 \theta} = \frac{\sin \theta_0}{1 - \cos^2 \theta_0 \sin^2 s}$$

This is again a separable equation. Use the substitution $\cot s = t$ to integrate:

$$d\phi = \frac{\sin \theta_0 ds}{1 - \cos^2 \theta_0 \sin^2 s} = -\frac{\sin \theta_0 dt}{t^2 + \sin^2 \theta_0} \Rightarrow \boxed{\sin \theta_0 \tan(\phi - \phi_0) = -\tan s}. \quad (2.8)$$

Eqs. (2.7) and (2.8) describe geodesics on a spherical surface – great circles.¹⁰ Note that geodesics corresponding to the larger arcs of great circles describe an optimal but not the shortest path.

The fact that geodesic lines on a spherical surface are large circles was clear for us even before the calculation. Imagine, however, 2D beings who live on this surface and have no idea about the existence of a third dimension. Still, exact measurements help them to learn the metric of their space, $ds^2 = d\theta^2 + \sin^2 \theta d\phi^2$. Then, they find exactly the same equations, (2.7) and (2.8), for geodesics. As θ and ϕ are natural coordinates for them, the geodesics would be drawn in these coordinates, and will be seen as “curved”.

Example 2.5. A surface of negative curvature.

Consider a surface with the metric $ds^2 = dx^2 + e^{-2x} d\phi^2$. The Lagrangian $\mathcal{L} = \dot{x}^2 + e^{-2x} \dot{\phi}^2$. The two first integrals are

$$\mathcal{L} = \dot{x}^2 + e^{-2x} \dot{\phi}^2 = 1 \text{ and } \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = 2\dot{\phi}e^{-2x} \equiv 2J$$

Excluding $\dot{\phi}$ leads to the equation $\dot{x}^2 = 1 - J^2 e^{2x} \equiv 1 - e^{2(x-x_0)}$, if¹¹ $J \neq 0$. For any $J \neq 0$, a geodesic is bounded by $e^x \leq J^{-1} \equiv e^{x_0}$, i.e. $x \leq x_0 \equiv -\ln J$. The equation is integrated with the help of the substitution $t = \sqrt{1 - e^{2(x-x_0)}} \Rightarrow e^{2(x-x_0)} = 1 - t^2$:

$$s = \int \frac{dx}{\sqrt{1 - e^{2(x-x_0)}}} = -\int \frac{dt}{1 - t^2} = \ln \frac{1 - t}{1 + t} \implies t = \frac{1 - e^s}{1 + e^s} = -\tanh \frac{s}{2}$$

¹⁰See Example 2.6 when the parametric equations for a great circle are derived geometrically.

¹¹The parameterization $J = e^{-x_0}$ does not work for $J = 0$. For $J = 0$, the geodesic line is given by $\dot{\phi} = 0$, i.e. $\phi = \text{const}$, while $\dot{x} = \pm 1$ gives $x = \pm s + \text{const}$; this is a family of straight lines parallel to the x axis.

As $0 \leq t < 1$, s should be negative (we have chosen only "+" sign in $\sqrt{}$, and put to zero the constant of integration which makes $s = 0$ to correspond to the point x_0). Finally, we arrive at the first parametric equation $e^{2(x-x_0)} = 1 - t^2 = 1/\cosh^2(s/2)$. Note that t parameterizes the geodesics 'no worse' than s does. The 2nd parametric equation, $d\phi = e^{2x} J ds = (1 - t^2) ds/J = -dt$, yields $\phi - \phi_0 = -t/J = J^{-1} \tanh \frac{s}{2}$. From these equations, t can be easily excluded. Thus, both families of geodesic lines are given in an explicit form as follows:

$$\boxed{x = x_0 + \frac{1}{2} \ln (1 - e^{x_0}(\phi - \phi_0)^2)} \quad \boxed{\phi = \text{const}} \quad (2.9)$$

2.4 Finding a particular geodesic line

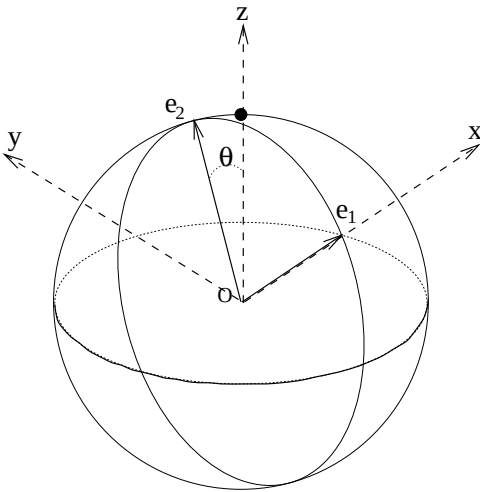
. The first axiom of geometry is valid for a non-flat metric as well: through any two **close** points one can draw only one geodesic line. Equally, a particular geodesic could be defined given a point it passes and a tangent vector at this point. Let's consider a family of geodesic on the trumpet and find a geodesic passing through two points at the same rim, x_1, ϕ_1 and x_2, ϕ_2 . We substitute these values into Eq. (2.9). It gives immediately $\phi_0 = \frac{1}{2}(\phi_1 - \phi_2)^2$, and x_0 may be found as a root of a cubic equation (there is only one real root). Note that uniqueness is no more if we identify points ϕ and $2\pi + \phi$ – obviously, this information is not in the metric. Then a "universe" of inhabitants of a trumpet surface is closed in one direction (and still open in another). A "signal" sent from ϕ will return along a proper geodesic line to $\phi + 2\pi$ which is the same point.

In general, to find a geodesic line through two arbitrary points is to solve a transcendent equation. However, we can always find a geodesic through two close points, or through one point with a given tangent, which is just a solution of parametric differential equations with initial values $x^\mu(0)$ and $\dot{x}^\mu(0)$.

Example 2.6. *Geodesics on a sphere – a geometric derivation.*

Consider a unit sphere in a 3D Euclidean space with Cartesian coordinates,

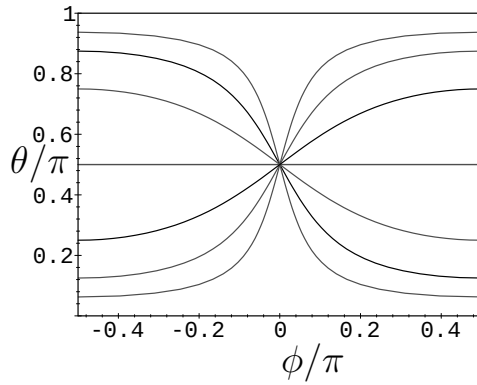
$$(x, y, z) = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta).$$



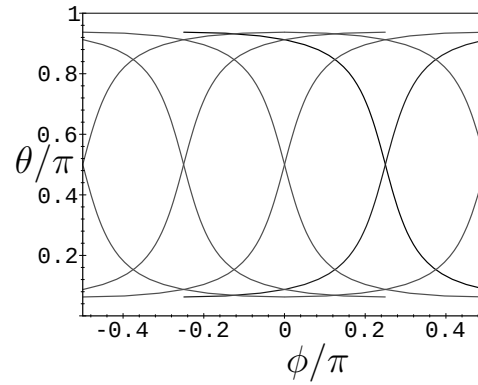
Any great circle can be parameterised as $e_1 \cos s + e_2 \sin s$ where e_1 and e_2 are mutually orthogonal unit vectors started at the origin. Choosing the two vectors as $e_1 = (1, 0, 0)$ and $e_2 = (0, \sin \theta_0, \cos \theta_0)$, we find the coordinates of a point on the great circle inclined at the angle θ_0 from the z axis as

$(x, y, z) = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta) = e_1 \cos s + e_2 \sin s = (\cos s, \sin \theta_0 \sin s, \cos \theta_0 \sin s)$ which leads to Eqs. (2.7) and (2.8) found by solving the Lagrangian problem (albeit with ϕ_0 shifted by $\pi/2$):

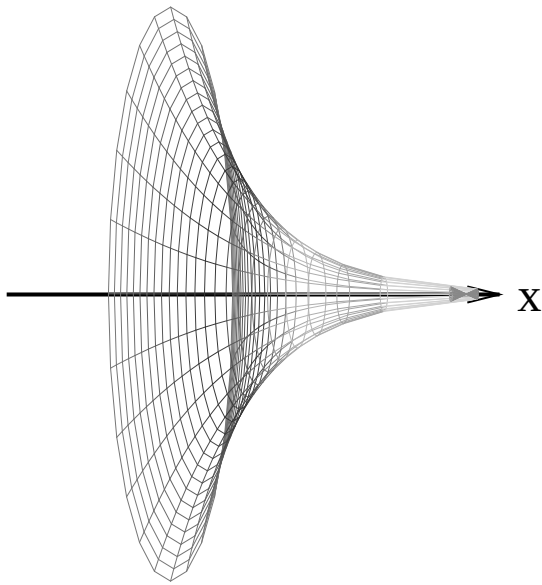
$$\cos \theta(s) = \cos \theta_0 \sin s, \quad \tan(\phi - \phi_0) = \sin \theta_0 \tan s$$

Geodesics on a sphere drawn without the knowledge about a third dimension

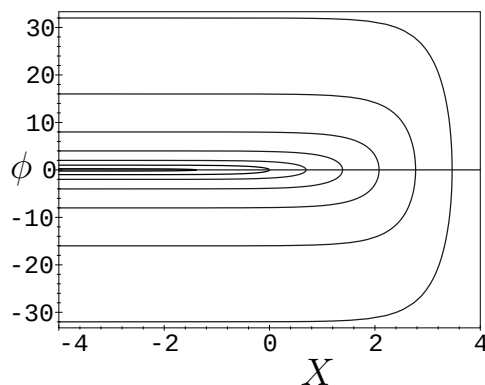
$\phi_0 = 0$, different θ_0



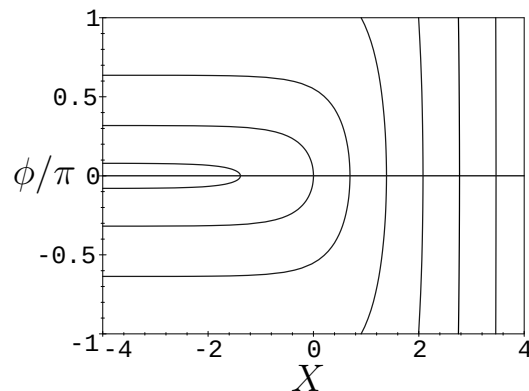
$\theta_0 = \pi/16$, different ϕ_0

Example 2.7. Geodesics on a concave surface

A curved 2D surface with the metric $ds^2 = dx^2 + e^{-2x} d\phi^2$ could be embedded into a 3D space. The metric similar to that will be induced by the 3D Euclidean metric on the surface described by $y^2 + z^2 = e^{-2x}$ in 3D space. It looks like an axial-symmetric trumpet which is exponentially expanded towards negative direction along the x axis. (You can show that the induced metric is, actually, a bit more complicated: $ds^2 = (1 + x^2 e^{-2x}) dx^2 + e^{-2x} d\phi^2$.) Here we made an unnecessary assumption that the points ϕ and $\phi + 2\pi$ are the same. This assumption refers to global properties of the curved surface (topology). However, local geometry on the surface would not change if we substituted y for ϕ , although the embedding into a 3D space would look differently).

Geodesic lines on a trumpet surface drawn without the knowledge about a third dimension

$\phi_0 = 0$, different x_0



the same geodesics within one "period"

Lecture 3. Particles and light in a weak gravitational field

The main postulate for particle's motion in a given gravitational field (described by a given metric of the space-time) that follows from the equivalence principle is that a particle moves along geodesic worldlines. The Einstein equivalence principle requires the same for light, albeit worldlines will be different. First we shall verify that this requirement correctly describes in a geometric way particle's and light motion in the flat Minkowski space. Then we generalise the Minkowski metric to describe motion in a 'weak' gravitational field without using the concept of forces.

We have formulated (lecture 1) the equivalence principle as the local equivalence between forces of gravitation and inertia; these forces exactly compensate each other in the vicinity of a free falling particle:

A freely falling framework is locally equivalent to an inertial one

Now we can postulate the geometric meaning a free falling particle:

World lines of freely falling particles in any frame are geodesics in spacetime

We need to specify the description of geodesics (given in the previous lecture for an arbitrary metric space) for the case of curved $3 + 1$ space-time. But first let us consider the flat Minkowski space.

3.1 Geodesic motion in Minkowski space

Minkowski space is described by the metric

$$ds^2 \equiv c^2 d\tau^2 = c^2 dt^2 - d\mathbf{r}^2 = \eta_{\mu\nu} dx^\mu dx^\nu, \quad \eta_{\mu\nu} \equiv \text{diag}(1, -1, -1, -1), \quad (3.1)$$

where Cartesian coordinates are chosen. As always in Relativity, we use the convention

$$x^0 \equiv ct, \quad x^1 \equiv x, \quad x^2 \equiv y, \quad x^3 \equiv z.$$

I remind that proper time, τ , is introduced only for time-like ($ds^2 > 0$) or null intervals, the latter corresponding to motion of light. Light propagates along null geodesics. For particle's motion, we need to optimise the interval. As we have done in the previous lecture, Eq. (2.2), but choosing the proper time τ as a geodesic parameter, we have for the interval between two events

$$\begin{aligned} s &= \int_{\tau_1}^{\tau_2} ds = \int_{\tau_1}^{\tau_2} \sqrt{\eta_{\mu\nu} dx^\mu dx^\nu} = \int_{\tau_1}^{\tau_2} \sqrt{c^2 dt^2 - d\mathbf{r}^2} = \int_{\tau_1}^{\tau_2} \sqrt{c^2 \left(\frac{dt}{d\tau}\right)^2 - \left(\frac{dx^\mu}{d\tau}\right)^2} d\tau \\ &\equiv \int_{\tau_1}^{\tau_2} \sqrt{c^2 \dot{t}^2 - \dot{\mathbf{r}}^2} d\tau \equiv \int_{\tau_1}^{\tau_2} \sqrt{\mathcal{L}} d\tau \end{aligned} \quad (3.2)$$

Here $\mathcal{L} = c^2 \dot{t}^2 - \dot{\mathbf{r}}^2 \equiv \eta_{\mu\nu} \dot{x}^\mu \dot{x}^\nu$ is the geometric Lagrangian corresponding to the Minkowski metric, and $\dot{x}^\mu \equiv \frac{dx^\mu}{d\tau}$. As $ds \equiv c d\tau$, the Lagrangian for particle motion is constant (as expected for the affine parameter τ) equal to c^2 . For the light motion, the proper time is not defined (as we cannot have a frame moving with speed of light), and from $ds^2 = 0$ follows $\mathcal{L} = 0$. Here,

$$\mathcal{L} = c^2 \dot{t}^2 - \dot{\mathbf{r}}^2 = \begin{cases} c^2 & \text{for timelike geodesics} \\ 0 & \text{for null geodesics} \end{cases} \quad (3.3)$$

As all the variables are cyclic we have four more conservation laws:

$$\dot{t} = \gamma, \quad \dot{\mathbf{r}} = \gamma \mathbf{v},$$

where both γ and $\mathbf{v}$ are constants of integration. (We could call $\dot{\mathbf{r}} = \mathbf{u}$, without introducing γ into the definition of $\mathbf{v}$). Substituting these constants into $\mathcal{L} = c^2$, we have

$$\gamma^2 c^2 - \gamma^2 v^2 = c^2 \implies \gamma^2 = \left(1 - \frac{v^2}{c^2}\right)^{-1},$$

which justifies the names of constants. Obviously, $\mathbf{v} = \gamma^{-1} \dot{\mathbf{r}} = d\mathbf{r}/dt$ is a three-velocity, and $t = \tau\gamma$ is the relation between the proper and laboratory time. Note that $(c\dot{t}, \dot{\mathbf{r}}) \equiv \dot{x}^\mu \equiv u^\mu = \gamma(c, \mathbf{v})$ is a usual definition of the 4-velocity in SR. Thus, as expected, a purely geometric requirement of moving along geodesic lines in the Minkowski space-time reproduces the laws of a free motion in the SR.

3.2 The interval and the action

We have exploited a close analogy between the interval and the mechanical action. Now we want a direct relation. In SR we have

$$ds \equiv \sqrt{\mathcal{L}} d\tau = \sqrt{c^2 \dot{t}^2 - \dot{\mathbf{r}}^2} d\tau = c \sqrt{dt^2 - d\mathbf{r}^2} = c \sqrt{1 - \frac{1}{c^2} \left(\frac{d\mathbf{r}}{dt}\right)^2} dt \equiv c\gamma^{-1} dt \quad (3.4)$$

With the accuracy up to a constant factor, this is indeed the action we have introduced in SR. Expanding this in the nonrelativistic limit, we find $s = \int (c - v^2/2c) dt$. In order to obtain the standard mechanical action we multiply this by $-m_0 c$ - up to a dimensionless factor, this is the only invariant that gives the correct dimensionality to the mechanical action $\hat{\mathcal{L}} \equiv -m_0 c \sqrt{\mathcal{L}}$:

$$S = -m_0 c \int \sqrt{g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu} d\tau \equiv -m_0 c \int \sqrt{\mathcal{L}} d\tau \equiv \int \hat{\mathcal{L}} dt \approx \int [-m_0 c^2 + \frac{1}{2} m_0 v^2] dt \quad (3.5)$$

The “-” sign in the action is quite convenient as we still have the Hamilton **least action principle**: when s has a maximum, S is minimal. Also, in the nonrelativistic limit it gives – up to a constant that does not effect equations of motion – the correct relation $\hat{\mathcal{L}} = \hat{T} - \hat{V} = \hat{T}$ in the absence of potential energy. A difference between the Lagrangian we use and the standard mechanical one is not only in the coefficient but also in an additional $\sqrt{}$. Let us recall that the $\sqrt{}$ has appeared in our derivation of Euler’s equations and we were able to get rid of it just using an affine geodesic parameter for which $\mathcal{L} = \text{const}$. As τ is obviously an affine parameter ($\mathcal{L} = c^2$) we shall use $\hat{\mathcal{L}}$ only for exploiting connections with Newtonian mechanics.

3.3 The metric in a weak field. Newtonian limit

We want to construct a proper geometric description of Newtonian motion of a particle in the gravitational potential. In other words, we want to find a metric in which geodesic lines correspond to lines of motion in a *weak* gravitational field. To this end, we compare the Newtonian Lagrangian to that generated by the metric.

In Newtonian mechanics, we know that $\hat{\mathcal{L}} = T - V$ and potential energy due to gravitation is $V = m_0 \varphi$ where φ is the gravitational potential. For a spherically symmetric mass distribution with the total mass M , we have $\varphi = -GM/r$ at the distance r from the mass centre.

In our geometric approach, we want to include φ into the metric via $ds = -(1/m_0c) dS = -(1/m_0c) \hat{\mathcal{L}} dt$. That's a crucial point: $\hat{\mathcal{L}}$ includes both $\hat{T}$ and $\hat{V}$ but the resulting ds is just an interval, without any external fields! For $v/c \ll 1$, using Eq. (3.5) and adding $-\hat{V} = m_0\varphi$ to $\hat{\mathcal{L}}$, we find

$$\frac{ds}{dt} = -\frac{1}{m_0c} \frac{dS}{dt} = -\frac{1}{m_0c} (-m_0c^2 + \frac{1}{2}m_0v^2 - m_0\varphi) = c - \frac{v^2}{2c} + \frac{\varphi}{c}. \quad (3.6)$$

Thus, neglecting in $(ds/dt)^2$ all the terms that vanish when $c \rightarrow \infty$, we find the metric as

$$\begin{aligned} ds^2 &= \left(\frac{ds}{dt}\right)^2 dt^2 = \left(c - \frac{v^2}{2c} + \frac{\varphi}{c}\right)^2 dt^2 \approx (c^2 - v^2 + 2\varphi) dt^2 \\ &= c^2 \left(1 + \frac{2\varphi}{c^2}\right) dt^2 - d\mathbf{r}^2 \equiv g_{\mu\nu} dx^\mu dx^\nu. \end{aligned} \quad (3.7)$$

Therefore, all we need to describe the Newtonian gravitation law is to choose the temporal component of $g_{\mu\nu}$ as

$$g_{00} = 1 + 2\varphi c^{-2}$$

so that in Cartesian coordinates

$$g_{\mu\nu} = \text{diag} (1 + 2\varphi c^{-2}, -1, -1, -1). \quad (3.8)$$

We arrive at a surprising conclusion:

A weak gravitational attraction can be described as a curvature of time in a flat space

We shall see later that the spatial curvature is of the same order as the temporal one. However, in a non-relativistic motion the temporal curvature is much more important as dt^2 is multiplied by c^2 .

3.4 Weak gravity in spherical polar coordinates

These are most suitable for considering gravitational fields of stars and planets. We know that in these coordinates Newtonian theory gives $\varphi = -GM/r$. It is convenient to rewrite post-Newtonian metric (3.7) in spherical coordinates as

$$ds^2 = c^2 \sigma(r) dt^2 - d\mathbf{r}^2 = c^2 \left(1 - \frac{r_s}{r}\right) dt^2 - dr^2 - r^2 d\Omega^2, \quad (3.9)$$

where $d\Omega^2 \equiv d\theta^2 + \sin^2 \theta d\phi^2$. We shall call this metric pseudo-Newtonian, while

$$r_s = \frac{2GM}{c^2} \quad \text{is called the Schwarzschild radius}$$

We can use the known values of G ($6.67 \times 10^{-11} \text{ Nm/kg}^2$), and the mass of the Sun ($2 \times 10^{30} \text{ kg}$) to find that for the Sun, $r_s \approx 3 \text{ km}$. As the radius of the Sun is $6 \times 10^5 \text{ km}$, the ratio r/r_s outside of the Sun's mass never exceed 4×10^{-6} . The appropriate ratio for the Earth is smaller than 1.4×10^{-9} . So using this as a small parameter is totally justified, at least within the Solar system. In general, the gravitational field of a star of radius R is weak if $r_s/R \ll 1$. It is also valid for an arbitrary spherically symmetric mass distribution of constant density ρ which leads to $GR^2\rho/c^2 \ll 1$, where $M \sim \rho R^3$. Thus, for a large enough scale and a constant density the gravitational field will become strong – that what happens at the centre of a typical galaxy.

3.5 Free fall in the radial direction

Let us check that metric (3.9) describes correctly a free radial fall in the field.¹ As t is a cyclic variable, we obtain the first integral $\sigma(r)\dot{t} = \tilde{\gamma}$. Here $\tilde{\gamma}$ is an arbitrary constant.² On substituting this into $\mathcal{L} = c^2$ (with $\dot{\Omega} = 0$ for radial motion), we obtain in the lowest order (*i.e.* neglecting **all** relativistic corrections):

$$\frac{c^2\gamma^2}{\sigma(r)} - \dot{r}^2 = c^2 \quad \Leftrightarrow \quad c^2\gamma^2 - \dot{r}^2 = c^2 \left(1 - \frac{r_s}{r}\right) \Rightarrow \ddot{r} = -\frac{r_sc^2}{2r^2} = -\frac{GM}{r^2},$$

where the second equation has been obtained by differentiating with respect to τ which, in the non-relativistic limit, $r_s/r \rightarrow 0$ and $\gamma \rightarrow 1$, is equivalent to differentiating with respect to t . Thus this equation is the standard Newtonian gravitation law. However, the metric Eq. (3.9) provides more than just a re-derivation of the Newtonian laws.

3.6 Gravitational time dilation and red shift

We have derived metric Eq. (3.9) in order to accommodate Newtonian gravity into a geometric description. However, the new theory is conceptionally different from Newton's one. We have already learned that there are no preferred frames in GR. This is in contrast not only to Newtonian absolute space but also to the special role of inertial frames in SR. It also means that there is no global (or universal, *i.e.* the same at scale of the Universe) time as the time scale is set by g^{00} which depends on particle's position in the field. Thus even this simple metric describes phenomena totally impossible in Newtonian physics, namely, the time dilation. Consider as an example the time-keeping in an earth-bound laboratory which is obviously not in a free fall and thus is not inertial. At any instant, one can choose a local geodesic frame (LGF)³ where $dr = v dt$, and thus

$$ds^2 = c^2 \left(1 - \frac{r_s}{r}\right) dt^2 - v^2 dt^2 = c^2 dt^2 \left(1 - \frac{r_s}{r} - \frac{v^2}{c^2}\right) \equiv c^2 d\tau^2.$$

Therefore, proper time $d\tau = ds/c$ is delayed by the generalised Lorentz factor, $\tilde{\gamma}$, as compared to an infinitely remote "stationary" observer:

$$\tilde{\gamma} \equiv \dot{t} \equiv \frac{dt}{d\tau} = \frac{1}{\sqrt{1 - r_s/r - v^2/c^2}} \quad (3.10)$$

Here the coordinate time t is the proper time for an infinitely remote observer. Apart from the relativistic time delay in a moving frame known from SR, $\tilde{\gamma}$ includes the gravitational time delay.⁴

We have treated an instantaneous velocity in a slightly offhand manner. The Earth rotates around its axis and makes the orbital motion around the Sun. The velocity of the orbital motion is about 30 km/s in the frame centred at the Sun; the velocity of the rotation is about 230 m/s in the frame

¹ Actually, it describes a classical orbital (elliptic or hyperbolic) motion when the mean radius of the orbit $R \gg r_s$ as we shall find the following lectures.

² $\tilde{\gamma}$ equals to the Lorentz factor γ only when gravitational corrections are much smaller than kinematic SR corrections, *i.e.* $r_s/r \ll v^2/c^2$. In many problems, both kinematic and gravitational relativistic corrections are of the same order.

³ It is a name for a particular free falling Cartesian frame with axes tangential to geodesic lines through a given point.

⁴ Although metric (3.9) is approximate, its temporal part coincides with that of the exact solution in the spherically-symmetric case. Therefore, the gravitational time delay given by γ^* is exact for an object at rest, when $dr = 0$. Moreover, it is correct when motion is non-relativistic when additional corrections from exact metric are of higher powers in $1/c^2$.

centred at the Earth's centre. The second one is much more important, however, as the difference due to Earth's rotation in velocities of two frames, one at the surface and another at altitude h , is quite noticeable. Let us compare the time-keeping in these two frames. Using $v_h = \omega(R + h)$ and $GM/R^2 = \omega^2 R$ for circular motion and $r_s = 2GM/c^2$, we find in the leading order in $1/c^2$ and h/R :

$$\frac{\tau_h}{\tau_o} - 1 = \frac{\tilde{\gamma}_0}{\tilde{\gamma}_h} - 1 = -\frac{1}{2} \left(\frac{v_h^2 - v_o^2}{c^2} + \frac{r_s}{R+h} - \frac{r_s}{R} \right) = -\frac{1}{2} \left(\frac{2\omega^2 R h}{c^2} - \frac{r_s h}{R^2} \right) \quad (3.11)$$

$$= -\frac{1}{2} \left(\frac{2GMh}{R^2 c^2} - \frac{r_s h}{R^2} \right) = \frac{h}{c^2} \left(g - \frac{v_o^2}{R} \right) \equiv \frac{h\tilde{g}}{c^2}. \quad (3.12)$$

Note that GR takes care: the effective gravitational acceleration, $\tilde{g}$, includes the centrifugal acceleration – as it should. Locally, it is indistinguishable from the acceleration due to “real” gravity. However, numerically it does not contribute much as $\omega^2 R \approx 0.03 m/s^2$ while $g \approx 10 m/s^2$. It makes no difference for a rough estimate of the time delay per metre, $\Delta\tau/\tau \sim 10^{-16}/m$. This seems very small but with time being measured with accuracy up to $10^{-19} s$, it is not just measurable but is used in the most precise altimetry (where g should also be known with a sufficient accuracy).

*We can see that this result is ‘classical’ by deriving it within Newtonian physics and the assumption that **light is heavy**, i.e. each photon of frequency ω and energy $\mathcal{E} = \hbar\omega$ has mass $\hbar\omega/c^2$. Its energy at altitude h is changed by $\Delta\mathcal{E} = mgh = \hbar\omega gh/c^2$. As the only way of changing $\mathcal{E}$ is to change frequency, $\Delta\mathcal{E} = \hbar\Delta\omega$. Comparing this two expressions, one has*

$$\frac{\Delta\omega}{\omega} = \frac{hg}{c^2}.$$

The time delay leads naturally to the **gravitational Doppler effect**. The wavelength $\lambda = cT$, where T is a period, i.e. time elapsed between two neighboring wave nodes. In the presence of the gravitational potential φ , the bigger is φ , the slower is T . We define the red shift as

$$Z = \frac{\Delta\lambda}{\lambda_{\text{emit}}} = \frac{\lambda_{\text{rec}}}{\lambda_{\text{emit}}} - 1 = \frac{T_{\text{rec}}}{T_{\text{emit}}} - 1 = \frac{d\tau_{\text{rec}}}{d\tau_{\text{emit}}} - 1. \quad (3.13)$$

Here we used that the period of wave either at the point of emitting the light signal or at the point of receiving it measures in proper time. In general, we should be careful when constructing a relation between proper times of two remote events (clocks must be synchronized!). However, if both events may be described in the same coordinate system with the static metric, we can use that $d\tau(\mathbf{r}) = \sqrt{g_{00}(\mathbf{r})} dt$ where t is a coordinate time. This obviously works for the quasi-Newtonian metric we have introduced above. Thus we have

$$Z = \sqrt{\frac{\sigma(r_{\text{rec}})}{\sigma(r_{\text{emit}})}} - 1 = \sqrt{\frac{1 - r_s/r_{\text{rec}}}{1 - r_s/r_{\text{emit}}}} - 1. \quad (3.14)$$

We shall see that for a spherically symmetric metric this relation is exact **for stationary receiver and emitter** even if the field is not small, as g_{00} is the same as in the exact (Schwarzschild) metric, and Z does not depend on other elements of the metric tensor. For a weak field, however, we can simplify this expression. If, as above, the source and the observer are only at slightly different altitudes in the gravitational field ($h \equiv r_{\text{rec}} - r_{\text{emit}} \ll r_{\text{emit}}$), we can use the formula above:

$$Z = \frac{\tau_h}{\tau_o} - 1 = \frac{h\tilde{g}}{c^2}.$$

This is a basis for experiments, as to measure Doppler effect is quite straightforward. Experiments by Pound, Rebka and Snider (1960, 1965) indeed confirmed this prediction (their h was only 22 m so that they measured a correction of order 10^{-15} which is not that small by standards of optical measurements).

We should stress, however, that this is not really a test of GR, as gravitational Doppler effect can be predicted even in Newtonian physics, provided that SR identification, $\mathcal{E} = m_0 c^2$, is taken.

3.7 Speed of light is not constant in noninertial frames

Let $ds^2 = A^2 c^2 dt^2 - B^2 d\mathbf{r}^2$. From $\hat{\mathcal{L}} = 0$ one finds the velocity of light $\frac{d\mathbf{r}}{dt} = \frac{A}{B}c \neq c$. When A and B are constants, this is a mere rescaling. Otherwise, **the speed of light is no longer constant**. We see from the above that it remain constant *only* when A and B are constants, *i.e.* when we are in a flat Minkowski space. Obviously, this is in agreement with Special Relativity. However, its cornerstone principle of $c = \text{const}$ is no longer a requirement in General Relativity. In the presence of the field, A (and B if the field is not weak) are coordinate-dependent and so is c .

Lecture 4. Orbital Motion in Schwarzschild metric.

In the previous lecture, we have considered the “pseudo-Newtonian” metric that, for distances $r \gg r_s \equiv 2GM/c^2$ reproduces results of Newtonian gravitational theory and predicts the gravitational time delay. Now we consider orbital motion in a weak field, for which we would use not this “pseudo-Newtonian” metric but the exact Schwarzschild solution of the Einstein field equation.

4.1 Schwarzschild solution

The “pseudo-Newtonian” metric considered in the previous lecture, contains correct features of the ‘real’ metric in an empty space outside of a spherically symmetric mass distribution. This metric is the famous Schwarzschild solution of the Einstein field equation; historically, it was the first solution where curvature (or a deviation of $g_{\mu\nu}$ from the flat Minkowski metric $\eta_{\mu\nu}$) was not considered as a small parameter:

$$ds^2 = c^2 \sigma(r) dt^2 - \sigma^{-1}(r) dr^2 - r^2 d\Omega^2, \quad (4.1)$$

$$\sigma(r) \equiv \left(1 - \frac{r_s}{r}\right), \quad d\Omega^2 \equiv d\theta^2 + \sin^2 \theta d\phi^2, \quad r_s = \frac{2GM}{c^2}$$

We shall derive this metric later on in the course as the solution to Einstein’s field equations. Now we consider “post-Newtonian” consequences of this metric in the weak field limit $r \gg r_s$. Note that in this limit we must keep corrections not only to g_{00} (as in quasi-Newtonian metric above), but also to g_{rr} :

$$ds^2 = c^2 (1 - r_s/r) dt^2 - (1 + r_s/r) dr^2 - r^2 d\Omega^2.$$

Although this metric is totally sufficient to consider only small **corrections** to Newtonian motion (as was shown by Einstein himself), using it will not make our considerations simpler than using the exact Eq. (4.1) which we will do even for $r \gg r_s$.

4.2 Geodesic equations in the Schwarzschild metric

A particle (e.g. a planet of the Solar system) or light is in a free fall in the gravitational field located at the centre of mass (the Sun) *i.e.* it moves along a timelike geodesic in the Schwarzschild coordinates (centered around the Sun if one considers motion in the solar system). So one can use the geodesic equation to describe a typical planetary orbit.

Let us first show that motion of particle (or light) in the Schwarzschild geometry is always planar. The Lagrangian corresponding to the Schwarzschild metric is

$$\mathcal{L} = c^2 \sigma(r) \dot{t}^2 - \sigma^{-1}(r) \dot{r}^2 - r^2 (\sin^2 \theta \dot{\phi}^2 + \dot{\theta}^2). \quad (4.2)$$

Hence, the Euler-Lagrange equation for θ becomes

$$0 = -\frac{d}{d\tau} \frac{\partial \mathcal{L}}{\partial \dot{\theta}} + \frac{\partial \mathcal{L}}{\partial \theta} = 2 \frac{d(r^2 \dot{\theta})}{d\tau} - 2r^2 \sin \theta \cos \theta \dot{\phi}^2.$$

Without limiting generality, we can choose initial conditions for this equation as $\theta = \pi/2$ and $\dot{\theta} = 0$. Using the first one, we find that $r^2 \dot{\theta} = \text{const}$, and the second means that $\text{const} = 0$, so that $\theta = \text{const} = \pi/2$ is the unique solution of this geodesic equation. Hence the motion is confined to the

equatorial plane for all τ . Note that σ did not enter this conservation law which is thus exactly the same as in Newtonian physics.

Because of the spherical symmetry the angular momentum is conserved, so that motion is always confined to a single plane perpendicular to the angular momentum.

The geodesic equations can be written down in the Euler-Lagrange form. Lagrangian (4.2) with $\theta = \pi/2$ reduces to

$$\mathcal{L} = c^2 \sigma \dot{t}^2 - \sigma^{-1} \dot{r}^2 - r^2 \dot{\phi}^2, \quad \sigma \equiv \left(1 - \frac{r_s}{r}\right). \quad (4.3)$$

Since both t and ϕ are cyclic variables, the first integrals of Euler's equations are

$$\begin{aligned} c^2 \sigma \dot{t} &= \mathcal{E} & \text{energy constant} \\ r^2 \dot{\phi} &= J & \text{angular momentum constant.} \end{aligned} \quad \mathcal{L} = \begin{cases} c^2 & \text{for a timelike geodesic} \\ 0 & \text{for a lightlike geodesic} \end{cases} \quad (4.4)$$

Here dot denotes the differentiation with respect to geodesic parameter τ which, **in the case of particle's motion**, we always choose as proper time. For a photon, neither proper time nor the standard Lorentz factor γ is defined. In this case, $c^2 \sigma \dot{t} = \mathcal{E}$ gives the definition of the affine geodesic parameter rather than a conservation law.

Sometimes it is more convenient to introduce the (r -dependent) time delay factor, $\dot{t} = \tilde{\gamma}$, instead of the energy constant, $c^2 \sigma \dot{t} = \mathcal{E}$, as above. Note the dimensions: $\mathcal{E}$ and J are, respectively, energy and angular momentum of a particle with unit mass.

substituting both $\dot{t}$ and $\dot{\phi}$ into $\mathcal{L} = \text{const}$, we find

$$\frac{\mathcal{E}^2}{c^2 \sigma} - \frac{J^2}{r^2} - \frac{\dot{r}^2}{\sigma} = C \quad \Rightarrow \quad \sigma^{-1} \left(\frac{\mathcal{E}^2}{c^2} - \dot{r}^2 \right) - \frac{J^2}{r^2} = C \quad \text{where } C = \begin{cases} c^2 & \text{timelike} \\ 0 & \text{lightlike} \end{cases}$$

Euler's equation for r is more cumbersome but, fortunately, redundant as the conservation laws in this case contain full information. We shall derive this 2nd order equation by differentiating Eq. (4.5) with respect to τ and using the chain rule, $\dot{\sigma} = \frac{d\sigma}{dr} \frac{dr}{d\tau} = \frac{\dot{r} r_s}{r^2}$:

$$\begin{aligned} \frac{\mathcal{E}^2}{c^2} - \dot{r}^2 &= \sigma \left(C + \frac{J^2}{r^2} \right) \\ \Rightarrow \quad -2\dot{r}\ddot{r} &= \dot{r} \left[\frac{r_s}{r^2} \left(C + \frac{J^2}{r^2} \right) - \frac{2J^2 \sigma}{r^3} \right] = \dot{r} \left[\frac{3J^2 r_s}{r^4} + \frac{C r_s}{r^2} - \frac{2J^2}{r^3} \right], \end{aligned} \quad (4.5)$$

Note that circular orbits, $\dot{r} = 0$, are exact solutions. (They are unstable both in the Newtonian and Einsteinian gravity, though). The remaining nontrivial equation we arrange (for $J \neq 0$) as¹

$$\boxed{\ddot{r} + J^2 \left(\frac{A}{r^2} - \frac{1}{r^3} + \frac{3 r_s}{2 r^4} \right) = 0}, \quad A \equiv \frac{C r_s}{2 J^2} = \begin{cases} \frac{GM}{J^2} & \text{timelike} \\ 0 & \text{lightlike} \end{cases} \quad (4.6)$$

The boxed equation has been derived for any motion in the Schwarzschild geometry **without any approximation**. It can be used for motion of light or particle in the Solar system, in the gravitational field of the Earth, or in the vicinity of a black hole.

¹For radial fall, $J = 0$, Eq. (4.5) reduces to $\ddot{r} = -GM/r^2$, which looks Newtonian; however, it is relativistic as we differentiate w.r.t. proper time. We consider this in detail in Lecture 6.

4.3 Orbital motion

We have already seen that circular orbits are exact in GR. Let us prove that **Kepler's third law** remains valid for circular orbits even within GR. In this case, $\dot{r} = 0$, so that the Lagrangian is reduced to $\mathcal{L} = c^2 \dot{\sigma}^2 - r^2 \dot{\phi}^2$. It is independent of $\dot{r}$ so that $\partial \mathcal{L} / \partial \dot{r} = 0$.

Hence we find

$$c^2 \dot{\sigma}^2 - r^2 \dot{\phi}^2 = 0 \Rightarrow \frac{\dot{\phi}^2}{\dot{\sigma}^2} \equiv \left(\frac{d\phi}{d\sigma} \right)^2 = \frac{c^2}{2r} \frac{d\sigma}{dr} = \frac{r_s c^2}{2r^3}$$

Noticing that $\frac{1}{2} r_s c^2 \equiv GM$, and introducing the orbital frequency $\omega \equiv d\phi/dt$, we see that this is equivalent to the Kepler 3rd law:

$$\boxed{\omega^2 r^3 = GM} \quad (4.7)$$

where in this case² r is the radius of a circular orbit, and $\omega = 2\pi/T$ where T is the orbital period. Let us stress again that this is exact: relativistic corrections are not neglected but simply absent.

Now we derive general orbital equations that relate r and ϕ for non-circular *stationary* orbits. We want to exclude the (proper) time τ by changing differentiation from that w.r.t. τ to that w.r.t. ϕ . Employing the first integral $\dot{\phi} = J/r^2$, Eq. (4.4), we find

$$\dot{r} \equiv \frac{dr}{d\tau} = \frac{dr}{d\phi} \frac{d\phi}{d\tau} = \frac{dr}{d\phi} \frac{J}{r^2} = -J \frac{d}{d\phi} \left(\frac{1}{r} \right) \Rightarrow \dot{r} = -J \frac{du}{d\phi} \equiv -J u'_\phi, \quad \text{where } u \equiv \frac{1}{r}.$$

Substituting this into Eq. (4.5) gives

$$\mathcal{E}^2/c^2 - J^2 (u'_\phi)^2 = (1 - r_s u) (\mathcal{C} + J^2 u^2)$$

Differentiating this once more with respect to ϕ , we find

$$-2J^2 u'_\phi u'' = -r_s u'_\phi (\mathcal{C} + J^2 u^2) + 2J^2 u u'_\phi (1 - r_s u) = u'_\phi J^2 (-A - r_s u^2 + 2u - 2r_s u^2)$$

where A is defined in Eq. (4.6). This reduces to either $u'_\phi = 0$ for circular orbits or to the general orbital equation:

$$\boxed{\frac{d^2 u}{d\phi^2} + u = A + \frac{3}{2} u^2 r_s} \quad (4.8)$$

This differential equation describes exactly all orbits in the Schwarzschild geometry (including circular, $u = \text{const}$). The only non-Newtonian term in this equation is $r_s u^3$ which vanishes as $c \rightarrow \infty$. Indeed, it is easy to see that $u''_\phi + u = A$ is exactly the equation for Kepler's orbit in Newtonian mechanics. Therefore, for $u r_s = r_s/r \ll 1$, the last term gives a small relativistic correction. It is relatively easy to find out how this correction effects planetary motion and light propagation in the Solar system. Experimental verification of appropriate results constitutes famous three classical tests of GR.

Before finding the effects of the correction, note that a general solution to Eq. (4.8) can be found exactly (to which end it is convenient to reduce it to the 1st order equation or, alternatively, exclude τ from the conservation law for the Lagrangian).

$$\phi - \phi_0 = \int_{u_0}^u \frac{du}{\sqrt{B + 2Au - u^2 + r_s u^3}}, \quad B \equiv \frac{(\mathcal{E}^2 - c^4)}{c^2 J^2}$$

²For a general elliptical orbit, r is the mean radius.

where ϕ_0 and u_0 are the constants of integration, as well as $\mathcal{E}$ and J which entered the problem by finding the first integrals of equations of motion (obviously, a point with the coordinates (u_0, ϕ_0) belongs to the orbit). This integral relation means that u is an elliptic function of ϕ (these functions have been introduced by Jacobi and are frequently called Jacobi functions). The fact that u is the elliptic function does not mean that the orbit's shape is elliptic. This is the case only in the nonrelativistic Newtonian limit, $r_s \rightarrow 0$. We do not need to know the theory of elliptic functions, however, to find the orbital shape with taking into account relativistic corrections – provided that $r \gg r_s$ (as in the case of our Solar system when $r_s \approx 3$ km – not really an astronomical distance). We can simply use the perturbation theory.

4.4 Kepler's orbits

Let us consider planetary and light motion in the Newtonian limit, $r_s \rightarrow 0$.

$$\frac{d^2 u}{d\phi^2} + u = A,$$

where $A = GM/J^2$ for a particle (J is a conserved projection of the angular momentum), and $A = 0$ for light, Eq. (4.6). This orbital equation is harmonic. The complementary function is $B \cos(\phi - \phi_0)$. The full solution is the sum of the complementary function and any particular solution. Obviously, $u = A$ is a particular solution. Thus, we have (with renaming $B \equiv Ae$):

$$u \equiv \frac{1}{r} = A + B \cos \phi \equiv A [1 + e \cos \phi]. \quad (4.9)$$

As usually, we choose $\phi_0 = 0$: the change of the phase is irrelevant (unless some specific orientation of the axes is implied!). This is the canonical equation (in spherical polar coordinates) for a second order curve – ellipse (when the eccentricity $|e| < 1$), parabola ($|e| = 1$), or hyperbola ($|e| > 1$).

To see this, one can change to Cartesian coordinates in the standard way thus reducing the above to

$$Ar + Br \cos \phi = 1 \Rightarrow A^2 r^2 = B^2 r^2 \cos^2 \phi - 2Br \cos \phi + 1 \Leftrightarrow (A^2 - B^2)x^2 + A^2 y^2 + 2Bx - 1 = 0,$$

(with $B \equiv Ae$) which is an almost canonical form of the 2nd order curve. To make it canonical, one completes the square and gets with straightforward notations

$$\frac{(x - x_0)^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{with} \quad a \equiv \frac{1}{A(1 - e^2)}, \quad b^2 \equiv \frac{1}{A^2(1 - e^2)}, \quad x_0 \equiv \frac{e}{A(1 - e^2)}.$$

For a material particle with $A \neq 0$, this is an ellipse for $|e| < 1$, parabola for $|e| = 1$, and hyperbola for $|e| > 1$, where A is given in Eq. (4.6) and e is defined by initial conditions. The 'orbital' equation for light is the same but with $A = 0$. The most general solution is $u \equiv 1/r = B \cos(\phi + \phi_0)$; in Cartesian coordinates it reduces to $1 = B \cos \phi_0 x - B \sin \phi_0 y$ which is a straight line as expected.

Note that the solution (4.9) in polar coordinates is actually much simpler than in Cartesian. After understanding the equivalence, we shall be using polar coordinates in what follows.

The next step is take into account the relativistic term in Eq. (4.8) and find Einstein's corrections.

Comparing small corrections predicted by GR with observations within the Solar system provided with the first four experimental tests of GR

We consider them in the next lecture.

Lecture 5. Classic Tests of GR

The gravitational time delay, which we have already discussed, is one of such tests performed more than half a century after the emergence of GR. In this lecture we consider three other classic tests of GR corrections: precession of orbital perihelion, known but unexplained before GR, deflection of light by the Sun, triumphantly discovered a few years after Einstein's breakthrough, and radar ranging of inter-planet distances which became possible in the last few decades.

These effects have been observed in a set of experiments with results that cannot be explained within the standard Newtonian theory¹ The most famous prediction by Einstein – deflection of light by the Sun (as well as the explanation of an 'extra' precession of Mercury perihelion) – made the general theory of relativity a media sensation in the gloomy atmosphere after the WW1.

5.1 Precession of orbital perihelion

The recipe of perturbation theory is to substitute the zero order solution, Eq. (4.9), for u^3 and find the first order correction (by iteration, one can find, in principle if not in reality, the entire series in powers of r_s , which in this case would be a representation of an elliptic function). *I remind that a circular orbit is a stable solution which is not affected by relativistic corrections.* We have $u = u_0 + u_1$ where u_0 is the solution of Kepler's orbit equation, and u_1 is a small relativistic correction (the dimensionless small parameter here is $u_0 r_s \ll 1$, which means that the scale of motion is much bigger than r_s). Then we have

$$\frac{d^2 u}{d\phi^2} + u = A + \frac{3}{2} r_s u^2 \quad \Rightarrow \quad \frac{d^2 u_1}{d\phi^2} + u_1 = \frac{3}{2} r_s u_0^2,$$

where we used that $u_0'' + u_0 = A$, and kept only u_0 in the r.h.s. as $r_s u_0$ is already small. Then substituting the solution for u_0 , Eq. (4.9), we find

$$\frac{d^2 u_1}{d\phi^2} + u_1 = \frac{3}{2} r_s A^2 (1 + e \cos \phi)^2 = \tilde{A} + 3 r_s A^2 e \cos \phi + \frac{3 r_s}{4} A^2 e^2 \cos 2\phi$$

Here $\tilde{A} \equiv \frac{3}{2} r_s A^2$ is a totally irrelevant constant: the particular solution corresponding to this will be absorbed as a small correction to the particular solution for u_0 , while the complementary function is simply the same as in Eq. (4.9). Thus we need to find two particular integrals corresponding to the second and third terms on the r.h.s. of this equation. A particular solution corresponding to the $\cos 2\phi$ term is $-\frac{1}{4} r_s A^2 e^2 \cos 2\phi$. This is periodic, and its only effect is a slight difference between two halves of the orbit, the first between 0 and ϕ and the second between ϕ and 2ϕ . This change, in the case of planets in the Solar system, are of order 3 km while the smallest diameter of the orbit (Mercury) is tens of millions of km – making this effect totally unmeasurable. The only important term here is $3 r_s A^2 e \cos \phi$, as this is in resonance with the original solution. Indeed, a particular integral to $u'' + u = C \cos \phi$ is $u = (C\phi/2) \sin \phi$. Therefore, one obtains the following correction to the Newtonian solution:

$$\delta u = \frac{3 r_s A^2 e}{2} \phi \sin \phi \quad (5.1)$$

¹ Although there exist alternative theories of gravitation other than GR which can also explain small deviation from the Newtonian theory, none of them – in contrast to GR – can explain all the available experiments.

This correction is very small (for the planets of the Solar system). As expected, it vanishes for a circular orbit ($e = 0$). If we add it to the Newtonian solution, we have (with accuracy up to the first order in $r_s A \ll 1$):

$$u = A + Ae \left[\cos \phi + \frac{3r_s A}{2} \phi \sin \phi \right] \approx A + Ae \cos \left(\phi - \frac{3r_s A}{2} \phi \right) \quad (5.2)$$

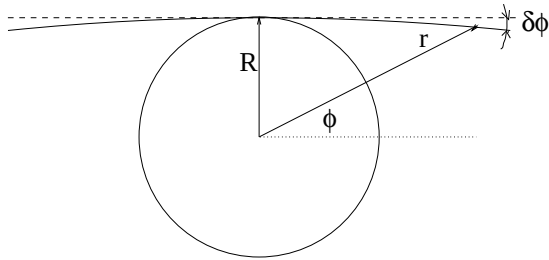
Thus, the orbit is no longer re-entrant: when ϕ is changed by 2π , the argument of $\cos$ has a phase shift. The function is still periodic but the period is slightly different from 2π ; the perihelion is the same but is reached at a different value of ϕ , *i.e.* it precesses.

Let us have a rough numeric estimate of $A \equiv GM/J^2$. The angular momentum constant, Eq. (4.4), $J = R^2\omega$ where R is the mean radius of the orbit, so that using Kepler's law, Eq. (4.7), we have $A = GM/R^4\omega^2 = 1/R$. For Mercury, $A^{-1} = R = 5.8 \times 10^7 \text{ km}$ so that (with the Solar $r_s \approx 3 \text{ km}$) the anomalous precession is about 3×10^{-6} radians per orbit, which is in excellent agreement with astronomical observations (back to Leverrier, 1845).

A normal precession which is much bigger than this one is due to influence of other planets of the solar systems; pre-Einsteinian attempts to explain the anomalous precession by ascribing it to the influence of yet unknown planet failed.

5.2 Deflection of light rays in the gravitational field

We have already seen that light is heavy in GR! It means that it no longer propagates along “straight” lines (but what are they?) but falls along null geodesics, $ds = 0$. We use the orbital equation derived in the precious lecture.



It has the same form as that describing timelike geodesics (planetary orbits) but with $A = 0$:

$$\frac{d^2 u}{d\phi^2} + u = 3r_s u^2 / 2. \quad (5.3)$$

In the limit $r_s \rightarrow 0$, it has only a homogeneous solution, $u = C \sin \phi$, where ϕ is counted as in the figure, and the constant $C = 1/R$ when we consider a ray grazing the sun (R is the sun's radius).

Then, $r \sin \phi = R$ describes, as expected, a straight line in the polar coordinates. A relativistic correction can again be found with the help of perturbation theory, substituting $u_0 = \sin \phi / R$ into the r.h.s. of Eq. (5.3):

$$\frac{d^2 u_1}{d\phi^2} + u_1 = \frac{3r_s}{4R^2} (1 - \cos 2\phi)$$

which has a particular integral

$$u_1 = \frac{3r_s}{4R^2} \left(1 + \frac{\cos 2\phi}{3} \right)$$

Therefore, in the lowest approximation in r_s one finds

$$u = u_0 + u_1 = \frac{\sin \phi}{R} + \frac{3r_s}{4R^2} \left(1 + \frac{\cos 2\phi}{3} \right)$$

For $r \gg R$ the angle ϕ is very small, and so $\sin \phi \approx \phi$ and $\cos \phi \approx 1$. Hence

$$u \equiv \frac{1}{r} \approx \frac{\phi}{R} + \frac{r_s}{R^2} \Rightarrow 1 = \frac{r\phi}{R} + \frac{rr_s}{R^2}, \quad (5.4)$$

while the absence of the GR correction, $r_s = 0$, we have

$$1 = r\phi_0/R \quad (5.5)$$

so that $y = R$ – the straight line grazing the Sun. Substituting this into Eq. (5.4) we have $\phi_0 = \phi + r_s/R \equiv \phi + \delta\phi$. The angle $\delta\phi$ measures the half-deflection as the picture is symmetric. Hence the full deflection angle is given by

$$2\delta\phi \equiv 2(\phi_0 - \phi) = \frac{2r_s}{R}. \quad (5.6)$$

It was the most famous first **prediction** of the GR theory. (The perihelion shift had been known for half a century, and the gravitational Doppler effect was not possible to measure until 1960). The prediction has been confirmed by the expedition of Arthur Eddington (1919) who measured the light deflection during the Sun eclipse. The prediction, just $1.75''$, was confirmed beyond any doubt.

Note that the statement “*light is heavy*”, i.e. ascribing mass $\mathcal{E}/c^2$ to a photon of energy $\mathcal{E}$ as an obvious consequence of the equivalence principle, could be applied in Newtonian theory as well, either by direct calculation, or using the quasi-Newtonian spatially- flat metric, Eq. (3.9). In this case, spectacularly, the deflection would be exactly 2 times smaller than predicted by GR. Thus, the confirmation was indeed a triumph of the Einstein’s theory!

5.3 Gravitational time delay and Doppler effect

Gravitational time delay in the Schwarzschild metric for a stationary observer will be the same as in the pseudo-Newtonian one, as g_{00} coincide in both cases. Therefore, considerations in Lecture 3.6 give all the necessary predictions for this effect. More general time delay for a moving observer we consider in Lecture 16. Experimentally, this time delay was measured to coincide with the GR predictions by Hafele and Keating in 1971, by comparing the reading of atomic clocks in flying twice around the world, first eastward and then westwards. The time-delay factor of Eq. (3.10) is

$$\gamma^* \equiv 1 + \frac{r_s}{2r} + \frac{v^2}{2c^2}.$$

At the Earth surface, $r = R \approx 6,400\text{km}$, while during the flight $r = R + h$, where $h \sim 5\text{km}$ or so. It is a small correction to R but a gravitational difference in time-keeping between the surface and the flight is about (omitting numerical factors) $r_s h/R^2 \sim GMh/(c^2 R^2) = gh/c^2$. In calculating the kinematic difference, we assume that our FFF is at the Earth centre, so that at the surface $v_0^2 = \omega^2 R^2$, and in the flight $v^2 = v^2 = (v_{\text{flight}} \pm \omega R)^2$ so that the difference in kinematic corrections is of order $\pm v_{\text{flight}} \omega R/c^2$. Here $\omega = 2\pi/T$ with $T = 24\text{h}$ is the period of the Earth rotation. Now, $gh \sim 5 \cdot 10^4 \text{m}^2/\text{s}^2$, so that $\sqrt{gh} \sim 220\text{m/s}$, while $\omega R = 2\pi R/T \approx 6 \cdot 6400/24 \sim 1600\text{km/h} \approx 450\text{m/s}$, and $v_{\text{flight}} \sim 150\text{m/s}$. So all the corrections are of the same order of magnitude, and the total time difference is of order of 10^{-7}s , i.e. or a hundred of nanoseconds. It was measured, and compared to a carefully calculated difference confirming the GR prediction with accuracy of 10%.

Note that for the satellite in a free fall in orbit of radius r around the Earth, the GR correction is twice bigger than the kinematic one. Indeed, from Kepler's law, Eq. (4.7), we find the kinematic

$$\frac{v^2}{c^2} = \frac{\omega^2 r^2}{c^2} = \frac{GM}{rc^2} = \frac{r_s}{2r}.$$

Of course, it is the satellite orbit which is geodesic, but when we calculate the difference it does not matter which of the two frames we choose as the standard one for time keeping.

5.4 Radar ranging

The fourth test of weak GR corrections within the Solar system – measuring the time delay of the reflected radar signal passing in the vicinity of the Sun – has been suggested by Irwin Shapiro in 1964 and performed by his team using the MIT radar antenna in 1966 (signal bounced from Venus) and 1967 (from Mercury). The idea is that the GR time delay is longest when the planets are at superior conjunction, *i.e.* when the target (Venus or Mercury) is about to pass behind the Sun, so that the radar signal passes in close proximity to the Sun. Note that even in the most favourable case, this delay is about 0.2 millisecond, and adding this to the total time of ranging (of order of half an hour) by itself is not noticeable. But one can watch the variation of the return time while the target planet passes behind the Sun, and single out the correction as the perihelion of the signal grazing the Sun varies.

To solve the problem, we rewrite Eq. (4.4) for a null geodesic in terms of the metric rather than the Lagrangian – as the parameter τ does not have a physical meaning for light propagation, we get rid of it from the start:

$$ds^2 = c^2 \sigma dt^2 - \sigma^{-1} dr^2 - r^2 d\phi^2 = 0, \quad \Leftrightarrow \quad c^2 dt^2 = \sigma^{-2} dr^2 + \sigma^{-1} r^2 d\phi^2. \quad (5.7)$$

The reason in this rewriting is that now we are interested in the time dependence rather than in the orbital shape. These two types of corrections are independent in the lowest order in $1/c^2$ so that we can consider the trajectory of light in the radar ranging as a straight line connecting the Earth with the target planet (Venus, in the first actual measurement) and (almost) grazing the Sun. The equation for this straight line, as in section 5.2, is $r \sin \phi = r_0$, where r_0 is the perihelion of the light path. Hence $r \cos \phi d\phi = -dr \sin \phi$ and

$$r^2 d\phi^2 = \tan^2 \phi dr^2 = \frac{\sin^2 \phi}{1 - \sin^2 \phi} dr^2 = \frac{r_0^2}{r^2 - r_0^2} dr^2.$$

Substituting this into Eq. (5.7), we have

$$c^2 \left[\frac{dt}{dr} \right]^2 = \sigma^{-2} + \sigma^{-1} \frac{r_0^2}{r^2 - r_0^2}. \quad (5.8)$$

Without GR corrections $\sigma = 1$, and the above is reduced to

$$c \frac{dT}{dr} = + \frac{r}{\sqrt{r^2 - r_0^2}}, \quad (5.9)$$

where T is the (classical) time for the light ray to reach the target. This can be immediately integrated to give $cT = \sqrt{r_{\text{ep}}^2 - r_0^2}$ with r_{ep} being the distance from the Earth to the target planet when it is on the opposite side of the Sun.

Now we use perturbation theory, as in the other classic tests. Writing $t = T + \delta T$, we obtain – expanding to the lowest powers of r_s/r and of δT – the following 1st order differential equation for $\delta \equiv c\delta T$:

$$\begin{aligned} c^2 \left[\frac{dT}{dr} \right]^2 + 2c \frac{dT}{dr} \frac{d\delta}{dr} &= \frac{r^2}{r^2 - r_0} + \frac{2r_s}{r} + \frac{r_s}{r} \frac{r_0^2}{r^2 - r_0} \\ \Rightarrow \frac{d\delta}{dr} &= \frac{r_s \sqrt{r^2 - r_0^2}}{r^2} + \frac{r_s r_0^2}{r^2 \sqrt{r^2 - r_0^2}} = \frac{r_s}{\sqrt{r^2 - r_0^2}}. \end{aligned}$$

where we have used Eq. (5.9). Now one can see that this small correction is most relevant when the r.h.s. of the last equation diverges as $r \rightarrow r_0$, which happens only when the target planet is at superior conjunction, and the light ray passes through the perihelion $r = r_0$. Hence, we should integrate with caution, first along the path of length r_e from the Earth to perihelion, and then along the path r_p from the perihelion to the target. Using $\int dx (x^2 - 1)^{-1/2} = \ln(x + \sqrt{x^2 - 1})$ (with $x \equiv r/r_0$), we find the correction to the full round-trip time as

$$2\delta \equiv 2c\delta T = 2r_s \left[\ln \frac{r_e + \sqrt{r_e^2 - r_0^2}}{r_0} + \ln \frac{r_p + \sqrt{r_p^2 - r_0^2}}{r_0} \right] \approx 2r_s \ln \frac{4r_e r_p}{r_0^2}, \quad (5.10)$$

where in the last step the inequality $r_{e,p} \gg r_0$ was used. There are some other corrections proportional to r_s - e.g., due to the time delay on the Earth surface, but none of them depends on r_0 and would be absorbed without trace into the classical result followed from Eq. (5.9), as we obviously do not know a “real” distance. The radar experiments give the best workable measurements of the distance between planets.

The distinct feature of Eq. (5.10) is its dependence on r_0 . By repeating the measurements in slightly different positions of the target would single out this dependence.

The first experiments (1966, 1967) confirmed the result of Eq. (5.10) with the accuracy of 20%. Experiments made after the Viking landers on Mars have provided with the best available check of GR in a weak gravitational field: the result was a vindication of the Einstein theory prediction with the accuracy of 0.1%!

However, there were a few alternative theories that also survived such classic tests. Where GR is really unique is in its prediction of strong effects where scale of relativistic motion is comparable to r_s , like in the vicinity of black holes whose existence was predicted by the Schwarzschild solution of Einstein's equations. But one needs to distinguish effects of gravity from those owing to the choice of reference frame. Some of the latter could be quite bizarre as we shall demonstrate in the next section.

Lecture 6. Motion down to and around the black hole

The existence of black holes is one of the most famous predictions of GR, now fully confirmed experimentally. In this lecture we will use the Schwarzschild metric (to be derived later in the course) to see that time longer than infinity may exist, and space and time could interchange their roles near the singularity.

6.1 Schwarzschild sphere as the event horizon

Today we consider motion in the Schwarzschild metric without the restriction $r \gg r_s$. The metric describes the empty space, that it is applicable when $r > R$, where R is the radius of the spherically symmetric mass distribution that causes the spacetime curvature described by this metric:

$$\boxed{ds^2 = c^2 \sigma(r) dt^2 - \sigma^{-1}(r) dr^2 - r^2 d\Omega^2} \quad \sigma \equiv \left(1 - \frac{r_s}{r}\right) \quad (6.1)$$

It has obvious singularities at $r = r_s$ as $\sigma(r = r_s) = 0$ while $\sigma^{-1}(r = r_s)$ is divergent.¹ Such a singularity might be relevant only if the point $r \gtrsim r_s$ is in empty space – not inside the massive body where the metric is not valid. Thus we come to the requirement $R < r_s$, which is a crucial assumption: **there exist mass trapped completely inside the Schwarzschild sphere**, $r = r_s$, so that the space is empty in the vicinity of r_s . Such a mass distribution is called the **black hole**.

The existence of such gravitational traps is not obvious as, e.g., for the Sun $r_s \approx 3\text{km}$. However, the concept of black hole has naturally arisen in the theory of stars evolution – a massive enough star will end its life as a black hole. Now there is an overwhelming astronomical evidence of the existence of the black holes. In particular, the recently observed *gravitational waves* have originated from a pair of merging black holes.

Let's make a straightforward estimation to show that $R < r_s$ is possible. For a spherically symmetric mass distribution with density ρ , the Schwarzschild radius increases as R^3 so that for the large enough distribution r_s would exceed R . Indeed, we find

$$R < \frac{2Gm}{c^2} = \frac{2G}{c^2} \frac{4\pi\rho R^3}{3} \Leftrightarrow \frac{8}{3} \frac{\rho\pi GR^2}{c^2} > 1 \implies R > \left(\frac{3c^2}{8\pi\rho G}\right)^{1/2}$$

If R is large enough, this could be valid for spheres of any density. The Schwarzschild radius of a typical galaxy of mass $\sim 10^{42}\text{ kg}$ is $\sim 10^{14}\text{ m}$ ($\approx 0.01\text{ l.y.}$). Such a galaxy has a radius $\sim 10^{20}\text{ m}$. Thus, a contraction by a factor of $\sim 10^6$ would bring it within its horizon. The stars would be very far from touching each other (the critical volume is 10^{42} m^3 so that the density at the moment of collapse is about 1 kg/m^3 - this is a density of air which is at least 3 orders of magnitude smaller than the stellar density). Thus, unless the galaxy has large enough kinetic energy – due to rotation, e.g., it will be sucked into the horizon by its own gravity. Usually, the kinetic energy is sufficiently large to prevent a total collapse but at the centre of a typical galaxy there is a supermassive black hole.

A gravitational collapse of a star is prevented only by the energy of thermonuclear reaction in it. The gravitational pull in a burnt-out star would first crash the atoms into nucleon-electron plasma making a white dwarf; for a more massive star the pull would eventually squeeze electrons into protons and pack the resulting neutrons very tightly, forming a neutron star with enormous density

¹Note, however, that $\det \text{diag}(\sigma, -\sigma^{-1}, -r^2, -r^2 \sin^2 \theta) = -r^4 \sin^2 \theta$ is not vanishing at $r = r_s$. We shall learn later that the only physical, unavoidable singularity of the Schwarzschild metric is at $r = 0$.

$\sim 3 \times 10^{17} \text{ kg/m}^3$. When the mass of a parent star exceeds 8 Solar mass, at the end of its evolution it would be a black hole instead of a neutron star.² So, what is the black hole?

Relativistic effects at a scale $r \gtrsim r_s$ are no longer small, and motion at this scale is entirely dominated by them. Some strange features can be seen even before solving equations of motion. When one approaches the vicinity of $r = r_s$, then troubles are inevitable: the metric is singular at this point. The coordinate spacelike intervals diverge in the radial direction, while – which is even more unpleasant – the coordinate time ‘stops’. If one can penetrate inside the sphere, situation should become yet worse! Assuming that the Schwarzschild solution can be analytically continued to $r < r_s$, one sees σ changing sign, which means that t **becomes spacelike** and r **becomes timelike** coordinates.³ Time and space are swapping completely! (They are mixed even in the Lorentz transformation but not that drastically). This makes the fall to the centre, $r = 0$, inevitable as goind along the time axis to the future means reducing r .

So let us consider the motion along the timelike and null geodesics in the space with the Schwarzschild metric for $r > r_s$ but not necessarily $r \gg r_s$. Considering motion in the equatorial plane, $\theta = \pi/2$, we write the first integrals of the Lagrangian as

$$\mathcal{L} = c^2 \sigma \dot{t}^2 - \sigma^{-1} \dot{r}^2 - r^2 \dot{\phi}^2 = \begin{cases} c^2 & \text{timelike geodesic} \\ 0 & \text{lightlike geodesic} \end{cases} \quad \begin{cases} c^2 \sigma \dot{t} = \mathcal{E} \equiv c^2 \sigma \gamma & \text{energy,} \\ r^2 \dot{\phi} = J & \text{angular momentum.} \end{cases} \quad (6.2)$$

Even in non-relativistic motion, only the presence of a large enough tangential component of velocity, *i.e.* of the angular momentum, may save a particle from the fall to the centre of gravity. For a particle (or light) with zero angular momentum, $J = 0$, its fall is inevitable. Let us consider it.

6.2 Radial Fall

Consider first a photon in a radial fall, when $J = r^2 \dot{\phi} = 0$. The energy conservation law is irrelevant as the proper time is not defined for light, τ does not have a clear physical meaning. We can use $ds^2 = 0$ directly. From $c^2 \sigma dt^2 - \sigma^{-1} dr^2 = 0$, we have $\left(\frac{dr}{dt}\right)^2 = c^2 \sigma^2$, *i.e.*

$$\frac{dr}{dt} = \begin{cases} -c\sigma & \text{ingoing (falling) photon} \\ +c\sigma & \text{outgoing (raising) photon} \end{cases} \implies \frac{r dr}{r - r_s} = \mp c dt \quad (6.3)$$

(The choice of signs: $dr/dt < 0$ when r is decreasing with time). Mind you that $\frac{dr}{dt}$ is the *coordinate* speed of light. As $\sigma \rightarrow 0$ for $r \rightarrow r_s$, we see even before integrating this equation that

The coordinate speed of light $\rightarrow 0$ on approaching the Schwarzschild sphere.

This means that something bad happens to the coordinate time. Indeed, on integrating this equation for the infalling photon with the initial condition $r = r_0 \gg r_s$ at $t = 0$, we obtain the time of the fall:

$$ct = r_0 - r + r_s \ln \frac{r_0}{r - r_s}. \quad (6.4)$$

²We postpone till Lecture 17 (or so) a brief consideration of stars evolution that ends up either in white dwarf, or in a neutron star, or in a black hole. Note that only a fraction of the star with mass of order 1.3 the Solar mass would make it into the black hole.

³The choice of the coordinate system is crucial: we shall see in the following lecture that event horizons can exist even in Minkowski space, without any gravitation. However, only in 1960, more than 40 years after the triumph of GR, it has become clear how to build good coordinates in the vicinity and inside of a black hole. We consider such coordinates much later in the course.

The coordinate time increases unlimitedly on approaching the Schwarzschild sphere, $r = r_s$.

Material particles do no better. Substituting $\dot{t}$ from energy conservation, $c^2 \dot{t} \sigma = \mathcal{E}$, into $\mathcal{L}$, we have

$$\frac{\mathcal{E}^2}{c^2 \sigma} - \frac{\dot{r}^2}{\sigma} = c^2 \quad \Rightarrow \quad \dot{r} = \mp c \sqrt{\mathcal{E}^2/c^4 - \sigma}, \quad (6.5)$$

where again the ‘ $-$ ’ sign is for the falling particle. Then we obtain the coordinate velocity dr/dt :

$$\dot{r} = \frac{dr}{dt} \frac{dt}{d\tau} = \frac{dr}{dt} \frac{\mathcal{E}}{c^2 \sigma} \quad \Rightarrow \quad \frac{dr}{dt} = \mp \frac{c^3 \sigma}{\mathcal{E}} \sqrt{\frac{\mathcal{E}^2}{c^4} - 1 + \frac{r_s}{r}} \rightarrow 0 \quad \text{when} \quad r \rightarrow r_s \quad (6.6)$$

We recall that $\mathcal{E}$ is just a constant of integration which is equal to the energy of the particle of unit rest mass $m_0 = 1$ at the initial moment. If the particle was initially at an ‘infinitely remote’ point $r_0 \gg r_s$, then $\mathcal{E} = c^2 \gamma$ where γ is the Lorentz factor; in particular, $\mathcal{E} = c^2$ for a particle (of mass 1 – as always unless otherwise is stated) initially at rest at $r_0 \rightarrow \infty$. Thus, as $\sigma \rightarrow 0$ for $r \rightarrow r_s$, the coordinate velocity of the particle also $\rightarrow 0$, while its proper velocity, Eq. (6.5), $|\dot{r}| \rightarrow c$. The particle’s trajectory is the same as that of the ingoing photon, Eq. (6.4), in the region where $\sigma \ll \mathcal{E}$ and can be found exactly (even for $J \neq 0$) in any region. Even before finding this solution, we want to stress is that

From the viewpoint of a remote freely falling observer, neither a material particle nor light ever reaches the Schwarzschild sphere.

Thus, for a remote observer $r = r_s$ is an **event horizon**: the observer (if patient enough) sees the falling particle forever approaching but never reaching it (that explains the origin of the term *horizon* in this context).

However, the proper velocity is approaching the speed of light: we shall see that for lucky observers sitting in a spacecraft approaching $r = r_s$, there is nothing special at this point and they will inevitably fall into the centre over a finite interval of proper time, which from our (remote) viewpoint are times “after” $+\infty$.

But what about the proper radial velocity of a particle falling from an infinitely remote point with initial energy $\mathcal{E} > c^2$? From Eq. (6.5) it follows that at $r \rightarrow r_s$, the radial velocity $|\dot{r}| \rightarrow c(\mathcal{E}/c^2) > c$. Hence, there is something unphysical in $\dot{r}$. Indeed, the proper radial distance between two points, r_1 and r_2 , is not equal to $r_1 - r_2$. It should be defined as an interval between these two points at the same coordinate time, $t_1 = t_2$, when $ds^2 = -\sigma^{-1} dr^2$. Hence, the proper radial distance between the two points is given by

$$|\Delta s| = \int_{r_1}^{r_2} \frac{dr}{\sqrt{1 - \frac{r_s}{r}}}$$

which is log-divergent when $r \rightarrow r_s$. Hence, $\dot{r}$ of Eq. (6.5) does not have physical meaning. A physical question: does it take a finite proper time of a particle falling from some r_0 to reach the horizon, $r = r_s$, finite? From Eq. (6.5) we find (with $\rho \equiv r/r_s$, $\rho_0 \equiv r_0/r_s \gg 1$ and $a = \mathcal{E}^2/c^4 - 1 \geq 0$)

$$c\tau_0 = \int_{r_s}^{r_0} \frac{dr}{\sqrt{\mathcal{E}^2/c^4 - 1 + r_s/r}} = r_s \int_1^{\rho_0} \frac{\sqrt{\rho} d\rho}{\sqrt{a\rho + 1}} \approx \begin{cases} \frac{2}{3} r_s \rho_0^{3/2}, & a = 0 \text{ (i.e. } \mathcal{E} = c^2 \text{)} \\ r_0/\sqrt{a}, & a\rho_0 \gg 1 \end{cases}.$$

In both cases, τ_0 is finite.⁴

As an exercise, let’s find the coordinate time (i.e. the time recorded by a remote observer) that takes for a particle falling from rest from a very remote point.⁵ $r = r_0 \gg r_s$, to reach the vicinity of the

⁴The integral can be calculated exactly to give $c\tau_0 = a^{-3/2} \left[\sqrt{a\rho} \sqrt{a\rho + 1} - \ln [\sqrt{a\rho} + \sqrt{a\rho + 1}] \right]_1^{\rho_0}$.

⁵An exact solution, without using $\mathcal{E}_0 \approx c^2$, is possible in parametric form.

horizon. At this point $\dot{r} = 0$ so that one finds from the first integrals, Eq. (6.2), that $\mathcal{E} = c^2 \sigma(r_0) \approx c^2$. Then Eq. (6.6) reduces to (with $\rho \equiv r/r_s$ and $u \equiv \rho^{1/2}$)

$$\begin{aligned} \frac{dr}{dt} &= -\sigma c \sqrt{\frac{r_s}{r}} \Leftrightarrow r_s \frac{d\rho}{dt} = -c \frac{1 - \rho^{-1}}{\sqrt{\rho}} \Rightarrow \\ -\frac{c dt}{r_s} &= \frac{\rho^{3/2} d\rho}{\rho - 1} = \rho^{1/2} d\rho + \frac{\rho^{1/2} d\rho}{\rho - 1} \Rightarrow \\ -\frac{ct}{r_s} &= \frac{2}{3} \rho^{3/2} + 2 \int \frac{u^2 du}{u^2 - 1} = \frac{2}{3} \rho^{3/2} + \int \left[2 + \frac{1}{u - 1} - \frac{1}{u + 1} \right] du \\ &= \frac{2}{3} \rho^{3/2} + 2\rho^{1/2} + \ln \frac{\rho^{1/2} - 1}{\rho^{1/2} + 1} + \text{const.} \end{aligned} \quad (6.7)$$

As $t = 0$ at $\rho = r_0/r_s \equiv \rho_0 \gg 1$, one finds the constant of integration (to which the log-term adds negligible contribution of order $\rho_0^{-1/2}$ that we omit) to obtain

$$\frac{ct}{r_s} \approx \frac{2}{3} \left(\rho_0^{3/2} - \rho^{3/2} \right) + 2 \left(\rho_0^{1/2} - \rho^{1/2} \right) + 2 \ln \frac{\rho^{1/2} + 1}{\rho^{1/2} - 1}.$$

When ρ_0 is very large, the first term could be extremely large but it is always finite. It is the last term which creates the same problems as for a photon: whatever is the starting point, it diverges when $\rho \rightarrow 1$ so that it takes infinitely long coordinate time t to reach $r = r_s$. Thus, we come to the conclusion that the **coordinate time behaves badly**.

However, a static observer waiting for a falling particle close to the Schwarzschild sphere (at $r = R > r_s$) will see an absolutely different picture. In the frame of this observer, proper time $d\hat{t} = \sqrt{\sigma} dt$, and a proper radial length is $d\hat{r} = dr/\sqrt{\sigma}$, so that the velocity of the falling particle becomes (with $\mathcal{E} = c^2$)

$$\frac{d\hat{r}}{d\hat{t}} = \frac{1}{\sigma} \left| \frac{dr}{dt} \right| = \frac{c^3}{\mathcal{E}} \sqrt{\frac{\mathcal{E}^2}{c^4} - 1} + \frac{r_s}{r} = c \sqrt{\frac{r_s}{r}} \rightarrow c \quad \text{as} \quad r \rightarrow r_s.$$

Therefore, the observer sees the falling particle approaching the Schwarzschild sphere almost at speed of light. And nothing special happens at $r = r_s$. From this, one can conclude that a particle does cross this 'horizon' in a finite interval of its proper time. Note that this expression is meaningless to analytically continue to $r < r_s$, as $\hat{t}$ is no longer has the meaning of time there (not speaking of the fact that this proper velocity would exceed c).

But before refreshing the meaning of different timekeeping in the Schwarzschild geometry, let's consider more general motion near $r \rightarrow r_s$.

6.3 Hovering around the black hole

We have seen that for the particle's with zero angular momentum, i.e. is fired **directly** to the centre of coordinates (the black hole), its eventual fall to the singularity is inevitable. On the contrary, with $J = 0$ and $\dot{r} > 0$, the remote particle will fly away from the centre. Now we consider general motion ($J \neq 0$) of a particle of unit mass in the Schwarzschild geometry. In particular, we shall find a minimal value of J at which the particle with $\dot{r}_0 < 0$ **can** avoid the singularity.

We recall the general 1st order equation of motion, Eq. (4.5), as

$$\dot{r}^2 = \frac{\mathcal{E}^2}{c^2} - (1 - ur_s) (C + J^2 u^2) \equiv \frac{\mathcal{E}^2}{c^2} - V(u), \quad C = \begin{cases} c^2 & \text{timelike} \\ 0 & \text{lightlike} \end{cases} \quad (6.8)$$

where $V(u)$ is the effective potential. Using the dimensionless variables, $\varepsilon \equiv \mathcal{E}/c^2$, $j \equiv J/(r_s c)$, $\chi \equiv r_s u \equiv r_s/r \in (0, 1)$, and the dimensionless potential⁶

$$\tilde{V}(\chi) \equiv V(\chi)/c^2 = \begin{cases} (1 - \chi)(1 + j^2 \chi^2), & \text{timelike} \\ j^2 \chi^2 - j^2 \chi^3, & \text{lightlike} \end{cases} \quad (6.9)$$

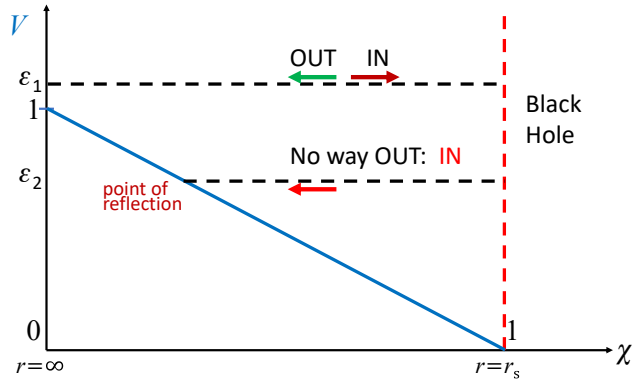
we rewrite Eq. (6.8) (without changing to χ in the r.h.s.) as

$$\frac{\dot{r}^2}{c^2} = \varepsilon^2 - \tilde{V}(\chi). \quad (6.10)$$

Since the l.h.s. of this equation is non-negative, the conserved energy ε of particle or light must obey the inequality $\varepsilon^2 \geq \tilde{V}(\chi)$, which we can relatively straightforwardly analyse thus finding possible trajectories.⁷

First, for a radial motion when $j = 0$, we have $\tilde{V} = 0$ for light. The absence of effective potential for light means that its radial worldlines are straight – going through the Schwarzschild horizon when $\dot{r} < 0$ and away from the black hole when $\dot{r} > 0$. For a particle, $\tilde{V} = 1 - \chi$, which is the straight line from infinitely remote points ($\chi = 0$) to the Schwarzschild sphere ($\chi = 1$).

When the energy $\varepsilon \geq 1$, the particle goes into the black hole when $\dot{r} < 0$, or away to infinity when $\dot{r} > 0$. However for $\varepsilon < 1$, the particle can only move on the right side of the line $V = 1 - \chi$: it cannot escape the gravitational pull even when its initial velocity $\dot{r} > 0$ (but not large enough to get ε above 1) – it will change the sign at the point of reflection.



For $j \neq 0$, we have $\tilde{V}$ as a cubic function of χ . It can be monotonic, or have min and max, depending on its derivative

$$\frac{d\tilde{V}}{d\chi} = \begin{cases} -1 + j^2(2\chi - 3\chi^2), & \text{timelike} \\ j^2(2\chi - 3\chi^2), & \text{lightlike} \end{cases} \quad (6.11)$$

The roots of $\tilde{V}' = 0$ for timelike motion are

$$\chi_{1,2} = \frac{1}{3} \left(1 \pm \sqrt{1 - \frac{3}{j^2}} \right). \quad (6.12)$$

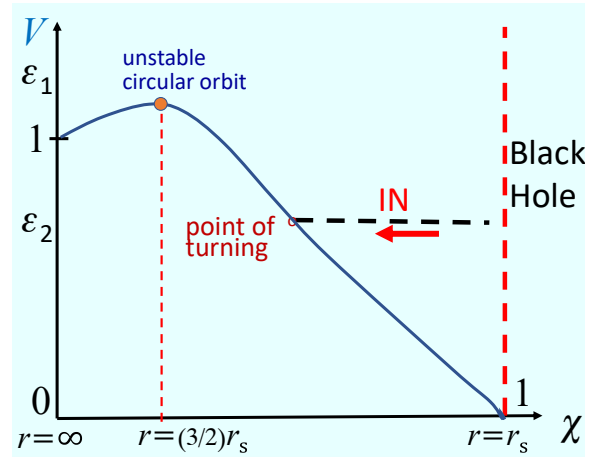
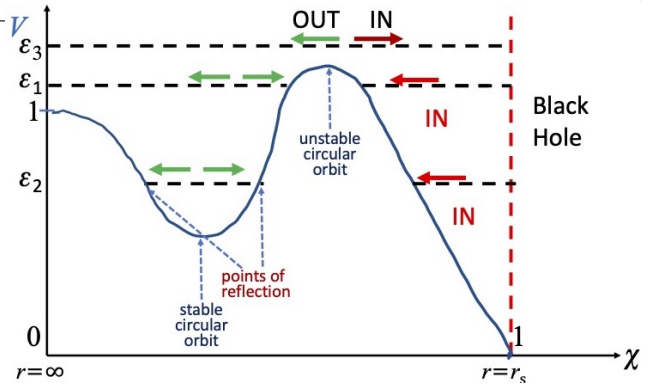
For $j < \sqrt{3}$, there are no real roots so that $\tilde{V}' < 0$ for all χ , as it is -1 when $\chi \rightarrow 0$, and cannot change sign. For such small angular momenta, $\tilde{V}$ is monotonically decreasing from 1 at $\chi = 0$ to 0 at $\chi = 1$. Hence the particle's motion remains qualitatively the same as radial, with no escape from the gravitational pull when $\varepsilon < 1$.

⁶The dimensional variables are convenient but they hide the difference between classical and GR contributions. So let us recall that, although only the $j^2 \chi^3$ term in $\tilde{V}$ has the relativistic origin, the l.h.s. of Eq. (6.8) is fully relativistic, as we have already seen investigating radial motion.

⁷The variable χ is convenient for describing hovering around the black hole, as all remote trajectories with $r \gg r_s$ are collapsed to an infinitesimal neighbourhood of $\chi = 0$.

For $j > \sqrt{3}$, two real roots correspond to min (the smaller one) and max in $\tilde{V}$. Note that the maximal value of $\tilde{V}$ can be either smaller or larger than 1 (which is the rest energy at infinity). Consider here the latter case, leaving the smaller for problem sheets. The particles with $\varepsilon \geq \tilde{V}_{\max}$, like ε_3 in figure on the right, either fall into the black hole or escape it altogether depending only on the sign of $\dot{r}$, similar to radial motion (although their trajectories are not straight lines as $j \neq 0$). Those with $1 < \varepsilon < \tilde{V}_{\max}$ (like ε_1) have two types of trajectories: “hyperbolic” ones with some min possible distance from the centre,⁸ and those with inevitable fall into the black hole, if too close to $\chi = 1$. This second type is qualitatively the same for $\varepsilon < 1$ (like ε_2), but they are “elliptic” ones when, with the same energy, they are further away from the centre. Finally, $\varepsilon = \tilde{V}_{\min(\max)}$ correspond to (un)stable circular orbit. For $\varepsilon < \tilde{V}_{\min}$ only one types of trajectories left, with inevitable fall into the singularity.

For light, zeros of $\tilde{V}'$ are independent of j and are equal to 0 and $2/3$. The minimum is trivial and corresponds to free motion at infinity, while the maximum that corresponds to an unstable circular motion is always at $r = \frac{3}{2}r_s$. All trajectories with $\varepsilon \leq 1$ will end up in the black hole, while for $1 < \varepsilon < \tilde{V}_{\max} = \frac{4}{27}j^2$ this is only true if motion starts at $\chi > \frac{2}{3}$ – for smaller χ (larger r), the trajectories are of “hyperbolic” type. Finally, when $\varepsilon > \frac{4}{27}j^2$, the trajectories go to the black hole when $\dot{r} < 0$ and escape it altogether when $\dot{r} > 0$.



6.4 Time delay revisited

In general, for observers in motion

$$\frac{d\tau_A}{d\tau_B} = \frac{\dot{t}_B}{\dot{t}_A},$$

where $\dot{t}_{A,B}$ are calculated for observers A and B . Let us consider one of the observers (say, A) as an infinitely remote (the metric is locally Minkowski) and stationary. Then the time delay is simply $\dot{t}_B$, as for A the coordinate and proper time coincide. Omitting the index B , we find from $\mathcal{L} = c^2$:

$$c^2 = c^2 \sigma \dot{t}^2 - \sigma^{-1} \dot{r}^2 - r^2 \dot{\phi}^2 = c^2 \sigma \dot{t}^2 \left[1 - \frac{v_{\perp}^2}{\sigma^2 c^2} - \frac{v_{\parallel}^2}{\sigma c^2} \right], \quad v_{\perp} \equiv \frac{dr}{dt}, \quad v_{\parallel} \equiv r \frac{d\phi}{dt}. \quad (6.13)$$

This gives the time delay factor in terms of the coordinate velocities. For observer B it would be preferable to have this in terms of proper velocities. To this end, imagine observer B' stationary at the same spatial point as B . Its proper time, and proper radial and tangential lengths are simply

$$d\tau_{B'} = \sigma^{1/2} dt, \quad dr_{\perp} = \sigma^{-1/2} dr, \quad dr_{\parallel} = r d\phi.$$

⁸They are hyperbolic in the Newtonian limit, and distorted by gravity hyperbolas when χ is not too close to 0.

Then, the components of proper velocity of B in the rest frame of B' are

$$u_{\perp} = \frac{dr_{\perp}}{d\tau_{B'}} = \frac{1}{\sigma} \frac{dr}{dt} = \frac{v_{\perp}}{\sigma}, \quad u_{\parallel} = \frac{dr_{\parallel}}{d\tau_{B'}} = \frac{r}{\sqrt{\sigma}} \frac{d\phi}{dt} = \frac{v_{\parallel}}{\sqrt{\sigma}}.$$

Therefore, the factor within square brackets in Eq. (6.13) becomes $1 - u^2/c^2$ with $u^2 = u_{\perp}^2 + u_{\parallel}^2$ so

that with the usual $\gamma = \left(1 - \frac{u^2}{c^2}\right)^{-1/2}$ we find the time delay factor as

$$\boxed{\dot{t} \equiv \frac{dt}{d\tau} = \frac{\gamma}{\sqrt{\sigma}}}. \quad (6.14)$$

From the viewpoint of the remote observer, the time is delayed by the combination of the GR factor, $1/\sqrt{\sigma}$, and the SR factor, γ . Note that the latter factor is the sole reason for the time dilation between observers B and B' who are at the same spatial point so that the gravitational field makes an equal impact on both of them. It does not mean that gravitational effects are mutually cancelled: γ does include GR effects as well, as we substitute there proper rather than coordinate velocity.

*In calculating the time dilation between observers at the same point, as well as other local relativistic effects, one can use SR relations **provided that** one substitutes proper (rather than coordinate) time, displacement, velocity etc.*

This is again a consequence of the equivalence principle. Remember that locally inertial frames are those which are in free fall, and effects of gravity are locally indistinguishable from those caused by the motion of the frame. In other words, coordinate time is our book-keeping device, while locally only proper time could be measure.

Note that Eq. (6.14) is exact: GR effects have not been considered as small corrections. This expression remains valid for any metric of the form $ds^2 = c^2\sigma(r)dt^2 - d\ell^2$, no matter what is a functional dependence of σ and $d\ell^2$ on r . For the Schwarzschild metric at $r \gg r_s$, i.e. in a weak field, keeping only the lowest order terms in $1/c^2$, one finds $\gamma/\sqrt{\sigma} = (1 - r_s/r - v^2/c^2)^{-1/2} \equiv \gamma^$ which is exactly the same as for the pseudo-Newtonian metric considered earlier.*

6.5 Red shift and Doppler effect

Let us stress that the Doppler effect, i.e. the change of frequency of an electromagnetic signal in the frame of a remote receiver, is not completely defined by the time dilation. Eq. (3.14) for the red shift, $Z = \sqrt{\sigma(r_{\text{rec}})/\sigma(r_{\text{emit}})} - 1$, remains valid for the stationary receiver and emitter, but if they are moving, it is not sufficient to multiply it also by the ratio of the γ factors. To see why it is so let us recall

6.5.1 Doppler effect in SR.

Let $u^\mu = (\gamma c, \gamma \mathbf{v})$ be the 4-velocity of the source in observer's frame Σ , and $k^\mu = (k^0, \mathbf{k})$ is the 4 wave-vector, with $k \equiv |\mathbf{k}| = k^0 = \omega/c$. In frame Σ' where the source is at rest $u'^\mu = (c, 0)$. Writing the invariant $g_{\mu\nu}u^\mu k^\nu$ with the Minkowski metric $g_{\mu\nu} = \text{diag}(1, -1, -1, -1)$ in Σ and Σ' , we have (with $k^{0'} \equiv \omega_0/c$, $\mathbf{v} \cdot \mathbf{k} = v(\omega/c) \cos \theta$, and $\gamma = (1 - \beta^2)^{-1/2}$):

$$g_{\mu\nu}u^\mu k^\nu = c\gamma k^0 - \gamma \mathbf{v} \cdot \mathbf{k} = ck^{0'} \Rightarrow \omega = \frac{\omega_0}{\gamma(1 - \beta \cos \theta)}$$

Here v is the standard 3 velocity of the source in observer's frame and ω_0 is the proper frequency of the light signal. Note that only the transverse Doppler effect ($\theta = \pi/2$) is entirely due to the relativistic time delay. This remains valid also in GR.

6.5.2 Doppler effect in GR

We cannot immediately use $g_{\mu\nu}k^\mu u^\nu = \text{inv}$ to find λ in the frame of the remote observer: a 4-vector in a curved space is defined only locally, and we have not (yet) learned how to transport it from one event to another. But we can repeat a trick by considering first the red (or blue) shift between the source of radiation and an observer who is stationary in a given metric placed at the same spatial point in the direction of observation by the remote receiver. Then the signal received by the latter could be considered as sent by this intermediate observer; as the latter is stationary, the wavelength change will be entirely due to the gravitational time delay.

In the first step, we can use the SR expressions according to the rule formulated above: the only change would be substituting u for v . (One can verify this by comparing the invariant $g_{\mu\nu}k^\mu u^\nu$ in the frames B and B'). Then, the signal obtained by the remote observer will be additionally shifted by the standard factor of $\sqrt{\sigma}$. Thus we obtain

$$Z + 1 \equiv \frac{\lambda_\infty}{\lambda_{\text{emit}}} = \frac{\omega_0}{\omega_\infty} = \frac{\gamma(1 - u \cos \theta/c)}{\sqrt{\sigma(r)}}, \quad \gamma \equiv \frac{1}{\sqrt{1 - u^2/c^2}}, \quad (6.15)$$

where r is the radial coordinate of the emitter, and u is its proper velocity measured at this coordinate. Note that there is no dependence on angular coordinates. The measured frequency will be the same for all stationary remote observers, whether they are on the same radius with the emitter or not.

6.6 Crossing the horizon: what happens *after* infinite time

Although the Schwarzschild metric was only derived for $r > r_s$, one can speculate about the events inside the Schwarzschild sphere, *i.e.* for $r < r_s$. Note that the singularity at $r = r_s$ is not real, since the determinant of the Schwarzschild metric remains finite and nonzero at $r = r_s$.

The time-keeping of the fall would be totally different on a clock attached to a falling particle which counts the proper time τ . To see this, we note that Eq. (6.5) is free of singularities (they appeared in t due to the extra factor of σ). In the absence of singularity τ must be finite! Indeed, Eq. (6.5) is easy to integrate; for the particle fall we consider only $\mathcal{E} = 1$ as above

$$-c d\tau/r_s = -\rho^{1/2} d\rho \quad \Rightarrow \quad c\tau/r_s = \frac{2}{3} \left(\rho_0^{3/2} - \rho^{3/2} \right) \quad \text{for particle.}$$

Nothing happens at $\rho = 1$. Therefore, the particle crosses the Schwarzschild sphere without noticing that something special is going on around and continues its motion inside the Schwarzschild sphere at the time t which is, from the viewpoint of the remote observer, is longer than $+\infty$. However, our description based on the Schwarzschild metric fails even for the proper frame, as the particle's proper velocity would exceed the speed of light.

As the time-reading in t and τ are infinitely different, t is a bad coordinate in the vicinity of the black hole (but a natural one for us – as we all are very remote observers). One can anticipate that there exists light or a particle which come from the inside of the Schwarzschild sphere as well – remember the second solution with $+$ sign which corresponds to an outward motion. But from the

viewpoint of a remote observer – i.e. us – such a particle started its motion “before” time $t = -\infty$ and crosses the Schwarzschild horizon “exactly” at $t = -\infty$.

Let us stress again: the Schwarzschild coordinates are meaningful and useful as they describe the events from the viewpoint of a remote observer for whom everything stops at the Schwarzschild sphere. But it takes only a finite proper time at the falling particle’s clock. Hence these coordinates are definitely not suitable for $r < r_s$.

We shall learn how to introduce “better coordinates” in the vicinity of the event horizon much later in the course, as we have travelled already for quite a distance without proper foundation – a serious mathematical description of the curved spacetime culminated by the Einstein’s equations describing how a mass distribution creates a metric. So it is *almost* time to learn some bits of the mathematical language suitable for describing nonlinear spaces – tensor analysis which comprises the very first steps into the differential geometry. However, before that we shall have a small digression to illustrate how a choice of coordinate system might lead to the appearance of the event horizon even in a flat Minkowski space completely devoid of gravitational forces.

Lecture 7. Rindler space: life in an accelerated frame

We have used the Schwarzschild metric $g_{\mu\nu}$ for describing motion in a weak gravitational field $r \gg r_s$. In the next lecture, we will be using in the vicinity of the *black hole*, i.e. approaching the region $r \gtrsim r_s$. We know that $g_{\mu\nu}$ is singular at $r = r_s$. We shall see in then that it leads to quite bizarre effects like the existence of the *event horizon* and, as a consequence, of the time 'after $t = +\infty$ ' or 'before $t = -\infty$ '. A natural feeling is that these bizarre effects with space and time are happening due to the presence of a strong gravitational field. However, we will show in this lecture that similar effects can happen in a flat Minkowski space viewed from an accelerated frame. This gives us a hint before going to the description of the black hole: sometimes, it is the choice of a reference frame (rather than gravity) that is responsible for the madness of the time.

7.1 Coordinate relations in an accelerated frame

Let us first recall what we know about an accelerated motion from SR. In an instantaneous rest frame ¹ Σ' attached to a particle accelerated along the x -axis, its 4-velocity $u^{\mu'} \equiv \dot{x}^{\mu'} = (c, 0, 0, 0)$ (as it is instantaneously at rest), and its 4-acceleration $w^{\mu'} \equiv \ddot{x}^{\mu'} = (0, a, 0, 0)$. In laboratory FFF (free falling frame) Σ , which is in standard configuration with Σ' , one has $w^\mu = \Lambda_{\mu'}^\mu w^{\mu'} = (\beta\gamma a, \gamma a, 0, 0)$, where $\Lambda_{\mu'}^\mu$ is the (inverse) Lorentz transformation matrix. From now on, we omit irrelevant y and z coordinates. We choose the time setting in Σ so that $t = 0$ at the moment when it was an instantaneous rest frame of the particle when, by definition, $v = 0$.

To find its 4-acceleration at a time t we use the Lorentz transformation for its first component, $w^1 = \gamma a$:

$$\gamma a = w^1 \equiv \ddot{x} = \frac{d\dot{x}}{d\tau} = \gamma \frac{d\dot{x}}{dt} \Rightarrow a = \frac{d\dot{x}}{dt} \Rightarrow at = \dot{x} \equiv \frac{dx}{d\tau} = \gamma \frac{dx}{dt} \equiv \frac{v}{\sqrt{1 - v^2/c^2}}.$$

Here τ is proper time measured by a clock at the rocket, and we used $dt = \gamma d\tau$ and $v(t=0) = 0$. Changing to the dimensionless variables $\zeta = at/c$ and $\beta = v/c$, we solve the algebraic equation $\zeta = \beta\gamma = \beta(1 - \beta^2)^{-1/2}$ which gives

$$\beta = \frac{\zeta}{\sqrt{1 + \zeta^2}} \quad \text{and} \quad \gamma = \sqrt{1 + \zeta^2}. \quad (7.1)$$

In the nonrelativistic limit, $\zeta \ll 1$, we have $\beta \approx \zeta$, i.e. $v = at$, while in the ultra-relativistic limit, $\zeta \gg 1$, we have $\beta \rightarrow 1$, as expected. Then, integrating the equation for $\frac{1}{c} \frac{dx}{dt} = \frac{a}{c^2} \frac{dx}{d\zeta} = \beta$, we find $x(t)$ or, more conveniently, $x(\zeta)$, in terms of the length-constant² $w \equiv c^2/a$:

$$\frac{a}{c^2} \frac{dx}{d\zeta} = \frac{\zeta}{\sqrt{1 + \zeta^2}} \Rightarrow x = w \left(\sqrt{1 + \zeta^2} - 1 \right) = w(\gamma - 1) \approx \begin{cases} \frac{1}{2} w \zeta^2 = \frac{1}{2} at^2, & \zeta \ll 1 \\ w(\zeta - 1) \equiv ct - w, & \zeta \gg 1 \end{cases}. \quad (7.2)$$

This describes a hyperbolic motion of an accelerated particle. The asymptote for this motion is $x = ct - w$, i.e. the null worldline of a photon. This is obvious: whatever is the value of acceleration – if one waits long enough, an accelerated rocket will be approaching the speed of light.

¹also known as LGF - local geodesic frame

²It is unfortunate that this traditional notation for the length constant, which is inversely proportional to a , uses the same letter as for the four-acceleration w^μ . Hopefully, this would lead to no confusion as they do not appear together.

The coordinate relation, $x(\zeta)$, describes this motion in the laboratory frame. To understand how it is viewed by the inhabitants of the constantly-accelerating rocket, we need to express x in terms of its proper time τ . Again, it is convenient to introduce the dimensionless proper time, $u \equiv a\tau/c$. Then $dt = \gamma d\tau \Leftrightarrow d\zeta = \gamma du$, i.e. $du = d\zeta/(1 + \zeta^2)$ so that $u = \sinh^{-1} \zeta$, or $\zeta = \sinh u$. For small ζ , there is no difference between the laboratory and proper time, while for $\zeta \gg 1$ the time in the laboratory frame (t) becomes exponentially longer than the time measured by the clock in the accelerated rocket (τ). Therefore, from the viewpoint of a free-falling observer in Σ , time almost stops at the accelerated rocket. One expects that it could be seen in a very different way by an inhabitant of the accelerated rocket.

Let us write down all the relations between coordinates in Σ in terms of the dimensionless proper time u . Noticing that $ct = w\zeta$ and $\gamma = \cosh u$, and shifting the coordinate origin by w so that Eq. (7.2) reduces to $x = w\gamma$, we have the hyperbolic motion parametrised as

$$ct = w \sinh u, \quad x = w \cosh u, \quad x^2 - (ct)^2 = w^2. \quad (7.3)$$

Now the motion starts at the event A with coordinates $t = \tau = 0$ and $x_0 = w$. On the ct - x plane, the hyperbolic worldline has asymptote (for $t > 0$) as $x = ct$, obtained (as usual for hyperbolas) by putting $w = 0$ in Eq. (7.3). Note that $\beta = dx/cdt = \tanh u$; hence, the lines of a constant proper time $u \propto \tau$ for the accelerated particle. Since $x/ct = \coth u = 1/\beta$, these are straight lines through the origin of Σ . Then the asymptote, $x = ct$, corresponds to the (unreachable) velocity $\beta = \tanh u = 1$, and to the proper time $u = \tau = \infty$: it will take an infinite proper time to approach the asymptote.

Thus we see that there exists **the event horizon** that the rocket people will never reach. Moreover, any particle approaching the event horizon, will do it forever from the viewpoint of the inhabitants of the rocket: any straight line through the origin measures their local time which slows down unlimitedly with approaching the external boundary of the light cone. However, for the people left at the starting point A the worldline $x = x_0 = w$ is vertical, so that it crosses the ‘horizon’ (without noticing it) at $t = w/c = c/a$ (note that for $a = g \approx 10\text{m/s}^2$ we have $t \approx 3 \cdot 10^7\text{s} \approx 1$ year). All events which happen around the starting point after this time correspond to the times exceeding ∞ for the rocket inhabitants – this is really beyond horizon! If they constantly accelerate with g , they will reach a very old age (and inevitably die) before just one year passes on the Earth (assuming that it was the starting point). From their viewpoint, all signals that they send to the Earth after this time would never reach the addressed.

How can we reconcile this with the famous twin paradox? It is well known that the travelling twin (Alice) will be younger than the Earth-dwelling twin (Bob) when she returns to the Earth. The point is that such an event horizon is only due to the choice of coordinates. In order to return to the Earth, Alice needs to stop her acceleration and reverse the spaceship, accelerating it in the opposite direction. (Note that – in contrast to an exercise in Problem Sheet 2, the deceleration and U-turn could be practically instantaneous in the spaceship’s frame³ in comparison to the total duration of the trip). Assuming that the U-turn happens at $x = x_1$, and $x_1 - x_0 \gtrsim w$ so that the travelled distance was at least about one light year, the return hyperbolic worldline will be

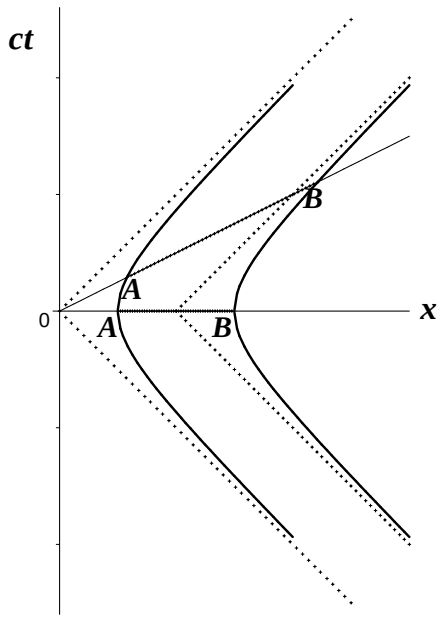
$$(x - x_1)^2 - c^2(t - t_1)^2 = w^2, \quad x < x_1 \text{ for the return motion}, \quad (7.4)$$

³Quite an interesting problem to consider is how long such an “instantaneous” U-turn takes in the Bob’s reference frame. Can you show that for $x_1 \gg w$ and $a = g \approx 10\text{m/s}^2$ this takes about 2 years from Bob’s viewpoint?

where x and t shifted, respectively, by x_1 and t_1 , the latter found from Eq. (7.3) as $ct_1 = \sqrt{w^2 + x_1^2}$. The asymptote is the solution of Eq. (7.4) at $w = 0$, i.e. $ct - ct_1 = x_1 - x$ so that it crosses the Earth worldline $x = 0$ at $ct_2 = ct_1 + x_1 = x_1 + \sqrt{x_1^2 + w^2}$. If $x_1 \gg w$, then $t_2 \approx 2x_1/c$, which is understandable: most of the travel was at ultrarelativistic velocities. Naturally, the proper time of travel for Alice, ζ in this case will be exponentially shorter than t_2 . In dimensionless units, $\zeta_2 \equiv at_2/c \approx 2ax_1/c^2 \equiv 2x_1/w$, and from $\zeta_2 = \sinh u_2$ follows $u_2 = \ln(\zeta_2 + \sqrt{\zeta_2^2 - 1}) \approx \ln 2\zeta_2 \approx \ln(4x_1/w)$.

7.2 Can a rigid finite-length spaceship be uniformly accelerated?

One needs to construct the coordinate system which would be suitable for a spaceship – which, in contrast to a material particle, has finite dimensions. Such a necessity arises from the impossibility of having all points on the solid (non-extendable) object to have the same acceleration for a long time. Let us illustrate this.



Let us draw the worldlines of particles A and B , starting at the same time $t = 0$ with the same acceleration a (in their instantaneous LGFs): A from $x = w$ and B from $x = x_0 + w$. Their worldlines are hyperbolas, separated by x_0 . Consider events in the frame of A . A straight line from the origin is the line of constant proper time $u_a = c\tau_A/a$. Clearly, particle B moves further than A at the same moment of τ_A , so that distance AB increases with time in this frame, despite starting simultaneously with the same acceleration. And what about frame B ? In this frame, particle A simply does not exist – it is sitting in “absolute elsewhere”, i.e. beyond the event horizon of B (as in this figure, we have chosen $x_0 > w$). Hence, it is impossible to uniformly accelerate a rigid body in the relativistic regime: to keep the rigidity points with different x -coordinates at $t = 0$ should be accelerated differently. This is essential only at a scale of order $w \equiv c^2/a$ which is about a light year long if $a \approx g$. A much shorter spaceship would be OK.

Then consider two particles moving with **different** proper accelerations a_1 and a_2 , both having been at rest in Σ at the instant $t = 0$, separated by $L_0 \equiv dx = x_1 - x_2 = w_1 - w_2 = dw$. At any other moment of time, we obtain by differentiating Eq. (7.3) at $t = \text{const}$:

$$dx = \frac{w dw}{x} = \frac{dw}{\cosh u} = \frac{L_0}{\gamma(\beta)}.$$

The distance between the particles is contracted by the factor $\gamma(\beta)$ as it should for a **rigid body** moving with instantaneous velocity β . Obviously, any small element of a rigid body **must** be contracted by the same factor – otherwise, elastic stresses arise. Therefore, if for any two points of the body separated at $t = 0$ by the proper distance dw , i.e. the difference in the inverse acceleration is equal to the initial separation, it will move as a whole. Then, we also see that Δw measures proper distances in a non-homogeneously accelerated rocket with w of different points chosen to keep it **rigid**. This choice means that $x_n = w_n \cosh u$ for any point n in the rocket, i.e. all the hyperbolas have the same centre at the origin ($ct = x = 0$), in contrast to the equidistant hyperbolas on the insert.

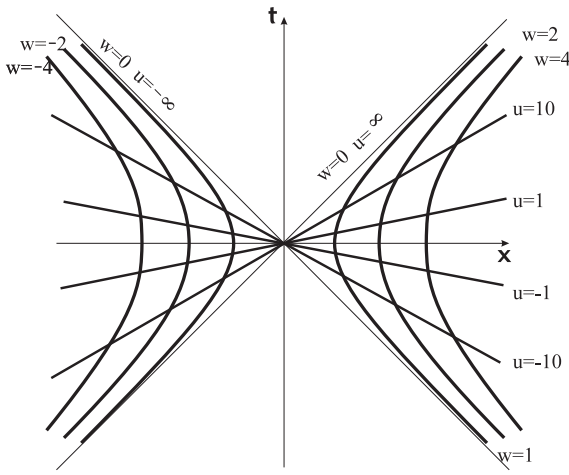
Now, the whole brave new world (which is usually called the Rindler space) can be constructed using w and u as a new coordinate grid. The absolute flexibility in the choice of a coordinate system

is indeed available in GR. Then (as you know from the problem sheet 1), the metric becomes

$$ds^2 = w^2 du^2 - dw^2 - dy^2 - dz^2 \quad (7.5)$$

7.3 A 'fake' event horizon

Let us sketch the new coordinate lines on the "old" ct - x plane. We choose the spatial origin ($x = 0$) to be in the focus of all hyperboles. The light cone centred at the origin gives asymptotes of the hyperboles, as the straight lines $x = \pm ct$ correspond to $w = 0$, i.e. to an infinite acceleration. The straight lines $u = \text{const}$ are the lines of a constant 'time' u in these coordinates, while the hyperboles $w = \text{const}$ represent the lines of constant 'distance' in this world.



Let us stress that the choice of w as a spacelike coordinate is quite reasonable: it remains unchanged at any given point of the rocket (so that locally the coordinate system is attached to the moving rocket); a ruler to measure this distance remains rigid at any time; it can be considered as a 'height' in our new world - the bigger the w , the smaller is the acceleration, and thus an effective gravitational force. On the other hand, u is proportional to the proper time at any 'fixed' spatial point $w = \text{const}$, so that it measures time in some units. Very important is the fact that the lines $u = \text{const}$ represent the same moments of time throughout the whole Rindler space, or the whole infinitely long rocket.

One can introduce coordinates which measure directly proper time and proper distance in a different way. Let us choose $x' = w - w_0$ as a new spacelike coordinate in this world where w_0 is a constant. As $dw = dx'$, the new coordinate measures distances in the same way as the old one, w . Suppose that we want to describe events in the vicinity of a small part of the accelerated rocket (small compared to $c^2/a \approx 9 \times 10^{12} \text{ km} \approx 1 \text{ l.y.}$ for $a = 9.8 \text{ m/s}^2$), or in a very small (compared to one light year) rocket. It moves along the line $w_0 = \text{const}$ so that its position in the Rindler world does not change. Then in $u \equiv c\tau/w_0$, the coefficient of proportionality between τ and u is constant, and we can choose τ itself to measure time: $du = (c/w_0) d\tau$. Then $w du = c(x'/w_0 + 1) d\tau \equiv c(1 + \kappa x') d\tau$ where we introduce $\kappa \equiv 1/w_0 \equiv a/c^2$ to distinguish between a variable w coordinate and a constant κ in the newest coordinates where the metric is equal to

$$ds^2 = (1 + \kappa x')^2 d\tau^2 - (dx')^2 - dy^2 - dz^2 \quad (7.6)$$

These coordinates are equally suitable for describing the Rindler space of the accelerated rocket. The Rindler world is a very strange place (but not impossible if some resources for maintaining a constant acceleration were available). Let us list its features.

- First, note that **the time slows down** when we move towards smaller w . By itself, it is not unusual as w^{-1} corresponds to the acceleration: by the equivalence principle, a large acceleration corresponds to large gravitation and we have already learned about the gravitational time dilation. However, at the point $w = 0$, the acceleration is infinite and time stops. For the rocket people

this is a real singularity - the beginning and the end of the world. However, in the inertial system Σ absolute nothing of interest happens at $w = 0$. The whole singularity is a consequence of the choice of coordinate system. But the rocket people have no choice (if they cannot switch off whatever maintains their crazy acceleration)! For them, their world and their coordinate system is a reality.

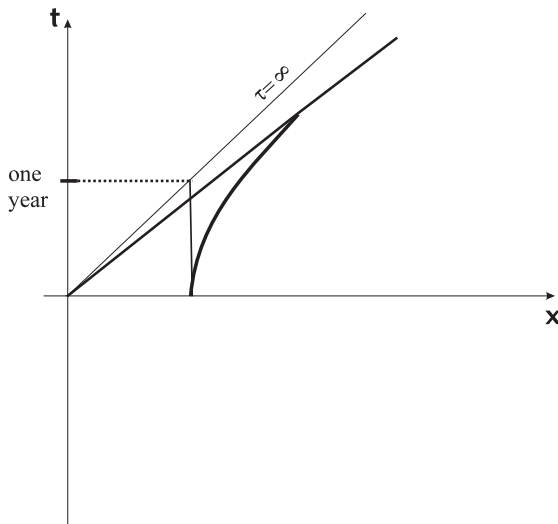
- The lines of constant time, $u = \text{const}$, correspond to $\beta = \tanh u = \text{const}$, where β is the instantaneous velocity of the rocket in Σ . The asymptote $\beta = 1$ is then not only the line of the infinite proper acceleration (photon!) but also the line of the infinite proper time. Thus we see that there exists **the event horizon** that the rocket people will never reach. Moreover, any particle approaching the event horizon, will do it forever from the viewpoint of the inhabitants of the rocket: any straight line through the origin measures their local time which slows down unlimitedly with approaching the external boundary of the light cone. Somebody who scratches his/her head at the line $w = 0$ will be doing this forever from the viewpoint of the Rindler people. However, from the viewpoint of an inertial observer in Σ , there is nothing specific or singular in this **event horizon**: any particle (or the observer, if in inertial motion) crosses it without paying a slightest attention to such a great event! Besides the **future event horizon** which is unreachable from the rocket world, there exist also the **past event horizon**, $u = -\infty$, which is unpenetrable **into of the rocket**.
- It's not only time which is weird. **Distance is behaving badly** in this system. The event horizon is unapproachable as the distance to it does never reduce! Indeed, it equals $\Delta w = w - 0 = w = \text{const}$ at any fixed point of the rocket. (One can approach it slightly by moving down in the rocket, if its length is comparable to one light year). If we consider two "small" rockets moving with different accelerations, w_1 and w_2 (or κ_1 and κ_2), the distance between them does not change at all, provided that **they both had zero velocities at $t = 0$** . (This is actually the condition for a "large" rocket to remain rigid which we have used from the very beginning). But it sounds crazy enough: two runners divided by some distance have started to run at the same instant and one with the higher acceleration will never catch another one! (The reason is that they are divided not by **some distance** but by the distance related to their acceleration: $\Delta x' = c^2/a_1 - c^2/a_2$. If the difference in their acceleration is at human scale, i.e. of order g , then the distance between them is of order of light year; if the distance were of order of several metres, then the difference in their acceleration would be $10^{-16} g$ - in both cases it is not that strange that they do not meet).

However, these features are still not most curious about the Rindler world. To see that some more surprises are inevitable, let us re-visit the pair of twins, Alice and Bob.

7.4 Twins paradox revisited: Alice, Bob, Anti-Bob and 'utterly elsewhere'

Alice stays on Earth and Bob travels in the Rindler rocket accelerating for 5 years, decelerating for the next 5, then accelerating and decelerating in the opposite direction in the same way. We have solved this problem a year ago within SR but now we want a more scrutinising look at it.

The time on the Earth and at the rocket are related by Eq. (7.3): $t/T_0 = \sinh(\tau/T_0)$ where $T_0 = w_0/c = c/g \approx 1$ l.y. Thus we conclude that after the first 5 years at the rocket clock, the time elapsed at the Earth is $\approx \frac{1}{2}e^5 T_0 \approx 75$ years, and thus the whole trip lasted 300 years (poor old Alice!), and took the rocket 150 light years apart from the Earth. The conclusions have nothing to do with GR as we used only the inertial system Σ for calculations. But how does the travel look for Bob? After 5 years of travel - according to the rocket τ clock - Bob reached a point of intersection with the line $\beta = ct/x = \tanh 5$ and stopped to accelerate.



But this is a line of constant time in the Rindler world - therefore, Alice's intersection with this line also happens in 5 years - according to Bob's clock, and less than in a year of the Alice's proper time. Therefore, Bob thinks that it is Alice who is younger at the moment when the rocket stopped to accelerate! Obviously, events simultaneous for Bob are not simultaneous for Alice - so here there is no paradox. But it is still interesting how Bob will be able to reverse the events - when they will finally meet, Alice will be indeed 280 years older than Bob!

Obviously, the crucial instant is the reversing of acceleration. At some short time, $a = 0$ so that $w = \infty$, and the whole Rindler world collapsed. The rocket instantly jumped to a point with the coordinate $w = \infty$, but it is still at the same event in the inertial $x-t$ plane. What then happens with the rocket moving with the opposite acceleration? We come back to the Minkowski-Rindler diagram, and see that it can include the **mirror image of the rocket world** - a second rocket that accelerates in the direction of the negative x axis. This is obvious, as w becomes negative. What is less obvious but becomes clear after looking at the picture is that **the coordinate time u - or τ - proceeds backwards**, i.e. from $+\infty$ to $-\infty$. The future and past event horizons are swapped between the two rockets. Thus, from the viewpoint of Bob, Anti-Bob who lives in the mirror rocket (forever) should be dead before he was born - if he moves accelerating rather than decelerating (i.e. its velocity in the negative x direction increases in its time). Fortunately, these two worlds are in utterly elsewhere with respect to one another, and will never communicate. (We still may assume that they had communicated before the start of travel, when Bob and Anti-Bob have started their trips in the opposite direction). However, it is very easy to jump from one to another - all is necessary is to stop for an instant, and then to reverse the acceleration.

This 'jump' takes place at the same event, x, t , in the inertial system Σ . As the rocket - a material body - could not jump, it is the whole $w-u$ mirror grid that jumps to this point. However, time does not start to move backwards for Bob, since in the mirror world he is decreasing rather than increasing its velocity. Thus, asymptotes swap with each other, and Bob moves from the lower to the upper part of the line $w = \text{const}$ which is still to the future. Its velocity, β , is decreasing but now $\beta = -\tanh u$ (choosing an appropriate starting point at the clock), and the lower asymptote $\beta = 1$ corresponds to $\tau = -\infty$. The **event horizon that existed along the line $x = ct$ before inverting the acceleration disappears!** (Actually, the new horizon appears instead but changing acceleration again will get rid of this as well). Now, from Bob's viewpoint, Alice's time is speeding up dramatically, and the next 5 years of his flight will clock up as approximately 149 years at the Earth. In total, Bob's worldline on the Minkowski diagram is **longer** than the straight Alice's world-line. Thus, as the **shortest worldline corresponds to the longest time**, Alice will be older, indeed.

A strange Rindler world is due to an unfortunate choice of coordinates in a perfectly flat world. (But inhabitants of the accelerating rocket are not free in their choice: they've got what they've got). Hence, a curved world in the vicinity of the black hole requires even more care. We will start this next time, and see a lot of analogies with the brave new Rindler world.

Lecture 8. Vectors and tensors in curved space

The most important immediate consequence of the equivalence principle is that particles move along geodesic lines – the shortest (and straightest) lines connecting two points in a curved space. We have learned how to find geodesic lines in the space with a given metric, and implied that the metric is somehow related to curvature. It is time to learn how to formalise these statements. In this way, we will eventually be able to derive Einstein's equations that allow one to find curvature (and metric) for a given mass distribution. All this requires knowledge of vectors and tensors in a curved space. They are necessary to formalise the General Relativity Principle – the laws of nature must be *covariant* (i.e. be of the same mathematical form) in all frames of reference.

8.1 Linear and nonlinear spaces.

In *Radiation and Relativity*, we were dealing with vectors in linear vector spaces. Our aim now is to build up vector (and tensor) analysis for curved spaces as we anticipate their importance for general relativity. As we associate vectors with directed straight-line segments, it is clear that they cannot belong to a *curved* space.¹ We shall see that vectors are naturally defined on a tangent hyperplane to a given point on the curved surface. For a $2D$ curved space it is a usual tangent plane. In many-dimensional cases, a tangent hyperplane is a linear space of the same dimensionality as the surface. It is immediately clear that vectors (and tensors) defined in this way are dependent on the point where they are attached – in contrast to vectors in linear vector spaces. Our first task is to make all necessary working definitions. In most cases, they will be identical to the definitions made in *Radiation and Relativity* and *Eigenphysics* for purely linear spaces.

8.2 Transformation rules define vectors and tensors

Notations:

For the appropriate coordinates in frame Σ' we use $x^{\mu'}$ (instead of more widespread x'^{μ}). We will be referring to x^{μ} and $x^{\mu'}$ as *old* and *new* coordinates, respectively. We assume that the new coordinates could be expressed as smooth functions of the old ones:

$$x^{\mu'} = F^{\mu'}(x^{\mu}) \equiv x^{\mu'}(x^{\mu}) \quad (8.1)$$

Above there are d equations with d functions $F^{\mu'}$ of d old variables. We use for these functions the same notations $x^{\mu'}$ as for the new coordinates. Although transformations (8.1) are, in general, nonlinear, infinitesimal displacement (i.e. differentials of position x^{μ}) transforms linearly. The chain rule gives (applying Einstein's summation agreement² and introducing notation $\partial_{\mu} \equiv \frac{\partial}{\partial x^{\mu}}$):

$$\begin{aligned} dx^{\mu'} &= \sum_{\mu=1}^d \frac{\partial x^{\mu'}}{\partial x^{\mu}} dx^{\mu} \equiv \sum_{\mu=1}^d \lambda_{\mu}^{\mu'} dx^{\mu} \equiv \lambda_{\mu}^{\mu'} dx^{\mu}, & \lambda_{\mu}^{\mu'} &\equiv \frac{\partial x^{\mu'}}{\partial x^{\mu}} \equiv \partial_{\mu} x^{\mu'} \\ \Rightarrow & \boxed{x^{\mu'} = \lambda_{\mu}^{\mu'} x^{\mu}} \end{aligned} \quad (8.2)$$

¹One can think of a curved d -dimensional space as a curved d -dimensional hypersurface embedded into a $(d+1)$ -dimensional linear space. To visualise, think of a $2D$ surface embedded in a $3D$ space

²**Reminder:** Einstein's summation agreement means summation over two repeated indices (one sub- and one super-index). Such repeated indices are called dumb since $\lambda_{\mu}^{\mu'} dx^{\mu} = \lambda_{\nu}^{\mu'} dx^{\nu}$ – the pair can be arbitrarily renamed. Non-repeated indices can only be renamed simultaneously on two sides of an equation; one can never rename primed to non-primed indices as they correspond to different frames.

Definition 8.1. Transformation matrix $\lambda_{\mu}^{\mu'} \equiv \partial_{\mu} x^{\mu'}$ is called the *Jacobi matrix*; $J = \det |\lambda_{\mu}^{\mu'}|$ is called *Jacobian*.

Definition 8.2.

Scalar (or invariant) is a quantity that remains the same in any frame.

Consider a **scalar field** which can be expressed as a function of either new or old coordinates: $\varphi = \varphi(x^{\mu}) = \varphi(x^{\mu'})$, where the coordinates are related by Eq. (8.1). Its magnitude changes from point to point in our d -dimensional space but at any given point it is the same in any frame. Its differential, $d\varphi$, also remains the same, whether expressed via x^{μ} or $x^{\mu'}$ so that

$$d\varphi(\{x^{\mu}\}) = \sum_{\mu=1}^d \frac{\partial \varphi}{\partial x^{\mu}} dx^{\mu} \equiv \partial_{\mu} \varphi dx^{\mu} = d\varphi(\{x^{\mu'}\}) = \partial_{\mu'} \varphi dx^{\mu'}.$$

Substituting here Eq. (8.2) we find the transform from the new to old coordinates:

$$\begin{aligned} \partial_{\mu} \varphi dx^{\mu} &= \partial_{\mu'} \varphi \lambda_{\mu}^{\mu'} dx^{\mu} \Leftrightarrow \sum_{\mu=1}^d \left(\partial_{\mu} \varphi - \partial_{\mu'} \varphi \lambda_{\mu}^{\mu'} \right) dx^{\mu} = 0 \\ \Rightarrow \quad \partial_{\mu} \varphi &= \lambda_{\mu}^{\mu'} \partial_{\mu'} \varphi. \end{aligned}$$

Renaming $\mu \rightleftharpoons \mu'$, which is possible since frames Σ and Σ' stand on equal footing, we find the transform from the old to new coordinates:

$$\partial_{\mu'} \varphi = \lambda_{\mu'}^{\mu} \partial_{\mu} \varphi. \quad (8.3)$$

Of course, this can be seen directly:

$$\begin{aligned} \partial_{\mu'} \varphi &\equiv \partial_{\mu'} \varphi(\{x^{\mu'}\}) = \partial_{\mu'} \varphi(\{x^{\mu}\}) \equiv \frac{\partial \varphi(\{x^{\mu}\})}{\partial x^{\mu'}} = \sum_{\mu=1}^d \frac{\partial \varphi(\{x^{\mu}\})}{\partial x^{\mu}} \frac{\partial x^{\mu}}{\partial x^{\mu'}} \\ &\equiv \partial_{\mu} \varphi \lambda_{\mu'}^{\mu}. \end{aligned}$$

By the chain rule, the transformation matrices $\lambda_{\mu}^{\mu'}$ and $\lambda_{\mu'}^{\mu}$ that enter the transformation laws for the displacement dx^{μ} , Eq. (8.2), and the gradient of a scalar $\partial_{\mu} \varphi$, Eq. (8.3), are the inverse of each other (assuming that these matrices are non-singular, i.e. the Jacobian $J \equiv \det \lambda_{\mu}^{\mu'} \neq 0$ so that the inverse does exist). Indeed, we have

$$\lambda_{\mu'}^{\mu} \lambda_{\mu}^{\mu'} \equiv \sum_{\mu'=1}^d \frac{\partial x^{\mu}}{\partial x^{\mu'}} \frac{\partial x^{\mu'}}{\partial x^{\mu}} = \frac{\partial x^{\mu}}{\partial x^{\mu}} = \delta^{\mu}_{\mu} \quad (8.4)$$

The displacement dx_{μ} and the gradient of a scalar, $\partial_{\mu} \varphi$, are vectors³ in a d -dimensional space. However, even in a linear Euclidean space their transformations between frames, Eq. (8.2) and Eq. (8.3), are totally different (unless we restrict considerations to Cartesian coordinates). These transformation laws define two types of vectors (and distinguishing them is crucial in *Relativity*).

³Strictly speaking they are vector components, but in Relativity they are referred to as vectors.

Definition 8.3.

A set of d functions, $\xi^\mu \equiv \{\xi^1 \dots \xi^d\}$ (or $\xi_\mu \equiv \{\xi_1 \dots \xi_d\}$) is called a contravariant (or, respectively, a covariant) vector if it transforms between frames as dx^μ (or $\partial_\mu \varphi$)

In general, ξ^μ and ξ_μ are the fields,⁴ i.e. they depend on the point in space. For brevity (and following traditions), we will often call contravariant vectors simply vectors, and covariant – covectors.

These are general definitions. We shall be mainly concerned with 4 dimensional space time when indices μ, ν , etc take values 0, 1, 2, 3. In this case, we shall also imply a convention $\mu \equiv (0, m)$, with $m = 1, 2, 3$ being the spatial indices. Sometimes, and especially in this introductory part, we shall use μ for arbitrary spaces of any dimensions.

8.2.1 Direct vector products and tensors

A direct product of two vectors (that is totally different from the scalar product defined in the next section) is defined straightforwardly:

Definition 8.4. A direct product of two vectors $\xi^\mu \zeta^\nu$ transforms between frames according to Eqs. (8.2)

Naturally, such a product has d^2 components. This can be straightforwardly generalised to a direct product of any number of vectors and covectors. The transformation rules for such products directly follow from Eqs. (8.2) and 8.3. For example,

$$\xi^{\mu'} \zeta^{\nu'} = \lambda_{\mu}^{\mu'} \lambda_{\nu}^{\nu'} \xi^{\mu} \zeta^{\nu}; \quad \xi^{\mu'} \zeta_{\nu'} = \lambda_{\mu}^{\mu'} \lambda_{\nu'}^{\nu} \xi^{\mu} \zeta_{\nu}.$$

This leads us to the mathematical concept of tensor central which is central in *Relativity*:

Definition 8.5.

A set of functions $\xi^{\mu_1 \dots \mu_m}$ is called contravariant tensor (of rank m) when it transforms as a direct product of m vectors; covariant tensor, $\xi_{\nu_1 \dots \nu_n}$, transforms as a direct product of n covectors; mixed tensor, $\xi^{\mu_1 \dots \mu_m}_{\nu_1 \dots \nu_n}$, transforms as a direct product of m vectors and n covectors.

Clearly, a tensor of rank n has n^d components in a d -dimensional space.⁵ In Special Relativity, you have already dealt with tensors such as the electromagnetic field tensor, $F_{\mu\nu}$, or the energy-momentum tensor, $T_{\mu\nu}$, but considered only the Lorentz transforms of them between different inertial frames, these transforms being a particular case of general transformation rules of Eq. (8.2).

8.2.2 Tensor contraction

According to the Einstein's summation agreement, the product $\xi^\mu \zeta_\mu$ is just a number. It is easy to see that this number is invariant (frame-independent), i.e. it is a scalar:

$$\xi^{\mu'} \zeta_{\mu'} = \lambda_{\mu}^{\mu'} \xi^{\mu} \lambda_{\mu'}^{\nu} \zeta_{\nu} = \underbrace{\lambda_{\mu'}^{\nu} \lambda_{\mu}^{\mu'}}_{=\delta^{\nu}_{\mu}} \xi^{\mu} \zeta_{\nu} = \delta^{\nu}_{\mu} \xi^{\mu} \zeta_{\nu} = \xi^{\mu} \zeta_{\mu}. \quad (8.5)$$

⁴Although I use the same letter, ξ , for both the covariant and contravariant vector, they become related only after a metric is introduced. Mathematically, one can consider abstract vector spaces not supplied with the metric.

⁵While a 2nd rank tensor can be represented as a square matrix, there is no useful generalisation of such a representation for higher ranks: the index notations is all we have in this course.

If we choose an upper and a lower indices to coincide in a mixed tensor, again according to the Einstein's summation agreement we need to sum over all values, resulting in a tensor with the rank lower by 2 than the original one.

$$\xi^{\mu\nu\alpha}{}_{\beta\alpha} \rightarrow \xi^{\mu\nu}{}_{\beta}.$$

Such an operation is called tensor contraction. For example, $\delta^\mu{}_\mu = \delta^1_1 + \delta^2_2 + \dots + \delta^d_d = d$ in a d -dimensional case.

The contraction of contra- and co-variant vectors, Eq. (8.5), looks like the scalar product of two vectors. It does become the scalar product in the metric space as we will see below, in section (9).

8.3 Linear vector spaces as hyperplanes in a curved space

In elementary vector analysis (including *Maths2* and *Eigenphysics* in Year 2), we have defined a vector space as set of objects obeying the linearity axioms that can be summarised as follows:⁶

Definition 8.6. V is a vector space if for any $\xi, \zeta \in V$ a linear combination $a\xi + b\zeta \in V$.

We need a few steps to see how this is related to Definition (8.3) adopted in Relativity. First, as the spacetime is nonlinear we need to see how to build relevant linear spaces. It can be done *locally*, at any point of a *curved* space:

Definition 8.7.

A linear vector space associated with any point of the underlying space is the 'hyperplane' tangent to this point, with vectors locally defined on this hyperplane.

Only in the (pseudo)Euclidean case, such a hyperplane coincides with the underlying space itself so that vectors belong to this space.

To define such a tangent hyperplane, we start with the vector tangent to a curve. A curve in a d -dimensional space can be represented as a set of d parametric equations, $x^\mu = x^\mu(s)$ (e.g., $x^1 \equiv x = \cos s$, $x^2 \equiv y = \sin s$, and $x^3 \equiv z = 0$ define a circle in 3D space). Then for some scalar field along the curve, we define derivative $d\varphi(x^\mu(s))/ds$ which is also a scalar. The chain rule gives

$$\frac{d\varphi(x^\mu(s))}{ds} = \frac{\partial\varphi}{\partial x^\mu} \frac{dx^\mu}{ds} \equiv \partial_\mu \varphi \dot{x}^\mu. \quad (8.6)$$

Note that here $\partial_\mu \varphi$ does not depend on the curve while $\dot{x}^\mu$ does not depend on the field φ . This illustrates a difference in geometrical meaning of contra- and co-variant vectors: $\frac{dx^\mu}{ds}$ is the tangent vector to a given curve, while $\partial_\mu \varphi$ is a rate of change of the field in the x^μ direction.

If we draw all possible curves through a given point, the appropriate tangent vectors make a linear vector space – the hyperplane attached to this point. Its dimensionality is the same as that of the underlying curved space. All tensors are constructions in this linear vector space attached to a given point. As this linear space changes from point to point, each of the d vector components in general depends on the point to which the vector is attached.

In $2D$ space, the hyperplane is a usual plane uniquely defined by **any** two linearly-independent, i.e. non-parallel, vectors tangent to **different** smooth curves crossing on the curved surface at the

⁶In the general definition, $a, b \in \mathbb{F}$, where $\mathbb{F}$ is some field, e.g., that of real, $\mathbb{R}$, or complex, $\mathbb{C}$, numbers. In the context of relativity, we need only $a, b \in \mathbb{R}$.

point considered. For a d -dimensional space, the tangent hyperplane is defined by any d (mutually non-parallel) vectors, tangent to different smooth curves crossing at this point. As in elementary vector analysis, where vector by itself does not depend on a choice of coordinates, any set of such LI vectors can be used as a basis to represent any given vector, ξ . Writing ξ in two different coordinate sets (i.e. in two different frames), we have (implying Einstein's summation agreement):

$$\xi = \xi^\mu e_\mu = \xi^{\mu'} e_{\mu'}, \quad (8.7)$$

where e_μ and $e_{\mu'}$ are sets of basic vectors⁷ in Σ and Σ' .

We want ξ^μ to transform as covariant vectors (more precisely – vector components), i.e. according to Eq. (8.2). Hence to make the vector ξ independent of the choice of coordinates like the scalar $\frac{d\phi}{ds}$ in Eq. (8.6), basis vectors, e_μ , must transform as gradients, $\partial_\mu \varphi$, Eq. (8.3).

In elementary vector analysis, basis vectors usually are normalised, i.e. all of unit length. In modern geometry (and in the theory of relativity), the transformation rules of Eqs. (8.2) and (8.3) are defined in the most simple (covariant) way which makes the basis vectors, in general, non-unit. The set of basis vectors must be **complete**, but we formalise this requirement later.

⁷Reminder: to simplify notations, I use $\xi^{\mu'}$ and $e_{\mu'}$ for vector components and basis vectors in Σ' , instead of ξ'^{μ} and e'_{μ} , as in the most of textbooks.

Lecture 9. Tensors in metric spaces

We have defined co- and contra-variant vector components independently of each other; they become related only in metric spaces. After introducing the Riemannian metric we will express the metric tensor via basis vectors and relate it to Jacobi transformation matrices.

9.1 Riemann metric space

Definition 9.1. Space is called Riemann when the metric corresponding to the infinitesimal interval is defined as

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu$$

where the metric tensor $g_{\mu\nu}$ is symmetric, $g_{\mu\nu} = g_{\nu\mu}$, and $\det g_{\mu\nu} \neq 0$.

As ds^2 is invariant and $dx^\mu dx^\nu$ is the 2nd rank contravariant tensor (by definition), then, indeed, $g_{\mu\nu}$ must be a covariant tensor. In general, $g_{\mu\nu}$ changes with coordinates. At any given point, it is just a symmetric matrix. Hence, the metric tensor can always be **locally** diagonalised and then rescaled so that each diagonal element is ± 1 . Such a diagonal metric tensor,

$$g_{\bar{\mu}\bar{\nu}} \equiv \eta_{\bar{\mu}\bar{\nu}} = \text{diag}\{s_\mu\} \equiv \text{diag}(s_1, \dots, s_d) \text{ where each } s_\mu = 1 \text{ or } -1$$

defines the (pseudo)Cartesian basis,¹ $\bar{\Sigma}$, in the linear (Euclidean or pseudo-Euclidean) space, which is the tangent hyperplane to this point.

We use the 'barred' symbols for such a (pseudo)Cartesian basis. In an arbitrary frame Σ centred at the same point the metric tensor can be **locally** expressed via $\eta_{\bar{\mu}\bar{\nu}}$ and the transformation matrices $\lambda_{\mu}^{\bar{\mu}}$ using the tensor transformation law:

$$g_{\mu\nu} = \lambda_{\mu}^{\bar{\mu}} \eta_{\bar{\mu}\bar{\nu}} \lambda_{\nu}^{\bar{\nu}} \quad (9.1)$$

The same transformation matrices relate the basis vectors in Σ to those in $\bar{\Sigma}$:

$$\mathbf{e}_\mu = \lambda_{\mu}^{\bar{\mu}} \mathbf{e}_{\bar{\mu}}, \quad (9.2a)$$

so that $\lambda_{\mu}^{\bar{\mu}}$ are just the components of the 'new' basis $\{\mathbf{e}_{\bar{\mu}}\}$ in the (pseudo)Cartesian basis $\mathbf{e}_{\bar{\mu}}$. As all the basis vectors are defined in the linear (tangent) space, we can use the standard (pseudo)Euclidean scalar product. Noticing that by definition of the (pseudo)Cartesian basis

$$\langle \mathbf{e}_{\bar{\mu}} \cdot \mathbf{e}_{\bar{\nu}} \rangle = \eta_{\bar{\mu}\bar{\nu}}, \quad (9.2b)$$

we find that the (pseudo)Euclidean scalar product, $\langle d\mathbf{x} \cdot d\mathbf{x} \rangle$, coincides with the metric,

$$\langle d\mathbf{x} \cdot d\mathbf{x} \rangle = dx^{\bar{\mu}} dx^{\bar{\nu}} \langle \mathbf{e}_{\bar{\mu}} \cdot \mathbf{e}_{\bar{\nu}} \rangle = dx^{\bar{\mu}} dx^{\bar{\nu}} \eta_{\bar{\mu}\bar{\nu}} \equiv ds^2,$$

we generalise this to any basis and expand each $d\mathbf{x}$ in a linear (tangent) space using Eq. (8.7), in this case $d\mathbf{x} = \mathbf{e}_\mu dx^\mu$:

$$ds^2 = \langle d\mathbf{x} \cdot d\mathbf{x} \rangle = dx^\mu dx^\nu \langle \mathbf{e}_\mu \cdot \mathbf{e}_\nu \rangle = g_{\mu\nu} dx^\mu dx^\nu.$$

¹In the context of GR, i.e. in the 4D spacetime, such a pseudo-Cartesian basis is called Lorentzian; there $s_0 = 1$ and $s_m = -1$, $m = 1, 2, 3$.

Hence we express the metric tensor $g_{\mu\nu}$ in a given basis as the scalar product of basis vectors; using Eqs. (9.2), we see that it coincides with Eq. (9.1):

$$g_{\mu\nu} = \langle \mathbf{e}_\mu \cdot \mathbf{e}_\nu \rangle = \eta_{\bar{\mu}\bar{\nu}} \lambda_{\mu}^{\bar{\mu}} \lambda_{\nu}^{\bar{\nu}}. \quad (9.3)$$

Now we can relate the scalar product and the tensor contraction, Eq. (8.5):

$$\langle \xi \cdot \zeta \rangle = \langle \xi^\mu \mathbf{e}_\mu \cdot \zeta^\nu \mathbf{e}_\nu \rangle = \xi^\mu \zeta^\nu \langle \mathbf{e}_\mu \cdot \mathbf{e}_\nu \rangle = \xi^\mu \zeta^\nu g_{\mu\nu} \equiv \xi^\mu \zeta_\mu, \quad (9.4)$$

where we have used the following

Definition 9.2. In any frame, covariant components, ζ_μ of vector ζ are expressed via contravariant, ζ^ν , with the help of metric tensor:

$$\zeta_\mu = g_{\mu\nu} \zeta^\nu. \quad (9.5)$$

Hence the metric tensor is lowering the index. We can use it for an arbitrary tensor, writing, e.g.

$$\xi_{\alpha\beta} \gamma^\delta = g_{\alpha\mu} \xi^\mu{}_\beta \gamma^\delta.$$

The inverse tensor, $g^{\mu\nu}$ is defined by

$$g_{\mu\nu} g^{\nu\lambda} = g^{\lambda\nu} g_{\nu\mu} = \delta^\lambda_\mu,$$

and can be used for raising the index:

$$\xi^\alpha = g^{\alpha\beta} \xi_\beta.$$

9.2 Metric tensor in curvilinear coordinates

The most well-known examples of non-Cartesian coordinates are polar coordinates in 2D and 3D.

9.2.1 Plane polar coordinates

Consider the 2D transformation from Cartesian coordinates, $x^{\bar{\mu}} \equiv (x, y)$, to polar ones, $x^\mu = (r, \phi)$. As $x = r \cos\phi$, $y = r \sin\phi$, we have:

$$\lambda_{\mu}^{\bar{\mu}} \equiv \partial_{\mu} x^{\bar{\mu}} = \begin{pmatrix} \lambda_r^x & \lambda_\phi^x \\ \lambda_r^y & \lambda_\phi^y \end{pmatrix} = \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \phi} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \phi} \end{pmatrix} = \begin{pmatrix} \cos\phi & -r \sin\phi \\ \sin\phi & r \cos\phi \end{pmatrix}$$

Basis vectors in the polar coordinates are obtained from Eq. (9.2), $\mathbf{e}_\mu = \lambda_{\mu}^{\bar{\mu}} \mathbf{e}_{\bar{\mu}}$; hence in this case

$$\mathbf{e}_r = \lambda_r^x \mathbf{e}_x + \lambda_r^y \mathbf{e}_y = \cos\phi \mathbf{e}_x + \sin\phi \mathbf{e}_y; \quad \mathbf{e}_\phi = \lambda_\phi^x \mathbf{e}_x + \lambda_\phi^y \mathbf{e}_y = -r \sin\phi \mathbf{e}_x + r \cos\phi \mathbf{e}_y.$$

Scalar products of all pairs of two vectors, $\mathbf{e}_r$ and $\mathbf{e}_\phi$, in the Cartesian coordinates define all the elements of the metric tensor in new, polar coordinates (using $\mathbf{e}_x^2 = \mathbf{e}_y^2 = 1$ and $\langle \mathbf{e}_x \cdot \mathbf{e}_y \rangle = 0$):

$$\begin{aligned} g_{rr} &= \langle \mathbf{e}_r \cdot \mathbf{e}_r \rangle = \cos^2\phi + \sin^2\phi = 1, \\ g_{r\phi} &= g_{\phi r} = \langle \mathbf{e}_r \cdot \mathbf{e}_\phi \rangle = -r \cos\phi \sin\phi + \sin\phi r \cos\phi = 0, \end{aligned}$$

$$g_{\phi\phi} = \langle \mathbf{e}_\phi \cdot \mathbf{e}_\phi \rangle = r^2 \sin^2 \phi + r^2 \cos^2 \phi = r^2$$

Hence $g_{\mu\nu} = \text{diag}(1, r^2)$. In curvilinear coordinates, off-diagonal elements of $g_{\mu\nu}$ always vanish while diagonal ones are given by the squares of length of basis vectors. **It is extremely important that they are not unit and, in general, are not constant.**

One can use directly Eq. (9.3), with $\eta_{\bar{\mu}\bar{\nu}} = \mathbb{1}$, so that $g_{\mu\nu} = \sum_{\bar{\mu}} \lambda_{\mu}^{\bar{\mu}} \lambda_{\nu}^{\bar{\mu}}$. This is not a matrix product as we add products of matrix elements from the same row. However, you don't need to use matrices; the sum is unambiguous and all elements there are known. Naturally, the results are the same:

$$\begin{aligned} g_{rr} &= \lambda_r^x \lambda_r^x + \lambda_r^y \lambda_r^y = \cos^2 \phi + \sin^2 \phi = 1, \\ g_{\phi\phi} &= \lambda_\phi^x \lambda_\phi^x + \lambda_\phi^y \lambda_\phi^y = r^2 \sin^2 \phi + r^2 \cos^2 \phi = r^2. \end{aligned}$$

Let us look in detail how vector components change in the transformation from Euclidean to polar coordinates. We call $\mathbf{i} \equiv \mathbf{e}_x$, $\mathbf{j} \equiv \mathbf{e}_y$. Let $\mathbf{r} = x\mathbf{i} + y\mathbf{j}$. We know how to transform (dx, dy) rather than (x, y) . More convenient to write the transformation for the velocity vector $v^\mu = \dot{x}^\mu \equiv dx^\mu/d\tau$ where we use “time” τ as parameter characterising some curve $x^\mu(\tau)$:

$$\dot{x}^{\bar{\mu}} = \lambda_{\mu}^{\bar{\mu}} \dot{x}^\mu = \begin{pmatrix} \cos \phi & -r \sin \phi \\ \sin \phi & r \cos \phi \end{pmatrix} \begin{pmatrix} \dot{r} \\ \dot{\phi} \end{pmatrix} = \begin{pmatrix} \dot{r} \cos \phi - \dot{\phi} r \sin \phi \\ \dot{r} \sin \phi + \dot{\phi} r \cos \phi \end{pmatrix}$$

Let us then consider the **vector** of velocity instead of its components:

$$\begin{aligned} \mathbf{v} &\equiv \frac{dx^{\bar{\mu}}}{d\tau} \mathbf{e}_{\bar{\mu}} \equiv (\dot{r} \cos \phi - \dot{\phi} r \sin \phi) \mathbf{i} + (\dot{r} \sin \phi + \dot{\phi} r \cos \phi) \mathbf{j} \\ &= \dot{r}(\cos \phi \mathbf{i} + \sin \phi \mathbf{j}) + \dot{\phi}(-r \sin \phi \mathbf{i} + r \cos \phi \mathbf{j}) \\ &= \dot{r} \mathbf{e}_r + \dot{\phi} \mathbf{e}_\phi. \end{aligned} \tag{9.6}$$

Therefore, our transformation formulae are consistent: contravariant components transform in such a way as to guarantee that the vector remains the same in new coordinates: we can equally write $\mathbf{v} = \dot{x}\mathbf{i} + \dot{y}\mathbf{j}$ or $\mathbf{v} = \dot{r} \mathbf{e}_r + \dot{\phi} \mathbf{e}_\phi$.

However, not only $\dot{\mathbf{r}}$ but also both $\mathbf{r}$ and $\ddot{\mathbf{r}}$ are ‘vectors’ in the ‘usual’ geometric sense. But not from the viewpoint of the coordinate transformations! Indeed, $\mathbf{r} = r \mathbf{e}_r$ rather than $r \mathbf{e}_r + \phi \mathbf{e}_\phi$, and the expression $\ddot{\mathbf{r}} = \ddot{r} \mathbf{e}_r + \ddot{\phi} \mathbf{e}_\phi$ is also wrong.

One can easily find $\ddot{\mathbf{r}}$ in the polar coordinates by direct differentiation of the last line of Eq. (9.6):

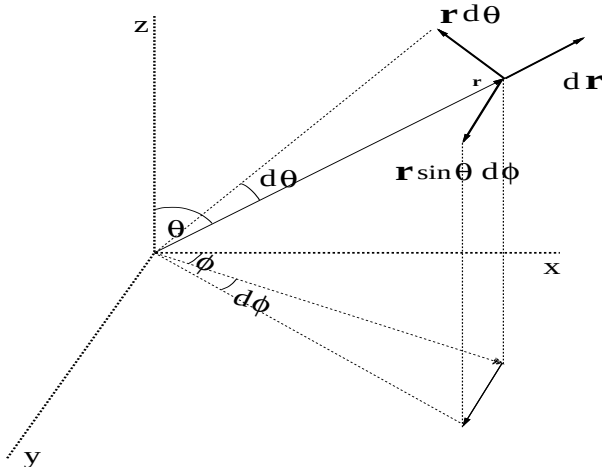
$$\begin{aligned} \ddot{\mathbf{r}} &= \ddot{r} \mathbf{e}_r + \ddot{\phi} \mathbf{e}_\phi + \dot{r} \dot{\mathbf{e}}_r + \dot{\phi} \dot{\mathbf{e}}_\phi \quad \text{where} \\ \dot{\mathbf{e}}_r &= \dot{\phi}(-\sin \phi \mathbf{i} + \cos \phi \mathbf{j}) = \frac{\dot{\phi}}{r} \mathbf{e}_\phi \\ \dot{\mathbf{e}}_\phi &= \dot{r}(-\sin \phi \mathbf{i} + \cos \phi \mathbf{j}) - r \dot{\phi}(\cos \phi \mathbf{i} + \sin \phi \mathbf{j}) = \frac{\dot{r}}{r} \mathbf{e}_\phi - r \dot{\phi} \mathbf{e}_r \\ \therefore \quad \ddot{\mathbf{r}} &= (\ddot{r} - r \dot{\phi}^2) \mathbf{e}_r + \left(\ddot{\phi} + 2 \frac{\dot{\phi} \dot{r}}{r} \right) \mathbf{e}_\phi. \end{aligned} \tag{9.7}$$

The lesson from here is that differentiation (and integration) in curvilinear coordinates is quite nontrivial, and we will need to learn how to make it *covariant*.

9.2.2 Spherical polar coordinates.

The transformation from $\{x^{\bar{\mu}}\} \equiv \{x, y, z\}$ to $\{x^\mu\} \equiv \{r, \theta, \phi\}$ is:

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta$$



The metric in both sets of coordinates could be found geometrically, as ds^2 is the distance between two close points, and for small displacements one can still use the Pythagoras theorem. In spherical coordinates (see the picture) $ds^2 = dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2$. Thus, $g_{ij} = \text{diag}(1, r^2, r^2 \sin^2 \theta)$. Similarly, in cylindrical coordinates $ds^2 = dz^2 + dr^2 + r^2 d\phi^2$ so that $g_{ij} = \text{diag}(1, 1, r^2)$.

We find the Jacobi matrix as follows:

$$\lambda_{\bar{\mu}}^{\bar{\mu}} \equiv \partial_{\bar{\mu}} x^{\bar{\mu}} \equiv \begin{pmatrix} \lambda_r^x & \lambda_\theta^x & \lambda_\phi^x \\ \lambda_r^y & \lambda_\theta^y & \lambda_\phi^y \\ \lambda_r^z & \lambda_\theta^z & \lambda_\phi^z \end{pmatrix} = \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \phi} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \phi} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial \phi} \end{pmatrix} = \begin{pmatrix} \sin \theta \cos \phi & r \cos \theta \cos \phi & -r \sin \theta \sin \phi \\ \sin \theta \sin \phi & r \cos \theta \sin \phi & r \sin \theta \cos \phi \\ \cos \theta & -r \sin \theta & 0 \end{pmatrix}$$

You can check that the Jacobian is $(r^2 \sin \theta)$ as should be. Now the elements of $g_{\mu\nu}$:

$$\begin{aligned} g_{rr} &= \langle \mathbf{e}_r \cdot \mathbf{e}_r \rangle = \sum_{\bar{\mu}} (\lambda_{\bar{\mu}}^r)^2 = \sin^2 \theta (\cos^2 \phi + \sin^2 \phi) + \cos^2 \theta = 1, \\ g_{\theta\theta} &= \langle \mathbf{e}_\theta \cdot \mathbf{e}_\theta \rangle = \sum_{\bar{\mu}} (\lambda_{\bar{\mu}}^\theta)^2 = r^2 \cos^2 \theta (\cos^2 \phi + \sin^2 \phi) + r^2 \sin^2 \theta = r^2, \\ g_{\phi\phi} &= \langle \mathbf{e}_\phi \cdot \mathbf{e}_\phi \rangle = \sum_{\bar{\mu}} (\lambda_{\bar{\mu}}^\phi)^2 = r^2 \sin^2 \theta (\sin^2 \phi + \cos^2 \phi) = r^2 \sin^2 \theta, \\ g_{r\theta} &= g_{\theta r} = \langle \mathbf{e}_r \cdot \mathbf{e}_\theta \rangle = \sum_{\bar{\mu}} \lambda_{\bar{\mu}}^r \lambda_{\bar{\mu}}^\theta = r \sin \theta \cos \theta [(\cos^2 \phi + \sin^2 \phi) - 1] = 0, \\ g_{r\phi} &= g_{\phi r} = \langle \mathbf{e}_r \cdot \mathbf{e}_\phi \rangle = \sum_{\bar{\mu}} \lambda_{\bar{\mu}}^r \lambda_{\bar{\mu}}^\phi = r \sin^2 \theta (-\cos \phi \sin \phi + \sin \phi \cos \phi) = 0, \\ g_{\theta\phi} &= g_{\phi\theta} = \langle \mathbf{e}_\theta \cdot \mathbf{e}_\phi \rangle = \sum_{\bar{\mu}} \lambda_{\bar{\mu}}^\theta \lambda_{\bar{\mu}}^\phi = r \sin \theta \cos \theta \cos \phi \sin \phi (-1 + 1) = 0; \\ \therefore \quad g_{\mu\nu} &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2 \sin^2 \theta \end{pmatrix} \equiv \text{diag}(1, r^2, r^2 \sin^2 \theta). \end{aligned}$$

To find g_{ij} geometrically is much easier (see the figure). Brute force would also do: $ds^2 = [d(r \sin \theta \cos \phi)]^2 + [d(r \sin \theta \sin \phi)]^2 + [d(r \cos \theta)]^2 = dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2$, without using the general formalism adopted above.

9.2.3 Curved 2D surfaces

Up to now, all our examples referred to a 3D Euclidean space where we elect to use non-Cartesian coordinates. Let us fix the radius, $r \rightarrow r_0$, either in spherical polar or cylindrical coordinates. Thus we arrive at a 2D *curved* space - a spherical or cylindrical surface.

The 2D metric on the spherical surface of radius r_0 is $ds^2 = r_0^2(d\theta^2 + \sin^2\theta d\phi^2)$. The appropriate metric tensor is $g_{ab} = \text{diag}(r_0^2, r_0^2 \sin^2\theta)$. Its determinant $g = r_0^2 \sin^2\theta$ which is nonzero everywhere but at $\theta = 0$. At this singular point, the inverse metric tensor g^{ab} does not exist.

It is typical for any curved space: one cannot choose coordinates which are non-singular everywhere.

However, a singularity could be just a property of a particular metric rather than that of the space.

Coordinates θ and ϕ (colatitude and longitude on a sphere) are non-Cartesian. Still, at any point one can introduce Cartesian coordinates locally. (Obviously, $\theta - \phi$ coordinates become Cartesian at $\theta = \pi/2$; thus we could introduce such coordinates at any point choosing it to be at the equator of the sphere).

The metric tensor on a cylindrical surface corresponding to the cylindrical coordinates in 3D space, $r = r_0, z, \phi$ is Euclidean *everywhere*: $g_{ab} = \text{diag}(1, r_0^2)$, where indices are either z or ϕ . By changing units of length along ϕ (i.e. choosing $r = 1$) we come to a standard unit metric. It tells us immediately that cylindrical surface is intrinsically flat (this is clear as it could be made from a sheet of paper without shrinks). Its metric properties are the same as of a flat sheet, and it differs from it only globally (topologically).

9.3 Pseudo-Euclidean and curved Riemannian spaces

9.3.1 Pseudo-Euclidean space

This is a flat space with an indefinite metric: $ds^2 = g_{\mu\nu}x^\mu x^\nu$ where $g_{\mu\nu} = \text{diag}(\pm 1)$. It is often called a $(n + m)D$ space, where n is the number of $+1$, and m is the number of -1 in $g_{\mu\nu}$. *Note that an overall change of sign of a metric tensor is irrelevant.* Thus, Minkowski space is a $(1 + 3)D$ flat space. Consider a $(1 + 2)D$ space: $ds^2 = dt^2 - dx^2 - dy^2$. We can introduce new coordinates

$$\left. \begin{array}{l} t = r \cosh \chi \\ x = r \sinh \chi \\ y = y \end{array} \right\} \text{pseudo-cylindrical :} \quad t^2 - x^2 = r^2 \quad \text{and} \quad \left. \begin{array}{l} t = r \cosh \chi \\ x = r \sinh \chi \cos \phi \\ y = r \sinh \chi \sin \phi \end{array} \right\} \text{pseudo-spherical :} \quad t^2 - x^2 - y^2 = r^2$$

The metric in both cases can be found by a brute force, i.e. substituting dt etc, into the Cartesian expression for ds^2 to get $ds^2 = dr^2 - r^2 d\chi^2$ for pseudo-polar coordinates, and $ds^2 = dr^2 - r^2 d\chi^2 + \sinh^2 \chi d\phi^2$ for pseudo-spherical coordinates.

The surface given by $t^2 - x^2 - y^2 = R^2$ is called a pseudo-sphere. (Its shape is independent of the metric, and in a usual 3D space - as well as in $(1 + 2)D$ space has a shape of a two-cavity hyperboloid). In the pseudo-spherical coordinates its equation is simply $r = R$, and any χ and ϕ - similarly to the equation of a sphere in the spherical coordinates. As $r = \text{const}$ on a pseudo-spherical surface, its metric is $ds^2 = -r^2(d\chi^2 + \sinh^2 \chi d\phi^2)$. Its always negative, so that vectors tangent to this surface are spacelike. (A pseudo-sphere is called sometimes the Lobachevskii plane, as geometry on this surface is equivalent to a non-Euclidean geometry introduced by Lobachevskii).

9.3.2 Curved Riemannian space

In general, the metric $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$ (with $g_{\mu\nu}$ being a function of coordinates and $\det\{g_{\mu\nu}\} \neq 0$ almost everywhere) defines a curved Riemannian space. Space may be intrinsically curved without any regards to its embedding into a higher-dimensional space. Even when a metric tensor is singular somewhere (e.g., at the origin of polar coordinates), the space could be flat. In the close neighbourhood of any non-singular point, $g_{\mu\nu}$ is almost a constant. Thus, as this matrix is symmetric, it can be

diagonalised with all eigenvalues being real, and reduced after rescaling to $\text{diag}(\pm 1, \dots, \pm 1)$. (The metric signature equal to $|n - m|$ remains the same in the entire space.) Such a diagonal metric defines locally (pseudo) Cartesian coordinates.

Note that it is not easy to judge whether the space is flat or curved just by looking at the metric tensor. For example, of the spaces with the following three metrics,

$$(du^2 + dv^2)e^{u-v}, \quad \frac{dr^2}{1-r^2} + r^2 d\phi^2, \quad dx^2 + e^{-2x} d\phi^2,$$

the first one is flat, the second has constant positive and the third constant negative curvature (we give exact meaning to this later).

Although we can not always easily decide whether a given metric tensor corresponds to flat or curved space, there is one exception: when all the elements of $g_{\mu\nu}$ are spatially independent, it describes a flat space as it can be diagonalised by a global transformation. The only question is a “signature” of this space, *i.e.* the number of positive and negative elements in the metric tensor. Consider, *e.g.*, $ds^2 = 2 du dv$. The diagonalization of $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ gives $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ so that this metric belongs to a pseudo-Euclidean space. We may find out whether a 2D space is Euclidean or pseudo-Euclidean without an explicit diagonalization: $\det g_{\mu\nu}$ is invariant under coordinate transformation; therefore, if its sign is negative, the space is pseudo-Euclidean. Obviously, for $D > 2$ the metric signature is not defined by $\det g_{\mu\nu}$.

Lecture 10. Covariant Differentiation and Parallel Transfer

We have seen in the previous lecture that straightforwardly differentiating the velocity vector gives acceleration whose components are not transformed as a contra- or co-variant vector. So differentiation is not straightforward, while fundamental laws of nature are typically represented by differential equations. In order to make these laws consistent with the general relativity principle, differentiation must be covariant. Now it is time to learn how to write it in the form valid in any frame.

10.1 Covariant derivatives of basis vectors. Christoffel Symbols.

A usual differentiation involves neighbouring points; hence *even when the vector field is constant* its components change from one point to another when basis vectors change. For example, a constant field in the 2D Euclidean space, e.g., $\mathbf{a} = \mathbf{i}$ in Cartesian coordinates, is written in polar coordinates r, θ as $\mathbf{a} = \cos \theta \mathbf{e}_r - (\sin \theta / r) \mathbf{e}_\theta$. Of course, $\frac{d\mathbf{a}}{dt} = 0$ in any coordinates, but to get this result correctly in polar coordinates we need to differentiate not only its components but also the basis vectors. We have seen in section (9.2.1) that the basis vectors (with renaming $\phi \rightarrow \theta$) are

$$\mathbf{e}_r = \lambda_r^x \mathbf{e}_x + \lambda_r^y \mathbf{e}_y = \cos \theta \mathbf{e}_x + \sin \theta \mathbf{e}_y; \quad \mathbf{e}_\theta = \lambda_\theta^x \mathbf{e}_x + \lambda_\theta^y \mathbf{e}_y = -r \sin \theta \mathbf{e}_x + r \cos \theta \mathbf{e}_y.$$

Hence,

$$\begin{aligned} \dot{\mathbf{e}}_r &= \dot{\theta} (-\sin \theta \mathbf{e}_x + \cos \theta \mathbf{e}_y) = \frac{\dot{\theta}}{r} \mathbf{e}_\theta, \\ \dot{\mathbf{e}}_\theta &= \dot{r} (-\sin \theta \mathbf{e}_x + \cos \theta \mathbf{e}_y) - r \dot{\theta} (\cos \theta \mathbf{e}_x + \sin \theta \mathbf{e}_y) = \frac{\dot{r}}{r} \mathbf{e}_\theta - r \dot{\theta} \mathbf{e}_r. \end{aligned} \quad (10.1)$$

An important lesson here is that $\dot{\mathbf{e}}_\mu$ is spanned by the set of $\{\mathbf{e}_\mu\}$. Coefficients are in general complicated, and always include coordinate velocities $\dot{x}^\mu$. The necessity to differentiate basis vectors explains why $\partial_\mu \xi_\nu$ is **not** a 2nd rank tensor. One may define how to transfer a vector from point to point before defining covariant differentiation. We will describe a parallel transfer later while using here an equivalent approach to introducing covariant differentiation: we define how basis vectors in any coordinate system changes in differentiation.

Let us consider a derivative $d\xi/ds$ which describes the rate of change of the vector field $\xi(x^\mu)$ along the curve $x^\mu = x^\mu(s)$ parameterized with the *affine* parameter s . By definition, both $\xi \equiv \xi^\mu \mathbf{e}_\mu$ and its rate of change along the curve ('velocity') are invariant with respect to the change of coordinates. We want to find **contravariant components** of velocity $\dot{\xi}$. When this system is non-Cartesian, basis vectors are coordinate-dependent as in the above example. Therefore, we should differentiate along the curve both ξ^μ and $\mathbf{e}_\mu$:

$$\frac{d\xi}{ds} = \frac{\partial(\xi^\mu \mathbf{e}_\mu)}{\partial x^\beta} \frac{dx^\beta}{ds} = (\partial_\beta \xi^\mu) \dot{x}^\beta \mathbf{e}_\mu + (\partial_\beta \mathbf{e}_\mu) \xi^\mu \dot{x}^\beta \equiv \xi^\mu_{,\beta} \dot{x}^\beta \mathbf{e}_\mu + \mathbf{e}_{\mu,\beta} \xi^\mu \dot{x}^\beta. \quad (10.2)$$

Here I have introduced yet another short notation for the partial derivative w.r.t. x^β :

$$\xi^\mu_{,\beta} \equiv \partial_\beta \xi^\mu \quad \text{and} \quad \mathbf{e}_{\mu,\beta} \equiv \partial_\beta \mathbf{e}_\mu \quad (\text{and similarly } \theta_{,\beta} \equiv \partial_\beta \theta), \quad (10.3)$$

i.e. any index (say β) sitting after the comma means differentiating w.r.t. the appropriate component, say x^β . The convenience of this new notation will become clear very soon.

For any given μ and β , the quantity $\mathbf{e}_{\mu,\beta}$ in Eqs. (10.2) and (10.3) is a vector in a linear space with basis set $\{\mathbf{e}_\alpha\}$. It can be spanned over this set with the coefficients depending on three indices:

$$\mathbf{e}_{\mu,\beta} = \Gamma_{\mu\beta}^\alpha \mathbf{e}_\alpha \quad (10.4)$$

Definition 10.1. The coefficients $\Gamma_{\alpha\beta}^\mu$ are called the *Christoffel symbols*, or alternatively *coefficients of affine connection*.

Let me stress: in spite of having upper and lower indices, $\Gamma_{\mu\nu}^\alpha$ is not a tensor. This set of symbols is specific for a chosen frame (coordinate system), and **does not** transform between frames with the help of transformation matrices $\lambda_{\mu'}^\mu$. In particular, in (pseudo)Cartesian coordinates (which we will call *Locally Cartesian Frames* – LCF – when used in a curved space), where $\{\mathbf{e}_\mu\}$ are all constant, all Γ 's vanish. Had it been a tensor, its components would be zero in any other frame according to the tensor transformation laws.

Note that as $\dot{\mathbf{e}}_\mu = \sum_\beta \frac{\partial \mathbf{e}_\mu}{\partial x^\beta} \dot{x}^\beta \equiv \mathbf{e}_{\mu,\beta} \dot{x}^\beta$, we can rewrite the general definition (10.3) as

$$\dot{\mathbf{e}}_\mu = \Gamma_{\mu\beta}^\alpha \dot{x}^\beta \mathbf{e}_\alpha \quad (10.5)$$

It is often more convenient to use this form.

Example 10.1. Let us compare this general definition with derivatives of polar basis vectors, Eq. (10.1). We will be using indices r and θ instead of 1 and 2., so that, e.g., $\Gamma_{r\theta}^r \equiv \Gamma_{12}^1$.

$$\begin{aligned} \dot{\mathbf{e}}_1 \equiv \dot{\mathbf{e}}_r &= \frac{1}{r} \dot{\theta} \mathbf{e}_\theta & \Leftrightarrow \dot{\mathbf{e}}_r &= \Gamma_{r\beta}^\alpha \dot{x}^\beta \mathbf{e}_\alpha = \Gamma_{rr}^r \dot{r} \mathbf{e}_r + \Gamma_{r\theta}^r \dot{\theta} \mathbf{e}_r + \Gamma_{rr}^\theta \dot{r} \mathbf{e}_\theta + \Gamma_{r\theta}^\theta \dot{\theta} \mathbf{e}_\theta \\ \dot{\mathbf{e}}_2 \equiv \dot{\mathbf{e}}_\theta &= -r\dot{\theta} \mathbf{e}_r + \frac{1}{r} \dot{r} \mathbf{e}_\theta & \Leftrightarrow \dot{\mathbf{e}}_\theta &= \Gamma_{\theta\beta}^\alpha \dot{x}^\beta \mathbf{e}_\alpha = \Gamma_{\theta r}^r \dot{r} \mathbf{e}_r + \Gamma_{\theta\theta}^r \dot{\theta} \mathbf{e}_r + \Gamma_{\theta r}^\theta \dot{r} \mathbf{e}_\theta + \Gamma_{\theta\theta}^\theta \dot{\theta} \mathbf{e}_\theta \end{aligned}$$

Comparing left and right sides, we find that only three Christoffel symbols are nonzero:

$$\Gamma_{r\theta}^\theta = \Gamma_{\theta r}^\theta = \frac{1}{r}, \quad \Gamma_{\theta\theta}^r = -r \quad (10.6)$$

Now, using Eq. (10.4) we can rewrite Eq. (10.2) in terms of the Christoffel symbols

$$\dot{\boldsymbol{\xi}} = (\xi_{\mu,\beta}^\mu \mathbf{e}_\mu + \Gamma_{\mu\beta}^\alpha \xi^\mu \mathbf{e}_\alpha) \dot{x}^\beta$$

It would be more convenient to rename dumb indices in the first sum, $\xi_{\mu,\beta}^\mu \mathbf{e}_\mu = \xi_{\beta,\mu}^\alpha \mathbf{e}_\alpha$, which would allow us to take $\mathbf{e}_\alpha$ out of the brackets. This gives

$$\dot{\boldsymbol{\xi}} = (\xi_{\beta,\mu}^\alpha + \Gamma_{\mu\beta}^\alpha \xi^\mu) \dot{x}^\beta \mathbf{e}_\alpha \quad (10.7)$$

Since $\dot{\boldsymbol{\xi}}$ is invariant under coordinate transformation, so is the expression in the r.h.s. of the boxed formula. The product $\dot{x}^\beta \mathbf{e}_\mu$ transforms as a mixed tensor ($\dot{x}^\beta$ is a contravariant vector while $\mathbf{e}_\mu$ transforms as a covariant vector). In order the entire r.h.s. be a scalar, **the expression in brackets must also be a mixed tensor**.

Definition 10.2. The covariant derivative of a contravariant vector is the following tensor:

$$\nabla_\beta \xi^\mu \equiv \xi^\mu{}_{;\beta} = \xi^\mu{}_{,\beta} + \Gamma^\mu_{\alpha\beta} \xi^\alpha$$

The semicolon notations are widely used; any index that follows a semicolon implies the operation of covariant differentiation, while any index (say, β) after **comma** implies a usual component differentiation $\xi^\mu{}_{,\beta} \equiv \partial_\beta \xi^\mu$.

The word *covariant* in the definition means having the same form in any frame. Note that ∇_β cannot be treated as a covariant vector (operator), as it has no sense without acting upon a contravariant or – see Eq. (10.9) – covariant vector.

10.2 Derivatives of tensors. Comma-goes-to-semicolon rule.

First, we need to introduce the covariant derivative of a covariant vector. Note that the usual derivative of a scalar is already covariant, since $\partial_\beta \theta$ is a covariant vector. Thus, $\theta_{,\beta} = \theta_{;\beta}$. Since $\xi_\mu \zeta^\mu$ is a scalar, we have

$$\begin{aligned} (\xi_\mu \zeta^\mu)_{;\alpha} &= (\xi_\mu \zeta^\mu)_{,\alpha} = \xi_{\mu,\alpha} \zeta^\mu + \xi_\mu \zeta^\mu{}_{,\alpha} = (\xi_{\mu,\alpha} \zeta^\mu) + \xi_\mu (\zeta^\mu{}_{,\alpha} + \Gamma^\mu_{\beta\alpha} \zeta^\beta) - \Gamma^\mu_{\beta\alpha} \xi_\mu \zeta^\beta \\ &= \zeta^\mu (\xi_{\mu,\alpha} - \Gamma^\beta_{\mu\alpha} \xi_\beta) + \xi_\mu \zeta^\mu{}_{;\alpha}. \end{aligned} \quad (10.8)$$

Here we have added and subtracted $\Gamma^\mu_{\beta\alpha} \xi_\mu \zeta^\beta$ to the second term in order to make the covariant derivative of ζ^μ . Then we relabelled the dummy indices: $\Gamma^\mu_{\beta\alpha} \xi_\mu \zeta^\beta = \Gamma^\beta_{\mu\alpha} \xi_\beta \zeta^\mu$, which allowed us to take ζ^μ out of the brackets. Now, the expression in the l.h.s. of Eq. (10.8) is the covariant vector with the free index α , and so is the second term in the r.h.s. Hence, the expression in brackets in the first term must be covariant tensor with indices μ and α , since contracted with ζ^μ it gives covariant vector with index α . Therefore, we arrive at the expression for the covariant derivative of covariant vector:¹

$$\nabla_\alpha \xi_\mu \equiv \xi_{\mu;\alpha} = \xi_{\mu,\alpha} - \Gamma^\beta_{\mu\alpha} \xi_\beta. \quad (10.9)$$

A generalisation for calculating the covariant derivative of any tensor is straightforward: one deals with each co- and contra-variant indices in turn, according to the rules (10.2) and (10.9), ignoring all the rest indices during each step. It results from the definition: a tensor transforms as a direct product of contra- and co-variant vectors. Thus,

The covariant derivative ∇_α of the tensor $A^{\dots\mu\dots}_{\dots\nu\dots}$, is made of the ordinary derivative, $\partial_\alpha A^{\dots\mu\dots}_{\dots\nu\dots}$, plus $\Gamma^\mu_{\beta\alpha} A^{\dots\beta\dots}_{\dots\nu\dots}$ for each contravariant index μ , minus $\Gamma^\beta_{\nu\alpha} A^{\dots\mu\dots}_{\dots\beta\dots}$ for each contravariant index ν .

For example, for a mixed tensor of the 2nd rank, we have

$$\nabla_\alpha \xi^\mu{}_\nu \equiv \xi^\mu{}_{\nu;\alpha} = \xi^\mu{}_{\nu,\alpha} + \Gamma^\mu_{\beta\alpha} \xi^\beta{}_\nu - \Gamma^\beta_{\nu\alpha} \xi^\mu{}_\beta.$$

Hence, covariant differentiation obeys the standard product rule, *etc*:

$$\nabla_\alpha (\xi^\mu \zeta^\nu) = \partial_\alpha (\xi^\mu \zeta^\nu) + \Gamma^\mu_{\beta\alpha} \xi^\beta \zeta^\nu + \Gamma^\nu_{\beta\alpha} \xi^\mu \zeta^\beta = (\xi^\mu{}_{,\alpha} + \Gamma^\mu_{\beta\alpha} \xi^\beta) \zeta^\nu + (\zeta^\nu{}_{,\alpha} + \Gamma^\nu_{\beta\alpha} \zeta^\beta) \xi^\mu$$

¹Naturally, since ∇_α is a legitimate tensor (or vector) operation, we can raise index here as well, introducing $\nabla^\nu = g^{\mu\nu} \nabla_\mu$. However, although such an operation is well defined, I will try to avoid it, keeping both comma and semicolon only in the lower indices.

$$\equiv \xi^\mu{}_{;\alpha} \zeta^\nu + \xi^\mu \zeta^\nu{}_{;\alpha}.$$

Now we can formulate a general recipe for differential operations in Riemannian space, which fully justifies the introduction of the *comma* and *semicolon* notations.

Theorem 10.1. Comma-goes-to-semicolon (CG2S) rule:

Any **first-order** differential identity proved in a locally Cartesian system is generalized to an arbitrary coordinate system by $\partial_\mu \rightarrow \nabla_\mu \iff \xi_{\gamma\delta,\mu} \rightarrow \xi_{\gamma\delta;\mu}$

The reason is that in a LCF, where all $\{e_\mu\}$ are constant, $\Gamma^\nu_{\alpha\beta} = 0$, so that $\partial_\mu = \nabla_\mu$. Then, as ∇ is a fully covariant operation, any rule formulated with the help of covariant derivatives is valid in any coordinate system. This recipe is very useful since calculations are much easier in a locally Cartesian coordinate system.

10.3 Covariant derivative of the metric tensor.

We have defined a locally flat space by the following construction: we think of a d -dimensional curved space as a d -dimensional surface embedded into $(d+1)$ -dimensional space (like a 2D sphere in a 3D space), and build the hyperplane tangent to a given point A of the curved space. The simplest way to define such a hyperplane is to span it over d mutually perpendicular lines, each tangent to A . These lines then define a (pseudo)Cartesian basis in the hyperplane. In such a LCF, the elements of the metric tensor are constant; hence, $g_{\mu\nu,\alpha} = 0$. This expression is made covariant by applying the above **comma-goes-to-semicolon** rule so that in any frame

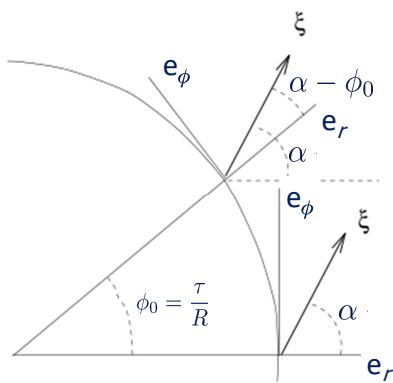
$$g_{\mu\nu;\alpha} = 0 \quad \text{in any coordinate frame}$$

Thus $g_{\mu\nu}$ also lowers (and $g^{\mu\nu}$ raises) tensor indices in the presence of the semicolon: indeed, using the product rule gives

$$\xi_\nu{}_{;\alpha} = (g_{\mu\nu} \xi^\mu)_{;\alpha} = g_{\mu\nu;\alpha} \xi^\mu + g_{\mu\nu} \xi^\mu{}_{;\alpha} = g_{\mu\nu} \xi^\mu{}_{;\alpha}.$$

10.4 Parallel Transfer

The reason for nontriviality of the covariant differentiation is the necessity to compare vectors attached to different (albeit close) points in space. Even in a flat space coordinates of the vector after *parallel transfer* to another point change (as illustrated in the figure), unless we use LCF.



The vector ξ is unchanged in a flat space after parallel transfer along a circular arc, $r = R$ through $\phi_0 = \tau/R$, and so are its Cartesian components (the difference of coordinates of its end and its origin). Not so are its polar components (where the basis vectors are $e_r = \cos \phi \mathbf{i} + \sin \phi \mathbf{j}$ and $e_\phi = -r \sin \phi \mathbf{i} + r \cos \phi \mathbf{j}$): its components before the transfer are defined by $\xi^r = |\xi| \cos \alpha$ and $r \xi^\phi = |\xi| \sin \alpha$ and after the transfer, $\xi^r = |\xi| \cos (\alpha - \phi_0)$ and $r \xi^\phi = |\xi| \sin (\alpha - \phi_0)$. Here the components depend on the final point but not on a curve along which the vector has been transferred, which, as we shall see, is a characteristic feature of a flat space.

In a curved space, when vector ξ is transferred to a neighbouring point (where it belongs to a different tangent hyperplane), its components do depend on the curve along which it was transferred.

Definition 10.3. The change $\delta\xi^\mu$ of contravariant components of a vector $\xi = \xi^\mu e_\mu$ in parallel transfer along the curve defined by $x^\mu = x^\mu(s)$ is given by

$$\delta\xi^\mu = \xi^\mu(s + \delta s) - \xi^\mu(s)$$

since s governs the value of coordinate x in $\xi^\mu(x)$ along the curve. The rate of change in LCF is $u^\mu \equiv \dot{\xi}^\mu \equiv \frac{d\xi^\mu}{ds} = \xi^\mu_{,\alpha} \dot{x}^\alpha$. In an arbitrary frame, the CG2S rule demands, in according with Eq. (10.7), changing the partial derivative by the covariant one, $\xi^\mu_{,\beta} \rightarrow \xi^\mu_{;\beta}$. Hence in any frame the rate of change of a (contravariant) component of the vector field ξ in parallel transfer along the curve $x^\mu = x^\mu(\tau)$ characterised by the velocity $u^\mu = \dot{x}^\mu$ is given by the '**covariant dot**' operation² ∇_u :

$$\left[\frac{d\xi^\mu}{ds} \right]_{\text{cov}} \equiv \xi^\mu_{;\beta} u^\beta = u^\beta \xi^\mu_{;\beta} + u^\beta \Gamma^\mu_{\alpha\beta} \xi^\alpha = u^\beta \nabla_\beta \xi^\mu \equiv \nabla_u \xi^\mu \text{ with } \nabla_u \equiv u^\beta \nabla_\beta \quad (10.10)$$

If vector ξ is **constant**, then its components remain unchanged in LCF: $\delta\xi^\mu = \dot{\xi}^\mu ds = 0$. In any frame, it's $\nabla_u \xi^\mu ds = 0$ so that the contravariant component change in parallel transfer³ is $\delta\xi^\mu = -\Gamma^\mu_{\nu\alpha} \xi^\nu dx^\alpha$. This allows the following interpretation of the two terms in ∇_u , Eq. (10.10):

$$\text{rate of change } \dot{\xi}^\mu = \text{rate of real change } \nabla_u \xi^\mu + \text{rate of parallel transfer}$$

Example 10.2. Let us formally reproduce the parallel transfer in the above figure.

The parametric equations for the circle in polar coordinates are $x^1 \equiv r = R$ and $x^2 \equiv \theta = \tau/R$ so that $\dot{r} = 0$ and $\dot{\theta} = 1/R$ (here $\tau = R\theta$ is the arc length). As $\dot{r} = 0$, Eq. (10.10) gives $\dot{\xi}^\mu = -\Gamma^\mu_{\nu\theta} \xi^\nu \dot{\theta}$, where we fixed $\xi^\alpha \rightarrow x^2 \equiv \theta$, while ν and μ take both values r and θ . However, the only nonzero Christoffel symbols, Eq. (10.6), are $\Gamma^\theta_{r\theta} = 1/R$ and $\Gamma^r_{\theta\theta} = -R$. Hence $\dot{\xi}^r = -\Gamma^r_{\mu\theta} \xi^\mu \dot{\theta} \rightarrow -\Gamma^r_{\theta\theta} \xi^\theta \dot{\theta} = R(1/R) \xi^\theta = \xi^\theta$, and $\dot{\xi}^\theta = -\Gamma^\theta_{\mu\theta} \xi^\mu \dot{\theta} \rightarrow -\Gamma^\theta_{r\theta} \xi^r \dot{\theta} = -(1/R)(1/R) \xi^r = -(1/R)^2 \xi^r$. Thus, $\ddot{\xi}^r = \dot{\xi}^\theta = -(1/R)^2 \xi^r$. This is a simple harmonic oscillator: as $\tau = R\theta$, changing $d^2/d\tau^2 = (1/R^2) d/d\theta^2$ gives $\frac{d^2 \xi^r}{d\theta^2} = -\xi^r$. Its solution with the initial (at $s = 0$) values of $\xi^r(0) = |\xi| \cos \alpha$ and $\xi^\theta(0) = (|\xi|/R) \sin \alpha$ coincides with that given at the figure.

²Note that $u^\beta \nabla_\beta$ is not a scalar: as well as ∇_β , it does not make sense without acting upon contra- or co-vector.

³In many books, the boxed formula is used as the definition of $\Gamma^\mu_{\nu\alpha}$, which is equivalent to Def. (10.1).

Lecture 11. Christoffel symbols and metric tensor

We need covariant derivatives, and thus Christoffel symbols, to write the laws of nature in covariant form. Instead of calculating $\Gamma_{\mu\nu}^\alpha$ via basis vectors, we learn how express them via derivatives of $g_{\mu\nu}$.

11.1 Index symmetry in Christoffel symbols.

All indices enter the definition of $\Gamma_{\mu\nu}^\beta$ in a different way. However, we shall prove now that Γ is symmetric in lower indices. In LCF we have $\varphi_{,\alpha\beta} = \varphi_{,\beta\alpha}$ (a usual repeated differentiation). On the other hand, $\varphi_{;\alpha} \equiv \varphi_{;\alpha}$ by the definition of a covariant vector. Thus we have $(\varphi_{,\beta})_{;\alpha} \equiv (\varphi_{;\beta})_{,\alpha}$. Introducing the notation $\xi_\alpha \equiv \phi_{;\alpha}$, we have for this particular vector ξ_α in LCF the identity $\xi_{\alpha;\beta} = \xi_{\beta;\alpha}$. Now we can use the **comma-goes-to-semicolon** rule to this identity we have $\xi_{\alpha;\beta} = \xi_{\beta;\alpha}$. (Note that such an identity is not valid for an arbitrary ξ_α but only for the covariant derivative of a **scalar**). Therefore, the following covariant relation is valid: $\varphi_{;\alpha\beta} = \varphi_{;\beta\alpha}$. Then

$$\begin{aligned} \varphi_{;\beta\alpha} &= (\varphi_{,\beta})_{;\alpha} = \varphi_{,\beta\alpha} - \Gamma_{\beta\alpha}^\nu \varphi_{,\nu} \\ \varphi_{;\alpha\beta} &= (\varphi_{,\alpha})_{;\beta} = \varphi_{,\alpha\beta} - \Gamma_{\alpha\beta}^\nu \varphi_{,\nu} \end{aligned} \implies \Gamma_{\beta\alpha}^\nu \varphi_{,\nu} = \Gamma_{\alpha\beta}^\nu \varphi_{,\nu} \implies \boxed{\Gamma_{\beta\alpha}^\nu = \Gamma_{\alpha\beta}^\nu} \quad (11.1)$$

This symmetry allows us to express the Christoffel symbols in terms of $g_{\mu\nu}$. To this end, let us take the scalar product of both sides of Eq. (10.4), $e_{\mu;\beta} = \Gamma_{\mu\beta}^\alpha e_\alpha$, with e_ν :

$$\langle e_{\mu;\beta} \cdot e_\nu \rangle = \Gamma_{\mu\beta}^\alpha \langle e_\alpha \cdot e_\nu \rangle = g_{\alpha\nu} \Gamma_{\mu\beta}^\alpha \equiv \Gamma_{\nu\mu\beta} \quad (11.2)$$

where we have defined the auxiliary symbol $\Gamma_{\nu\mu\beta}$, since $g_{\alpha\nu} \equiv \langle e_\alpha \cdot e_\nu \rangle$ acts as a lowering operator on any tensor.¹ Using the symmetry in the last two indices of $\Gamma_{\mu\nu}^\alpha$, Eq. (11.1), we rewrite this as

$$\Gamma_{\nu\mu\beta} = \frac{1}{2} (\langle e_\nu \cdot e_{\mu;\beta} \rangle + \langle e_\nu \cdot e_{\beta;\mu} \rangle). \quad (11.3)$$

Now we want to express $\langle e_{\mu;\beta} \cdot e_\nu \rangle$, etc, in terms of derivatives of the metric tensor $g_{\mu\nu}$, which – in contrast to covariant derivatives of $g_{\mu\nu}$ – vanish only in LCF. To this end, note that

$$\begin{aligned} \langle e_\nu \cdot e_{\mu;\beta} \rangle &\equiv \langle e_\nu \cdot \partial_\beta e_\mu \rangle = \partial_\beta \langle e_\nu \cdot e_\mu \rangle - \langle \partial_\beta e_\nu \cdot e_\mu \rangle = g_{\nu\mu,\beta} - \langle e_\mu \cdot e_{\nu;\beta} \rangle \\ \langle e_\nu \cdot e_{\beta;\mu} \rangle &\equiv \langle e_\nu \cdot \partial_\mu e_\beta \rangle = \partial_\mu \langle e_\nu \cdot e_\beta \rangle - \langle \partial_\mu e_\nu \cdot e_\beta \rangle = g_{\nu\beta,\mu} - \langle e_\beta \cdot e_{\nu;\mu} \rangle \end{aligned}$$

and hence rewrite Eq. (11.3) as follows:

$$\Gamma_{\nu\mu\beta} = \frac{1}{2} (g_{\nu\mu,\beta} - \langle e_\mu \cdot e_{\nu;\beta} \rangle + g_{\nu\beta,\mu} - \langle e_\beta \cdot e_{\nu;\mu} \rangle).$$

Using again the symmetry $\langle e_\mu \cdot e_{\nu;\beta} \rangle = \langle e_\mu \cdot e_{\beta;\nu} \rangle$, etc, gives

$$\langle e_\beta \cdot e_{\nu;\mu} \rangle + \langle e_\mu \cdot e_{\nu;\beta} \rangle = \langle e_{\mu,\nu} \cdot e_\beta \rangle + \langle e_\mu \cdot e_{\beta,\nu} \rangle = \partial_\nu \langle e_\mu \cdot e_\beta \rangle = g_{\mu\beta,\nu},$$

so that we arrive at

$$\Gamma_{\nu\mu\beta} = \frac{1}{2} (g_{\nu\mu,\beta} + g_{\nu\beta,\mu} - g_{\mu\beta,\nu}) \quad (11.4)$$

¹ $\Gamma_{\mu\beta}^\alpha$ is not a tensor, but the last identity in Eq.(11.2) is just the definition to ensure consistent notations.

This expression is obviously symmetric w.r.t. swapping the last two indices, but the first index enters in a clearly different way. Raising the first index of Γ , we finally express the Christoffel symbols in terms of derivatives of $g_{\mu\nu}$:

$$\Gamma_{\mu\beta}^{\alpha} = \frac{1}{2}g^{\alpha\nu} (g_{\nu\mu,\beta} + g_{\nu\beta,\mu} - g_{\mu\beta,\nu}) \quad (11.5)$$

This will be the main expression for calculations. However, sometimes it is useful to reverse it and express derivatives of $g_{\mu\nu}$ in terms of Γ . To this end, we could use the fact that $g_{\mu\nu}\xi^{\mu}\eta^{\nu}$ is a scalar so that $(g_{\mu\nu}\xi^{\mu}\eta^{\nu})_{,\alpha} = (g_{\mu\nu}\xi^{\mu}\eta^{\nu})_{;\alpha}$. As $g_{\mu\nu;\alpha} = 0$, the covariant derivative becomes just $g_{\mu\nu}(\xi^{\mu}\eta^{\nu})_{;\alpha}$. Then we have

$$g_{\mu\nu,\alpha}\xi^{\mu}\eta^{\nu} + g_{\mu\nu}(\xi^{\mu}\eta^{\nu})_{,\alpha} = +g_{\mu\nu}(\xi^{\mu}\eta^{\nu})_{;\alpha} + g_{\mu\nu}(\Gamma_{\beta\alpha}^{\mu}\xi^{\beta}\eta^{\nu} + \Gamma_{\beta\alpha}^{\nu}\xi^{\mu}\eta^{\beta}) .$$

After relabelling $\mu \leftrightarrow \beta$ in the first Γ term, and $\nu \leftrightarrow \beta$ in the second, we have

$$g_{\mu\nu,\alpha}\xi^{\mu}\eta^{\nu} = g_{\beta\nu}\Gamma_{\mu\alpha}^{\beta}\xi^{\mu}\eta^{\nu} + g_{\beta\mu}\Gamma_{\nu\alpha}^{\beta}\xi^{\mu}\eta^{\nu} \Rightarrow \boxed{g_{\mu\nu,\alpha} = \Gamma_{\mu\nu\alpha} + \Gamma_{\nu\mu\alpha}} \quad (11.6)$$

11.2 Calculating Christoffel symbols: examples

• **Is it simple?** How many components has Γ ? In d dimensional space, the total number of components is d^3 . The number of components with the same lower indices is d^2 . Among the $d^3 - d^2$ off-diagonal terms only the half are independent because of the symmetry Eq. (11.1). Thus, the number of independent components $= d^2 + \frac{1}{2}(d^3 - d^2) = \frac{1}{2}(d^3 + d^2)$ which is 6 for $d = 2$, 18 for $d = 3$, and 40 for $d = 4$. Each of the component is given by Eq. (11.3). Usually, it is much easier to calculate Γ from the metric tensor, Eq. (11.5) rather than from the expression Eq. (10.4). In Eq. (11.5) all the symmetries are explicit, and a fortunate fact that many of the components vanish can be exploited more straightforwardly.

In 2D space, the three-symbol $\Gamma_{\mu\nu}^{\alpha}$ has 8 components; because of the symmetry in the lower indices, no more than 6 of them could be independent. Let us calculate them for the examples we have considered before, using the coordinate names for indices, like in $\Gamma_{xx}^x \equiv \Gamma_{11}^1$, $\Gamma_{xy}^x \equiv \Gamma_{12}^1$, etc.

Example 11.1. Polar coordinates again

Now we calculate $\Gamma_{\mu\nu}^{\alpha}$ directly from $g_{\mu\nu} = \text{diag}(1, r^2)$ with $x^1 \equiv r$, $x^2 \equiv \phi$. Of possible derivatives of nonzero (diagonal) elements of the metric tensor with respect to r and ϕ , only one does not vanish: $\partial_1 g_{22} \equiv g_{22,1} = 2r$. The expression for $\Gamma_{\beta\mu\nu}$ shows that this derivative could only enter $\Gamma_{212} = \Gamma_{221}$ and Γ_{122} , i.e. the symbols that contain permutations of indices of $g_{22,1}$. All other symbols contain only vanishing derivatives so that $\Gamma_{111} = \Gamma_{112} = \Gamma_{121} = \Gamma_{211} = \Gamma_{222} = 0$. For nonvanishing components, we find²

$$\begin{aligned} \Gamma_{221} = \Gamma_{212} &= \frac{1}{2}(g_{22,1} + g_{21,2} - g_{21,2}) = \frac{1}{2}g_{22,1} = r \\ \Gamma_{122} &= \frac{1}{2}(g_{12,2} + g_{12,2} - g_{22,1}) = -\frac{1}{2}g_{22,1} = -r \end{aligned}$$

Raising the first index with the help of $\Gamma_{\mu\nu}^{\alpha} = g^{\alpha\beta}\Gamma_{\beta\mu\nu}$ and using $g^{\mu\nu} = \text{diag}(1, r^{-2})$, we finally obtain

$$\Gamma_{12}^2 = \Gamma_{21}^2 = g^{22}\Gamma_{221} = \frac{1}{r}, \quad \Gamma_{22}^1 = g^{11}\Gamma_{122} = -r, \quad (11.7)$$

²Note in passing that for any **diagonal** metric $\Gamma_{aab} = \Gamma_{aba} = -\Gamma_{baa}$ (no summation!), while this is not valid for an off-diagonal metric.

which coincides with our earlier result, Eq. (10.6).

Example 11.2. Spherical surface of radius r .

Let us identify $x^1 \equiv \theta$, $x^2 \equiv \phi$. The metric tensor is $g_{\mu\nu} = \text{diag}(r^2, r^2 \sin^2 \theta)$, the inverse $g^{\mu\nu} = \text{diag}(r^{-2}, r^{-2} \sin^{-2} \theta)$. There is only one nonzero derivative of $g_{\mu\nu}$: $\partial_1 g_{22} \equiv g_{22,1} = \frac{\partial r^2 \sin^2 \theta}{\partial \theta} = r^2 \sin(2\theta)$. Thus, again only Γ 's with indices that are permutations of {221} do not vanish:

$$\Gamma_{212} = \Gamma_{221} = -\Gamma_{122} = \frac{1}{2} g_{22,1} = \frac{1}{2} r^2 \sin(2\theta) \Rightarrow \Gamma_{12}^2 = \Gamma_{21}^2 = \cot \theta, \quad \Gamma_{22}^1 = -\frac{1}{2} \sin(2\theta).$$

Note that the Christoffel symbols do not depend on radius of the spherical surface.

Example 11.3. A trumpet surface.

The metric $ds^2 = dx^2 + e^{-2x} d\phi^2$ so that $g_{\mu\nu} = \text{diag}(1, e^{-2x})$ and $g^{\mu\nu} = \text{diag}(1, e^{2x})$. Again, only $\Gamma_{221} = \Gamma_{212}$ and Γ_{122} do not vanish, since $g_{22,1} = -2e^{-2x}$:

$$\Gamma_{221} = \Gamma_{212} = -\Gamma_{122} = -e^{-2x} \Rightarrow \Gamma_{21}^2 = \Gamma_{12}^2 = -1, \quad \Gamma_{22}^1 = e^{-2x}.$$

Example 11.4. Something else

Let's take the metric $ds^2 = y^2 dx^2 - x^2 dy^2$. Here $g_{11} = y^2$, and $g_{22} = -x^2$, so that $g_{11,2} = 2y$, and $g_{22,1} = -2x$. Now we have 6 could-be-non-zero elements: $\Gamma_{112} = \Gamma_{121}$, Γ_{211} , $\Gamma_{221} = \Gamma_{212}$, and Γ_{122} while only Γ_{111} and Γ_{222} vanish. Using Eq. (11.4) and the inverse matrix $g^{\mu\nu} = \text{diag}(y^{-2}, x^{-2})$, we find:

$$\begin{aligned} \Gamma_{221} = \Gamma_{212} = -\Gamma_{122} = \frac{1}{2} g_{22,1} = -x & \Rightarrow \Gamma_{21}^2 = \Gamma_{12}^2 = g^{22} \Gamma_{221} = \frac{1}{x}, \quad \Gamma_{22}^1 = g^{11} \Gamma_{122} = \frac{x}{y^2}; \\ \Gamma_{112} = \Gamma_{121} = -\Gamma_{211} = \frac{1}{2} g_{11,2} = y & \Rightarrow \Gamma_{12}^1 = \Gamma_{21}^1 = g^{11} \Gamma_{112} = \frac{1}{y}, \quad \Gamma_{11}^2 = g^{22} \Gamma_{211} = \frac{y}{x^2}. \end{aligned}$$

Although this looks nontrivial, we will show later (in Lec. 14) that this metric describes a **flat** surface in a pseudo-Euclidean space. I remind that $\Gamma_{\mu\nu}^\alpha$ vanish only in LCF, while in a flat space (either Euclidean or pseudo-Euclidean) with non-Cartesian coordinates some components are always nonzero.

Example 11.5. A non-diagonal metric.

Consider the following metric:

$$g_{\mu\nu} = \begin{pmatrix} y^2 & x \\ x & 0 \end{pmatrix} \Rightarrow g^{\mu\nu} = \frac{1}{\det g} \begin{pmatrix} 0 & -x \\ -x & y^2 \end{pmatrix} = \begin{pmatrix} 0 & 1/x \\ 1/x & -y^2/x^2 \end{pmatrix}$$

The nonzero derivatives of $g_{\mu\nu}$ are $g_{11,2} = 2y$ and $g_{12,1} = g_{21,1} = 1$. Therefore, all nonzero $\Gamma_{\alpha\beta\gamma}$ have indices which are permutations of {112}. However, $\Gamma_{112} = -\Gamma_{211}$ is no longer true.

$$\Gamma_{112} = \Gamma_{121} = \frac{1}{2} (g_{11,2} + g_{12,1} - g_{21,1}) = y, \quad \Gamma_{211} = \frac{1}{2} (2g_{21,1} - g_{11,2}) = 1 - y.$$

Moreover, the nonzero components of $\Gamma_{\beta\gamma}^\alpha$ have different indices:

$$\begin{aligned} \Gamma_{11}^1 &= g^{11} \Gamma_{111} + g^{12} \Gamma_{211} = \frac{1-y}{x}, & \Gamma_{12}^1 &= g^{11} \Gamma_{112} + g^{12} \Gamma_{212} = 0, & \Gamma_{22}^1 &= 0 \\ \Gamma_{12}^2 &= g^{21} \Gamma_{112} + g^{22} \Gamma_{212} = \frac{y}{x}, & \Gamma_{11}^2 &= g^{21} \Gamma_{111} + g^{22} \Gamma_{211} = -\frac{(1-y)y^2}{x^2}, & \Gamma_{22}^2 &= 0. \end{aligned}$$

The elements in the last column vanish since $\Gamma_{22}^\mu = g^{\mu\nu} \Gamma_{\nu 22}$ but all the components of $\Gamma_{\beta\mu\nu}$ with $\mu = \nu = 2$ vanish. Naturally, $\Gamma_{21}^2 = \Gamma_{12}^2$, so that there are four nonvanishing components of $\Gamma_{\mu\nu}^\alpha$.

11.3 Covariant differential operators

In the elementary vector analysis, we used covariant operations like $\text{div } \xi \equiv \nabla \cdot \xi$, $\Delta \xi \equiv \nabla^2 \xi$, $\text{curl } \xi \equiv \nabla \times \xi$. We have introduced the symbol ∇_β for the covariant derivative. Naturally, it is related to the ∇ symbol in the above expressions. In particular,

$$\text{div } \xi = \nabla \cdot \xi = \nabla_\beta \xi^\beta \equiv \xi^\beta_{;\beta} = \xi^\beta_{,\beta} + \Gamma^\beta_{\mu\beta} \xi^\mu,$$

in the agreement with the CG2S rule.

One can express $\text{div } \xi$ directly in the terms of the metric tensor. First, it follows from Eq. (11.5) that

$$\Gamma^\beta_{\mu\beta} = \frac{1}{2} g^{\beta\nu} (g_{\nu\mu,\beta} + g_{\nu\beta,\mu} - g_{\mu\beta,\nu}) = \frac{1}{2} g^{\beta\nu} g_{\nu\beta,\mu}, \quad (11.8)$$

since the last term in the brackets above cancels the first one on renaming $(\beta, \nu) \rightarrow (\nu, \beta)$. Next, we prove the following identity

$$\text{Tr } g^{-1} \partial_\mu g = |g|^{-1} \partial_\mu |g| \quad \text{where} \quad |g| \equiv |\det g|, \quad (11.9)$$

valid for any nonsingular **matrix** g , (not only for the metric). Eq. (11.9) follows immediately from applying the differential operator ∂_μ to the remarkable identity,

$$\boxed{\ln \det g = \text{Tr } \ln g},$$

and using that ∂_μ and Tr commute, and $\text{Tr } \partial_\mu \ln g = \text{Tr } g^{-1} \partial_\mu g$. The latter identity can be directly checked for any diagonalisable matrix (the set which includes all symmetric matrices) in the diagonal representation, $g = U^{-1} g_{\text{diag}} U$, applying the cyclic permutations both inside $\det$ and inside Tr .

Now, noticing that if $g = g_{\mu\nu}$, then $g^{-1} = g^{\nu\mu}$, we rewrite Eq. (11.9) in matrix notations as $g^{\beta\nu} \partial_\mu g_{\nu\beta} = |g|^{-1} \partial_\mu |g|$. Substituting this into Eq. (11.8), we obtain

$$\boxed{\Gamma^\beta_{\mu\beta} = \frac{|g|_{,\mu}}{2|g|} = \frac{1}{2} \partial_\mu \ln |g| = \partial_\mu \ln \sqrt{|g|}}$$

Finally,

$$\text{div } \xi \equiv \xi^\mu_{;\mu} = \xi^\mu_{,\mu} + \Gamma^\beta_{\mu\beta} \xi^\mu = \xi^\mu_{,\mu} + \left(\partial_\mu \ln \sqrt{|g|} \right) \xi^\mu = \frac{1}{\sqrt{|g|}} \partial_\mu \left(\sqrt{|g|} \xi^\mu \right).$$

This is a very useful formula since calculating $|g|$ is much simpler than calculating $\Gamma^\alpha_{\mu\nu}$.

• **The Laplacian operator** in a curved space, or just in curvilinear coordinates can be found immediately. Indeed, as

$$\nabla^2 \psi = \text{div} \cdot \text{grad } \psi \quad \text{and} \quad (\text{grad } \psi)^\mu = \psi^{;\mu} = \psi^{,\mu} = g^{\mu\nu} \psi_{,\nu}$$

one can apply the above expression for the divergence to the contravariant vector $\psi^{;\mu}$ to obtain

$$\boxed{\nabla^2 \psi = (g^{\mu\nu} \psi_{,\nu})_{;\mu} = \frac{1}{\sqrt{|g|}} \partial_\mu \left(\sqrt{|g|} g^{\mu\nu} \partial_\nu \psi \right)}$$

• **Examples.** As usual, a general formula saves a lot of work. We have in spherical polar coordinates $g_{\mu\nu} = (1, r^2, r^2 \sin^2 \theta)$, so that $\sqrt{|g|} = r^2 \sin \theta$. Using also $g^{\mu\nu} = \text{diag}(1, 1/r^2, 1/r^2 \sin^2 \theta)$, we immediately obtain

$$\begin{aligned}\nabla^2 &= \frac{1}{r^2 \sin \theta} \left[\sin \theta \partial_r (r^2 \partial_r) + \partial_\theta (\sin \theta \partial_\theta) + \frac{1}{\sin \theta} \partial_\phi^2 \right] \\ &= \frac{1}{r^2} \left[\partial_r (r^2 \partial_r) + \frac{1}{\sin \theta} \partial_\theta (\sin \theta \partial_\theta) + \frac{1}{\sin^2 \theta} \partial_\phi^2 \right]\end{aligned}$$

11.4 Volume Integration³

The appearance of $\sqrt{|g|}$ above is not incidental: it is the Jacobian of the coordinate transformation from Cartesian (or pseudo-Cartesian) to given coordinates. Let's transform the integral of a scalar over some volume from Cartesian to general coordinates:

$$\int_V \phi(x^\nu) \prod_1^d dx^{\bar{\mu}} = \int_V \phi(x^\nu) \left| \frac{\partial(x^{\bar{1}}, \dots, x^{\bar{d}})}{\partial(x^1, \dots, x^d)} \right| \prod_1^d dx^\mu \equiv \int_V \phi(x^\nu) |\det\{\lambda_{\bar{\beta}}^\alpha\}| \prod_1^d dx^\mu.$$

This is the standard formula for the coordinate transformations. The Jacobian, J , is the determinant of the transformation matrix $J = \det\{\lambda_{\bar{\mu}}^\mu\}$. When we change coordinates, we can calculate it directly. However, we can also define it **intrinsically**, for volume integration in any coordinate system with a known metric tensor. Indeed, from $g_{\mu\nu} = \lambda_{\bar{\mu}}^\mu g_{\bar{\mu}\bar{\nu}} \lambda_{\bar{\nu}}^\nu$ follows

$$g \equiv \det\{g_{\mu\nu}\} = \det\{\lambda_{\bar{\mu}}^\mu g_{\bar{\mu}\bar{\nu}} \lambda_{\bar{\nu}}^\nu\} = \underbrace{\det\{\lambda_{\bar{\mu}}^\mu\}}_{=J} \underbrace{\det\{g_{\bar{\mu}\bar{\nu}}\}}_{=\pm 1} \underbrace{\det\{\lambda_{\bar{\nu}}^\nu\}}_{=J} = \pm J^2 \quad \implies \quad J = \sqrt{|g|}.$$

Therefore, the element of volume integration in an arbitrary coordinate system may be expressed as $dV = \sqrt{|g|} dx^1 \dots dx^d$.

³Not to be shown in the class

Lecture 12. Geodesic equations in canonical form

We have formally defined covariant differentiation in terms of the Christoffel symbols, $\Gamma_{\mu\nu}^{\alpha}$, and we have expressed these symbols via the metric tensor. First we shall see how the Christoffel symbols arise naturally in the framework of the Lagrangian description of geodesics. Then we show that canonical form of geodesic equations classically means representing Newton's equations of motion geometrically. This representation provides us with a bridge to go from Newtonian universal gravity to a fully relativistic description.

12.1 Euler – Lagrange equations and Christoffel symbols

Let us call the r.h.s. of Euler – Lagrange equations $2f_{\beta}$:

$$2f_{\beta} \equiv \frac{d}{ds} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^{\beta}} \right) - \frac{\partial \mathcal{L}}{\partial x^{\beta}}, \quad \mathcal{L} = g_{\mu\nu} \dot{x}^{\mu} \dot{x}^{\nu} \quad (12.1)$$

As $\mathcal{L}$ is a scalar, f_{β} is a covector. A curve along which $f_{\beta} = 0$ satisfies Euler – Lagrange equations and is therefore a geodesic. Thus a nonzero f_{β} indicates how strongly a given curve (along which one takes the derivative with respect to s) deviates from a geodesic. **We call f_{β} the line curvature vector.** Note that it represents the curvature of the line (not of the space where this curved line lies!).

Let us get a more explicit expression for f_{β} . Since $\partial_{\beta} x^{\nu} = \delta^{\nu}_{\beta}$ and $\partial_{\dot{\beta}} \dot{x}^{\nu} \equiv \partial \dot{x}^{\nu} / \partial \dot{x}^{\beta} = \delta^{\nu}_{\beta}$, we have

$$\frac{d}{ds} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^{\beta}} \right) = \frac{d}{ds} (2g_{\beta\nu} \dot{x}^{\nu}) = 2 \left(g_{\beta\nu} \ddot{x}^{\nu} + \frac{dg_{\beta\nu}}{ds} \dot{x}^{\nu} \right) = 2 (g_{\beta\nu} \ddot{x}^{\nu} + g_{\beta\nu,\mu} \dot{x}^{\nu} \dot{x}^{\mu})$$

and

$$\frac{\partial \mathcal{L}}{\partial x^{\beta}} = \frac{\partial g_{\mu\nu}}{\partial x^{\beta}} \dot{x}^{\nu} \dot{x}^{\mu} \equiv g_{\mu\nu,\beta} \dot{x}^{\nu} \dot{x}^{\mu},$$

we have

$$f_{\beta} = g_{\beta\nu} \ddot{x}^{\nu} + (g_{\beta\nu,\mu} - \frac{1}{2} g_{\mu\nu,\beta}) \dot{x}^{\nu} \dot{x}^{\mu} = g_{\beta\nu} \ddot{x}^{\nu} + \frac{1}{2} (g_{\beta\nu,\mu} + g_{\beta\mu,\nu} - g_{\mu\nu,\beta}) \dot{x}^{\nu} \dot{x}^{\mu}$$

where we have mutually relabelled the dummy pairs ν and μ . We recognise here

$$f_{\beta} = g_{\beta\nu} \ddot{x}^{\nu} + \Gamma_{\beta\mu\nu} \dot{x}^{\nu} \dot{x}^{\mu}.$$

Raising the index of f_{β} (i.e. writing $f^{\alpha} = g^{\alpha\beta} f_{\beta}$), we finally obtain

$$f^{\alpha} = \ddot{x}^{\alpha} + \Gamma_{\mu\nu}^{\alpha} \dot{x}^{\nu} \dot{x}^{\mu} = u^{\alpha}{}_{,\nu} u^{\nu} + \Gamma_{\mu\nu}^{\alpha} u^{\nu} u^{\mu} = u^{\nu} u^{\alpha}{}_{;\nu} \equiv \nabla_u u^{\alpha}, \quad (12.2)$$

where $u^{\mu} \equiv \dot{x}^{\mu}$, and we use the standard chain rule for $\dot{u}^{\alpha} = u^{\alpha}{}_{,\nu} \dot{x}^{\nu} = u^{\alpha}{}_{,\nu} u^{\nu}$ and the covariant dot, Eq. (10.10),

$$\nabla_u \equiv u^{\nu} \nabla_{\nu}.$$

As $f^{\alpha} = 0$ is the set of the Euler – Lagrange equations, we rewrite them as the **geodesic equations in 'canonical form'**:

$$\boxed{f^{\alpha} = \nabla_u u^{\alpha} = \ddot{x}^{\alpha} + \Gamma_{\mu\nu}^{\alpha} \dot{x}^{\nu} \dot{x}^{\mu} = 0} \quad (12.3)$$

Hence **geodesics is the straightest line** (or simply straight in a flat space) since its velocity (tangent vector) remains unchanged along it. Now I remind that dx^{α} (rather than x^{α}) is contravariant vector by definition so that

$$u^{\alpha} \equiv \dot{x}^{\alpha} \equiv \frac{dx^{\alpha}}{d\tau} = \nabla_u x^{\alpha}, \quad (12.4)$$

i.e. the covariant rate of change of x^μ is velocity itself, in contrast to the rate of change of an arbitrary ξ^μ , for which the covariant generalisation of *dot*, Eq. (10.10), is nontrivial. Hence Eq. (12.3) can be rewritten as

$$f^\alpha = \nabla_u^2 x^\alpha \quad \Rightarrow \quad f^\alpha = 0 \quad \text{along a geodesic,}$$

which is a direct generalisation of the Newton's law of inertia: in the absence of a non-gravitational force, a material point moves along a straight(est) line – geodesics.

In a flat space the Christoffel symbols vanish, and these equations becomes just $\ddot{x}^\alpha = 0$ - the 2nd Newton law in the absence of an external force (or the Galileo principle). We are using a mechanical analogy for describing a geometrical problem but it is more than justified. In general, Eq. (12.3) could be considered as a Newtonian form of Euler – Lagrange equations in a curved space. To make it more visible, note that

Therefore, the curvature of the curve, f^α , has a physical meaning of the “acceleration” along the curve which could be nonzero even in the absence of an external force due to intrinsic curvature of the space (in the Newtonian terminology, due to the presence of gravity). We have started with the statement that $f^\alpha = 0$ when Euler – Lagrange equations are satisfied, *i.e.* when a particle moves along a geodesic line. So, this is indeed a free motion - without acceleration.

It could be instructive to re-derive Eq. (12.2) without using Euler – Lagrange equations. In a LGF, we have $\ddot{x}^\alpha = \frac{d\dot{x}^\alpha}{d\tau} = \dot{x}^\alpha_{;\nu} \dot{x}^\nu = 0$, which in Minkowski space-time is a standard law of free motion: 4-acceleration vanishes. Now we use the “comma-goes-to-semicolon” rule to obtain the law of a free motion in an arbitrary non-geodesic frame:

$$\ddot{x}^\alpha = \dot{x}^\alpha_{;\nu} \dot{x}^\nu = 0 \quad \Rightarrow \quad \dot{x}^\alpha_{;\nu} \dot{x}^\nu = \underbrace{\dot{x}^\alpha_{;\nu} \dot{x}^\nu}_{\ddot{x}^\alpha} + \Gamma^\alpha_{\beta\nu} \dot{x}^\beta \dot{x}^\nu = 0.$$

In the absence of an external force, a particle moves along the geodesic, and this coincides with Eq. (12.2). Again, from a pure geometry we are able to see things that describe correct (although, up to now trivial) mechanics.

12.2 Calculating Christoffel symbols using Euler – Lagrange equations

This is a less straightforward but sometimes much quicker way to get $\Gamma^\alpha_{\nu\mu}$. We shall compare the canonical form of geodesic equations, (12.1), with the Euler-Lagrange form, (12.2).

Example 12.1. *Polar coordinates.* $ds^2 = dr^2 + r^2 d\phi^2$ hence $\mathcal{L} = \dot{r}^2 + r^2 \dot{\phi}^2$. Therefore, Euler – Lagrange equations are:

$$\begin{aligned} \frac{d}{ds} \left(\frac{\partial \mathcal{L}}{\partial \dot{r}} \right) - \frac{\partial \mathcal{L}}{\partial r} &= 2\ddot{r} - 2r \dot{\phi}^2 \quad \Rightarrow \quad \ddot{r} - r \dot{\phi}^2 = 0 \quad \Leftrightarrow \quad \ddot{r} + \Gamma^r_{\phi\phi} \dot{\phi}^2 = 0 \\ \frac{d}{ds} \left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}} \right) - \frac{\partial \mathcal{L}}{\partial \phi} &= 2r^2 \ddot{\phi} + 4r \dot{r} \dot{\phi} \quad \Rightarrow \quad \ddot{\phi} + \frac{2}{r} \dot{r} \dot{\phi} = 0 \quad \Leftrightarrow \quad \ddot{\phi} + (\Gamma^\phi_{r\phi} + \Gamma^\phi_{\phi r}) \dot{r} \dot{\phi} = 0 \end{aligned}$$

Comparing the canonical expression with the Euler – Lagrange form gives all nonzero components of $\Gamma^\alpha_{\mu\nu}$, in this case $\Gamma^r_{\phi\phi} = -r$ and $\Gamma^\phi_{r\phi} = \Gamma^\phi_{\phi r} = 1/r$.

Example 12.2. Spherical polar coordinates: $\mathcal{L} = \dot{r}^2 + r^2(\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2)$; hence we have

$$\begin{aligned} \frac{d}{ds} \left(\frac{\partial \mathcal{L}}{\partial \dot{r}} \right) - \frac{\partial \mathcal{L}}{\partial r} &= 2\ddot{r} - 2r\dot{\theta}^2 - 2r\sin^2 \theta \dot{\phi}^2 \\ &\Rightarrow \ddot{r} + \underbrace{(-r)}_{=\Gamma_{\theta\theta}^r} \dot{\theta}^2 + \underbrace{(-r\sin^2 \theta)}_{=\Gamma_{\phi\phi}^r} \dot{\phi}^2 = 0 & \therefore \Gamma_{\theta\theta}^r = -r, \quad \Gamma_{\phi\phi}^r = -r\sin^2 \theta \\ \frac{d}{ds} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) - \frac{\partial \mathcal{L}}{\partial \theta} &= 2r^2\ddot{\theta} + 4r\dot{r}\dot{\theta} - \sin(2\theta) r^2 \dot{\phi}^2 \\ &\Rightarrow \ddot{\theta} + \underbrace{(2/r)}_{=\Gamma_{\theta r}^\theta + \Gamma_{r\theta}^\theta} \dot{r}\dot{\theta} + \underbrace{(-\frac{1}{2}\sin 2\theta)}_{=\Gamma_{\phi\phi}^\theta} \dot{\phi}^2 = 0 & \therefore \Gamma_{r\theta}^\theta = \Gamma_{\theta r}^\theta = 1/r, \quad \Gamma_{\phi\phi}^\theta = -\frac{1}{2}\sin 2\theta \\ \frac{d}{ds} \left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}} \right) - \frac{\partial \mathcal{L}}{\partial \phi} &= 4r\sin^2 \theta \dot{r}\dot{\phi} + 2r^2\sin^2 \theta \ddot{\phi} + 2r^2\sin(2\theta) \dot{\theta}\dot{\phi} \\ &\Rightarrow \ddot{\phi} + \underbrace{(2/r)}_{=\Gamma_{r\phi}^\phi + \Gamma_{\phi r}^\phi} \dot{r}\dot{\phi} + \underbrace{(2\cot \theta)}_{=\Gamma_{\theta\phi}^\phi + \Gamma_{\phi\theta}^\phi} \dot{\theta}\dot{\phi} = 0 & \therefore \Gamma_{r\phi}^\phi = \Gamma_{\phi r}^\phi = 1/r, \quad \Gamma_{\theta\phi}^\phi = \Gamma_{\phi\theta}^\phi = \cot \theta \end{aligned}$$

All Christoffel's symbols, which did not appear in the equation above, vanish.

Example 12.3. Spherical surface. $\mathcal{L} = r^2(\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2)$; hence we have

$$\begin{aligned} \frac{d}{ds} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) - \frac{\partial \mathcal{L}}{\partial \theta} &= r^2(2\ddot{\theta} - \sin(2\theta) \dot{\phi}^2) & \Rightarrow & \ddot{\theta} + \underbrace{(-\frac{1}{2}\sin 2\theta)}_{=\Gamma_{\phi\phi}^\theta} \dot{\phi}^2 = 0 \\ \frac{d}{ds} \left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}} \right) - \frac{\partial \mathcal{L}}{\partial \phi} &= 2r^2(\sin^2 \theta \ddot{\phi} + \sin(2\theta) \dot{\theta}\dot{\phi}) & \Rightarrow & \ddot{\phi} + \underbrace{(2\cot \theta)}_{=\Gamma_{\theta\phi}^\phi + \Gamma_{\phi\theta}^\phi} \dot{\theta}\dot{\phi} = 0 \end{aligned}$$

Results for nonzero components are the same as in example (11.2): $\Gamma_{\theta\phi}^\phi = \Gamma_{\phi\theta}^\phi = \cot \theta$, $\Gamma_{\phi\phi}^\theta = -\frac{1}{2}\sin 2\theta$; the other five components of Γ do not appear and hence equal zero.

Example 12.4. Trumpet surface with the metric $ds^2 = dx^2 - e^{-2x}d\phi^2 \Rightarrow \mathcal{L} = \dot{x}^2 - e^{-2x}\dot{\phi}^2$, where we have deliberately changed the sign of $d\phi^2$ contribution to make space non-Euclidean.

$$\begin{aligned} \frac{d}{ds} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) - \frac{\partial \mathcal{L}}{\partial x} &= 2\ddot{x} - 2e^{-2x}\dot{\phi}^2 & \Rightarrow & \ddot{x} + \underbrace{(-e^{-2x})}_{=\Gamma_{\phi\phi}^x} \dot{\phi}^2 = 0 \\ \frac{d}{ds} \left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}} \right) - \frac{\partial \mathcal{L}}{\partial \phi} &= -2e^{-2x}\ddot{\phi} - 4e^{-2x}\dot{\phi}\dot{x} & \Rightarrow & \ddot{\phi} + \underbrace{(2)}_{=\Gamma_{x\phi}^\phi + \Gamma_{\phi x}^\phi} \dot{x}\dot{\phi} = 0 \end{aligned}$$

So we obtain almost the same expression as in example (11.3),

$$\Gamma_{\phi\phi}^x = -e^{-2x}, \quad \Gamma_{x\phi}^\phi = \Gamma_{\phi x}^\phi = 1.$$

The signs of all $\Gamma_{\mu\nu}^\alpha$ are opposite now. Later we shall see that this does not change the curvature, although the present metric corresponds to a locally pseudo-Euclidean space, while the example of Lec. (11.2) referred to a locally Euclidean space.

Example 12.5. The non-diagonal metric of example (11.5) corresponds to the Lagrangian $\mathcal{L} = y^2\dot{x}^2 + 2x\dot{x}\dot{y}$.

The Euler–Lagrange equations:

$$f_1 = \frac{1}{2} \frac{d}{ds} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) - \frac{1}{2} \frac{\partial \mathcal{L}}{\partial x} = \ddot{x}y^2 + 2y\dot{x}\dot{y} + \dot{x}\ddot{y} + x\ddot{y} - \dot{x}\dot{y} = 0$$

$$f_2 = \frac{1}{2} \frac{d}{ds} \left(\frac{\partial \mathcal{L}}{\partial \dot{y}} \right) - \frac{1}{2} \frac{\partial \mathcal{L}}{\partial y} = x\ddot{x} + \dot{x}^2 - y\dot{x}^2 = 0$$

None of these equations has canonical form. This is natural: we have calculated a covariant component of the acceleration vector, $f_\beta = (f_1, f_2)$, where $f_\beta = g_{\beta\nu}\ddot{x}^\nu + \Gamma_{\beta\mu\nu}\dot{x}^\nu\dot{x}^\mu$. For a non-diagonal metric, it mixes second derivatives of x^μ . We get canonical form, Eq. (12.3), by raising indices: $f^\alpha = g^{\alpha\beta}f_\beta$, i.e. $f^1 = g^{11}f_1 + g^{12}f_2$ and $f^2 = g^{21}f_1 + g^{22}f_2$. As the metric tensor and its inverse here are

$$g_{\mu\nu} = \begin{pmatrix} y^2 & x \\ x & 0 \end{pmatrix} \quad \text{and} \quad g^{\mu\nu} = \begin{pmatrix} 0 & 1/x \\ 1/x & -y^2/x^2 \end{pmatrix}$$

we find:

$$f^1 = g^{12}f_2 = \ddot{x} + \frac{1-y}{x}\dot{x}^2 \quad \therefore \quad \Gamma^1_{11} = \frac{1-y}{x},$$

$$f^2 = g^{21}f_1 + g^{22}f_2$$

$$= \frac{y^2}{x}\ddot{x} + \frac{2y}{x}\dot{x}\dot{y} + \ddot{y} - \frac{y^2}{x}\ddot{x} - \frac{(1-y)y^2}{x^2}\dot{x}^2$$

$$= \ddot{y} - \frac{(1-y)y^2}{x^2}\dot{x}^2 + \frac{2y}{x}\dot{x}\dot{y} \quad \therefore \quad \Gamma^2_{11} = -\frac{(1-y)y^2}{x^2}, \quad \Gamma^2_{21} = \Gamma^2_{12} = \frac{y}{x},$$

in agreement with the results of Example (11.5).

Example 12.6. Let us consider a different non-diagonal metric:

$$ds^2 = (1+y^2)dx^2 + 2y dx dy + dy^2 \quad \Rightarrow \quad \mathcal{L} = (1+y^2)\dot{x}^2 + 2y\dot{x}\dot{y} + \dot{y}^2.$$

Covariant components of the Euler – Lagrange line curvature:

$$f_1 \equiv \frac{1}{2} \frac{d}{ds} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) - \frac{1}{2} \frac{\partial \mathcal{L}}{\partial x} = \ddot{x}(1+y^2) + 2y\dot{x}\dot{y} + y\ddot{y} + \dot{y}^2$$

$$f_2 \equiv \frac{1}{2} \frac{d}{ds} \left(\frac{\partial \mathcal{L}}{\partial \dot{y}} \right) - \frac{1}{2} \frac{\partial \mathcal{L}}{\partial y} = \ddot{y} + y\ddot{x} + \dot{x}\dot{y} - y\dot{x}^2 - \dot{x}\dot{y} = \ddot{y} + y\ddot{x} - y\dot{x}^2 \quad (12.5)$$

None of these equations has canonical form. This is natural: we have calculated a covariant component of the force vector, $f_\beta = (f_1, f_2)$, where $f_\beta = g_{\beta\nu}\ddot{x}^\nu + \Gamma_{\beta\mu\nu}\dot{x}^\mu\dot{x}^\nu$. For a non-diagonal metric, it mixes second derivatives of x^μ . We get canonical form, Eq. (12.3), by raising indices: $f^\alpha = g^{\alpha\beta}f_\beta$, i.e. $f^1 = g^{11}f_1 + g^{12}f_2$ and $f^2 = g^{21}f_1 + g^{22}f_2$. As the metric tensor and its inverse here are

$$g_{\mu\nu} = \begin{pmatrix} 1+y^2 & y \\ y & 1 \end{pmatrix}, \quad g^{\mu\nu} = \begin{pmatrix} 1 & -y \\ -y & 1+y^2 \end{pmatrix},$$

we find:

$$f^1 = f_1 - yf_2 = \ddot{x} + y^2\dot{x}^2 + 2y\dot{x}\dot{y} + \dot{y}^2$$

$$f^2 = -yf_1 + (1+y^2)f_2 = \ddot{y} - y(1+y^2)\dot{x}^2 - 2y^2\dot{x}\dot{y} - y\dot{y}^2.$$

Hence,

$$\Gamma^1_{11} = y^2, \quad \Gamma^1_{12} = y, \quad \Gamma^1_{22} = 1; \quad \Gamma^2_{11} = -y(1+y^2), \quad \Gamma^2_{12} = -y^2, \quad \Gamma^2_{22} = -y.$$

One can obtain the equations for $\ddot{x}^\mu$ without any explicit knowledge of $g^{\mu\nu}$, but just by finding two combinations of equations for f_β , in this case Eqs. (12.5), that leave only one second derivative in each. Reducing the coefficient attached to the second derivative to 1, one obtains geodesic equations in canonical form. However, $f^\alpha = g^{\alpha\beta} f_\beta$ gives automatically the necessary row operations that allows one to get the equations containing 2nd derivatives of one of the variables with the coefficient 1.

We have exploited the situation where $f^\alpha = 0$, which describes geodesic motion. When $f^\alpha \neq 0$, it can be considered as a non-gravitational force acting on a particle of unit mass in the presence of gravity: it is this force that makes it deviate from geodesics. If we consider a finite-size body falling in a inhomogeneous gravitational field, only its mass centre moves along a geodesics while other points are made to move along non-geodesic lines by forces of rigidity. From Newtonian viewpoint, such forces are compensated by *tidal* forces. Considering geodesic deviations and tidal forces will lead us to Einstein's field equations.

Lecture 13. Geodesic deviation and Riemann curvature

We have considered particle's and light motion in curved spacetime with a given metric. Now it is time to find out what metrics are allowed by the principles of the general theory of relativity. First we look at empty space – that is points (events) in spacetime that are away from mass distribution – and give arguments in favour of Einstein's equations.

The centerpiece of GR is the principle of equivalence. It states (loosely) that gravitation is locally equivalent to inertia. We have seen that the mathematical meaning of this statement is in the possibility to choose a local geodesic frame where SR is valid. Now, having in mind the fact that inertia and gravity which is in they are equivalent **only locally** we shall focus on the *difference* between them. For any finite-size particle only the centre of mass moves along a geodesics while other parts of the body, kept together by non-gravitational forces (e.g., elasticity), moves along lines equidistant from geodesics. Such an equidistant line with necessity **deviates** from geodesics in a curved spacetime as any gravitational field is non-uniform. Our aim is to describe this geodesic deviation and relate it in the 4-dimensional spacetime to Newtonian tidal forces. Although all considerations in the next section are valid for geodesic deviation in an arbitrary curved space, we would focus on the 4-dimensional spacetime, assuming all indices to be $\alpha \equiv \{0, i\} \equiv \{0, 1, 2, 3\}$.

13.1 Geodesic deviation

The rate of change of a vector ξ^α parallel-transferred along a worldline $x^\mu = x^\mu(\tau)$ characterised by velocity $u^\mu = \dot{x}^\mu$ is given by the **covariant dot** operation, Eq. (10.10), $\nabla_u \xi^\alpha \equiv u^\mu \xi^\alpha_{;\mu}$. If the worldline is geodesics, the velocity itself does not change along it, as $\nabla_u u^\alpha = 0$ is equivalent to geodesic equation (12.3).¹ If the worldline is not geodesic, particle moves along it with the acceleration

$$f^\alpha = \nabla_u^2 x^\alpha = \nabla_u u^\alpha \equiv u^\mu u^\alpha_{;\mu} = u^\mu u^\alpha_{;\mu} + \Gamma^\alpha_{\nu\mu} u^\nu u^\mu = \ddot{x}^\alpha + \Gamma^\alpha_{\mu\nu} \dot{x}^\mu \dot{x}^\nu.$$

Let us introduce a family of 'close' **equidistant** worldlines (curves) labelled by a certain continuous parameter η : $x^\alpha = x^\alpha(\tau; \eta)$. We label the **geodesic** curve in this family, say P , by η_0 so that $\nabla_u^2 x^\alpha(\tau; \eta_0) = 0$. Let Q be an infinitesimally close curve with $\eta = \eta_0 + d\eta$. As Q is not geodesic in a nonuniform field, a particle moves along it with acceleration f^α (in other words, a non-gravitational force f^α acts on it to keep it on a non-geodesic worldline):

$$f^\alpha = \nabla_u^2 x^\alpha(\tau; \eta) = \nabla_u^2 x^\alpha(\tau; \eta_0 + d\eta) - \underbrace{\nabla_u^2 x^\alpha(\tau; \eta_0)}_{=0} = \nabla_u^2 \left[\frac{\partial x^\alpha(\tau; \eta)}{\partial \eta} d\eta \right] \equiv \nabla_u^2 \left[v^\alpha(\tau; \eta) \right] d\eta \quad (13.1)$$

Changing η in $x^\alpha(\tau_0; \eta)$ with a fixed τ_0 describes motion '**across**' the family along the η -dependent geodesics R , $x^\alpha = x^\alpha(\tau_0, \eta)$, with velocity $v^\alpha = \frac{\partial x^\alpha}{\partial \eta}$. R is intersecting with the τ -geodesics P at point A with $x^\alpha(\eta_0, \tau_0)$. In Eq. (13.1) we have implicitly used the equidistance of P and Q by writing $dx(\tau, \eta) = \frac{\partial x(\tau, \eta)}{\partial \eta} d\eta$: we did not include $\frac{d}{d}\tau$ as the 'distance', i.e. the interval $dx^\alpha dx_\alpha$ is τ -independent.

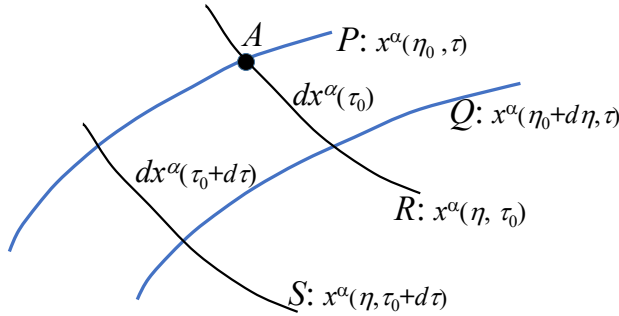
¹Let us stress again: geometrically, this velocity is a tangent vector to the curve; the fact that it does not change when parallel-transferred means that the geodesics is not only the shortest but also the straightest line.

The linear Minkowski space tangential to spacetime at A includes all geodesics passing through it. Choosing a LGF in this space, we have $\nabla_u \equiv \partial_\tau$ and $\nabla_v \equiv \partial_\eta$, so that

$$u^\mu v^\alpha{}_{;\mu} = v^\alpha{}_{;\mu} \frac{\partial x^\mu}{\partial \tau} = \frac{\partial v^\alpha}{\partial \tau} = \frac{\partial}{\partial \tau} \left(\frac{\partial x^\alpha}{\partial \eta} \right) = \frac{\partial^2 x^\alpha}{\partial \tau \partial \eta} = \frac{\partial}{\partial \eta} \left(\frac{\partial x^\alpha}{\partial \tau} \right) = \frac{\partial u^\alpha}{\partial \eta} = u^\alpha{}_{;\mu} \frac{\partial x^\mu}{\partial \eta} = v^\mu u^\alpha{}_{;\mu}.$$

As this is a 1st order differential relation, apply the CG2S rule to find that at the intersection point

$$u^\mu v^\alpha{}_{;\mu} = v^\mu u^\alpha{}_{;\mu} \quad \Leftrightarrow \quad \boxed{\nabla_u v^\alpha = \nabla_v u^\alpha} \quad (13.2)$$



The differential $dx^\alpha(\tau) = v^\alpha(\tau) d\eta$ measures the spacelike interval between equidistant P and Q . Equidistant means that the interval remains constant with changing τ . In a flat spacetime, this does not require any effort as the worldline parallel to geodesic is also a geodesic. In a curved space, it requires force f^α which thus characterises the curvature.

Using Eq. (13.2), we transform Eq. (13.1) in the vicinity of the P – R intersection:

$$\begin{aligned} \frac{\partial f^\alpha}{\partial \eta} &= \nabla_u \nabla_u v^\alpha = \nabla_u (\nabla_v u^\alpha) - \nabla_v (\nabla_u u^\alpha) \equiv \underbrace{(u^\mu \nabla_\mu v^\nu \nabla_\nu - v^\mu \nabla_\mu u^\nu \nabla_\nu)}_{=0} u^\alpha \\ &= v^\nu u^\mu \nabla_\mu u^\alpha{}_{;\nu} + \underbrace{u^\mu v^\nu{}_{;\mu} u^\alpha}_{\equiv \nabla_u v^\nu}{}_{;\nu} - v^\mu u^\nu \nabla_\mu u^\alpha{}_{;\nu} - \underbrace{v^\mu u^\nu{}_{;\mu} u^\alpha}_{\equiv \nabla_v u^\nu}{}_{;\nu} \\ &= u^\mu v^\nu [\nabla_\mu, \nabla_\nu] u^\alpha + \underbrace{(\nabla_u v^\nu - \nabla_v u^\nu)}_{=0, \text{ Eq. (13.2)}} u^\alpha{}_{;\nu} = (u^\alpha{}_{;\nu\mu} - u^\alpha{}_{;\mu\nu}) u^\mu v^\nu. \quad (13.3) \end{aligned}$$

The l.h.s. of this expression, $df^\alpha/d\eta$, is a contravariant vector, and so must be the r.h.s. Hence, the expression in $(\dots)$ must be a tensor with the index structure ${}^\alpha{}_{\mu\nu}$. But this tensor must be linearly proportional to the velocity u . Hence we arrive at

Definition 13.1. The Riemann curvature tensor (or simply the Riemann) is defined by the commutator of two covariant derivatives of the four-velocity along a geodesics:

$$\boxed{u^\alpha{}_{;\nu\mu} - u^\alpha{}_{;\mu\nu} \equiv [\nabla_\mu, \nabla_\nu] u^\alpha = R^\alpha{}_{\beta\mu\nu} u^\beta}$$

Now, substituting this definition into Eq. (13.3) and multiplying it by $d\eta$ (which is unaffected by derivatives), we obtain at any point of the equidistant from geodesics curve Q , $\eta = \eta_0 + d\eta \in Q$, the **equation of geodesic deviation**:

$$\boxed{f^\alpha = \nabla_u^2 x^\alpha(\tau; \eta) = \nabla_u u^\alpha = R^\alpha{}_{\beta\mu\nu} u^\beta u^\mu \xi^\nu} \quad (13.4)$$

Here $u^\mu = dx^\mu/d\tau$ is the four-velocity along geodesics, $x^\mu = x^\mu(\tau; \eta_0)$, while $\xi^\nu \equiv v^\nu d\eta \equiv x^\nu(\tau; \eta_0 + d\eta) - x^\nu(\tau; \eta_0)$ defines the spacelike interval between it and the equidistant curve.² In the Minkowski space (*i.e.* spacetime without any source of gravity) such an equidistance curve is also a geodesics, *i.e.* $R^\alpha_{\beta\mu\nu} = 0$. Hence the Riemann, $R^\alpha_{\beta\mu\nu}$, characterises curvature.³

The Riemann by itself is independent of a geodesics and is expressible via Christoffel symbols and its derivatives, *i.e.* via first and second derivatives of the metric tensor:

$$R^\alpha_{\beta\mu\nu} = -\Gamma^\alpha_{\beta\mu,\nu} + \Gamma^\alpha_{\beta\nu,\mu} - \Gamma^\sigma_{\beta\mu}\Gamma^\alpha_{\sigma\nu} + \Gamma^\sigma_{\beta\nu}\Gamma^\alpha_{\sigma\mu} \quad (13.5)$$

We will use this expression regularly, starting from the next lecture. I give a full derivation in section 13.2.⁴ Note that, being independent of geodesics, the Riemann acts in the same way not only on u^β but on any contravariant vector:

$$R^\alpha_{\beta\mu\nu}\xi^\beta = [\nabla_\mu, \nabla_\nu]\xi^\alpha \equiv \xi^\alpha_{;\nu\mu} - \xi^\alpha_{;\mu\nu}. \quad (13.6)$$

From the Newtonian viewpoint, spacial components of f^α are components of the force that keeps particle of unit mass along the non-geodesic curve Q . This force counter-balances forces created by inhomogeneity of gravitational field, *i.e.* the tidal forces. Writing f^α as $f^\alpha = \Delta^\alpha_\nu \xi^\nu$, we define the tidal tensor Δ^α_ν . Comparing with Eq. (13.4), we express it via the Riemann:

$$\Delta^\alpha_\nu \equiv \frac{\partial f^\alpha}{\partial x^\nu} = R^\alpha_{\beta\mu\nu} u^\beta u^\mu \quad (13.7)$$

We shall use this expression and its Newtonian counterpart to establish Einstein's equations.

13.2 Expressing the Riemann via Christoffel symbols

We start from the definition, Eq. (13.1), $R^\alpha_{\beta\mu\nu}u^\beta = (u^\alpha_{;\nu\mu} - u^\alpha_{;\mu\nu})$ and apply the general formulae for the covariant derivative, Eq. (10.2), to the 2nd covariant derivatives of the 2nd rank tensors $u^\alpha_{;\mu}$ and $u^\alpha_{;\nu}$:

$$\begin{aligned} R^\alpha_{\beta\mu\nu}u^\beta &= (u^\alpha_{;\nu})_{;\mu} - (u^\alpha_{;\mu})_{;\nu} = (u^\alpha_{;\nu})_{;\mu} + \Gamma^\alpha_{\beta\mu}u^\beta_{;\nu} - \Gamma^\beta_{\nu\mu}u^\alpha_{;\beta} \\ &\quad - (u^\alpha_{;\mu})_{;\nu} - \Gamma^\alpha_{\beta\nu}u^\beta_{;\mu} + \Gamma^\beta_{\mu\nu}u^\alpha_{;\beta} \\ &= (u^\alpha_{;\nu})_{;\mu} - (u^\alpha_{;\mu})_{;\nu} + \Gamma^\alpha_{\beta\mu}u^\beta_{;\nu} - \Gamma^\alpha_{\beta\nu}u^\beta_{;\mu}, \end{aligned}$$

where we have used the symmetry $\Gamma^\beta_{\nu\mu} = \Gamma^\beta_{\mu\nu}$. Now we use Eq. (10.2) for the remaining covariant derivatives:

$$\begin{aligned} R^\alpha_{\beta\mu\nu}u^\beta &= (u^\alpha_{;\nu} + \Gamma^\alpha_{\beta\nu}u^\beta)_{;\mu} - (u^\alpha_{;\mu} + \Gamma^\alpha_{\beta\mu}u^\beta)_{;\nu} \\ &\quad + (u^\beta_{;\nu} + \Gamma^\beta_{\lambda\nu}u^\lambda)\Gamma^\alpha_{\beta\mu} - (u^\beta_{;\mu} + \Gamma^\beta_{\lambda\mu}u^\lambda)\Gamma^\alpha_{\beta\nu} \\ &= u^\alpha_{;\nu\mu} + \Gamma^\alpha_{\beta\nu,\mu}u^\beta + \Gamma^\alpha_{\beta\nu}u^\beta_{;\mu} - u^\alpha_{;\mu\nu} - \Gamma^\alpha_{\beta\mu,\nu}u^\beta - \Gamma^\alpha_{\beta\mu}u^\beta_{;\nu} \\ &\quad - \Gamma^\alpha_{\beta\nu}\Gamma^\beta_{\lambda\mu}u^\lambda - \Gamma^\alpha_{\beta\mu}\Gamma^\beta_{\lambda\nu}u^\lambda + \Gamma^\alpha_{\beta\mu}\Gamma^\beta_{\lambda\mu}u^\lambda + \Gamma^\alpha_{\beta\mu}u^\beta_{;\nu} \end{aligned}$$

²Let us stress that the interval is spacelike as both events on P and Q are at the same proper time τ .

³Equation (13.4) also describes a relative acceleration between particles going along close geodesics separated by a constant vector ξ^ν . Obviously, geodesics are not equidistant in a curved space.

⁴For intellectually curious, I will also give an alternative derivation based on parallel transfer around a closed loop but will not show this in the class.

Here all terms containing derivatives of u^β cancel each other. After mutually renaming the dummy pairs $\lambda \rightleftharpoons \beta$, we find

$$R^\alpha_{\beta\mu\nu} u^\beta = [-\Gamma^\alpha_{\beta\mu,\nu} + \Gamma^\alpha_{\beta\nu,\mu} - \Gamma^\sigma_{\beta\mu} \Gamma^\alpha_{\sigma\nu} + \Gamma^\sigma_{\beta\nu} \Gamma^\alpha_{\sigma\mu}] u^\beta.$$

Cancelling u^β on both sides leads to Eq. (13.5) for the Riemann.

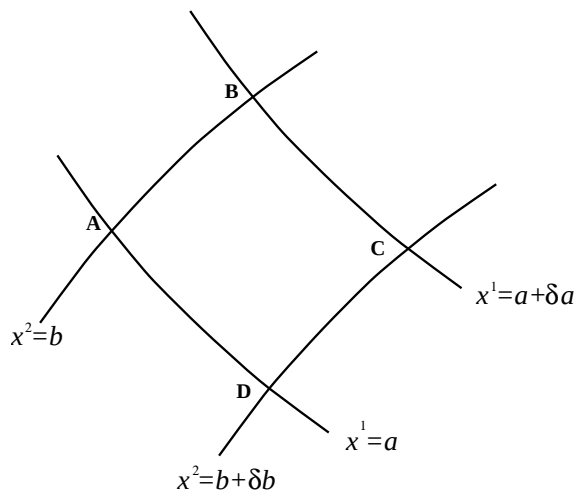
13.3 Parallel transfer along a closed loop⁴

Here I give an alternative definition (and derivation) of the Riemann, which is more longwinded but clearly shows its geometric meaning and is adopted in the majority of GR textbooks.⁵

The idea: we parallel-transfer vector ξ along a closed small loop (e.g., a small parallelogram) belonging to a curved space (eg, lying on a curved surface). The vector comes back changed by its journey, this change being totally due to the space curvature, since in a flat space it will return unchanged, independently of the choice of a loop or of coordinates. In a curved space, its change depends on the choice of the loop but is covariant, i.e. has the same form in any coordinates. The simplest choice of two sides of the parallelogram is to take them along coordinate lines in a given coordinate system. Then a covariant expression for the change in a (contravariant) vector component must involve a tensor of 4th rank. Thus, we anticipate that

$$\delta\xi^\mu = R^\mu_{\alpha\beta\gamma} x^\alpha x^\beta \xi^\gamma$$

where x^α and x^β are two (coordinate) vectors defining the closed loop. Now we need to express this tensor in terms of Christoffel symbols, i.e. to relate it to the metric $g_{\mu\nu}$.



Consider an infinitesimal loop whose four sides are the coordinate lines,

$$\begin{aligned} AD : x^1 &= a & BC : x^1 &= a + \delta a \\ AB : x^2 &= b & DC : x^2 &= b + \delta b. \end{aligned}$$

We transfer vector ξ first from A to B:

$$\begin{aligned} \xi^\mu(B) &= \xi^\mu(A) - \int_A^B \Gamma^\mu_{\beta 1} \xi^\beta dx^1 \\ &= \xi^\mu(A) - \int_a^{a+\delta a} \Gamma^\mu_{\beta 1} \xi^\beta dx^1. \end{aligned}$$

Repeating such an operation along three sides we obtain the total change in ξ^μ after the return to A:

$$\begin{aligned} \delta\xi^\mu(A) &\equiv \xi^\mu(A_f) - \xi^\mu(A_0) = \\ &= - \left(\int_a^{a+\delta a} \underbrace{\Gamma^\mu_{\beta 1} \xi^\beta dx^1}_{\text{along } x^2=b} + \int_{a+\delta a}^a \underbrace{\Gamma^\mu_{\beta 1} \xi^\beta dx^1}_{x^2=b+\delta b} \right) - \left(\int_b^{b+\delta b} \underbrace{\Gamma^\mu_{\beta 2} \xi^\beta dx^2}_{x^1=a+\delta a} + \int_{b+\delta b}^b \underbrace{\Gamma^\mu_{\beta 2} \xi^\beta dx^2}_{x^1=a} \right). \end{aligned}$$

⁵Not to be shown in the class

Note that this would cancel in pairs if we were in a flat space (both Γ and ξ would be constant on the loop in this case). Now, to lowest order we have (with $f \equiv \Gamma\xi$)

$$\begin{aligned} \int_{a+\delta a}^a f(x^2=b+\delta b) dx^2 &= \int_{a+\delta a}^a f(x^2=b) dx^2 + \int_{a+\delta a}^a \partial_2 f(x^2) \bigg|_{x^2=b} \delta b dx^2 \\ &= - \int_a^{a+\delta a} f(x^2=b) dx^2 + \delta a \delta b \partial_2 f(x^2) \bigg|_{x^2=b}, \end{aligned}$$

where $\partial_2 \equiv \partial/\partial x^2$. It gives, on substituting into the previous expression

$$\delta\xi^\mu(A) = \delta a \delta b \left[(\Gamma^\mu_{\beta 1} \xi^\beta)_{,2} - (\Gamma^\mu_{\beta 2} \xi^\beta)_{,1} \right] = \delta a \delta b \left[\Gamma^\mu_{\beta 1,2} \xi^\beta + \Gamma^\mu_{\beta 1} \xi^\beta_{,2} - \Gamma^\mu_{\beta 2,1} \xi^\beta - \Gamma^\mu_{\beta 2} \xi^\beta_{,1} \right]$$

Now we can use that, by definition, in parallel transform $\xi^\beta_{;\mu} = 0$, i.e. $\xi^\beta_{,\mu} = -\Gamma^\beta_{\alpha\mu} \xi^\alpha$. Therefore,

$$\begin{aligned} \delta\xi^\mu(A) &= \delta a \delta b \left[\Gamma^\mu_{\beta 1,2} \xi^\beta - \Gamma^\mu_{\beta 2,1} \xi^\beta + \Gamma^\mu_{\beta 2} \Gamma^\beta_{\alpha 1} \xi^\alpha - \Gamma^\mu_{\beta 1} \Gamma^\beta_{\alpha 2} \xi^\alpha \right] = \\ &= - \left[\Gamma^\mu_{\beta 1,2} - \Gamma^\mu_{\beta 2,1} + \Gamma^\mu_{\alpha 2} \Gamma^\alpha_{\beta 1} - \Gamma^\mu_{\alpha 1} \Gamma^\alpha_{\beta 2} \right] \xi^\beta \delta a \delta b \end{aligned}$$

A privilege role of x^1 and x^2 can now be abandoned, and we can rewrite the expression for a parallel transfer around any infinitesimal loop. Note that $\delta a \delta b$ is the coordinate area of the loop. Important that the expression in brackets is antisymmetric in 1 and 2. Thus, it should be antisymmetric in any indices σ and λ .

Fortunately, in general the area of a two-dimensional coordinate loop in a space of higher dimensions is given by the antisymmetric second rank tensor: $\delta S^{\sigma\lambda} = dx^\sigma dx^\lambda - dx^\lambda dx^\sigma$. This is easy to check using the area of a parallelogram in a flat 2D space as an example. We have used in SR that area is usually a vector, whose direction is normal to the surface. However, such a vector is easy to introduce only in 3D space by $S_\mu = \varepsilon_{\mu\sigma\lambda} S^{\sigma\lambda}$. In general, a tensor expression for the area of a loop made as a product of two vectors is more natural, and it is antisymmetric as it should vanish when these vectors are parallel.

Thus, we obtained in general:

$$\delta\xi^\mu(A) = \left[\Gamma^\mu_{\beta\sigma,\lambda} - \Gamma^\mu_{\beta\lambda,\sigma} + \Gamma^\alpha_{\beta\sigma} \Gamma^\mu_{\alpha\lambda} - \Gamma^\alpha_{\beta\lambda} \Gamma^\mu_{\alpha\sigma} \right] \xi^\beta dS^{\sigma\lambda}$$

Expression on the l.h.s. is a contravariant vector component attached to A (important, that it does not involve any other points in space). A product outside the brackets on the r.h.s. is a 3d rank tensor. Therefore, the expression within the brackets gives components of the tensor. Obviously, up to the names of indices it coincides with Eq. (13.5) for the Riemann:

$$\boxed{R^\mu_{\alpha\beta\gamma} = -\Gamma^\mu_{\alpha\beta,\gamma} + \Gamma^\mu_{\alpha\gamma,\beta} - \Gamma^\nu_{\alpha\beta} \Gamma^\mu_{\nu\gamma} + \Gamma^\nu_{\alpha\gamma} \Gamma^\mu_{\nu\beta}}$$

Thus, the Riemann governs a change of (contravariant) vector after parallel transfer along a closed loop.

A useful mnemonic rule: covariant indices of R are repeated in the same order in both negative terms on the r.h.s. The positive terms are obtained by the interchange of the last two indices.

There is no overall agreement about the sign of the Riemann tensor and even about the placement of its indices. Always check the convention in a book you read!

We can *simplify* (which is the synonym of *contract*) the Riemann over a pair indices:

13.4 Ricci tensor and scalar curvature

The Riemann is rather complicated; we will see that in the $4D$ spacetime it may contain up to 20 independent components. Its simplifications by contracting one or two pairs of indices are simpler and quite important on their own.

Definition 13.2. The contraction of the Riemann tensor, $R_{\mu\nu} \equiv R^\alpha_{\mu\alpha\nu} \equiv g^{\alpha\beta} R_{\alpha\mu\beta\nu}$, is called the Ricci curvature tensor.

This second rank tensor could be also used in mixed representation, $R^\beta_{\nu} = g^{\beta\mu} R_{\mu\nu}$. Further contracting the Ricci leads us to the simplest characteristic of the curved space:

Definition 13.3. The scalar curvature is defined by $R = g^{\alpha\beta} R_{\alpha\beta}$.

We shall see that in the $2D$ case R coincides with the standard (for some people) Gaussian curvature.

13.5 Symmetries of the Riemann tensor

It is clearly seen from the expression for the Riemann, Eq. (13.5), that it is antisymmetric in the last pair of indices:

$$R^\mu_{\nu\alpha\beta} = -R^\mu_{\nu\beta\alpha} \quad (13.8)$$

We are looking for other symmetries. To this end, it is convenient to consider the Riemann in LCF and use the CG2S rule, Theorem (10.1). In LCF, Christoffel symbols (and first derivatives of g) vanish. Thus, in Eq. (13.5) all the Γ 's vanish but their derivatives remain:

$$R^\mu_{\nu\alpha\beta} = -\Gamma^\mu_{\nu\alpha,\beta} + \Gamma^\mu_{\nu\beta,\alpha}.$$

Using Eq. (11.5) for Γ and keeping only 2nd derivatives of $g_{\mu\nu}$, we have

$$\Gamma^\mu_{\nu\alpha,\beta} = \frac{1}{2} g^{\lambda\mu} \Gamma_{\lambda\nu\alpha,\beta} = \frac{1}{2} g^{\lambda\mu} (g_{\lambda\nu,\alpha\beta} + g_{\lambda\alpha,\nu\beta} - g_{\nu\alpha,\lambda\beta}).$$

Therefore,

$$R^\mu_{\nu\alpha\beta} = \frac{1}{2} g^{\lambda\mu} (-g_{\lambda\nu,\alpha\beta} - g_{\lambda\alpha,\nu\beta} + g_{\nu\alpha,\lambda\beta} + g_{\lambda\nu,\beta\alpha} + g_{\lambda\beta,\nu\alpha} - g_{\nu\beta,\lambda\alpha}).$$

The first and the fourth terms cancel and, lowering the first index, we obtain

$$R_{\mu\nu\alpha\beta} = \frac{1}{2} (-g_{\mu\alpha,\nu\beta} + g_{\nu\alpha,\mu\beta} + g_{\mu\beta,\nu\alpha} - g_{\nu\beta,\mu\alpha}). \quad (13.9)$$

It is immediately obvious that in the LCF

$$R_{\mu\nu\alpha\beta} = -R_{\nu\mu\alpha\beta} \quad \text{and} \quad R_{\mu\nu\alpha\beta} = R_{\alpha\beta\mu\nu} \quad (13.10)$$

Since this is a tensor equation, it should remain valid in any frame so that the Riemann is antisymmetric in the first pair of indices as well, and symmetric w.r.t. swapping the first and second pairs of indices.

Now notice that we can lower or raise indices simultaneously in both sides of any tensor relation. Therefore, from Eq. (13.6) follows that

$$\xi_{\alpha;\nu\mu} - \xi_{\alpha;\mu\nu} = R_{\alpha\beta\mu\nu} \xi^\beta = -R_{\beta\alpha\mu\nu} \xi^\beta = -R^\beta_{\alpha\mu\nu} \xi_\beta = R^\beta_{\alpha\nu\mu} \xi_\beta$$

Let's now use this for symmetries choosing $\xi_\beta \equiv \varphi_{;\beta}$. For an arbitrary scalar, $\varphi_{,\alpha} = \varphi_{;\alpha}$ so that from $\varphi_{,\alpha\beta} = \varphi_{,\beta\alpha}$ valid in any LCF follows $(\varphi_{;\alpha})_{;\beta} = (\varphi_{;\beta})_{;\alpha}$. This is the 1st order differential relation for the contravariant vector $\xi_\beta \equiv \varphi_{;\beta}$, i.e. $\xi_{\alpha;\beta} = \xi_{\beta;\alpha}$, valid in any LCF. Then the CG2S rule ensures that in the arbitrary frame $\varphi_{;\alpha\beta} - \varphi_{;\beta\alpha} = 0$, i.e. (changing the names of indices)

$$0 = (\phi_{;\rho\sigma} - \phi_{;\sigma\rho})_{;\nu} = \phi_{;\rho\sigma\nu} - \phi_{;\sigma\rho\nu}$$

On the other hand,

$$\phi_{;\sigma\rho\nu} - \phi_{;\sigma\nu\rho} = (\phi_{;\sigma})_{;\rho\nu} - (\phi_{;\sigma})_{;\nu\rho} = R^\alpha_{\sigma\rho\nu}\phi_{;\alpha}$$

Performing in the last equation all cyclic permutations and adding these together, we find that all the terms on the l.h.s. cancel in pairs, which leaves

$$0 = \phi_{;\alpha} (R^\alpha_{\rho\nu\sigma} + R^\alpha_{\nu\sigma\rho} + R^\alpha_{\sigma\rho\nu})$$

As ϕ is arbitrary, this leads to the permutation symmetry

$$R^\alpha_{\beta\mu\nu} + R^\alpha_{\mu\nu\beta} + R^\alpha_{\nu\beta\mu} = 0 \quad (13.11)$$

13.5.1 Bianchi identities

We have up to now seen some symmetries between different components of R . Our next step is to derive extremely important for the future differential identities.

We differentiate R with respect to x^γ in a LCF to obtain

$$\begin{aligned} R_{\mu\nu\alpha\beta;\gamma} &= \frac{1}{2} (\mathfrak{g}_{\nu\beta,\mu\alpha\gamma} + \mathfrak{g}_{\mu\alpha,\nu\beta\gamma} - \mathfrak{g}_{\nu\alpha,\mu\beta\gamma} - \mathfrak{g}_{\mu\beta,\nu\alpha\gamma}) \\ R_{\mu\nu\beta\gamma;\alpha} &= \frac{1}{2} (\mathfrak{g}_{\nu\gamma,\mu\beta\alpha} + \mathfrak{g}_{\mu\beta,\nu\gamma\alpha} - \mathfrak{g}_{\nu\beta,\mu\gamma\alpha} - \mathfrak{g}_{\mu\gamma,\nu\beta\alpha}) \\ R_{\mu\nu\gamma\alpha;\beta} &= \frac{1}{2} (\mathfrak{g}_{\nu\alpha,\mu\gamma\beta} + \mathfrak{g}_{\mu\gamma,\nu\alpha\beta} - \mathfrak{g}_{\nu\gamma,\mu\alpha\beta} - \mathfrak{g}_{\mu\alpha,\nu\gamma\beta}) \end{aligned}$$

Adding this together and using that the partial derivatives commute and $\mathfrak{g}_{\mu\nu}$ is symmetric, we obtain that

$$R_{\mu\nu\alpha\beta;\gamma} + R_{\mu\nu\beta\gamma;\alpha} + R_{\mu\nu\gamma\alpha;\beta} = 0.$$

Then we use the CG2S rule to get the following tensorial (i.e. fully covariant) Bianchi identities:

$$R^{\mu\nu}_{\alpha\beta;\gamma} + R^{\mu\nu}_{\beta\gamma;\alpha} + R^{\mu\nu}_{\gamma\alpha;\beta} = 0. \quad (13.12)$$

Here we raised the first two indices to stress that they are idle in this identity.

13.5.2 Curvature is complicated

The curvature tensor is quite complicated. The total number of elements of $R^\mu_{\nu\alpha\beta}$ is d^4 , i.e. 16 for 2D case, and 256 for 4D case. Because of the symmetries, the number of independent components is significantly reduced. I show below that it equals to $d^2(d^2 - 1)/12$ which is only one component in the 2D case but quite nontrivial 20 components in the relevant 4D case. Each component is defined by Eq. (13.5) that, in 4D case, contains (summation convention!) 10 terms on each side. If you remember that each of this term includes Γ 's which by itself are not that easy to calculate, you see why calculations in GR could be extremely bulky even when straightforward (which is also usually not the case!)

While proving the general relation in the next section, let us consider $d = 2, 3$, and 4 , as most important for us. For $d = 2$, all indices are either 1 or 2. Because of the antisymmetry in the first and the second pairs, they must alternate. (Obviously, $R_{\alpha\beta 11} = 0$). Thus, we can have only R_{1212} , as all other nonzero elements can be obtained by symmetry (e.g., $R_{2112} = -R_{1212}$, etc). Note that although all covariant components could differ only by sign, R^2_{112} could be quite different from R^1_{212} , if $g^{22} \neq g^{11}$.

For $d = 4$, there are ${}^2C_4 = 6$ ways to make a pair of different indices. Then, there are 6 components of R that have the same first and second pair of indices, and $15 (= 6 \times 5/2)$ components where these pairs are different. Three out of these 15 components, namely those which have all indices different, are related by

$$R^0_{123} + R^0_{231} + R^0_{312} = 0.$$

Therefore, there are $6 + 15 - 1 = 20$ **independent** components. In a curved space, some (but not necessarily all) of them are nonzero. In $d = 3$, you easily check that there are 6 independent components.

We will show later that Einstein's equations in empty space can be written in terms of the Ricci tensor, Def. (13.2), simply as $R_{\mu\nu} = 0$. It has only two indices and is clearly symmetric as $R_{\mu\nu} = g^{\alpha\beta} R_{\alpha\mu\beta\nu} = g^{\alpha\beta} R_{\beta\nu\alpha\mu} = R_{\nu\mu}$. Therefore, it has no more than $\frac{1}{2}d(d-1) + d = \frac{1}{2}d(d+1)$ components. In $d = 2$ this gives 3 but since only one component of the Riemann is independent, so is for the Ricci. In $d = 3$, there is no more than 6, which is the same number as for the Riemann. Therefore, if $R_{\alpha\beta} = 0$, so is $R^\mu_{\nu\alpha\beta} = 0$, i.e. Einstein's equations can be satisfied only in the absence of curvature, i.e. in the absence of gravity. Hence, $d = 4$ is the minimal dimensionality for the GR gravitational theory to exist. Thus, although we can use lower-dimensional spaces for modelling the motion in a curved space-time, it could be no direct analogue of the field equations in $d < 4$.

A few more words on the construction of locally (pseudo)Cartesian frame – LCF where we choose the basis coordinate set of mutually normal tangent vectors. Each tangent defines a geodesic line along which it could be transferred. So we can say that we choose a set of mutually normal geodesic lines for a bases: given any coordinate set $\{x^\mu\}$ in the vicinity of the point $x = 0$, we choose a new set as $y^\mu = x^\mu + \frac{1}{2}\Gamma^\mu_{\nu\lambda}x^\nu x^\lambda$. Then, $\ddot{y}^\mu = 0$ so that all Γ vanish at this point in new coordinates. Their derivatives would vanish, however, only if the underlying space is flat, so that $R^\mu_{\nu\alpha\beta} = 0$.

13.5.3 How many independent components in the Riemann?⁶

The Riemann can have two distinct indices (like $R_{\alpha\beta\alpha\beta}$), three (like $R_{\alpha\beta\alpha\mu}$), or four (like $R_{\alpha\beta\mu\nu}$). The number of independent non-zero components with just two distinct indices is equal to the number of ways of choosing 2 out of d indices, i.e. $\frac{1}{2}d(d-1)$, and all the components with the same two pairs of indices are related by the symmetries. The number of independent non-zero components with 3 distinct indices is equal to the number of ways of choosing these indices, dC_3 multiplied by 3 for the number of choices of the repeated index. This leaves the components with 4 distinct indices, which is $4! \times {}^dC_4$. However, three binary symmetries of Eqs. (13.8) and (13.10) reduces this number by the factor of $\frac{1}{8}$, and the skew symmetry of Eq. (13.11) by the factor of $\frac{2}{3}$. Hence we obtain

$$\frac{d(d-1)}{2} + 3 \frac{d(d-1)(d-2)}{3!} + \frac{4!}{8} \frac{2}{3} \frac{d(d-1)(d-2)(d-3)}{4!} = \frac{d^2(d^2-1)}{12}.$$

⁶Not to be shown in the class

Lecture 14. Riemann tensor and curvature

To understand all the symmetries of the Riemann, we formulate how to write it in locally (pseudo)Euclidean space (tangential to a given curved space). Then we have a training session of calculating the Riemann and the scalar curvature in a 2D space.

14.1 Main steps in calculating the curvature. Examples

- (i) calculate $g_{\mu\nu}$ and $g^{\mu\nu}$
- (ii) find all non-vanishing derivatives $g_{\mu\nu,\alpha}$ (Alternatively, reduce the Euler – Lagrange equations to canonical form)
- (iii) calculate the Christoffel symbols
- (iv) calculate some components of the Riemann tensor $R^\mu_{\nu\alpha\beta}$
- (v) reduce the Riemann $R^\alpha_{\mu\beta\nu}$ to the Ricci $R_{\mu\nu}$ and the curvature R .

In 2D case, as we saw, there is only one independent component of the Riemann tensor. So, calculations should be much easier than in $(1 + 3)D$ space (up to 20 independent components!).

Example 14.1. The 2D spherical surface. For completeness, we repeat here the calculation of Γ 's performed in Lecture 11.2. The metric tensor is $g_{\mu\nu} = \text{diag}(r^2, r^2 \sin^2 \theta)$, the inverse is $g^{\mu\nu} = \text{diag}(r^{-2}, r^{-2} \sin^{-2} \theta)$ (with the standard identification $x^1 \equiv \theta$, $x^2 \equiv \phi$). There is only one derivative of $g_{\mu\nu}$ that does not vanish: $g_{22,1} = \frac{\partial r^2 \sin^2 \theta}{\partial \theta} = r^2 \sin(2\theta)$. Thus, only the combination 122 is possible in the nonvanishing Christoffel symbols:

$$\Gamma_{122} = -\frac{1}{2}g_{22,1}, \quad \Gamma_{212} = \Gamma_{221} = \frac{1}{2}g_{22,1}.$$

Raising indices, we have

$$\Gamma^1_{22} = g^{11}\Gamma_{122} = -\frac{1}{2}\sin(2\theta), \quad \Gamma^2_{12} = \Gamma^2_{21} = g^{22}\Gamma_{212} = \cot \theta.$$

For the Riemann tensor, we also need the derivatives:

$$\Gamma^1_{22,1} = -\cos 2\theta, \quad \Gamma^2_{12,1} = \Gamma^2_{21,1} = -\frac{1}{\sin^2 \theta}.$$

We already know that there is only one independent component of R . All the rest could be expressed via any one. Obviously, it contains the indices 1212. So we calculate

$$\begin{aligned} R^1_{212} &= -\Gamma^1_{21,2} + \Gamma^1_{22,1} - \Gamma^\beta_{21}\Gamma^1_{\beta 2} + \Gamma^\beta_{22}\Gamma^1_{\beta 2} = \Gamma^1_{22,1} - \Gamma^2_{21}\Gamma^1_{22} \\ &= -\cos 2\theta - \cot \theta (-1/2) \sin 2\theta = -\cos 2\theta + \cos^2 \theta = \sin^2 \theta \end{aligned}$$

We can immediately get any other component:

$$R^2_{121} = g^{22}R_{2121} = g^{22}R_{1212} = g^{22}g_{11}R^1_{212} = \frac{1}{r^2 \sin^2 \theta} \times r^2 \times \sin^2 \theta = 1.$$

Let's verify this by the direct calculation.

$$R^2_{121} = \Gamma^2_{11,2} - \Gamma^2_{12,1} + \Gamma^\beta_{11}\Gamma^2_{\beta 2} - \Gamma^\beta_{12}\Gamma^2_{\beta 1} = -\Gamma^2_{12,1} - \Gamma^2_{12}\Gamma^2_{21} = \frac{1}{\sin^2 \theta} - \cot^2 \theta = 1,$$

indeed. The other two non-zero components are different by the sign only. Now, the Ricci tensor is

$$R_{\mu\nu} = R^\alpha_{\nu\alpha\mu} = R^1_{\nu 1\mu} + R^2_{\nu 2\mu} \Rightarrow \begin{aligned} R_{11} &= R^1_{111} + R^2_{121} = R^2_{121}, & R_{12} &= R^1_{112} + R^2_{122} = 0 \\ R_{22} &= R^1_{212} + R^2_{222} = R^1_{212}, & R_{21} &= R^1_{211} + R^2_{221} = 0 \end{aligned}$$

As expected, there is only one independent component. Then the curvature is

$$R = g^{\mu\nu} R_{\mu\nu} = g^{11} R_{11} + g^{22} R_{22} = g^{11} R^2_{121} + g^{22} R^1_{212} = 2g^{11} R^2_{121} = 2g^{22} R^1_{212}$$

which is valid for any **diagonal** 2D metric. In the present case, it gives $R = 2/r^2$.

Example 14.2. Spherical polar coordinates in a 3D flat space. Obviously, we anticipate that $R = 0$. Calculations are quite similar to the previous example, as the metric tensor is $g_{\mu\nu} = \text{diag}(1, r^2, r^2 \sin^2 \theta)$, the inverse $g^{\mu\nu} = \text{diag}(1, r^{-2}, r^{-2} \sin^{-2} \theta)$ with the identification $x^0 \equiv r, x^1 \equiv \theta, x^2 \equiv \phi$. Now there are a few derivatives of $g_{\mu\nu}$ that do not vanish:

$$g_{22,1} = r^2 \sin 2\theta, \quad g_{22,0} = 2r \sin^2 \theta, \quad \text{and} \quad g_{11,0} = 2r.$$

The number of nonvanishing Christoffel symbols then is also bigger, as the combinations (110), (220), and (221) are possible. The symbols with (221) are still the same as in the previous example. Then, in agreement with Example (12.2),

$$\begin{aligned} \Gamma^2_{20} = \Gamma^2_{02} &= \frac{1}{2} g^{22} g_{22,0} = \frac{1}{r}, & \Gamma^0_{22} &= -\frac{1}{2} g^{00} g_{22,0} = r \sin^2 \theta \\ \Gamma^1_{10} = \Gamma^1_{01} &= \frac{1}{2} g^{11} g_{11,0} = \frac{1}{r}, & \Gamma^0_{11} &= -\frac{1}{2} g^{00} g_{11,0} = r \end{aligned}$$

Their nonvanishing derivatives are:

$$\begin{aligned} \Gamma^2_{20,0} = \Gamma^2_{02,0} &= -\frac{1}{r^2}, & \Gamma^0_{22,0} &= \sin^2 \theta, & \Gamma^0_{22,1} &= r \sin 2\theta \\ \Gamma^1_{10,0} = \Gamma^1_{01,0} &= -\frac{1}{r^2}, & \Gamma^0_{11,0} &= 1 \end{aligned}$$

We have more possibilities for nonvanishing components of R which are not related to each other by the symmetries.

$$R^1_{010}, \quad R^2_{121}, \quad R^2_{020}, \quad R^1_{021}, \quad R^1_{020}, \quad R^2_{120}$$

Let's calculate any of them:

$$R^1_{010} = \Gamma^1_{00,1} - \Gamma^1_{01,0} + \Gamma^{\beta}_{00} \Gamma^1_{\beta 1} - \Gamma^{\beta}_{01} \Gamma^1_{\beta 0} = -\Gamma^1_{01,0} - \Gamma^1_{01} \Gamma^1_{10} = \frac{1}{r^2} - \frac{1}{r} \frac{1}{r} = 0.$$

You may check that the other 5 components also vanish, as expected.

Example 14.3. A flat surface in disguise. Consider the metric

$$ds^2 = y^2 dx^2 + x^2 dy^2.$$

The Christoffel symbols for this metric have been calculated in Lecture 11.2:

$$\Gamma^y_{yx} = \frac{1}{x}, \quad \Gamma^x_{yy} = -\frac{x}{y^2}, \quad \Gamma^x_{xy} = \frac{1}{y}, \quad \Gamma^y_{xx} = -\frac{y}{x^2}$$

Now we have:

$$R^y_{xyx} = \Gamma^y_{xx,y} - \Gamma^y_{xy,x} + \underbrace{\Gamma^{\sigma}_{xx} \Gamma^y_{\sigma y}}_{=0} - \Gamma^{\sigma}_{xy} \Gamma^y_{\sigma x} = -\frac{1}{x^2} - \left(-\frac{1}{x^2}\right) - (\Gamma^y_{xy})^2 - \Gamma^x_{yx} \Gamma^y_{xx} = 0.$$

Naturally, a calculation of any other component of $R^\mu_{\nu\alpha\beta}$ gives the same result.

Example 14.4. A non-diagonal metric. Let us consider the metric a of example (11.5):

$$ds^2 = y^2 dx^2 + 2x dx dy \Leftrightarrow g_{\mu\nu} = \begin{pmatrix} y^2 & x \\ x & 0 \end{pmatrix}, \quad g^{\mu\nu} = \begin{pmatrix} 0 & 1/x \\ 1/x & -y^2/x^2 \end{pmatrix}.$$

We have already calculated all the Γ 's:

$$\Gamma^2_{12} = \Gamma^2_{21} = \frac{y}{x}, \quad \Gamma^1_{11} = -\frac{y-1}{x}, \quad \Gamma^2_{11} = \frac{(y-1)y^2}{x^2}.$$

Now, $R^1_{212} = \Gamma^1_{22,1} - \Gamma^1_{21,2} + \Gamma^\mu_{22}\Gamma^1_{\mu 2} - \Gamma^\mu_{21}\Gamma^1_{\mu 2} = 0$, since Γ^1_{11} is the only nonzero symbol with the upper index 1. However, it would be premature to conclude that this metric refers to a flat space. Indeed, in this particular case, as a consequence of $g_{22} = 0$ we find

$$R^1_{212} = g_{2\mu}R^{1\mu}_{12} = g_{21}\underbrace{R^{11}_{22}}_{=0} + \underbrace{g_{22}}_{=0}R^{12}_{12} = 0,$$

no matter what is the value of R^{12}_{12} . So we better try to calculate a different component of $R^\mu_{\nu\alpha\beta}$. As we need the curvature, $R = 2R^2_{21} = 2R^{12}_{12}$, we can calculate either $R^{12}_{12} = g^{21}R^1_{112} + g^{22}R^1_{212} = g^{21}R^1_{112}$, or $R^2_{21} = g^{12}R^2_{221} + g^{11}R^2_{121} = g^{12}R^2_{221}$. The results must obviously coincide. We have:

$$R^1_{112} = \underbrace{\Gamma^1_{12,1}}_{=0} - \Gamma^1_{11,2} + \underbrace{\Gamma^\mu_{12}\Gamma^1_{\mu 1} - \Gamma^\mu_{11}\Gamma^1_{\mu 2}}_{=0} = \frac{1}{x} \implies R = g^{21}R^1_{112} = \frac{2}{x^2};$$

or, equivalently

$$R^2_{221} = \Gamma^2_{21,2} - \underbrace{\Gamma^2_{22,1}}_{=0} + \underbrace{\Gamma^\mu_{21}\Gamma^2_{\mu 2} - \Gamma^\mu_{22}\Gamma^2_{\mu 1}}_{=0} = \frac{1}{x} \implies R = g^{12}R^2_{221} = \frac{2}{x^2}.$$

We can derive a general expression for R in terms of a single component of $R^\alpha_{\beta\mu\nu}$. We find, using the symmetries of $R^{\mu\nu}_{\alpha\beta}$, that $R = R^{\mu\nu}_{\mu\nu} = R^{12}_{12} + R^{21}_{21} = 2R^{12}_{12}$. Then

$$R^1_{212} = g_{2\mu}R^{1\mu}_{12} = g_{22}R^{12}_{12} + g_{21}R^{11}_{12} = g_{22}R^{12}_{12} \implies R = \frac{2}{g_{22}}R^1_{212},$$

since $R^{11}_{12} = 0$ as two indices in the same pair coincide. However, this would not work if $g_{22} = 0$ (in which case, with necessity, $R^1_{212} = 0$). Alternatively, one can use $R = 2R^2_{121}/g_{11}$. But both formulae would fail simultaneously if $g_{\mu\nu}$ is off-diagonal. In this case, one can write $R^1_{112} = g_{\mu 1}R^{1\mu}_{12} = g_{21}R^1_{212}$ so that $R = 2R^1_{112}/g_{12}$.

Lecture 15. Tidal tensor and Einstein's equations

We have introduced the Riemann and Ricci tensors as characteristics of curvature. In Lecture 15, by comparing tidal forces in the Newtonian limit with the GR tidal forces, where they result from the geodesic deviation, we finally arrive at Einstein's equation: the Ricci=0 in empty space. This still might correspond to nontrivial curvature reflected in nonzero components of the Riemann. Then we make the first steps in calculating the Schwarzschild metric that has been extensively used in this course. We finish this calculation in the lecture 16 and show that this metric is the static solution of Einstein's equations in empty space around a spherically-symmetric mass distribution.

In the first 8 lectures, we have considered geodesic motion in a given metric, including the Schwarzschild. Today we find how the Einstein's equations in empty space give the only metric that is allowed by the spherical symmetry. These equations, $R^\mu{}_\nu = 0$, make a set of up to ten 2nd order PDEs for (up to 10) independent components of $g_{\mu\nu}$. As usual, PDEs should be solved subject to the boundary conditions – a difficult task for any 2nd order PDE, and next to impossible in GR – unless we employ physical symmetries of spacetime (when present) that go well beyond mathematical symmetries of the Riemann.

15.1 Newtonian tidal forces

Tidal effects in the Newtonian physics describe inhomogeneity of the gravitational acceleration. The acceleration (*i.e.* the force acting on a body of unit mass) is equal to the gradient of the gravitational potential, $\mathbf{f} = -\nabla\varphi$, or in components $f^i = -\frac{\partial\varphi}{\partial x_i}$. The inhomogeneity of the field is reflected in spatial dependence of this force expressed via its derivative:

$$\frac{\partial f^i}{\partial x^j} = -\frac{\partial^2\varphi}{\partial x_i \partial x^j} \equiv \tilde{\Delta}^i{}_j.$$

Here $\tilde{\Delta}^i{}_j$ is the Newtonian tidal tensor, which contains only spatial components. As we do not use $\frac{\partial}{\partial x_i}$ in GR, we rewrite this using differentiation only w.r.t. x^n :

$$\tilde{\Delta}^j{}_n = g^{mj} \tilde{\Delta}_{mn} = -g^{mj} \varphi_{,mn}, \quad (15.1)$$

using here Minkowski metric tensor,¹ $g^{\mu\nu} \equiv \eta^{\mu\nu} = \text{diag}(1, -1, -1, -1)$.

Tidal forces stretch a body in a central (rotationally symmetric) gravitational field, as it is inhomogeneous, and actually in any gravitational field. In a weak gravity, they reveal itself only at a large enough scale. Ocean tides (which give the name to these forces) are due to the field inhomogeneity due to a gravitational pull of the Moon. For a (relatively) small body in a free fall, e.g., in the gravitational field of the Earth these forces are tiny but still need to be compensated by non-gravitational forces to keep the body intact. Hence they are classical geodesic deviation forces. Although there is no GR analogy to the classical $\mathbf{f} = -\nabla\varphi$ in a homogeneous field (as we have just a free fall), the classical homogeneous $\mathbf{f}$ does not contribute to the tidal tensor of Eq. (15.1), as its derivative in tangential directions vanishes. The derivative is to be compared with that of the geodesic deviation (force), for which we have built a full geometric description of geodesic deviation in Lecture 13. It

¹The only role of g^{mj} in Eq. (15.1) is to change the sign of the 2nd derivatives. We keep this for correspondence with strong gravity where raising or lowering indices is a rather nontrivial operation.

must work in the Newtonian limit if we substitute the post-Newtonian metric, Eq. (3.8), into the general expressions for geodesic deviation. Conversely, we can build a full covariant generalisation of the Newtonian tidal tensor by considering inhomogeneity of the fully covariant geodesic deviation force of Eq. (13.4) in any metric.

15.2 GR tidal tensor

The covariant generalisation of the tidal tensor of Eq. (15.1), Δ^α_μ , is the tidal tensor in 1 + 3-dimensional spacetime describing the inhomogeneity of the gravitational field which is compensated by inhomogeneity of the geodesic deviation force. The latter is given by Eq. (13.4):

$$f^\alpha = \nabla_u u^\alpha = R^\alpha_{\beta\mu\nu} u^\beta u^\mu \xi^\nu. \quad (15.2)$$

We had quite a few names for it: line curvature, 4-acceleration, geodesic deviation force. Now we can call it relativistic tidal force. Taking its derivative w.r.t. $\xi^\nu \equiv dx^\nu$ gives the relativistically covariant tidal tensor mentioned previously, Eq. (13.7):

$$\Delta^\alpha_\nu \equiv \frac{\partial f^\alpha}{\partial x^\nu} = R^\alpha_{\beta\mu\nu} u^\beta u^\mu \quad (15.3)$$

Let us verify directly that its spatial components go over to the Newtonian limit, Eq. (15.1), in weak gravity where $g_{\mu\nu}$ is reduced to the post-Newtonian metric tensor, given by Eq. (3.7):

$$g_{\mu\nu} = \text{diag} (1 + 2\varphi c^{-2}, -1, -1, -1) \equiv \eta_{\mu\nu}. \quad (15.4)$$

The only nonzero derivatives of $g_{\mu\nu}$ are $g_{00,n}$. As $g_{\mu\nu}$ is diagonal, we have

$$\begin{aligned} \Gamma_{00n} &= \Gamma_{0n0} = -\Gamma_{n00} = \frac{1}{2} g_{00,n} = c^{-2} \varphi_{,n}, \quad n = 1, 2, 3 \\ \Rightarrow \Gamma^0_{0n} &= \Gamma^0_{n0} = g^{00} \Gamma_{00n} \approx c^{-2} \varphi_{,n}; \quad \Gamma^n_{00} = g^{nm} \Gamma_{m00} = g^{nm} c^{-2} \varphi_{,m}. \end{aligned} \quad (15.5)$$

where we have neglected corrections coming from g^{00} as they will be of a higher power in $1/c^2$. Note that in the post-Newtonian limit spatial contra- and co-variant components differ only by the sign, $\xi^n = -\xi_n$ (which is not a tensor identity but an equation that holds for any particular component).

To verify the correspondence to the Newtonian limit, Eq. (15.1), we need only the spatial components of Δ^α_ν , i.e. only Δ^n_m . Then, on the r.h.s. of Eq. (15.3) we need to keep only $u^0 = c$ in $u^\mu = \text{diag} (c, -\mathbf{v})$ neglecting $(v/c)^2$ corrections coming from the spatial components of the 4-velocity. Hence in Eq. (15.3) we need to keep only R^a_{00m} . Using Eq. (13.5), we have

$$R^a_{00m} = -\Gamma^a_{00,m} + \Gamma^a_{0m,0} - \Gamma^\alpha_{00} \Gamma^a_{\alpha m} + \Gamma^\alpha_{0m} \Gamma^a_{\alpha 0}$$

As there is no Christoffel symbol with two spatial indices, $\Gamma^a_{\alpha m} = 0$, while the nonvanishing part of the last term $\Gamma^0_{0m} \Gamma^a_{00} \propto 1/c^4$. Hence (with $a \rightarrow n$ for convenience)

$$R^n_{00m} = -\Gamma^n_{00,m} = -c^{-2} g^{\ell n} \varphi_{,\ell m}.$$

Substituting this into Eq. (15.3), we find $\Delta^n_m = g^{\ell n} \varphi_{,\ell m}$ in agreement with the Newtonian expression (15.1). Hence, Eq. (15.3) is, indeed, a proper covariant generalisation of the 3D tidal tensor.

15.3 Einstein's equations

The Newtonian gravitation potential obeys the Poisson equation,

$$\nabla^2 \varphi \equiv -\Delta^i_i = 4\pi G \rho(\mathbf{r}),$$

where $\rho(\mathbf{r})$ is the mass density. In empty of masses points of space (usually referred to as empty space), it is just the Laplace equation for φ :

$$\nabla^2 \varphi = 0 \quad \Longleftrightarrow \quad \tilde{\Delta}^i_i = 0, \quad (15.6)$$

In words, *the Newtonian tidal tensor is traceless* in empty space. To make this relativistically covariant, we need to demand that the tidal tensor 15.3 is traceless – with the trace now including a temporal component alongside with spatial ones. Thus we arrive at

$$\Delta^\alpha_\alpha = 0 \quad \Longleftrightarrow \quad R^\alpha_{\mu\alpha\nu} u^\mu u^\nu \equiv R_{\mu\nu} u^\mu u^\nu = 0 \quad \text{in empty space}$$

Here u^μ is non-zero velocity along an **arbitrary** chosen geodesics. Hence, for this expression to be equal to zero the contraction of the Riemann tensor, i.e. the Ricci tensor, must vanish.

Thus, we obtain that in empty space

$$R_{\mu\nu} = 0 \quad (15.7)$$

This is Einstein's field equations for the curvature tensor in empty space. Let me stress: we have not derived it but argued that this is a GR covariant expression that reproduces Newtonian physics for weak gravity. It is an *almost* unique way to express the equivalence principle in a fully mathematical way.

The word “almost” refers to the fact that a more general tensor equation, which also reduces to the Newtonian limit at the scale of the Solar system, is possible:

$$R_{\mu\nu} = -\Lambda g_{\mu\nu}.$$

Here Λ is a universal **cosmological** constant, and $\Lambda g_{\mu\nu}$ is called the cosmological term. It is assumed that Λ is extremely small, so that its effect could be observable only at cosmological scale much larger than the size of a typical galaxy. Einstein has introduced it in order to make possible a stationary cosmological solution of his equations, which is absent for $\Lambda = 0$. Later, accepting that the Universe is not stationary, he said that introducing Λ was the biggest blunder in his life. It is a little bit too harsh from today's viewpoint. This term is not only possible but necessary: all modern cosmological models really need it. The consensus reached over the last two decades or so suggests that the Universe expansion is accelerating rather than decelerating; the nonzero cosmological term (related to a rather mysterious **dark energy**) gives the most straightforward explanation of this, which I shall briefly illustrate in the penultimate lecture. So the “biggest blunder” turned out to be one of the biggest triumphs of Einstein's general relativity.

Equation (15.7) looks deceptively simple: in reality it comprises up to 10 independent 2nd order differential equations for the metric tensor $g_{\mu\nu}$. The spacetime is not flat as long as some components of $R^\mu_{\nu\alpha\beta}$ are nonzero while $R_{\mu\nu} = 0$. We have seen that this could only happen when the spacetime dimensionality ≥ 4 . Solving Eqs. (15.7) and matching the solutions at the boundaries of any mass distribution to the solutions in non-empty space allows one to find the metric generated by this distribution.

Lecture 16. Schwarzschild metric

We shall finally prove that the Schwarzschild metric is the static solution of Einstein's equations in empty space around a spherically-symmetric mass distribution centred at the origin. To do this, we shall calculate the Christoffel symbols and the Riemann for a general spherically-symmetric metric and show that equations $R_{\mu\nu} = 0$ produce the Schwarzschild solution.

16.1 Euler – Lagrange equations in the spherically-symmetric case

At this stage the Schwarzschild radius, r_s , will appear merely as a constant of integration. To prove directly that $r_s = 2GM/c^2$, we will (later) formulate Einstein's equations in the presence of masses and solve them **inside** the spherically-symmetric mass distribution.

First we need to go through quite a tedious and elaborate calculation of all the nonvanishing Christoffel symbols. The second step is calculating all the independent components of the Riemann, $R^\mu_{\nu\alpha\beta}$. Finally we shall get the Ricci, $R_{\mu\nu}$, and solve Einstein's equations in empty space. This is a very rare example of a calculation where Einstein's equations can be solved exactly. So it is worth doing it in detail.

We consider the static metric spherically symmetric metric. "Static" means all the coefficient are t -independent. As the $c^2 dt^2$ term is invariant w.r.t. time inversion, $t \rightarrow -t$, the static metric must remain invariant as well, which excludes cross-terms such as $dt dx^\mu$. Spherical symmetry implies that the angular dependence in the $d\Omega^2$ term must remain the same as in the Minkowski space, i.e. $d\Omega^2 = d\theta^2 + \sin^2\theta d\phi^2$, and off-diagonal spatial terms are also excluded. After that, the metric is like the Minkowski one but with all the three terms multiplied by arbitrary functions of r , as this does not break the symmetry. Finally, the coefficient attached to $r^2 d\Omega^2$ can always be chosen = 1 by rescaling r . Hence we can write the metric and Lagrangian as follows:

$$ds^2 = e^\alpha c^2 dt^2 - e^\beta dr^2 - r^2 d\Omega^2 \quad \Leftrightarrow \quad \mathcal{L} = e^\alpha c^2 \dot{t}^2 - e^\beta \dot{r}^2 - r^2 \dot{\theta}^2 - r^2 \sin^2\theta \dot{\phi}^2, \quad (16.1)$$

where e^α and e^β are functions of r only. As these coefficients must be non-negative (to preserve timelike and spacelike character of dt^2 and dx^2 , parameterising them as the exponent does not reduce generality.

The shortest way to calculate $\Gamma^\sigma_{\mu\nu}$ is to derive the Euler-Lagrange equations and to cast them into canonical form,

$$f^\sigma = \frac{1}{2} g^{\sigma\mu} \left(\frac{d}{d\tau} \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} - \frac{\partial \mathcal{L}}{\partial x^\mu} \right) = \ddot{x}^\sigma + \Gamma^\sigma_{\mu\nu} \dot{x}^\mu \dot{x}^\nu.$$

Note that $\frac{d\alpha}{d\tau} = \frac{d\alpha}{dr} \dot{r} \equiv \alpha' \dot{r}$, and similar for β . With the standard identification of the variables, $ct \equiv x^0$, $r \equiv x^1$, $\theta \equiv x^2$, $\phi \equiv x^3$, we have $g^{\mu\nu} = \text{diag}(e^{-\alpha}, -e^{-\beta}, -r^{-2}, -(r \sin\theta)^{-2})$. Hence the canonical Euler – Lagrange forces (or four-accelerations, which are the same as forces for the unit-mass particle) are:¹

$$f^0 = \frac{1}{2} e^{-\alpha} \frac{d}{d\tau} \left[\frac{\partial}{\partial \dot{t}} (e^\alpha \dot{t}^2) \right] = \ddot{t} + \alpha' \dot{t} \dot{r} \quad \therefore \Gamma^0_{01} = \Gamma^0_{10} = \frac{1}{2} \alpha' \quad (16.2a)$$

$$f^1 = \frac{1}{2} e^{-\beta} \left[\frac{d}{d\tau} [\partial_{\dot{r}} (e^\beta \dot{r}^2)] + \partial_r (e^\alpha \dot{t}^2 - e^\beta \dot{r}^2 - r^2 \dot{\theta}^2 - r^2 \sin^2\theta \dot{\phi}^2) \right]$$

¹We put $c = 1$ in all the calculations, restoring it in the end results.

$$= \ddot{r} + \frac{1}{2}\beta' \dot{r}^2 + \frac{1}{2}\alpha' e^{\alpha-\beta} (\dot{t})^2 - r e^{-\beta} (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2)$$

$$\therefore \Gamma_{00}^1 = \frac{1}{2}\alpha' e^{\alpha-\beta}, \Gamma_{11}^1 = \frac{1}{2}\beta', \Gamma_{22}^1 = -r e^{-\beta}, \Gamma_{33}^1 = -r \sin^2 \theta e^{-\beta}. \quad (16.2b)$$

$$f^2 = \frac{1}{2r^2} \left(\frac{d}{d\tau} [2r^2 \dot{\theta}] - \partial_\theta [r^2 \dot{\phi}^2] \right) = \ddot{\theta} + \frac{2}{r} \dot{r} \dot{\theta} - \frac{1}{2} \sin 2\theta \dot{\phi}^2$$

$$\therefore \Gamma_{12}^2 = \Gamma_{21}^2 = \frac{1}{r}, \Gamma_{33}^2 = -\frac{1}{2} \sin 2\theta \quad (16.2c)$$

$$f^3 = \frac{1}{2r^2 \sin^2 \theta} \frac{d}{d\tau} (r^2 \sin^2 \theta \dot{\phi}^2) = \ddot{\phi} + \frac{2}{r} \dot{r} \dot{\phi} + 2 \cot \theta \dot{\theta} \dot{\phi}$$

$$\therefore \Gamma_{13}^3 = \Gamma_{31}^3 = \frac{1}{r}, \Gamma_{23}^3 = \Gamma_{32}^3 = \cot \theta. \quad (16.2d)$$

16.2 Christoffel symbols and the Riemann components

Thus we have found 13 nonzero Christoffel symbols (only 7 of them are independent). Now we calculate all the independent elements of the Riemann ,

$$R^\mu_{\nu\alpha\beta} = \Gamma^\mu_{\nu\beta, \alpha} - \Gamma^\mu_{\nu\alpha, \beta} + \Gamma^\sigma_{\nu\beta} \Gamma^\mu_{\sigma\alpha} - \Gamma^\sigma_{\nu\alpha} \Gamma^\mu_{\sigma\beta}$$

It could be up to twenty independent elements in 4D. In the spherically symmetric case considered, however, only six independent elements survive, each having two identical pairs of indices. The nontrivial elements are:

$$R^0_{101} = -\Gamma^0_{10,1} + \Gamma^0_{11,0} - \underbrace{\Gamma^\nu_{10} \Gamma^0_{\nu 1}}_{\nu=0} + \underbrace{\Gamma^\nu_{11} \Gamma^0_{\nu 0}}_{\nu=1} = -\Gamma^0_{10,1} - \Gamma^0_{10} \Gamma^0_{01} + \Gamma^1_{11} \Gamma^0_{10} =$$

$$= -\frac{1}{2}\alpha'' - \frac{1}{4}(\alpha')^2 + \frac{1}{4}\alpha'\beta'$$

$$R^0_{202} = \Gamma^\nu_{22} \Gamma^0_{\nu 0} - \Gamma^\nu_{20} \Gamma^0_{\nu 2} = \Gamma^1_{22} \Gamma^0_{10} = -\frac{1}{2}\alpha' r e^{-\beta}$$

$$R^0_{303} = \Gamma^\nu_{33} \Gamma^0_{\nu 0} - \Gamma^\nu_{30} \Gamma^0_{\nu 3} = \Gamma^1_{33} \Gamma^0_{10} = -\frac{1}{2}\alpha' r e^{-\beta} \sin^2 \theta$$

$$R^1_{212} = \Gamma^1_{22,1} + \Gamma^\nu_{22} \Gamma^1_{\nu 1} - \Gamma^\nu_{21} \Gamma^1_{\nu 2} = \Gamma^1_{22,1} + \Gamma^1_{22} \Gamma^1_{11} - \Gamma^2_{21} \Gamma^1_{22} =$$

$$= -e^{-\beta} + \beta' r e^{-\beta} - \frac{1}{2}\beta' r e^{-\beta} + e^{-\beta} = \frac{1}{2}\beta' r e^{-\beta}$$

$$R^1_{313} = \Gamma^1_{33,1} + \Gamma^\nu_{33} \Gamma^1_{\nu 1} - \Gamma^\nu_{31} \Gamma^1_{\nu 3} = \Gamma^1_{33,1} + \Gamma^1_{33} \Gamma^1_{11} - \Gamma^3_{31} \Gamma^1_{33} =$$

$$= (\text{as above}) \times \sin^2 \theta = \frac{1}{2}\beta' r e^{-\beta} \sin^2 \theta$$

$$R^2_{323} = \Gamma^2_{33,2} + \Gamma^\nu_{33} \Gamma^2_{\nu 2} - \Gamma^\nu_{32} \Gamma^2_{\nu 3} = \Gamma^2_{33,2} + \Gamma^1_{33} \Gamma^2_{12} - \Gamma^3_{32} \Gamma^2_{33} =$$

$$= -\cos 2\theta - \sin^2 \theta e^{-\beta} + \cos^2 \theta = (1 - e^{-\beta}) \sin^2 \theta$$

Let us illustrate that the elements of $R^\mu_{\nu\alpha\beta}$ with different pairs of indices vanish. We show only one example of this:

$$R^1_{332} = -\Gamma^1_{33,2} - \Gamma^\nu_{33} \Gamma^1_{\nu 2} + \Gamma^\nu_{32} \Gamma^1_{\nu 3} = -\Gamma^1_{33,2} - \Gamma^2_{33} \Gamma^1_{22} + \Gamma^3_{32} \Gamma^1_{33} =$$

$$= r \sin 2\theta e^{-\beta} - \frac{1}{2} r \sin 2\theta e^{-\beta} - \frac{1}{2} r \sin 2\theta e^{-\beta} = 0$$

We assemble the independent nonzero elements of $R^\mu_{\nu\alpha\beta}$ into the following table:

$R^0_{101} = -\frac{1}{2}\alpha'' - \frac{1}{4}(\alpha')^2 + \frac{1}{4}\alpha'\beta'$	$R^1_{212} = \frac{1}{2}\beta' r e^{-\beta}$	(16.3)
$R^0_{202} = -\frac{1}{2}\alpha' r e^{-\beta}$	$R^1_{313} = \frac{1}{2}\beta' r e^{-\beta} \sin^2 \theta$	
$R^0_{303} = -\frac{1}{2}\alpha' r e^{-\beta} \sin^2 \theta$	$R^2_{323} = (1 - e^{-\beta}) \sin^2 \theta$	

In total, there are 24 nonzero elements of $R^\mu_{\nu\alpha\beta}$. Other 18 elements can be obtained from the six elements in the table by exploiting the symmetry relations. To this end and thus for calculating the Ricci, it is more convenient to consider the Riemann of the $2 + 2$ rank, *i.e.* to raise the second index: $R^{\mu\lambda}_{\alpha\beta} = g^{\nu\lambda}R^\mu_{\nu\alpha\beta}$, As $g^{\mu\nu}$ is diagonal, this is straightforward for all the components, *e.g.* $R^{01}_{01} = g^{11}R^0_{101} = -e^{-\beta}R^0_{101}$. The appropriate table is given below:

$$\begin{array}{ll}
 R^{01}_{01} = e^{-\beta} \left[\frac{1}{2}\alpha'' + \frac{1}{4}(\alpha')^2 - \frac{1}{4}\alpha'\beta' \right] & R^{12}_{12} = -\frac{\beta'e^{-\beta}}{2r} \\
 R^{02}_{02} = \frac{\alpha'e^{-\beta}}{2r} & R^{13}_{13} = -\frac{\beta'e^{-\beta}}{2r} \\
 R^{03}_{03} = \frac{\alpha'e^{-\beta}}{2r} & R^{23}_{23} = -\frac{1 - e^{-\beta}}{r^2}
 \end{array} \quad (16.4)$$

This table reveals an additional symmetry which was not obvious in the table for $R^\mu_{\nu\alpha\beta}$: there are only four different elements of the Riemann.

Now we can calculate the Ricci. It is clear from the above table that only its diagonal elements are nonzero. The calculation is straightforward. For example,

$$\begin{aligned}
 R^0_0 = R^{\nu 0}_{\nu 0} &= R^{10}_{10} + R^{20}_{20} + R^{30}_{30} = R^{01}_{01} + R^{02}_{02} + R^{03}_{03} = \\
 &= e^{-\beta} \left[\frac{1}{2}\alpha'' + \frac{1}{4}(\alpha')^2 - \frac{1}{4}\alpha'\beta' \right] + \frac{\alpha'e^{-\beta}}{r}.
 \end{aligned}$$

Thus, one obtains three differential equations (of which only two are independent):

$$\begin{array}{l}
 R^0_0 = e^{-\beta} \left[\frac{1}{2}\alpha'' + \frac{1}{4}(\alpha')^2 - \frac{1}{4}\alpha'\beta' + \frac{\alpha'}{r} \right] = 0 \\
 R^1_1 = e^{-\beta} \left[\frac{1}{2}\alpha'' + \frac{1}{4}(\alpha')^2 - \frac{1}{4}\alpha'\beta' - \frac{\beta'}{r} \right] = 0 \\
 R^2_2 = R^3_3 = \frac{(\alpha' - \beta')e^{-\beta}}{2r} - \frac{1 - e^{-\beta}}{r^2} = 0
 \end{array} \quad (16.5)$$

16.3 Solutions to Einstein's equations in empty space

Finally we shall obtain the Schwarzschild solution which we have already used many times, but with one constant (r_s) to be left undetermined – until we consider Einstein's equations with the mass-energy terms. There are three different components of $R_{\mu\nu}$ above but only two of them are independent because of the Bianchi identities, Eq. (13.12). Hence the two equations should determine the unknown functions $\alpha(r)$ and $\beta(r)$. It is convenient to use the combination that does not contain the second derivative:

$$R^0_0 - R^1_1 = \frac{(\alpha' + \beta')e^{-\beta}}{r} = 0 \quad \Rightarrow \quad \alpha + \beta = \text{const} \equiv D.$$

Substituting $\alpha' = -\beta'$ into R^2_2 we find

$$\begin{aligned} \frac{\beta' e^{-\beta}}{r} + \frac{1 - e^{-\beta}}{r^2} = 0 &\Rightarrow \frac{e^{-\beta} d\beta}{1 - e^{-\beta}} = -\frac{dr}{r} \Rightarrow \ln(1 - e^{-\beta}) = -\ln r + \text{const} \equiv \ln \frac{r_s}{r} \\ \Rightarrow e^{-\beta} = \left(1 - \frac{r_s}{r}\right) &\equiv \sigma(r). \end{aligned} \quad (16.6)$$

Here r_s is just a constant of integration,² and we have implicitly assumed $\beta > 0$, i.e. $r > r_s$.

The constant D and thus α can be found from the requirement that in the limit $r \rightarrow \infty$ the metric goes over to the Minkowski solution, i.e. both $\beta \rightarrow 0$ and $\alpha \rightarrow 0$. For the $\beta(r)$ given by Eq. (16.6), this condition is fulfilled: $\lim_{r \rightarrow \infty} \beta(r) = 0$, i.e. $B \rightarrow 1$. Then the requirement $\alpha \rightarrow 0$ leads to $D = 0$ so that $\alpha = -\beta$. Thus, $A = e^\alpha = e^{-\beta} = \sigma(r)$ and $B = e^\beta = \sigma^{-1}(r)$ which gives upon substituting into Eq. (16.1) the Schwarzschild solution:

$$ds^2 = c^2 \sigma dt^2 - \sigma^{-1} dr^2 - r^2 d\Omega^2 \quad (16.7)$$

Clearly, $r = r_s$ and $r = 0$ are singular points of this metric, derived for $r > r_s$. We will extend this metric also to $r < r_s$ considering this as an analytic continuation through the singularity.

Einstein's equations for empty space mean that the Ricci and (therefore) the scalar curvature vanish. However, the 24 components of $R^\mu_{\nu\alpha\beta}$ are still nonzero: substituting the values of α and β to the table above shows that only one independent combination left:

$$R^{01}_{01} = R^{23}_{23} = -\frac{r_s}{r^3}, \quad R^{02}_{02} = R^{03}_{03} = R^{12}_{12} = R^{13}_{13} = \frac{r_s}{2r^3}. \quad (16.8)$$

The solution describes a curved empty spacetime around a spherically-symmetric mass distribution.

There is no singularity in $R^\mu_{\nu\alpha\beta}$ at $r = r_s$; thus, this singularity in $g_{\mu\nu}$ is coordinate, not real

The meaning of this statement is that one can find coordinates in which metric is nonsingular at $r = r_s$. We shall do this in the next Lecture.

16.4 Christoffel symbols for the Schwarzschild metric

We still need this for the kinematics of the Schwarzschild space, e.g. to find the acceleration components of a moving body (see Problem Sheet 3). All we need is to substitute the values of $e^{\alpha(r)} = e^{-\beta(r)} = \sigma(r)$ and $\alpha' = -\beta' = \frac{r_s}{r^2\sigma}$ into Eqs. (16.2):

$$\begin{aligned} \Gamma^0_{01} &= \Gamma^0_{10} = \frac{r_s}{2r^2\sigma}; \\ \Gamma^1_{00} &= \frac{r_s\sigma}{2r^2}, \quad \Gamma^1_{11} = -\frac{r_s}{2r^2\sigma}, \quad \Gamma^1_{22} = -r\sigma, \quad \Gamma^1_{33} = -r \sin^2\theta \sigma; \\ \Gamma^2_{12} &= \Gamma^2_{21} = \frac{1}{r}, \quad \Gamma^2_{33} = -\frac{1}{2} \sin 2\theta \\ \Gamma^3_{13} &= \Gamma^3_{31} = \frac{1}{r}, \quad \Gamma^3_{23} = \Gamma^3_{32} = \cot\theta. \end{aligned} \quad (16.9)$$

Note that $\Gamma^2_{\mu\nu}$ and $\Gamma^3_{\mu\nu}$ are the same for any spherically symmetric metric.

²We know that $r_s = 2GM/c^2$ from comparing with quasi-Newtonian metric that geometrically reproduces Newtonian universal gravity. In the next lecture, we derive and solve Einstein's equations inside the spherically symmetric mass distribution and match this with the present soliton. Note in advance that the coefficients in Einstein's equations inside a mass distribution are also obtained by correspondence with Newtonian case.

16.5 How large are the tidal forces?

Consider a finite size body in free fall around (or to) the centre of gravity. In its own frame, all the spatial components of u^α vanish and only $u^0 \equiv \dot{c}t$ contributes to Eqs. (15.2) and (15.3). Hence,

$$f^\alpha = c^2 R^\alpha_{00\nu} \dot{t}^2 \xi^\nu = g_{00} R^{\alpha 0}_{0\nu} \dot{t}^2 \xi^\nu = -c^2 \sigma R^{0\alpha}_{0\nu} \dot{t}^2 \xi^\nu.$$

Hence $f^0 = 0$, and the spatial components are given by

$$f^n = -c^2 \sigma R^{0n}_{0\nu} \dot{t}^2 \xi^\nu = \begin{cases} \frac{c^2 r_s}{r^3} \xi^r & \text{for } f^r \\ -\frac{c^2 r_s}{2r^3} \xi^{\theta,\phi} & \text{for } f^{\theta,\phi} \end{cases}$$

where we have used the conservation law in a free falling frame, $\mathcal{L} = c^2 \sigma \dot{t}^2 = c^2$, and substituted the nonzero components of the Riemann, Eq. (16.8) (only those with the same pairs of indices survive).

So the tidal forces create tension in the radial direction and compression in the transverse directions.

Let us make some estimates. Recalling $c^2 r_s = 2GM$ and restoring the body mass m , we find

$$|f| \sim \frac{GMm}{r^2} \frac{a}{r},$$

i.e. it gives correction of order a/r to "normal" gravitational force. However, the normal force is not felt by a free falling body so that this correction is a "real" gravity. For a spherical body of radius $a \sim 1\text{m}$ in a free fall around the Earth, the tidal "correction" is about 10^{-7} of its weight, which is tiny but (quite accurately) measurable. Note that there is nothing relativistic in it as should be clear by our derivation of the tidal tensor and by the fact that there is no c in the result.

Now assume that the body falls into (or around) the black hole of radius r_s . A smallest black hole that originated from the gravitational collapse of a star have mass of order 1.3 the solar mass, so that $r_s \approx 3\text{km}$. Rewriting the tidal force as $|f| \sim k^3 (mc^2 a / r_s^2)$, where $k \equiv r_s / r$, we find the magnitude of this force for $m = 1\text{kg}$ at the Schwarzschild radius ($k = 1$) as

$$|f| \sim 10^{17} / 10^7 \text{N} \sim 10^{10} \text{N}.$$

Let us say that the body can withstand a force up to 10^4N (which is rather tough). Then one should have $k \lesssim 10^{-2}$ – at a distance smaller than $100 r_s$, the body would be torn apart by tidal forces.

On the other hand, a smooth journey (if one does not think about the final destination) into the black hole at the centre of a galaxy is quite possible: there $r_s \sim 10^{14} \text{m}$ so that the tidal force at the Schwarzschild sphere is only 10^{-11}N .

Lecture 17. Kruskal diagram for Schwarzschild geometry

Calculating the Riemann confirmed that $r = r_s$ is only a coordinate singularity rather than the real one. Hence some of peculiarities in the vicinity of the black hole were due to the choice of coordinates, as the comparison between the Schwarzschild and the Rindler Spaces had shown us (recall that the event horizon exists in the Rindler world of accelerated rocket in a perfectly flat spacetime). Our present task is to choose coordinates that are free from coordinate singularities.

17.1 Crossing the horizon?

In deriving the Schwarzschild metric, we did not have any limitations but the spherical symmetry. The solutions to the Einstein equations in vacuum are valid for $r > R$, where R is the size of a spherically symmetric mass distribution, and match the solutions with matter at $r = R$. Therefore, they should remain valid $r < r_s$ provided that the entire mass distribution is inside the Schwarzschild radius.¹ However, the temporal and radial components of the metric do not behave nicely, and particles and light trajectories are not really defined for $r < r_s$.

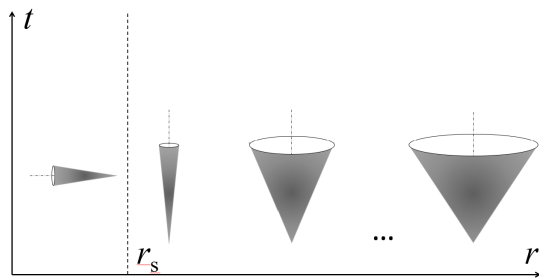
To see this, let us recall what are the radial null worldlines in the Schwarzschild geometry (Lecture 6). The metric for such a worldline $ds^2 = c^2 \sigma dt^2 - \sigma^{-1} dr^2 = 0$. Hence we have $\left(\frac{dr}{dt}\right)^2 = c^2 \sigma^2$, i.e.

$$\frac{dr}{dt} = \begin{cases} -c\sigma & \text{ingoing (falling) photon} \\ +c\sigma & \text{outgoing (raising) photon} \end{cases} \quad (17.1)$$

This means that the coordinate speed of light goes to zero as $r \rightarrow r_s$. The time dependence for the falling photon found from solving (17.1) is

$$ct = -r - r_s \ln \left(\frac{r}{r_s} - 1 \right) + r_0 \equiv r_0 - r^*, \quad \text{where } \frac{dr^*}{dr} = \sigma^{-1}. \quad (17.2)$$

Here constant r_0 is the radial position at $t = 0$, and r^* is called the tortoise coordinate.² Hence the coordinate time (which is the proper time for a remote static observer) of reaching the Schwarzschild sphere diverges. We cannot switch to the non-existent proper time for light. Instead, let us look at the



coordinate future light cones along the falling photon's trajectory. The cones become narrower when approaching $r = r_s$ as their tilt is defined by the coordinate speed of light which reduces proportional to σ , albeit this reduction is only noticeable close enough to $r = r_s$. At $r < r_s$ the sign of σ changes so that r becomes timelike and t spacelike coordinate. This means that the light cone should instantly rotate, which is not acceptable: we have solved Einstein's equations only for $r > r_s$, and the analytical continuation through the singularity is dubious.

The only real singularity in the Schwarzschild solution is at $r \rightarrow 0$. Both physical considerations and details of the derivation (we have introduced r_s in Eq. (16.6) by equating $\sigma(r)$ to the positive $e^{-\beta}$)

¹We shall see that if the entire mass is inside, it is actually all at the singularity $r = 0$.

²The name comes from Zeno's paradox: Achilles would never catch a tortoise. Indeed, if the time dependence would be as $t(r^*)$, he would not. But this slowing down is noticeable only when r becomes comparable with r_s .

show that this metric cannot be trusted in describing the space-time inside the Schwarzschild sphere. As we know from considerations in the frame of a falling particle, it will pass through the “horizon” without noticing it. Although the naïve analytic continuation to $r < r_s$ as above gives us some clues, we would need a better coordinate system to follow the doomed journey beyond the Schwarzschild horizon.

17.2 Eddington-Finkelstein coordinates

We have seen that a particle can easily get inside of the Schwarzschild sphere by examining its trajectory in terms of the proper rather than coordinate time. We do not have proper time for light, though. And we need to construct coordinates in which both null and timelike geodesics do not misbehave. We shall do it in a few steps. Let us rewrite Eq. (17.2) as follows:

$$V = ct + r^* \equiv ct + r + r_s \ln \frac{r\sigma}{r_s}, \quad (17.3)$$

where the constant of integration renamed from r_0 in Eq. (17.2) to V becomes a new timelike coordinate that labels events on the worldlines instead of t . Then $c dt = dV - dr^* = dV - \sigma^{-1} dr$, since $\frac{dr^*}{dr} = \sigma^{-1}$. This gives the metric in the new coordinates, $\{V, r, \theta, \phi\}$:

$$\begin{aligned} ds^2 &= \sigma (\sigma^{-1} dr - dV)^2 - \sigma^{-1} dr^2 - r^2 d\Omega^2 \\ &= \sigma dV^2 - 2 dV dr - r^2 d\Omega^2. \end{aligned} \quad (17.4)$$

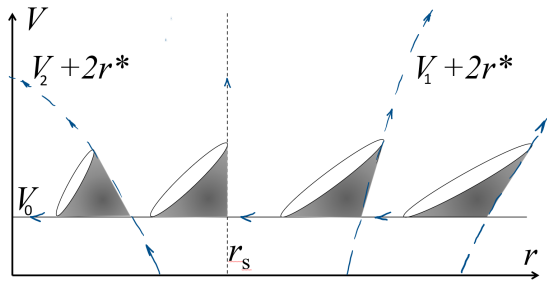
This metric is as good the solution to Einstein’s equations as the Schwarzschild metric, since these tensor equations remain valid under coordinate transformations from spherical polar to these new coordinates, which are called the Eddington-Finkelstein ingoing coordinates. As there is no divergence at $r = r_s$, it should remain valid for $r < r_s$. However, while the worldline of the falling (ingoing) photon is free of singularities, it is not true for the outgoing photon.

The radial null worldlines, $ds^2 = 0$, obey either $dV = 0$ or $dV = 2\sigma^{-1} dr$. The former one, $V = \text{const}$, corresponds to the falling photon worldline: as seen from the definition, (17.3), for V to remain constant r^* must decrease with increasing time. Both r^* and ct diverge at the horizon, but V is not – this is the whole point of introducing it: the worldline $V = \text{const}$ goes uninterrupted to the real singularity, $r = 0$. Thus V is a pseudo-time which **stands still** for falling down photons.

In contrast, the radial null geodesics corresponding to the outgoing (raising) photon is not smooth:

$$dV = 2\sigma^{-1} dr = 2 dr^* \Rightarrow V = 2r^* + V_0 \equiv 2r + 2r_s \ln\left(\frac{r}{r_s} - 1\right) + V_0, \quad (17.5)$$

where constant V_0 is defined by the initial conditions. Whatever is this constant, the time-like coordinate V diverges at the event horizon.



Let us inspect the future light cones in these coordinates. At each point the generating lines of the light cone are worldlines of the photon falling or raising from its vertex. One of the generating lines is horizontal, $V = V_0$ at any r , while the other is $V = V_n + r^*$ with the tangent $dV/dr = 2 dr^*/dr = 2\sigma$. At $r = r_s$ this worldline becomes vertical, and it is tilted to the left for $r < r_s$. After this point there is no return: both falling and “raising” worldlines end at the singularity at $r = 0$.

The **ingoing** Eddington-Finkelstein coordinates describe spacetime which is continuous for the ingoing worldlines. In a similar manner, one can construct the **outgoing** Eddington-Finkelstein coordinates that correspond to the raising null worldline in (17.1), $U = ct - r^*$. With $c dt = dU + \sigma^{-1} dr$ we find

$$ds^2 = \sigma dU^2 + 2 dU dr - r^2 d\Omega^2. \quad (17.6)$$

Thus, a photon raising from $r = 0$ along a constant U will cross the Schwarzschild sphere from inside without any problem, in no U -time, and in a finite proper time. In the Schwarzschild coordinates this event happens at $t = -\infty$, and the whole travelling before the crossing happens “before $t = -\infty$ ”, i.e. below the **past event horizon**.³ On the other hand, ingoing null radial geodesics are to be found from $dr/dU = -\frac{1}{2}\sigma$. Their trajectories differ only by the sign from those of outgoing photons, Eq. (17.5), in the V - r coordinates, thus being singular at the future event horizon.

Note in passing that the Eddington-Finkelstein coordinates could be convenient for some problems, e.g. calculating a blue shift of a signal sent from a remote emitter to a spacecraft falling down into the black hole. Let us define the blue shift, Z , as $Z = \lambda_{\text{rec}}/\lambda_{\text{emit}} - 1 \equiv \frac{1}{\alpha} - 1$ with

$$\alpha \equiv \frac{\lambda_{\text{emit}}}{\lambda_{\text{rec}}} = \frac{d(c\tau_{\text{emit}})}{d(c\tau_{\text{rec}})} = \frac{dt}{d\tau_{\text{rec}}}$$

where we use that the coordinate time t is the proper time at the emitter. As the emitter does not change spatial coordinates, $c dt = dV$. On the other hand, V does not change along the ingoing signal so that

$$\alpha = \frac{1}{c} \left(\frac{dV}{d\tau} \right)_{\text{rec}} \equiv \frac{1}{c} \dot{V}_{\text{rec}}.$$

Hence, the blue shift turns out to be a function of the trajectory of the falling receiver in the V, r coordinates.

17.3 Kruskal coordinates

Although none of the above coordinates solved the problem of coordinate singularity completely, they showed that the structure of the black hole is quite complicated. There exist two inner regions: one beyond the future event horizon, corresponding to times “after $t = +\infty$ ”; another below the past event

³This is often referred to as the white hole. In contrast to the black hole that develops in the process of the star or galaxy evolution, the white hole – essential singularity at $r = 0$ – should have existed “before times” so that it is somewhat unphysical and certainly was never observed.

horizon, corresponding to times “before $t = -\infty$. Again, it resembles Minkowski space as seen from the Rindler world of an accelerated rocket. The regions in the past and future light cones, totally normal for inhabitants of Minkowski space, are lying before the $-\infty$ or after the $+\infty$ event horizons for inhabitants of the Rindler world. Analogy is still not complete, since there are no singularities in Minkowski space – in contrast to the inner world of the black hole.

Now, we are ready to approach this inner world. The last step is to choose the coordinates free of singularities everywhere but at $r = 0$. This step has been done by Kruskal (and finalised by Kruskal and Szekeres, 1961). Kruskal coordinates can be obtained combining in- and out-going Eddington-Finkelstein coordinates. The idea is that lines $V = \text{const}$ will describe rising photons and $U = \text{const}$ will describe falling photons allowing to get rid of singularities at all. Let us start with the UV coordinates defined only for $r > r_s$ in terms of the tortoise coordinate $r^* \equiv r + \ln(r\sigma/r_s)$, Eq. (17.2):

$$\begin{aligned} U = ct - r^* & \Leftrightarrow \frac{1}{2}(V + U) = ct & \Leftrightarrow c dt = \frac{1}{2}(dV + dU) \\ V = ct + r^* & \Leftrightarrow \frac{1}{2}(V - U) = r^* & \Leftrightarrow \sigma^{-1} dr = \frac{1}{2}(dV - dU) \end{aligned}$$

On substituting the expressions for $c dt$ and dr into the Schwarzschild metric, we obtain

$$ds = \sigma dU dV - r^2 d\Omega^2,$$

where r is an implicit function of U and V . This is still not entirely free of problems as $\sigma = 0$ at the event horizon. This can be eliminated if we exponentiate these coordinates

$$u \equiv r_s \exp(-U/2r_s) \quad \text{and} \quad v \equiv r_s \exp(V/2r_s) \Rightarrow dU = -2r_s \frac{du}{u} \quad \text{and} \quad dV = 2r_s \frac{dv}{v} \quad (17.7)$$

This allows us to remove zero from $dU dV$, using

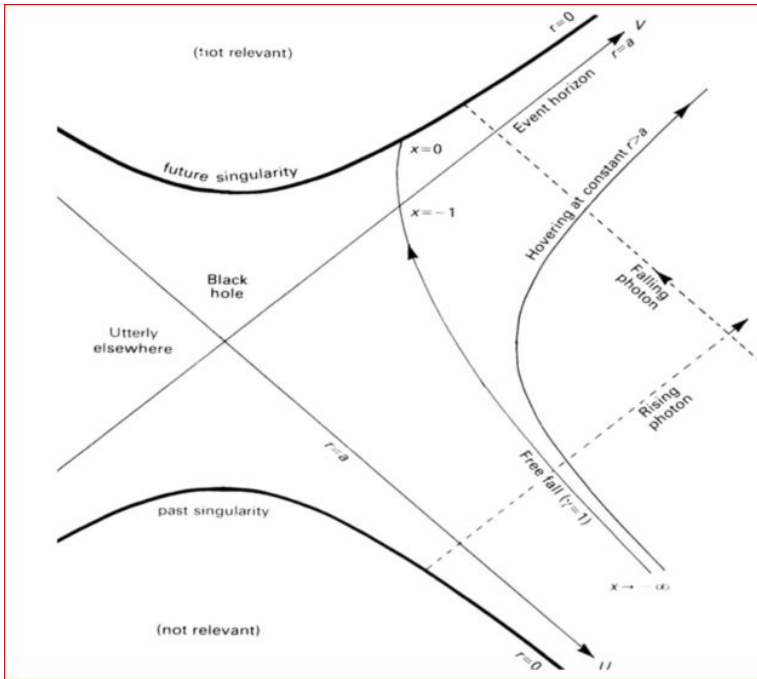
$$\begin{aligned} uv &= r_s^2 e^{\frac{V-U}{2r_s}} = r_s^2 e^{\frac{r^*}{r_s}} = \frac{r\sigma e^{r/r_s}}{r_s} \Rightarrow \sigma dU dV = -4\sigma \frac{du dv}{uv} = -4\rho^{-1} e^{-\rho} du dv \\ \frac{v}{u} &= e^{\frac{V+U}{2r_s}} = e^{ct/r_s} \end{aligned} \quad (17.8)$$

Here the dimensionless function $\rho \equiv r/r_s$ is defined implicitly as a function of uv by the 1st of Eqs. (17.8) which becomes $uv/r_s^2 = (\rho - 1)e^\rho$. Although originally u and v were positive, their analytical continuation to the negative values is straightforward. The coefficient attached to $du dv$ is obviously non-singular at $r = r_s$, and the full metric becomes

$$ds^2 = -4 \frac{e^{-\rho}}{\rho} du dv - r_s^2 \rho^2 d\Omega^2, \quad (17.9)$$

The null radial geodesics of metric (17.9) are just $u = \text{const}$ and $v = \text{const}$, corresponding to the raising and falling photons, respectively. At the Schwarzschild event horizon, $\rho = 1$ (which splits into

the lines $u = 0$ and $v = 0$, they simply change sign.



The lines $u = 0$ and $v = 0$ correspond to the past and future event horizons, respectively. The only singularity in this metric is at $\rho = 0$ which is not possible to rid of, as this one is completely physical. In the uv coordinates, the singular point becomes the hyperbola $uv = -1$. Now the coordinates cover the entire physical space. The left sector in these coordinates corresponds to the **utterly elsewhere** where (in a full analogy to the mirror rocket in the Rindler space, Lecture (7.3)) time runs in the opposite direction. For us, there is no option to communicate with this world, or get any evidence in favour or against of these utter speculations.

The final step is to make the metric Eq. (17.9) diagonal. This is relatively straightforward, with $u = \frac{1}{2}(w + \mathcal{T})$, and $v = \frac{1}{2}(w - \mathcal{T})$ one has

$$ds^2 = r_s^2 \left[\frac{e^{-\rho}}{\rho} (d\mathcal{T}^2 - dw^2) - \rho^2 d\Omega^2 \right], \quad (17.10)$$

This corresponds to rotating the $u-v$ axes by 45° , so that now $\mathcal{T}$ is the timelike, and w is the spacelike coordinate. The region $\rho \gg 1$ is asymptotically flat, corresponding to our (Schwarzschild) spacetime, $w > 0$, and “utterly elsewhere”, $w < 0$.

Lecture 18. Gravitational collapse

The Schwarzschild geometry is built as static. Nevertheless, using coordinates that are nonsingular at the horizon allows us to consider such a dynamical process as the gravitational collapse of a star. Before doing this, we shall briefly formulate conditions under which gravitational pull can be resisted by electron or neutron Fermi pressure.

The start of the evolution of a star is the gravitational contraction of a cloud of interstellar gas made mostly of hydrogen and helium. If the mass of the cloud is sufficiently large, the adiabatic compression raises the inside temperature high enough to ignite thermonuclear fusion of hydrogen. A further evolution is defined by balancing the gravitational pull against the internal pressure forces due to the expansion of gases created by the fusion. A steady state is reached and maintained for as long as hydrogen burns. The next stage comes when most of the hydrogen was spent, the internal pressure falls and the gravitational contraction resumes. The further compression raises the core temperature enough to ignite fusion of helium, which might be gradual, or a sudden explosion. The explosion creates a *Red Giant* with a size two or three orders in magnitude larger than the Sun (i.e. up to a billion km). The temperature of the explosion is enough to ignite hydrogen fusion in the outer shell, which is less intense and emits mostly in the red part of the spectrum. Eventually, the helium in the core runs out and its fusion stops. The star shrinks again, igniting more helium fusion. This ignition blows off outer layers of the star creating huge clouds of a relatively cold gas (nebulae), while the core continues its collapse to ignite fusion of more and more heavy elements (if the star mass is large enough). Eventually, only the elements remain that cannot be further fused like iron that has the highest binding energy per nucleon (although usually the fusion stops earlier leaving inert masses of carbon and oxygen in the core).

The following stage of the evolution is defined by balancing, if possible, of the gravitational pull by non-thermal outward pressure – namely, Fermi pressure of electrons or neutrons due to the Pauli principle. If the total mass of a star is not too large, such a pressure might be sufficient to lead to a cold equilibrium between it and gravitational pressure. This leads to the formation of either a White Dwarf maintained by the electronic Fermi pressure or, after a dramatic second explosion (supernova), a neutron star maintained by neutron's Fermi pressure if the mass is not too large. For a yet larger mass, the gravitational pull of the inner core is too strong and leads to the eventual gravitational collapse into a black hole where all the core mass ends up at the singularity at $r = 0$.

Below we semi-quantitatively estimate the maximal star masses¹ under which the equilibrium – either as the white dwarf or as the neutron star – is possible. Although this should be done using the formalism of GR, we will make this estimation using the Newtonian expression for gravitational energy. Surprisingly, this gives the parametrically correct result; obviously, numerical coefficients cannot be obtained in this way. Then we briefly consider the final gravitational collapse into the black hole of a spherically symmetric star that was too massive to stop at the equilibrium.

18.1 Fermi pressure

When the thermonuclear fusion fizzles out, atomic forces are nowhere near sufficient to resist the gravitational pull, so that atoms crushed and the star interior collapses to the nuclei-electron plasma.

¹When making estimates below, we only consider the mass M of the core, i.e. the part of the star that ends its life as a white dwarf, a neutron star or a black hole. The initial mass of the progenitor star was much higher but mostly shed away in the explosions.

Let us assume that a spherically symmetric star contains N electrons and μN nucleons in the core (here $(\mu - 1)N$ is the number of neutrons, and N is the number of protons), so that the core mass $M = \mu N m_p$. The Fermi pressure arises due to the Pauli principle that does not allow two fermions to be in the same quantum state. The uncertainty principle requires a fermion with momentum p to occupy space with the volume of order $(h/|p|)^3$. At zero temperature, all the fermionic states are filled up to the maximum (Fermi) momentum p_F , related to the number density $n = N/V$, where N is the total electron number in a sphere with volume $V = \frac{4}{3}\pi R^3$, by

$$n = \frac{N}{V} = 2 \int_{p < p_F} \frac{d^3 p}{h^3} = \frac{2}{h^3} 4\pi \int_0^{p_F} dp p^2 = \frac{p_F^3}{3\pi^2 \hbar^3}. \quad (18.1)$$

When the star is compressed, V decreases causing p_F to increase. This increases the full electron energy, given by

$$E = 2V \int_{p < p_F} \frac{\varepsilon(p) d^3 p}{h^3} \text{ where } \varepsilon(p) = \sqrt{p^2 c^2 + m_e^2 c^4}, \quad (18.2)$$

since the integral decrease with volume faster than V^{-1} . This integral can be analytically calculated in the nonrelativistic limit, $p \ll m_e c$, when² $\varepsilon \approx m_e c^2 + p^2/2m_e \rightarrow p^2/2m_e$, and in the ultrarelativistic limit, $p \gg m_e c$, when $\varepsilon \approx pc$:

$$E = \begin{cases} \frac{4\pi V}{h^3 m_e} \int_0^{p_F} dp p^4 = \frac{V p_F^5}{10\pi^2 \hbar^3 m_e} = \frac{\hbar^2 (3\pi^2 N)^{5/3}}{10\pi^2 m_e V^{2/3}}, & p_F \ll m_e c \\ \frac{8\pi V c}{h^3} \int_0^{p_F} dp p^3 = \frac{V p_F^4 c}{4\pi^2 \hbar^3} = \frac{(3\pi^2 N)^{4/3} \hbar c}{4\pi^2 V^{1/3}}, & p_F \gg m_e c. \end{cases} \quad (18.3)$$

The outward pressure at zero temperature is found from the 1st law of thermodynamics, $dE = -P dV$, so that using $n \equiv N/V = 3N/4\pi R^3$ with a constant N we find the Fermi pressure as

$$P_F = -\frac{dE}{dV} = \begin{cases} \frac{(3\pi^2)^{2/3}}{5} \frac{\hbar^2}{m_e} n^{5/3} = \frac{9}{40} \left(\frac{3}{2\pi}\right)^{1/3} \frac{N^{5/3} \hbar^2}{m_e R^5}, & \hbar n^{1/3} \ll m_e c \\ \frac{(3\pi^2)^{1/3}}{4} \hbar c n^{4/3} = \frac{3}{16\pi} \left(\frac{9}{4\pi^2}\right)^{1/3} \frac{\hbar c N^{4/3}}{R^4}, & \hbar n^{1/3} \gg m_e c \end{cases} \quad (18.4)$$

In the non-relativistic limit, it is clear why we have focused on the electron Fermi pressure: it is inversely proportional to the mass so that is at least by three order in magnitude bigger than the nucleon pressure. In the relativistic limit, pressure does not depend on mass but when the high enough density is reached for electrons to become relativistic, the nuclei are still in the nonrelativistic regime with a negligible Fermi pressure.

The nonrelativistic and ultrarelativistic limits in (18.4) produce the results that differ only by a numerical factor at the boundary density, $n_0 = (m_e c / \hbar)^3 \sim 3 \cdot 10^{31} \text{cm}^{-3}$ (which transpires to be up to 10^8 nucleons per atomic volume), namely $P \sim m_e^4 c^5 / \hbar^3 = m_e c^2 n_0$. This means that the kinetic energy density is of order the rest energy in the volume n_0^{-1} occupied by a single electron, and the corresponding pressure is of order 10^{20}atm .

²The rest energy is V -independent and hence do not contribute to pressure when V is changing.

18.2 Possible equilibria: white dwarfs and neutron stars

To estimate whether the equilibrium is possible, we use the Newtonian expression for the gravitational energy that result in an estimate that differs from a fully GR one only by a numerical factor.³ In this case, the gravitational energy of a sphere of radius R is unaffected by a spherically symmetric mass distribution outside of the sphere, i.e. is equal to $E_{\text{gr}} = -GM^2/R = -G\mu^2 N^2 m_p^2/R$ which gives the inward pressure

$$P_{\text{gr}} = -\frac{dE_{\text{gr}}}{dV} = -\frac{1}{4\pi R^2} \frac{dE_{\text{gr}}}{dR} = -\frac{G\mu^2 N^2 m_p^2}{4\pi R^4}. \quad (18.5)$$

The most important fact when comparing with the electron Fermi pressure, (18.3), is the higher power of the number of protons (or electrons), N , in (18.5). Hence, with large enough mass of the star, $M = \mu m_p N$, the gravitational pull inevitably wins.⁴

Let us first check conditions for the equilibrium when the compressed star is still in the nonrelativistic regime. Equating (by the absolute values) the gravitational pressure of (18.5) with the nonrelativistic Fermi pressure of (18.3) and dropping all the numerical factors and universal constants, we find that such an equilibrium is reached at radius $R \propto N^{-1/3} \propto M^{-1/3}$.

As the equilibrium radius is decreasing with M increasing, the particle density n is increasing so that for large enough $M = \mu N m_p$ it inevitably reaches the condition for the ultrarelativistic limit of (18.3). Let us write the equilibrium condition in this case, omitting this time only numerical prefactors. Since both the ultrarelativistic Fermi pressure and gravitational pressure have the same dependence on radius R , it drops out of the estimate, which gives the maximal possible number of nucleons, μN_0 , for which the Fermi pressure will withstand the gravitational pressure. For $N < N_0$, equilibrium is possible and results in the creation (after shedding some extra mass in the process of collapse) of a white dwarf – a star consisted of nuclei-electron plasma. The atoms crushed, and the density of this plasma could be 6 orders of magnitude higher than the normal star (atomic) density. For $N > N_0$ (after shedding the mass in the initial stage of collapse) gravitational pressure is unsustainable and the collapse goes to the next stage.

Comparing P_F and P_{gr} gives this N_0 (with accuracy up to a numerical factor) as follows:⁵

$$N_0 \sim \left(\frac{\hbar c}{G}\right)^{\frac{3}{2}} \frac{1}{(\mu m_p)^3} \equiv \left(\frac{M_P}{\mu m_p}\right)^3. \quad (18.6)$$

Here is the first (and, probably, only) appearance in this course of the Planck mass, $M_P \equiv \sqrt{\hbar c/G}$, that defines the energy scale ($M_P c^2$) at which gravitation must become quantum. Unfortunately, the quantum theory of gravitation is still (very much) wanted. But the appearance of m_P in Eq. (18.6) is more humble: it simply defines the number of nucleons, N_0 , and hence the mass, $M_0 = \mu m_p N_0$, such that at $M > M_0$ the electron Fermi pressure cannot sustain the gravitational pressure. Numerically, $M_P \approx 1.2 \times 10^{19} \text{ GeV}/c^2$ is huge. The resulting limiting mass (restoring the GR numerical

³Hence, I omit numerical coefficients in further estimates. For completeness, I append the GR derivation to the next lecture but will not go through this in the class.

⁴See footnote ¹.

⁵In this qualitative analysis, we have totally ignored electrostatic forces in the nuclei-electron plasma. The reason is that in the nonrelativistic quantum limit the kinetic energy depends on density as $p_F^2/m \propto n^{2/3}$, while electrostatic forces $\sim e^2 n^{1/3}$ so that with density increasing their role is reduced. In the ultrarelativistic limit, both kinetic and electrostatic energy increase with density $\propto n^{1/3}$; however, this limit is reached when the kinetic energy is of order $m_e c^2$, while $e^2 n^{1/3} \sim (e^2/\hbar c) m_e c^2$, where $e^2/\hbar c \approx 1/137$ is the fine structure constant.

factors), $M_0 \approx 1.44M_\odot$, is called the Chandrasekhar limit ($M_\odot$ is the Solar mass). Any star with a residual mass (after shedding its outer part) $M > M_0$ will not end its evolution as a white dwarf, as gravitation makes it to shrink further.

This further collapse makes all nuclei to disintegrate, and crushes electrons into protons resulting in the matter consisting only neutrons – this is the neutron star. As neutrons are also fermions, they also have Fermi pressure which does not depend on fermionic mass and is given by the second line in Eq. (18.4). However, it is reached at much higher densities, $\hbar n^{1/3} \gtrsim m_p c$. The GR considerations slightly change the limit at which gravitational pressure is unsustainable to $M_{0n} \sim 2M_\odot$ (the Oppenheimer–Volkoff limit).⁶ If the collapsing core has a mass higher than M_{0n} , it continues to collapse to its final destination – the black hole.⁷

18.3 Collapse to a black hole

When all resistance exhausted, the remnants of a supergiant star will end up at the singularity – $r = 0$. As the interior of the collapsing star is described in coordinates extended for $r < r_s$, like the ingoing Eddington–Finkelstein, or Kruskal coordinates, it is actually not one point but an entire 3D surface in spacetime, ($uv = -1$ in Kruskal coordinates, with arbitrary θ and ϕ).

Since the Fermi pressure was not enough to resist the gravitational pull, in the simplest model it can be ignored altogether. Then the only force acting at the surface of the collapsing star is a gravitational one, so that the surface moves along a geodesic line in the Schwarzschild geometry, which is convenient to describe in the ingoing Eddington–Finkelstein coordinates, where the new variable V is timelike.

Due to spherical symmetry, it is sufficient to describe the radial fall governed by the metric (17.4) which reduces to

$$ds^2 = \sigma dV^2 - 2 dV dr \quad \Rightarrow \quad \hat{\mathcal{L}} = \sigma \dot{V}^2 - 2\dot{V}\dot{r}. \quad (18.7)$$

The conservation laws for timelike (particle's) geodesics are

$$\sigma \dot{V}^2 - 2\dot{V}\dot{r} = c^2 \quad \text{and} \quad \sigma \dot{V} - \dot{r} = \text{const} \equiv v_0. \quad (18.8)$$

We need to find worldlines $r(V)$ for motion started at $r = r_0 > r_s$ at the “moment” $V = 0$. Substituting $\sigma \dot{V} = \dot{r} + v_0$ into $\hat{\mathcal{L}} = c^2$, we have $\dot{r}^2 = v_0^2 - \sigma c^2$, i.e. $\dot{r} = -\sqrt{v_0^2 - \sigma c^2}$ for the ingoing worldline. Hence we find the coordinate “velocity”, dr/dV :

$$\frac{dV}{dr} = \frac{\dot{V}}{\dot{r}} = \sigma^{-1} \left(1 + \frac{v_0}{\dot{r}} \right) \quad \Rightarrow \quad \frac{dr}{dV} = \sigma \left(1 - \frac{v_0}{\sqrt{v_0^2 - \sigma c^2}} \right)^{-1}. \quad (18.9)$$

The initial condition for this 1st order differential equation is $r(V = 0) = r_0$ so that $\sigma_0 = 1 - r_s/r_0$. Initially, $v_0^2 > v_0^2 - \sigma_0 c^2 > 0$ as $\dot{r}$ must be real. Hence the expression in the round brackets in Eq. (18.9) is negative, and so is the velocity dr/dV - with increasing time V radius is decreasing. At the moment of crossing the Schwarzschild sphere $\sigma = 0$ but so is the expression in the round brackets. Inside the sphere dr/dV remains negative so that the fall to the centre proceeds uninterrupted,

⁶The mass of the progenitor supergiant star should have been exceeding $10M_\odot$.

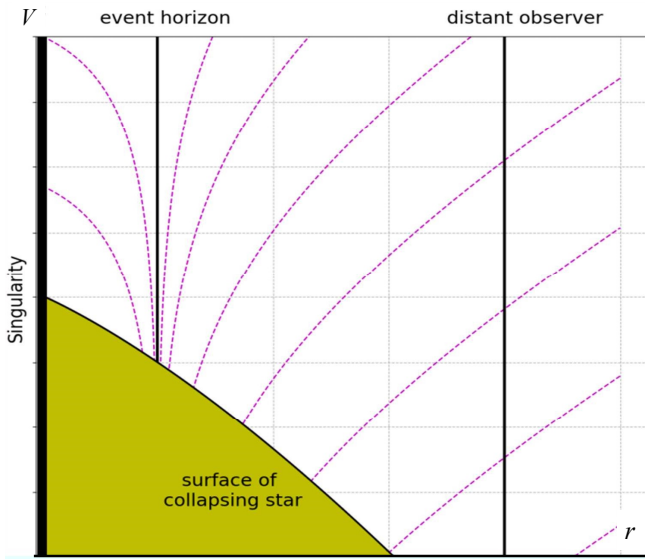
⁷All considerations have been made for non-rotating stars. The angular momentum conservation typically makes a neutron star spinning fast, and all more realistic estimations must include this spin.

as expected. Of course, there is no singularity at $r = r_s$: expanding in the vicinity of this point in small $\sigma c^2/v_0^2$, one finds $dr/dV = 2v_0^2/c^2$ depending only on the initial condition.

The outgoing (for $r > r_s$) timelike trajectory, where $\dot{r} = +\sqrt{v_0^2 - \sigma c^2}$, looks very different when $r \rightarrow r_s$, as in this case

$$\frac{dr}{dV} = \sigma \left(1 + \frac{v_0}{\sqrt{v_0^2 + \sigma c^2}} \right)^{-1} \xrightarrow{r \rightarrow r_s} \frac{\sigma}{2} \rightarrow 0 \quad (18.10)$$

It is the same (up to the corrections in $\sigma c^2/v_0^2$, which we have neglected) as the null worldline of outgoing photon, Eq. (17.5). And this stopping light or particle is entirely physical. From the surface of the falling star particles or light can still be ejected to escape the singularity, but the velocity tends to zero when the emitter at the surface approaches the event horizon. For $r < r_s$ the velocity becomes negative: there are no more outgoing worldlines as both $V = \text{const}$ and $V = \sigma/2$ lead to the singularity making the final collapse unstoppable. The worldline of a single surface point is below.



This point is in the Schwarzschild geometry, in contrast to those on the left of this worldline that are inside the star at any V . Hence only the part of this E–F diagram on the right of the worldline is meaningful. The light signals or particles ejected from the surface of a collapsing star **before the event horizon** propagate to the distant observer along the purple curves shown on the picture. The propagation time V diverges as $r \rightarrow r_s + 0$, but nothing can escape from $r < r_s$. As classically light is a harmonic signal, its pulses arrive separated by longer and longer intervals, so that the light becomes more and more dim, and more and more redshifted.

Equation (18.9) is straightforward (but cumbersome) to integrate to find $V(r)$. However, we have already done the equivalent calculation in Lecture 7, finding $t(r)$ in the Schwarzschild coordinates, Eq. (6.7), which was log-divergent as $r \rightarrow r_s$. But this coordinate divergence has been removed (for ingoing worldlines) by introducing the timelike Eddington-Finkelstein variable $V = ct + r^*$ instead of t , Eq. (17.3), where $r^* \equiv r + r_s \ln(r/r_s - 1)$ cancels the divergence in $t(r)$. Combining these equations, we find $V(r)$ -dependence for a falling worldline applicable for any point on the surface of the collapsing star as

$$V(r) = r - r_s \left[\frac{2}{3} \rho^{3/2} + 2\rho^{1/2} - 2 \ln(\rho^{1/2} + 1) + \text{const} \right], \quad \rho \equiv r/r_s, \quad (18.11)$$

which is valid all the way down to the real singularity.

To conclude, the Schwarzschild distant observer will wait forever to see the formation of the black hole completed, albeit it takes a finite proper time, or a finite coordinate time in the E–F or Kruskal coordinates. We are distant observers of the fully collapsed black holes – a lot are known and catalogued. But we are not the Schwarzschild observers, as the spherically symmetric geometry is (approximately) valid in some relative proximity to the black hole but not in the entire Universe. And one does not need to have the exact spherical symmetry for the gravitational collapse which is robust against symmetry violation – the proof of this fact earned the 2020 Nobel Prize to Sir Roger Penrose.

Lecture 19. Einstein's Equations and the Matter Tensor

In Lecture 15 we used Einstein's equations in empty space to find how the mass distribution curves the spacetime outside. The full set of Einstein's equations would allow us to find what is going on inside the mass distribution, thus complementing a brief history of stars in Lecture 17 by consistently using GR, rather than Newtonian gravity, to find the equilibria conditions. And history of the entire Universe, to which we will turn in the remaining lectures, requires using full Einstein's equations from the very beginning. We derive them in this Lecture by covariant generalization of Newtonian expressions.

19.1 Newton's gravity and the matter tensor in SR

Our reasoning for the Einstein's equation in empty space was based on the relativistically covariant generalisation of Newtonian equations in empty space:

$$\nabla^2 \varphi = 0 \quad \Rightarrow \quad \tilde{\Delta}_i^i = 0 \quad \rightarrow \quad \Delta_\alpha^\alpha = R^\alpha_{\mu\alpha\nu} u^\mu u^\nu = \Delta_0^0 + \Delta_i^i = 0 \quad \Rightarrow \quad R_{\mu\nu} = 0.$$

In a similar reasoning for the field equations in the presence of matter one should start from Newton's gravitation law in form of the Poisson equation,

$$\nabla^2 \varphi = \tilde{\Delta}_i^i = 4\pi G \rho(\mathbf{r})$$

where $\rho(\mathbf{r})$ is the mass density (in a frame where the centre of mass is at rest). In SR, the mass of a particle is a temporal component of the energy-momentum 4-vector. The mass density is a mass per unit volume. In a moving frame, mass increases by the Lorentz factor γ and volume is contracted by γ , so that the mass density increases by the factor of γ^2 under the LT. It means that it should be the temporal component of a second rank tensor, $T_{\mu\nu}$. Therefore, we anticipate that in the presence of matter the equation $R_{\mu\nu} = 0$ should become something like

$$R_{\mu\nu} - \lambda R g_{\mu\nu} = \kappa T_{\mu\nu}, \quad (19.1)$$

with λ and κ being some constant coefficients. We allow for the possibility of $g_{\mu\nu}$ entering the fields equations directly; in empty space, if $R_{\mu\nu} = 0$, so is R , and the l.h.s. seems to be the most general form consistent with the equations in empty space. As both $R_{\mu\nu}$ and $g_{\mu\nu}$ are symmetric tensors, $T_{\mu\nu}$ should be symmetric as well. As its temporal component contains mass density, this should be nothing else as the energy-momentum tensor (a.k.a. matter tensor).

However, we shall see that Eq. (19.1) they do not allow static solution at the scale of the entire Universe. Realising this, Einstein suggested to add another term, $\Lambda g_{\mu\nu}$, to the l.h.s. of Eq. (19.1), coming up with

$$R_{\mu\nu} - \lambda R g_{\mu\nu} - \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}, \quad (19.2)$$

As $T_{\mu\nu} = 0$ in the absence of matter, $R_{\mu\nu} = 0$ is no longer the solution in empty space. However, if this term is tiny at a scale of, say, the Solar system, this is still in correspondence with the Newtonian gravity (see Lecture 15.1 for discussions). Although this "blunder" is now one of the cornerstones of modern cosmology, we will not include it into considerations until the end of Lecture 21.

19.1.1 The dust model. The energy-momentum conservation

Let us first consider the matter tensor in SR for the simplest possible case of a **dust** which could be thought of as a distribution of small particles separated by empty space *which are all at rest in a certain frame*. In a sense, this is a perfect gas at zero temperature as in the rest frame particles do not move and thus exercise no pressure. Its proper energy density is obviously $\rho_0 c^2$ where ρ_0 is the mass density. The simplest 2nd rank tensor which has $T^{00} = \rho_0 c^2$ in the rest frame is

$$T^{\mu\nu} = \rho_0 u^\mu u^\nu, \quad u^\mu = \dot{x}^\mu = \gamma(c, \mathbf{v}).$$

In another inertial frame, $T^{00} = \rho_0 c^2 \gamma^2$ as expected. The time-space components, $T^{0j} = \rho_0 v^j c \gamma^2$, describe the energy flux through a moving surface. This can be seen from the particles number conservation law (a.k.a. the continuity equation):

$$T^{0\mu}_{,\mu} = 0 \iff \frac{1}{c} \frac{\partial T^{00}}{\partial t} + \text{div } T^{0j} = 0 \implies \frac{\partial}{\partial t} \int \rho \, dV = - \oint \rho \mathbf{v} \, d\mathcal{A}$$

where we have used Gauss' theorem (here $\rho = \rho_0 \gamma^2$, and the integration is carried out in any particular coordinate system). We see from the above relation that the energy density in a given volume is changed only due to the momentum flux out of the volume, as $\rho \mathbf{v}$ is the momentum density. (This is totally analogous to the current continuity equation, $\dot{\rho} + \text{div } \mathbf{j} = 0$). Similarly, the conservation law is valid for T^{0j} which is the density of energy flux and, equivalently, the momentum density. It has the same structure, $T^{\mu j}_{,\mu} = 0$ which gives the meaning to the space-space component: they describe the flux of the momentum density. (Again, they have the second meaning describing the components of the 3 tensor of angular momentum). Thus, we see that the energy-momentum tensor (or, for brevity, the matter tensor) obeys the condition

$$T^{\mu\nu}_{,\mu} = 0 \quad \text{SR,} \quad \text{or} \quad T^{\mu\nu}_{;\mu} = 0 \quad \text{GR, due to CG2SR.} \quad (19.3)$$

In the dust model, the former follows from the fact that $u^\mu_{,\nu} = 0$ as in SR we consider only the transformation with constant velocities.

19.1.2 The perfect fluid model

Before generalizing for GR, let us find $T_{\mu\nu}$ for a perfect gas (or for a perfect fluid which is the same for our purposes). In contrast to a dust, it has nonzero temperature and thus some internal pressure. We cannot have a frame where all gas particles are simultaneously at rest, and choose as a proper coordinate system one where the centre mass of the gas is at rest. For an observer in this frame, gas should look homogeneous and isotropic, at least locally. Let's consider $T_{\mu\nu}$ for the gas as the sum of tensors representing its particles (in SR, at least, this is possible). They can be grouped according to their velocities. Each can be obtained by transforming T^{00} from a rest frame of this particular group to the CMRF (centre-mass RF) where it has the form as above (assuming that densities are the same in all groups, although this assumption is not necessary for what follows):

$$T_{\mu\nu} = \rho_0 u^\mu u^\nu = \rho_0 \gamma^2 \begin{pmatrix} c^2 & cv_x & cv_y & cv_z \\ cv_x & v_x^2 & v_x v_y & v_x v_z \\ cv_y & v_x v_y & v_y^2 & v_z v_y \\ cv_z & v_x v_z & v_z v_y & v_z^2 \end{pmatrix}$$

Now we can average over velocities in different groups. For any isotropic motion, they are uncorrelated in different directions, and we shall obtain a diagonal tensor with $T^{00} = \rho c^2$ and $T^{(ii)} = \langle \rho v^i v^i \rangle$ (no summation over i !). Suppose that we consider a classical Maxwell gas. Then

$$\langle v_x v_x \rangle = \frac{1}{3} \langle v^2 \rangle = \frac{2}{3m} \langle \mathcal{E} \rangle = \frac{2}{3m} \frac{3k_B T}{2} \Rightarrow T^{11} = \frac{\rho}{m} k_B T = n k_B T = P,$$

where n is the number density, and P is pressure. In general, whatever is the equation of state, we call the quantity $\langle \rho_0 \gamma^2 (v^i)^2 \rangle$ the pressure along the i^{th} direction.

Therefore, one has $T^{\mu\nu} = \text{diag}(\rho c^2, P, P, P)$, or $T^\mu_\nu = \text{diag}(\rho c^2, -P, -P, -P)$, in the centre-of-mass frame. This can be cast into the Lorentz covariant form,

$$T^{\mu\nu} = (\rho + P/c^2) u^\mu u^\nu - P \eta^{\mu\nu} \Leftrightarrow T^\mu_\nu = (\rho + P/c^2) u^\mu u_\nu - P \delta^\mu_\nu \quad (19.4)$$

where $\eta^{\mu\nu} = \text{diag}(1, -1, -1, -1)$ is the Minkowski metric (*i.e.* $g^{\mu\nu}$ in an inertial frame). It is easy to check that this tensor obeys the conservation law of Eq. (19.3). The relation between P and ρ is defined by the equation of state for the gas (or liquid) under consideration. Note that the diagonal form of $T^{\mu\nu}$ in CMRF is valid for a perfect fluid (or gas). If there is viscosity in the fluid, in any reference frame $T^{\mu\nu}$ will be off-diagonal, with T^{ij} being the components of the stress 3-tensor. (We still can choose a frame where $T^{0j} = 0$ on average so that there are no net flows of liquid or gas).

19.1.3 Maxwell's stress tensor¹

In general, finding $T_{\mu\nu}$ (or the equation of state) is an extremely difficult problem which can be solved exactly only for a few idealized situations, like perfect gases. One case of particular interest for cosmology is a perfect ultra-relativistic gas, where in any frame $\langle v^2 \rangle \approx c^2$ (or more precisely, $\gamma \gg 1$).

Another example of the matter tensor is the energy-momentum tensor of the electromagnetic field,

$$T^{\mu\nu} = \frac{1}{8\pi} g^{\nu\beta} (F^{\mu\alpha} F_{\alpha\beta} + \mathcal{F}^{\mu\alpha} \mathcal{F}_{\alpha\beta}).$$

We have proved (a year ago) that it also obeys the conservation law Eq. (19.3). This conservation law is the most fundamental property of the matter tensor.

To summarize, any material object (dust, gas, liquid, a collection of rigid bodies, radiation, *etc*) is characterized by its energy-momentum tensor which obey Eq. (19.3). In the non-relativistic limit, this equation embraces separately all the conservation laws of energy, momentum, and angular momentum. Note that $T^{\mu\nu}$ could usually only be found approximately and it is model-dependent.

19.2 The matter tensor as the source of gravity in GR: Einstein's equations

In GR, $T_{\mu\nu}$ is the only possible candidate for the source of gravity. The conservation law of Eq. (19.3) should be generalise with the help of the standard CG2S rule:

$$\boxed{T^{\mu\nu}_{;\mu} = 0} \iff \boxed{T^{\mu\nu}_{,\mu} + \Gamma^\mu_{\beta\mu} T^{\beta\nu} + \Gamma^\nu_{\beta\mu} T^{\mu\beta} = 0} \quad (19.5)$$

Now we return to Eq. (19.1) to formulate finally the field equations. The r.h.s. of this equation is zero on contraction, due to Eq. (19.5). One says that such a tensor is divergenceless. Therefore, the l.h.s.

¹Not to be shown in the class

should also be divergenceless: this is the Einstein tensor,

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R$$

We have used Bianchi identities to prove that this tensor is divergenceless: $G^\alpha{}_{\beta;\alpha} = 0$. Therefore, one should choose $\lambda = \frac{1}{2}$ in Eq. (19.1). The constant κ should be found by reducing these equations to the Newtonian limit which we shall do later to prove that $\kappa = 8\pi G/c^4$. Thus, we finally arrive at the Einstein's field equations:

$$G_{\mu\nu} = \kappa T_{\mu\nu}. \quad (19.6)$$

All properties of the particles/light motion we have used up to now are derivable from this equation which summaries all the main principles of GR in concise mathematical form. Thus, the assumption that particles move along timelike geodesic is also a consequence of this equation.

It is useful to cast the Einstein's equations into an alternative form. We contract it by $g^{\mu\nu}$:

$$g^{\mu\nu} G_{\mu\nu} = g^{\mu\nu} R_{\mu\nu} - \frac{1}{2}Rg^{\mu\nu}g_{\mu\nu} = R - \frac{1}{2}R\delta^\mu{}_\mu = -R \quad \Rightarrow \quad -R = \kappa g^{\mu\nu} T_{\mu\nu} = \kappa T^\mu{}_\mu$$

Thus we obtain

$$R_{\mu\nu} = \kappa \left(T_{\mu\nu} - \frac{1}{2}g_{\mu\nu} T^\alpha{}_\alpha \right) \Leftrightarrow R^\mu{}_\nu = \kappa \left(T^\mu{}_\nu - \frac{1}{2}\delta^\mu{}_\nu T^\alpha{}_\alpha \right) \quad (19.7)$$

The last form is more convenient as it does not include an unknown $g_{\mu\nu}$.

19.2.1 Newtonian limit

To establish the value of κ , we look at the limit of a weak field. In this limit, $g_{\mu\nu}$ is reduced to the post-Newtonian metric tensor, given by Eq. (3.7):

$$g_{\mu\nu} = \text{diag} (1 + 2\varphi c^{-2}, -1, -1, -1) \equiv \eta_{\mu\nu} + h_{\mu\nu},$$

where $\eta_{\mu\nu}$ is the Minkowski metric tensor, Eq. (3.1), and $h_{\mu\nu} \equiv \text{diag} (2\varphi/c^2, 0, 0, 0)$ is the GR correction.

The dominant part of the matter tensor is the energy density $T^{00} = \rho_0 c^2$ (as all the other components contain the lower powers of c , independently of the model we use $T_{\mu\nu}$). Hence its trace $T^\mu{}_\mu = g_{\mu\nu} T^{\mu\nu} = \eta_{\mu\nu} T^{\mu\nu} = T^{00}$ in this limit so that the most relevant part of Eq. (19.7) reduces to

$$R_{00} = \frac{1}{2}\kappa T_{00} \quad \text{with} \quad T_{00} = \rho_0 c^2. \quad (19.8)$$

Thus we need to know only R_{00} . We expect from the correspondence between the GR and Newtonian tidal tensor that $\Delta^\alpha{}_\alpha = R_{\mu\nu} u^\mu u^\nu \rightarrow R_{00} c^2 = \nabla^2 \varphi$, but it is better to see this directly.

In evaluating R_{00} we can use a LGF where all Christoffel symbols vanish, and the only nonvanishing 2nd derivatives of $g_{\mu\nu}$ are $g_{00;ii} = 2\varphi_{,ii}/c^2$. In the LGF, we find using Eq. (13.9)

$$R_{00} = g^{\mu\nu} R_{\mu 0 \nu 0} = \frac{1}{2} \eta^{\mu\nu} (h_{\mu 0, 0 \nu} + h_{0 \mu, \nu 0} - h_{00, \nu \mu} - h_{\nu \mu, 00}) = -\frac{1}{2} \eta^{ij} h_{00, ij} = \frac{1}{2} \sum_{i=1}^3 h_{00, ii} = \frac{\nabla^2 \varphi}{c^2}.$$

Substituting this to Eq. (19.8) we find

$$\nabla^2 \varphi = \frac{1}{2} \kappa \rho_0 c^4 = 4\pi G \rho_0 \quad \Rightarrow \quad \kappa \equiv \frac{8\pi G}{c^4}.$$

19.3 The absence of the energy-momentum conservation in GR ²

An immediate consequence of the GR generalization (comma-goes-to-semicolon) is that the energy-momentum conservation is no longer valid. Indeed, in SR it was based on Gauss' theorem which is applied after integrating the differential relations over volume and valid for $T^{\mu\nu}_{;\mu}$ but not for $T^{\mu\nu}_{,\mu}$.

We could anticipate this by considering a collection of particles. We have seen that for a single particle in the Schwarzschild field the conservation laws are valid (some components of its 4 momentum are conserved due to the symmetry). However, if we have even only two particles, we do not know how to apply the conservation laws. The reason is that in curved space-time one cannot define the total energy-momentum of two particles, since one cannot add vectors defined at different points; **the total 4-momentum is ill-defined**. For the distribution of (many) particles, one can come up with this or that definition of the energy-momentum tensor; however, as it could not be based on the sum of constituent particles, there is no reason to expect the conservation laws to be valid.

Physically, the reason for the absence of the conservation laws is clear: one should care about the gravitational self-energy (a negative quantity equal to the work one gains by assembling the system from isolated particles at infinity). This energy cannot be assigned to separate particles but resides in geometry itself. Clearly, it is not local. One can say that this energy should represent the difference between the total energy (mass) of the system and the sum of energies of separate particles. Since such a sum is not well defined, neither is the gravitational potential energy.

In principle, there are some ways to "correct" this flaw. The most accepted one is to introduce the energy-momentum pseudo-tensor of the gravitational field which is well-defined for an isolated system whose total energy is conserved. However, it is tensor only with respect to linear transformations, and this concept is not entirely satisfactory.

19.4 Calculating the Schwarzschild radius

We calculate the gravitational field created by a spherically symmetric massive star using the simplest model for the matter tensor, i.e. is Eq. (19.4) for a perfect nonviscous isotropic liquid. (In the absence of an initial angular momentum of the mass distribution, its spherical symmetry is guaranteed by gravitational forces). The components of $g^{\mu\nu}$ are unknown functions of the radial coordinate r . Pressure and density are related by some unspecified equations of states $P = P(\rho)$, not that for a perfect gas as both P and ρ depends on r , distance from the centre.

Einstein's field equations in the presence of matter,³

$$G^\mu{}_\nu \equiv R^\mu{}_\nu - \frac{1}{2}\delta^\mu{}_\nu R = \kappa T^\mu{}_\nu,$$

should be solved self-consistently, together with the equation of state $P = P(\rho)$ to produce both the density-pressure distribution and the metric tensor. Quite surprisingly, the Schwarzschild radius can be finally calculated *without any knowledge of the equation of state* by matching the solutions in empty and filled spaces at the surface boundary. We have already calculated all the components of $R^\alpha{}_{\beta\mu\nu}$ for the spherically symmetric metric,

$$ds^2 = e^\alpha c^2 dt^2 - e^\beta dr^2 - r^2 d\Omega^2 \quad (19.9)$$

²Not to be shown in the class

³It is clear that the mixed component of the matter tensor are less cumbersome, as $T^\mu{}_\nu$, Eq. (19.4), does not contain $g_{\mu\nu}$, and thus equals to $\text{diag}(\rho_0 c^2, -P, -P, -P)$ in the centre-of-mass frame we consider.

in terms of yet unknown (in non-empty space) functions $\alpha(r)$ and $\beta(r)$, Eq. (16.5). It is the functions $\alpha(r)$ and $\beta(r)$ that we should calculate in the presence of matter, and then match them at the boundaries with the empty-space solutions found in Lecture 15.

We know the components of the Riemann and Ricci tensors (Lecture 15) and need to express the Einstein tensor, $G^\mu_\nu = R^\mu_\nu - \frac{1}{2}\delta^\mu_\nu R$, via α and β (a mixed representation is convenient as it does not involve $g_{\mu\nu}$). R^μ_ν is diagonal, and so is G^μ_ν :

$$\begin{aligned} G^0_0 &= R^0_0 - \frac{1}{2}R^\mu_\mu = \frac{1}{2}(R^0_0 - R^1_1 - R^2_2 - R^3_3) = \frac{\beta'e^{-\beta}}{r} + \frac{1 - e^{-\beta}}{r^2} \\ G^1_1 &= R^1_1 - \frac{1}{2}R^\mu_\mu = \frac{1}{2}(R^1_1 - R^0_0 - R^2_2 - R^3_3) = -\frac{\alpha'e^{-\beta}}{r} + \frac{1 - e^{-\beta}}{r^2} \\ G^2_2 &= G^3_3 = -\frac{1}{2}(R^0_0 + R^1_1) = -e^{-\beta} \left[\frac{1}{2}\alpha'' + \frac{1}{4}(\alpha')^2 - \frac{1}{4}\alpha'\beta' + \frac{\alpha' - \beta'}{2r} \right] \end{aligned}$$

To begin with, we restrict ourselves only to finding r_s . As G^0_0 depends only on β , we have

$$G^0_0 = \frac{\beta'e^{-\beta}}{r} + \frac{1 - e^{-\beta}}{r^2} = \kappa T^0_0 = \kappa \rho_0 c^2.$$

As this is the 1st order differential equation, it is immediately solvable:

$$\begin{aligned} \frac{1}{r^2} \frac{d}{dr} [r(1 - e^{-\beta})] &= \kappa \rho_0 c^2 \Rightarrow r(1 - e^{-\beta}) = \frac{8\pi G}{c^2} \int_0^r r^2 \rho_0(r) dr \equiv a(r) \Rightarrow \\ e^\beta &= \left(1 - \frac{a(r)}{r}\right)^{-1} \end{aligned} \quad (19.10)$$

This should match the solution in the exterior of the star (which is empty space as both P and ρ vanish) at the star surface, $r = R$, so that $r_s = a(R)$:

$$r_s = a(R) = \frac{2G}{c^2} \underbrace{\int_0^R \rho(r) 4\pi r^2 dr}_{=\int \rho_0 dV = M} \equiv \frac{2GM}{c^2},$$

so that we have finally confirmed by calculation the expression for the Schwarzschild radius. Here we have introduced the star mass by $M = 4\pi \int_0^R r^2 \rho(r) dr$ which is perfectly reasonable in Newtonian physics. The analogy is a bit deceptive and there are some subtle issues involved but the result is exact and correct: properly written, this integral would indeed give the mass of the spherically symmetric distribution.

19.5 The interior structure of a spherically symmetric star⁴

19.5.1 The Oppenheimer-Volkoff and Chandrasekhar equations

A full solution for the exterior requires to consider two more equations. Let us rewrite $G^1_1 = -\kappa P$:

$$\begin{aligned} -\frac{\alpha'e^{-\beta}}{r} + \frac{1 - e^{-\beta}}{r^2} &= \kappa P \iff -\frac{\alpha'}{r} \left(1 - \frac{a(r)}{r}\right) + \frac{a(r)}{r^3} = -\kappa P \iff \\ \alpha' &= \frac{a(r) + r^3 \kappa P}{r^2 - ra(r)}. \end{aligned} \quad (19.11)$$

⁴Not to be shown in the class

Next, instead of rearranging the equation for G^2_2 , we use the “conservation law” for the radial component of the matter tensor, $T^\mu_{1;\mu} = 0 \iff T^\mu_{1;\mu} + \Gamma^\mu_{\alpha\mu} T^\alpha_1 - \Gamma^\alpha_{1\mu} T^\mu_\alpha = 0$:

$$T^1_{1,1} + (\Gamma^0_{01} + \Gamma^1_{11} + \Gamma^2_{21} + \Gamma^3_{31}) T^1_1 - \Gamma^0_{01} T^0_0 - \Gamma^1_{11} T^1_1 - \Gamma^2_{21} T^2_2 - \Gamma^3_{31} T^3_3 = 0$$

Two terms containing Γ^1_{11} cancel identically; taking into account that $T^0_0 = \rho c^2$ and $T^1_1 = T^2_2 = T^3_3 = -P$, one finds that only three terms remain (for the case of isotropic pressure we consider):

$$T^1_{1,1} + \Gamma^0_{01} (T^1_1 - T^0_0) = 0 \iff P' + \Gamma^0_{01} (P + \rho c^2) = 0.$$

Substituting Γ^0_{01} from the table, we obtain for the remainder:

$$P' + (P + \rho c^2) \frac{1}{2a'} = 0.$$

Excluding α' with the help of Eq. (19.11) leads to the **Oppenheimer-Volkoff** equation:

$$\boxed{\frac{dP}{dr} = -\frac{(P + \rho c^2)(a + r^3 \kappa P)}{2(r^2 - ra)}} \quad (19.12)$$

The second differential equation follows from the definition of $a(r)$, Eq. (19.10):

$$\boxed{\frac{da}{dr} = \kappa r^2 \rho c^2} \quad (19.13)$$

Note the Newtonian limit of these equations. The metric is almost flat so that $a \ll r$. The integral of Eq. (19.10) has always the meaning of mass, and we can substitute $2Gm(r) = 2G \int \rho dV$ for ac^2 . On the other hand, $P \ll \rho c^2$ in this limit, so that $\kappa P r^3 / a \sim P / (\rho c^2) \ll 1$, and Eq. (19.12) is reduced to **the Chandrasekhar equation**,

$$\frac{dP}{dr} = -\frac{\rho m(r)}{r^2},$$

Equations (19.12) and (19.13) together with the equation of state, $P = P(\rho)$ (which is, generally speaking, unknown), define the interior structure of the star. One boundary condition, $a(0) = 0$, follows from the definition of $a(r)$, Eq. (19.10). It presumes, however, that the coordinate system is well defined down to $r = 0$, *i.e.* this point is contained in the spacetime. Thus we preclude singularity similar to those inside the black hole discussed earlier, and rather look at the structure of an “ordinary” star. For such a star, at any point $r > r_s(r)$ where $r_s(r) = 2GM(r)/c^2$ include only the mass ‘inside’. The second can be chosen, *e.g.*, as $P = \rho = 0$ at $r = R$ where R is the radius of the sphere. In the case when there is no sharp boundary between the mass distribution and empty space, it could be more convenient to choose $P = P_0$ at $r = 0$ as the boundary condition (although the pressure in the centre of star is definitely beyond an experimental observation). When such a boundary condition is chosen, the surface of the star is defined by the requirement $P = 0$. One can make a less restrictive choice, defining the second boundary condition at $r = 0$ instead of $r = R$ by, *e.g.*,

$$P(r=0) = P_0. \quad (19.14)$$

Then the set of two differential equations (19.12) and (19.13) has a unique solution dependent only on one unknown parameter, pressure at the star centre P_0 . For a given equation of state, $P = P(\rho)$, the set of all spherically symmetric static stellar models forms a one-parameter sequence, the parameter being central pressure (or density). However, for almost any realistic equation of state solving equations (19.12) and (19.13) analytically is hard. We consider just the simplest model below that has been used by Schwarzschild for the star interior.

19.5.2 A model of incompressible star

The simplest possible assumption about the equation of state is $\rho = \text{const.}$ This is obviously unphysical: for example, the speed of sound $\propto (dP/d\rho)^{1/2} = \infty$. However, for a neutron star the density is nearly constant (and the sound speed $\rightarrow c$). This approximation, being the simplest, leads to the solution which is qualitatively similar to solutions of more realistic models for the neutron star. In this approximation, one integrates in Eq. (19.13) to obtain

$$a(r) = \frac{\kappa r^3 \rho c^2}{3} \equiv \frac{2G}{c^2} \frac{4\pi \rho r^3}{3} \equiv \frac{2Gm(r)}{c^2}, \quad a(R) \equiv r_s.$$

Substituting this into Eq. (19.12), one reduces it to

$$\frac{dP}{dr} = -\kappa r \frac{(P + \rho c^2)(3P + \rho c^2)}{3 - \kappa \rho c^2 r^2}.$$

Integrating with the boundary condition 19.14, one finds

$$\frac{3P + \rho c^2}{P + \rho c^2} = \frac{3P_0 + \rho c^2}{P_0 + \rho c^2} \left[1 - \frac{a(r)}{r} \right]^{1/2}. \quad (19.15)$$

The star radius, R , and hence P_0 in terms of R and r_s are found by putting in (19.15) $P = 0$ at $r = R$ where $a(R)/R = r_s/a \equiv 2GM/Rc^2 = 8\pi GR^2\rho/c^2$:

$$R^2 = \frac{3c^2}{8\pi G\rho} \left[1 - \frac{(P_0 + \rho)^2}{(3P_0 + \rho)^2} \right] \Rightarrow P_0 = \rho c^2 \frac{1 - \sqrt{\sigma(R)}}{3\sqrt{\sigma(R)} - 1}$$

At the surface, we can legitimately use Schwarzschild notations, so that $\sigma(R) \equiv 1 - r_s/R$. As $P_0 \rightarrow \infty$ when $\sigma(R) \rightarrow 1/3$, we have the limitation on the M/R ratio: $r_s/R < 8/9$, i.e. $M/R < 4c^2/9G$. Surprisingly, this is a real limitation rather than the limitation of the constant-density model, as it arises in more realistic models of the star interior.

Next, we can exclude P_0 from (19.15) which gives the effective equation of state in terms of the star mass and radius:

$$P = \rho c^2 \frac{\sqrt{\tilde{\sigma}(r)} - \sqrt{\sigma(R)}}{3\sqrt{\sigma(R)} - \sqrt{\tilde{\sigma}(r)}}, \quad \tilde{\sigma}(r) \equiv 1 - \frac{r^2 r_s}{R^3}. \quad (19.16)$$

Finally, substituting this into (19.11) we solve this 1st order ODE to find $\alpha(r)$ and thus fully determine metrics (19.9) for the star interior:

$$e^\alpha = \frac{3}{2} \sqrt{\sigma(R)} - \frac{1}{2} \sqrt{\tilde{\sigma}(r)}.$$

This metric can be used for a more accurate calculation of the Oppenheimer–Volkoff limit for the mass neutron stars.

Lecture 20. The metric for the Universe

The size of our species is about 2 m. The radius of Earth's orbit is 1.5×10^{11} m. An astronomical yardstick for measuring distances to closest stars uses parsec, $1 \text{ pc} \approx 3 \times 10^{16} \text{ m} \approx 3.3 \text{ l.y.}$ – by definition, at this distance the radius of Earth's orbit is a mere angular second. A typical galaxy has a radius of order 10 kpc, contains about 10^{11} stars, and separated from its nearest neighbours by 1 Mpc. A “visible” size of the Universe (*i.e.* within reach of our telescopes) is about $3 \times 10^5 \text{ Mpc} \sim 10^{27} \text{ m}$. For comparison, a typical atomic size ($\sim 1 \text{ \AA} = 10^{-10} \text{ m}$) or even a nucleon size ($\sim 1 \text{ F} = 10^{-15} \text{ m}$) is much closer to our own scale. The fact that we cannot grasp such a scale was convincingly illustrated in a very nice and elaborated model of the Universe by Douglas Adams (*The Hitch Hiker's Guide to the Galaxy: a trilogy in five parts*). Somewhere in the Universe, there exists an ultimate torture machine called the Total Perspective Vortex which drives any human (and inhumane) mad just by making him/her/it to realise his/her/its scale: “an invisible dot on an invisible dot, infinitely small”. The Vortex has proved conclusively that if life is going to exist in the Universe, then the one thing it cannot afford to have is a sense of proportion.

However, the modern humankind (or at least, its scientific species) lost the sense of awe of the Universe. Being not able to reach physically much beyond the Moon (and possibly never ever able to reach beyond the Solar system), we cannot stop our curiosity and always look for answers to our questions about Life, Universe, and Everything (see the above quoted source for the ultimate answer). Our belief in our ability to comprehend something which is so much beyond our grasp is amazing but understandable. And it is based on a firm ground, namely, on our other belief that our place in the Universe is by no means exceptional but rather typical. Then, experiencing only an infinitesimal vicinity of the “invisible dot” we are still able (or believe in being able) to reconstruct the entire picture of the whole of creation. This has been explained very clearly in the above mentioned source:

Since every piece of matter in the Universe is in some way affected by every other piece of matter in the Universe, it is in theory possible to extrapolate the whole of creation – every sun, every planet, their orbits, their composition . . . – from, say, one small piece of fairy cake

20.1 Cosmological principle: homogeneous and isotropic Universe

The whole Universe as seen from Earth is fairly isotropic (and isotropically expanding) on large scales (beyond 100 Mpc). The assumption that we are not exceptional means that the Universe should be seen as isotropic from any other place. It implies also homogeneity (at the same large scale) – the dust made of galactic clusters fills homogeneously the whole Universe. This is complicated by the fact that, because of the finite speed of light, we observe the history of the Universe rather than the Universe “contemporary” to us at every point. Then the simplest possible **assumption** about the large-scale structure of the Universe can be formulated as the so called **cosmological principle**:

The entire Universe viewed at a fixed time from any point is homogeneous and isotropic at large scale.

A nontrivial notion here is that of a “fixed time” - we need to introduce a “cosmic” time coordinate universal for the entire Universe. We also imply that there exist **fundamental observers**, associated with coordinate frames with the cosmic time. These are the frames where Universe is viewed as

homogeneous and isotropic, assuming that it is coarse-grained in such a way that the "point" is of a galaxies scale.

The concept of cosmic time is not trivial in the **expanding** Universe, as fundamental observers in different frames *in principle* cannot compare their observation at a fixed time. The cosmic time can be formally introduced with the following imaginable procedure. Fundamental observers at different points (galaxies) log all their observation throughout all the time their galaxies existed. Because of the homogeneity, these logs must be identical (with the accuracy up to a chosen time scale). It is expanding universe itself which synchronize the cosmic time at all points. This time is also the proper time for each galaxy.

Obviously, not every coordinate system with the cosmic time coordinate is the system of a fundamental observer. We shall see later that any fundamental observer is at rest with respect to the matter tensor, $T^{\mu\nu}$, of the Universe. It means that locally (after averaging over distances of order of 10 Mpc) the momentum density is zero, and pressure is isotropic. It does not mean the return of the absolute space in disguise: although fundamental coordinates are more convenient for describing the Universe as a whole (as well as heliocentric coordinates are more convenient for describing the Solar system), one could use any coordinate system sacrificing, however, a relative simplicity of the most suitable system. Naturally, fundamental coordinates are not unique, and fundamental observers move with respect to each other. Their relative motion is due to the fact that the matter filling the universe is expanding at 'every point' (meaning by a point something of meta-galactic size). However, an inertial observer which is not 'co-moving' together with the expanding Universe would not see the Universe as homogeneous and isotropic (e.g., red shift of the light from distant galaxies would not be the same from different directions). Therefore, another crucial requirement for fundamental observers is that they are **co-moving** with the expanding Universe, *i.e.* their coordinate frames partake in the uniform isotropic expansion of the matter in the Universe. The short formulation of this:

*The **fundamental observer** is in free fall in the metric of the Universe*

Later, we will make a mathematically precise definition of co-moving.

We cannot verify or check whether the cosmological principle is correct - it is just the summary of our beliefs. There is no observation which contradicts to the cosmological principle, however, so that – as this is the simplest assumption about the large-scale Universe – all attempts to build the picture of Universe should be based on it (the Occam razor principle). More complicated assumptions could lead to different predictions which will remain totally unobservable. It could not be excluded, naturally, that when bigger (or more ancient) part of the Universe becomes available to observation, this principle would be violated. At the moment, however, there is no reason to abandon it.

Considering the uniform, isotropic, and expanding Universe, we should try to explain its structure within GR. We have already made estimations that the importance of gravity for a homogeneous mass distribution increases with the scale of this distribution. The measure of this is the ratio $r_s/R = GM/c^2R \propto G\rho R^2$ which increases with R whatever is the average density ρ . With the density which is of order 10^{-28} kg/m^3 at the scale where the Universe is homogeneous ($\gtrsim 10^3$ Mpc), the description of gravitation must be relativistic.¹

Now we need to construct the class of metrics which obey the cosmological principle: these are

¹The Universe is not described by the Schwarzschild metric, and the meaning of r_s is different: it is a natural scale beyond which relativistic effects in the theory of gravitation are no longer weak.

the metrics for fundamental observers. In the next lecture, modelling in an appropriate way the matter tensor for the Universe, we shall solve Einstein's equations to find the coefficients of such a metric.

20.2 The Friedmann–Robertson–Walker metric

The metric that obeys the cosmological principle should have the following form:

$$ds^2 = c^2 dt^2 - dS^2,$$

where dS^2 is the spatial part of metric which could be time-dependent. This metric must have the following special features (some of which are explicit and some impose limitations on its spatial part):

- *The space-time terms proportional to $dt dx^i$ are excluded since their presence would imply that $+x^i$ and $-x^i$ directions are not equivalent, in the violation of CP*
- *$g_{00} = 1$ which implies the existence of the **cosmic** time*
- *There exist **co-moving** coordinates in which the ratios of spatial distances remain unchanged at any t – otherwise space would not remain isotropic. In co-moving coordinates each point (a galaxy) initially at rest remains forever at rest.*
- *At any given cosmic time the spatial curvature should be the same everywhere*

To introduce co-moving coordinates, one just needs to factorize out the time dependence in dS^2 as this should be the same for all coordinates.

$$ds^2 = c^2 dt^2 - [\mathcal{R}(t)]^2 d\ell^2, \quad (20.1)$$

where $\mathcal{R}(t)$ is the **expansion function** of the Universe (aka **the history function**), and $d\ell$ is the dimensionless time-independent function of dx^i . The proper distance between any two points, $dS = \sqrt{-ds^2}$ (calculated at $dt=0$), is given by $\Delta S_{12} = \int_1^2 dS = \mathcal{R}(t) \int_1^2 d\ell$. Therefore, the ratio S_{12}/S_{34} is time-independent, as required for co-moving coordinates. However, each ΔS by itself is changed in time at the rate $\dot{\mathcal{R}}(t)$ so that the relative change in the distance is²

$$\frac{\Delta \dot{S}}{\Delta S} = \frac{\dot{\mathcal{R}}}{\mathcal{R}} \equiv H(t) \quad (20.2)$$

$H(t)$ is just the Hubble 'constant'. It follows from the definition that its dimensionality is inverse second. The current value is known to be about $2.2 \times 10^{-18} \text{s}^{-1}$. However, it is not very telling in this units. As $\dot{S}$ is the speed at which galaxies separated by S are receding from one another, $\dot{S} = HS$ means that farther the galaxies, the faster they run away from each other. So it is agreed to measure H in $(\text{km/s})/\text{Mpc}$, and in these units $H \approx 67.5 (\text{km/s})/\text{Mpc}$, i.e. the galaxy which is at the distance 1Mpc from the Milky Way is receding from us at 67.5 km/s.

If H were, indeed, a constant, the straightforward solution of Eq. (20.2) would be $\mathcal{R} = \exp(Ht)$ – the exponentially expanding Universe. Our Universe is well known to be expanding. Until (relatively) recently, astronomers believed that mutual gravitational attraction of receding galaxies would inevitably lead to the deceleration of this expansion, and to quantify it the deceleration parameter, q , was introduced:

$$q \equiv -\frac{\mathcal{R}\ddot{\mathcal{R}}}{\dot{\mathcal{R}}^2} \quad (20.3)$$

²In this and following lectures we use *dot* for the differentiation w.r.t. cosmic time: $\dot{\mathcal{R}} \equiv \frac{d\mathcal{R}}{dt}$.

The thrust of cosmological discussion in the 20th century was about whether q is large enough – or not – to change eventually the sign of H so that the expansion would be followed by contraction. The modern observations, however, proved beyond any doubt that q is negative: the Universe expansion is actually accelerating rather than slowing down. We shall briefly review these discussions in the following lectures. Now, let us specify $\mathcal{R}$ a bit more.

The fourth requirement of the cosmological principle, spatially constant curvature, allows one to specify the length element $d\ell^2$ in Eq. (20.1). A $2D$ space of a constant positive curvature is a sphere embedded into a $3D$ space. We can build a coordinate system for $3D$ space of constant positive curvature, considering it as a 3-sphere embedded into a $4D$ space (*i.e.* the surface of a 4-dimensional ball). The scalar curvature of the sphere is always $2/\mathcal{R}^2$, whatever is the space dimensionality. In a $4D$ space the distance between two points is

$$dS^2 = dx^2 + dy^2 + dz^2 + dw^2.$$

Here w is an auxiliary 4th coordinate. If the points are on the 3-surface (*i.e.* in our usual $3D$ space), dw can be excluded from dS using the equation for a 3-sphere of radius $\mathcal{R}$:

$$x^2 + y^2 + z^2 + w^2 = \mathcal{R}^2 \quad \Rightarrow \quad \begin{cases} \mathcal{R}^2 = r^2 + w^2 \\ dS^2 = dw^2 + dr^2 + r^2 d\Omega^2 \end{cases} \quad \Rightarrow \quad dw = -\frac{r dr}{w} = \frac{-r dr}{\sqrt{\mathcal{R}^2 - r^2}}.$$

where r and $\Omega \equiv (\theta, \phi)$ are standard spherical polar coordinates. Then, the length element on the 3D surface (*i.e.* in the $3D$ space of constant spatial curvature) becomes

$$dS^2 = dr^2 + \frac{r^2 dr^2}{\mathcal{R}^2 - r^2} + r^2 d\Omega^2 = \frac{\mathcal{R}^2 dr^2}{\mathcal{R}^2 - r^2} + r^2 d\Omega^2 = \mathcal{R}^2 \left[\frac{d\tilde{r}^2}{1 - \tilde{r}^2} + \tilde{r}^2 d\Omega^2 \right] \equiv \mathcal{R}^2 d\ell^2,$$

where we have introduced the dimensionless radial coordinate, $\tilde{r} \equiv r/\mathcal{R}$. It follows from the derivation that $2/\mathcal{R}^2(t)$ is just the time-dependent scalar spatial curvature. Note in advance that it has nothing to do with the scalar curvature of the **space-time** described by the metric 20.1; the spacetime Riemann curvature should be found from solving Einstein's equations for the Universe.

In a similar way, one can work out the metric for the space of constant negative curvature. By analogy with the $2D$ case where a hyperboloid (a pseudo-sphere) has a constant negative curvature, we have a $3D$ pseudo-sphere in a $4D$. Procedure is similar to the above, the only difference being the sign of w^2 in the equation for the pseudo-sphere, $w^2 - r^2 = \mathcal{R}^2$. This leads to

$$dS^2 = \mathcal{R}^2 \left[\frac{d\tilde{r}^2}{1 + \tilde{r}^2} + \tilde{r}^2 d\Omega^2 \right] \equiv \mathcal{R}^2 d\ell^2.$$

Thus, we finally arrive at the Robertson-Walker metric which is constructed in accordance with the cosmological principle:

$$ds^2 = c^2 dt^2 - \mathcal{R}^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 (d\theta^2 + \sin^2\theta d\phi^2) \right] \quad (20.4)$$

Here $2k/\mathcal{R}^2(t)$ is the constant scalar curvature of the space (not the spacetime): $k = 1$ and $k = -1$ describing the space of positive or negative curvature respectively. We can also consider the case $k = 0$ which is obviously spatially flat: dS^2 at any moment of time reduces to the Euclidean metric. Any value of $k \neq 0$ could be reduced either to $k = 1$ or to $k = -1$ by rescaling the radial coordinate.

Let us stress again: this is a suitable metric for **any** fundamental observer: $r = 0$ coordinate is not the centre of creation. Fundamental observers are not inside but on the surface of 3D sphere – or pseudo-sphere – where any point could be chosen as the coordinate centre

Note that this metric was first suggested by Friedmann in the context of general relativity. Later, Robertson and (independently) Walker proved that Eq. (20.4) (or equivalent to this) is the most general expression for the metric that obeys the CP. Today we have followed this line of reasoning. This means that the expansion function (and the curvature k) remains undetermined. In the framework of general relativity, both $\mathcal{R}$ and k should be found by solving Einstein's equations for a proper model of the matter distribution (described by T^μ_ν) in the Universe. Modelling T^μ_ν and solving the EE, one can find different scenarios for the evolution of the Universe in GR.

20.3 The Riemann for the FRW Metric

$$g_{\mu\nu} = \text{diag} \left(1, -\mathcal{R}^2 \eta^{-1}, -\mathcal{R}^2 r^2, -\mathcal{R}^2 r^2 \sin^2 \theta \right), \quad g^{\mu\nu} = \text{diag} \left(1, \frac{-\eta}{\mathcal{R}^2}, \frac{-1}{\mathcal{R}^2 r^2}, \frac{-1}{\mathcal{R}^2 r^2 \sin^2 \theta} \right).$$

Here $\eta \equiv 1 - kr^2$. Then, with the dot for $\partial_0 = \frac{1}{c} \partial_t$, and $x^0 \equiv ct$, $x^1 \equiv r$, $x^2 \equiv \theta$, $x^3 \equiv \phi$, let's consider nonzero derivatives of the RW metric tensor.

$$g_{11,0} = -2 \frac{\dot{\mathcal{R}} \mathcal{R}}{\eta} = 2g_{11} \frac{\dot{\mathcal{R}}}{\mathcal{R}}, \quad g_{22,0} = -2\dot{\mathcal{R}} \mathcal{R} r^2 = 2g_{22} \frac{\dot{\mathcal{R}}}{\mathcal{R}}, \quad g_{33,0} = -2\dot{\mathcal{R}} \mathcal{R} r^2 \sin^2 \theta = 2g_{33} \frac{\dot{\mathcal{R}}}{\mathcal{R}}$$

Therefore, for any i one can write $g_{ii,0} = 2g_{ii} \dot{\mathcal{R}}/\mathcal{R}$.

*Here and below there is no summation over repeated **Latin** indices*

Similarly, for $i \neq 1$, $g_{ii,1} = 2g_{ii}/r$. The last two nonzero derivatives are $g_{33,2} = 2g_{33} \cot \theta$, and $g_{11,1} = (2kr/\eta)g_{11}$. This form is convenient, as for any diagonal matrix $g^{ii}g_{ii} = 1$, so that we can anticipate some simplifications in writing all the $\Gamma^\alpha_{\mu\nu}$. Note that $\Gamma_{0ij} = 0$, $\Gamma_{\mu 00} = 0$, and in Γ_{ijk} at least two indices must be the same. The remaining:

$$\begin{aligned} \Gamma_{i0} = \Gamma_{0i} = -\Gamma_{0ii} = g_{ii} \frac{\dot{\mathcal{R}}}{\mathcal{R}} &\implies \Gamma^i_{i0} = \Gamma^i_{0i} = \frac{\dot{\mathcal{R}}}{\mathcal{R}} & \Gamma^0_{ii} = -g_{ii} \frac{\dot{\mathcal{R}}}{\mathcal{R}} \quad (\text{any } i) \\ \Gamma_{ii1} = \Gamma_{1ii} = -\Gamma_{1ii} = \frac{g_{ii}}{r} &\implies \Gamma^i_{i1} = \Gamma^i_{1i} = \frac{1}{r} & \begin{aligned} \Gamma^1_{22} &= -r\eta \\ \Gamma^1_{33} &= -r\eta \sin^2 \theta \end{aligned} \quad (i \neq 1) \\ \Gamma_{332} = \Gamma_{323} = -\Gamma_{233} = g_{33} \cot \theta &\implies \Gamma^3_{32} = \Gamma^3_{23} = \cot \theta & \Gamma^2_{33} = -\frac{1}{2} \sin 2\theta \\ \Gamma_{111} = g_{11} \frac{kr}{\eta} &\implies \Gamma^1_{11} = \frac{kr}{\eta} \end{aligned}$$

Explicitly, we have all the nonzero symbols (without repeating $\Gamma^1_{01} = \Gamma^1_{10}$, etc):

$$\begin{aligned} \Gamma^0_{11} &= \eta^{-1} \dot{\mathcal{R}} \mathcal{R}, & \Gamma^0_{22} &= r^2 \dot{\mathcal{R}} \mathcal{R}, & \Gamma^0_{33} &= r^2 \sin^2 \theta \dot{\mathcal{R}} \mathcal{R} \\ \Gamma^1_{01} &= \dot{\mathcal{R}} \mathcal{R}^{-1}, & \Gamma^1_{11} &= kr\eta^{-1}, & \Gamma^1_{22} &= -r\eta, & \Gamma^1_{33} &= -r\eta \sin^2 \theta \dot{\mathcal{R}} \mathcal{R} \\ \Gamma^2_{02} &= \Gamma^3_{03} = \dot{\mathcal{R}} \mathcal{R}^{-1}, & \Gamma^2_{12} &= \Gamma^3_{13} = r^{-1}, & \Gamma^2_{33} &= -\frac{1}{2} \sin 2\theta, & \Gamma^3_{23} &= \cot \theta. \end{aligned}$$

Then we calculate all the nonvanishing spatial-temporal components of $R^\mu_{\nu\alpha\beta}$ in one go:

$$R^i_{0i0} = \underbrace{\Gamma^i_{00,i}}_{=0} - \Gamma^i_{0i,0} + \underbrace{\Gamma^\mu_{00}\Gamma^i_{\mu i}}_{=0} - \underbrace{\Gamma^\mu_{0i}\Gamma^i_{\mu 0}}_{=0 \text{ if } \mu \neq i} = -\Gamma^i_{0i,0} - \Gamma^i_{0i}\Gamma^i_{i0} = -\frac{\ddot{\mathcal{R}}}{\mathcal{R}} + \frac{\dot{\mathcal{R}}^2}{\mathcal{R}^2} - \frac{\dot{\mathcal{R}}^2}{\mathcal{R}^2} = -\frac{\ddot{\mathcal{R}}}{\mathcal{R}}$$

Since $g^{00} = 1$, we have $R^{i0}_{i0} = R^i_{0i0}$. The two more nonzero components are

$$R^i_{1i1} = \Gamma^i_{11,i} - \Gamma^i_{1i,1} + \Gamma^\mu_{11}\Gamma^i_{\mu i} - \Gamma^\mu_{1i}\Gamma^i_{\mu 1} = -\Gamma^i_{1i,1} + \Gamma^0_{11}\Gamma^i_{0i} + \Gamma^1_{11}\Gamma^i_{1i} - \Gamma^i_{1i}\Gamma^i_{i1} = \frac{1}{r^2} - g_{11} \left(\frac{\dot{\mathcal{R}}}{\mathcal{R}} \right)^2 + \frac{k}{\eta} - \frac{1}{r^2} = -g_{11} \left(\frac{\dot{\mathcal{R}}}{\mathcal{R}} \right)^2 + \frac{k}{\eta}$$

Raising the second index, we obtain

$$R^{i1}_{i1} = g^{11}R^i_{1i1} = - \left(\frac{\dot{\mathcal{R}}}{\mathcal{R}} \right)^2 + g^{11} \frac{k}{\eta} = -\frac{\dot{\mathcal{R}}^2 + k}{\mathcal{R}^2}, \quad \text{obviously } i \neq 1.$$

The last possibly an independent component, R^{23}_{23} is calculated directly, as given in the handout, to coincide with the above. All components that contain different pairs of indices vanish, as for any spherically symmetric metric.

The Ricci and Einstein Tensors and the scalar curvature

Convoluting $R^\mu_{\nu\alpha\beta}$, one defines the Ricci and Einstein tensors. As $R^\alpha_\beta = R^{\alpha\nu}_{\beta\nu}$, it is clear that only diagonal components of the Ricci are nonzero. We have

$$R^0_0 = R^{10}_{10} + R^{20}_{20} + R^{30}_{30} = -\frac{3\ddot{\mathcal{R}}}{\mathcal{R}}$$

$$R^1_1 = R^{01}_{01} + R^{21}_{21} + R^{31}_{31} = -\frac{\ddot{\mathcal{R}}}{\mathcal{R}} - \frac{2(\dot{\mathcal{R}}^2 + k)}{\mathcal{R}^2} = R^2_2 = R^3_3$$

The scalar curvature of spacetime

$$R = R^0_0 + R^1_1 + R^2_2 + R^3_3 = -6 \left(\frac{\ddot{\mathcal{R}}}{\mathcal{R}} + \frac{\dot{\mathcal{R}}^2 + k}{\mathcal{R}^2} \right)$$

It must not be confused with the purely spatial curvature equal to $2k/\mathcal{R}^2$. Even for the flat expanding Universe (when $k = 0$), the scalar Riemann curvature does not vanish (except from a peculiar case when $\mathcal{R} \propto \sqrt{t}$ so that $\mathcal{R}\ddot{\mathcal{R}} + \dot{\mathcal{R}}^2 = 0$). The word “flat” refers in this case to a purely spatial structure.

The components of the Einstein tensor, $G^\mu_\nu = R^\mu_\nu - \frac{1}{2}\delta^\mu_\nu R$, are given by

$$G^0_0 = \frac{1}{2}(R^0_0 - R^1_1 - R^2_2 - R^3_3) = -\frac{1}{2}(R^0_0 - 3R^1_1) = 3\frac{\dot{\mathcal{R}}^2 + k}{\mathcal{R}^2},$$

$$G^1_1 = \frac{1}{2}(R^1_1 - R^0_0 - R^2_2 - R^3_3) = -\frac{1}{2}(R^0_0 + R^1_1) = \frac{2\ddot{\mathcal{R}}}{\mathcal{R}} + \frac{\dot{\mathcal{R}}^2 + k}{\mathcal{R}^2}, \text{ etc,}$$

so that finally (restoring c , i.e. considering the *dot* as ∂_t rather than ∂_{ct}) we have

$$G^0_0 = \frac{3(\dot{\mathcal{R}}^2 + kc^2)}{\mathcal{R}^2 c^2}, \quad G^1_1 = G^2_2 = G^3_3 = \frac{2\ddot{\mathcal{R}}}{\mathcal{R} c^2} + \frac{\dot{\mathcal{R}}^2 + kc^2}{\mathcal{R}^2 c^2} \quad (20.5)$$

Lecture 21. Friedman models of the Universe

We have have constructed the FRW metric that obeys the cosmological principle in terms of the unknown parameters, the curvature, $k = 1, 0$ or -1 , and the history function, $\mathcal{R}(t)$. We find them by solving Einstein's equations for different models of the matter tensor, T^μ_ν , and express in terms of observable parameters - the Hubble constant H and the deceleration parameter q .

21.1 Perfect gas of galaxies and radiation: solving Einstein's equations

In the present epoch, the simplest model of cosmic **dust** is quite appropriate. Galaxies could definitely be considered as non-interacting (but for the gravity) dust particles with zero pressure and temperature, and so is a very sparse intergalactic gas. The dust model does not cover all the matter. Background radiation at $T = 2.7$ K which fill the whole Universe is also a source of gravity. The same is true about the neutrinos radiation (especially if they have non-zero rest mass, as assumed in the "standard model").

More generally, one can consider the perfect-gas model of T^μ_ν given by Eq. (19.4):

$$T^\mu_\nu = \left(\rho + \frac{P}{c^2} \right) u^\mu u_\nu - P \delta^\mu_\nu$$

Here the pressure P is due to radiation, and the density ρ is due to matter. In the comoving coordinates (in which we have written the FRW metric), one has $T^\mu_\nu = \text{diag}(\rho c^2, -P, -P, -P)$. Hence, substituting the components of the Einstein tensor, $G^\mu_\nu = R^\mu_\nu - \frac{1}{2} \delta^\mu_\nu R$, Eq. (20.5), we obtain the Einstein equations (without the cosmological term!), $G^\mu_\nu = \frac{8\pi G}{c^4} T^\mu_\nu$, in the Friedman form:¹

$$G^0_0 = \frac{8\pi G \rho}{c^2} \Rightarrow \frac{3(\dot{\mathcal{R}}^2 + kc^2)}{\mathcal{R}^2} = 8\pi G \rho(t) \Rightarrow \frac{\dot{\mathcal{R}}^2}{\mathcal{R}^2} \equiv H^2 = \frac{8\pi G \rho}{3} - \frac{kc^2}{\mathcal{R}^2} \quad (21.1a)$$

$$G^1_1 = -\frac{8\pi G P(t)}{c^4} \Rightarrow \frac{2\ddot{\mathcal{R}}}{\mathcal{R}} + \frac{\dot{\mathcal{R}}^2 + kc^2}{\mathcal{R}^2} = -\frac{8\pi G P(t)}{c^2} \Rightarrow \frac{\ddot{\mathcal{R}}}{\mathcal{R}} = -\frac{4\pi G}{c^2} \left(P + \frac{\rho c^2}{3} \right), \quad (21.1b)$$

where $H \equiv \dot{\mathcal{R}}/\mathcal{R}$ is the Hubble constant. As usual, they should be complemented with the equation of state, $P = P(\rho)$. Then we will have a set of three equations for three unknown time-dependent functions, $\mathcal{R}$, ρ , and P . As H^2 is positive, so must be the r.h.s. of Eq. (21.1a). Dividing both sides by H^2 , we rewrite it as

$$\frac{kc^2}{H^2 \mathcal{R}^2} = \frac{8\pi G \rho}{3H^2} - 1 \equiv \Omega - 1 \quad \text{where } \Omega \equiv \frac{\rho}{\rho_{\text{crit}}} \quad \text{and } \rho_{\text{crit}} \equiv \frac{3H^2}{8\pi G}. \quad (21.2)$$

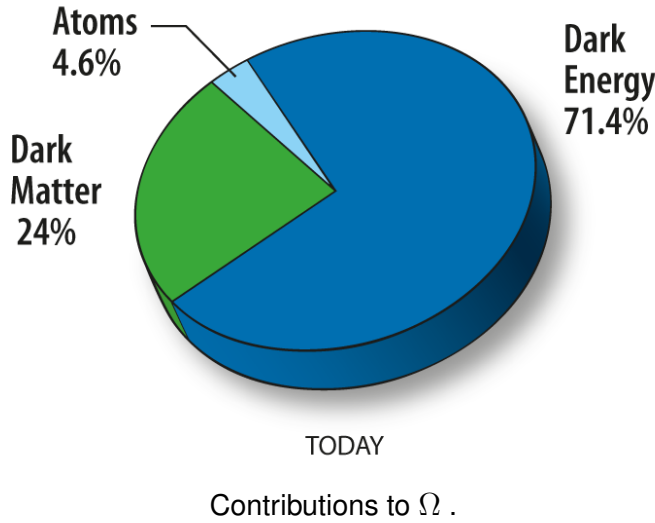
The importance of the **density parameter** Ω is hard to overestimate. Even before solving the Friedman equations, we see that a choice between the three types of the FRW metric is governed by it:

$$\begin{aligned} \Omega > 1 &\Leftrightarrow \rho > \rho_{\text{crit}} \Rightarrow k > 0 : \text{positively curved Universe,} \\ \Omega = 1 &\Leftrightarrow \rho = \rho_{\text{crit}} \Rightarrow k = 0 : \text{flat Universe,} \\ \Omega < 1 &\Leftrightarrow \rho < \rho_{\text{crit}} \Rightarrow k < 0 : \text{negatively curved Universe.} \end{aligned} \quad (21.3)$$

¹Friedman was the first to derive the "FRW" metric, and he suggested (in 1922) these equations for $P = 0$. They have been generalised to the above form by Le Maître (1927). These equations without the cosmological Λ -term do not have a static solution so that P and ρ are necessarily changing with time. Note that even in this simplest form the equations can account for the hidden (dark) mass but not for *dark energy*. According to the current understanding of observational data, the latter makes the dominant contribution to Ω – see the figure on the next page – so that the Friedman models should be amended to be applicable for our Universe. We add the Λ contribution in the last section of this lecture.

Moreover, both parameters defining Ω , i.e. the Hubble “constant” H and the known matter density ρ , as well as Ω itself, can be independently measured.

Remarkably, Ω is very close to 1 so that we live in an almost flat Universe.



The matter density in the present epoch is estimated as $\rho \sim 4 \times 10^{-31} \text{ g/cm}^3$ (and is dominated by the inter-galaxy gas), which is about one hydrogen atom per four cubic metre of space. The Hubble constant is about $70(\text{km/s})/\text{Mpc} \approx 2 \times 10^{-18} \text{ s}^{-1}$. This would give Ω the value of 0.04. The radiation adds a totally negligible contribution. A hidden, dark (*i.e.* not luminous) mass contributes considerably more, giving Ω the value about 0.25. The dominant contribution comes from **dark energy** that can be accounted for by the cosmological term.

However, before briefly discussing the dominant contribution of Λ , let us consider in more detail the standard Friedman equations with $\Lambda = 0$.

By itself, $\mathcal{R}$ is unmeasurable. However, we can express all the equations in terms of H and the **deceleration parameter** q defined by $q \equiv -\ddot{\mathcal{R}}/\dot{\mathcal{R}}^2$ so that $\dot{H} = -H^2(1 + q)$. Then, we combine Eqs. (21.1) to express them in terms of quantities, q , H , P , and ρ , which are at least in principle) observable:

$$\frac{4\pi G}{3c^2} (3P + \rho c^2) = -\frac{\ddot{\mathcal{R}}}{\mathcal{R}} = -\frac{\ddot{\mathcal{R}}}{\dot{\mathcal{R}}^2} \frac{\dot{\mathcal{R}}^2}{\mathcal{R}} \equiv qH^2 \quad \Rightarrow \quad q = \frac{\rho_M + \rho_R}{2\rho_{\text{crit}}} \equiv \frac{\Omega_M}{2} + \frac{\Omega_R}{2}, \quad (21.4)$$

where we rename the density parameter of Eq. (21.2) into Ω_M , and introduce a similar contribution from the radiation density, $\Omega_R \equiv P/(3c^2\rho_{\text{crit}}) = \rho_R/\rho_{\text{crit}}$. We will see that while ρ_R is negligible in the current epoch, it was inevitably dominating in the early Universe. Since P and ρ are non-negative, $q > 0$, *i.e.* the Universe expansion (or contraction) is **decelerates** in the Friedman models without the cosmological term, so that the Hubble constant could only be bigger in the past.

In the evolving Universe the equation of state does not stay the same. Thus, to write down the Friedman equations is not sufficient. One needs to model the dynamics of the Universe by trying different **scenarios**. That is why we are talking about the Friedman models in plural.

21.2 Matter-dominated and radiation dominated Universe

In the present epoch, the density of visible (luminous) matter, $\rho \sim 10^{-31} \text{ g/cm}^3$, should be compared to the radiation contribution $3P/c^2$. You might remember (from *Radiation and Relativity* and/or *Statistical Physics*) the equation of state for ultra-relativistic matter (or light): $P = \rho_R c^2/3$, where $\rho_R c^2$ is the radiation energy density. Using also the Stefan–Boltzmann law, we have

$$\rho_R = \frac{3P}{c^2} = \frac{4\sigma T^4}{c^3} = \frac{\pi^2}{15} \frac{(k_B T)^4}{\hbar^3 c^5} \approx 5 \times 10^{-34} \frac{\text{g}}{\text{cm}^3} \quad \text{for } T \approx 2.7 \text{ K}$$

So **in the current epoch** radiation energy is negligible and the simplest dust model works.

Such a model of the Universe is called **matter-dominated**. In the comoving coordinates one has $T^\mu_\nu = \text{diag}(\rho c^2, 0, 0, 0)$. The equation of state is trivial, $P = 0$, and only ρ evolves with time. To relate ρ and $\mathcal{R}$, we multiply Eq. (21.1a) by $\mathcal{R}^3$ which gives

$$(\dot{\mathcal{R}}^2 + kc^2)\mathcal{R} = \frac{8\pi G}{3}\rho\mathcal{R}^3.$$

Differentiating with respect to t gives²

$$\frac{d}{dt}\left(\frac{8\pi G}{3}\rho\mathcal{R}^3\right) = \frac{d}{dt}\left(\dot{\mathcal{R}}^2\mathcal{R} + kc^2\mathcal{R}\right) = \dot{\mathcal{R}}\left(\underbrace{2\ddot{\mathcal{R}}\mathcal{R} + \dot{\mathcal{R}}^2 + kc^2}_{=0 \text{ from Eq. (21.1b)}}\right) = 0 \Rightarrow \boxed{\frac{8\pi G\rho(t)\mathcal{R}^3}{3c^2} = R_0} \quad (21.5)$$

Important here is a sort of the matter conservation law in a comoving sphere:

$$\rho(t) = \frac{\text{const}}{\mathcal{R}^3}, \quad (21.6)$$

so that the density uniformly decreasing for the expanding Universe. This means that at earlier stages it was much bigger, with inevitable divergence at the beginning - *the Big Bang*. However, the equations contained only H^2 so that its sign remains undetermined. Hence, these equations would equally well describe a contracting Universe going to the Big Crunch. I show in the next section that this would inevitably happen for a universe (not ours!) which is currently matter-dominated with $\Omega_M > 1$ (unlike ours!) and is described by the Friedman equations without the cosmological term.

The constant³ R_0 introduced in Eq. (21.5) is convenient for writing expressions for Ω_M and q . Comparing the definitions of R_0 , Eq. (21.5), and Ω_M , Eq. (21.2), we see that $\Omega_M/R_0 = c^2/H^2\mathcal{R}^3$. Then we rewrite Eq. (21.2) as a relation between the mass density parameter Ω_M and $\mathcal{R}/R_0$:

$$\Omega_M - 1 = \frac{kc^2}{H^2\mathcal{R}^2} = \frac{k\Omega_M\mathcal{R}}{R_0} \Rightarrow \frac{R_0}{\mathcal{R}} = \frac{k\Omega_M}{\Omega_M - 1} \Leftrightarrow \Omega_M = 2q = \frac{1}{1 - k\mathcal{R}/R_0}. \quad (21.7)$$

In this scenario, if we assume $k = -1$ for a matter-dominated Universe (like ours where $\Omega_M \ll 1$), $q = \Omega_M/2$ becomes smaller and smaller with cosmic time, so such a universe will be forever expanding with the rate of expansion asymptotically decreasing as $1/t$ (as follows from $\dot{H} = -(1+q)H^2$).

However, such a universe (as well as the one with $k = 1$ that we consider in the next section) was not always matter-dominated: for earlier epoch the radiation was more important. To see this, we consider the radiation-only case when $P \neq 0$ but $\rho_M = 0$. The under-brace in Eq. (21.5) would be equal in this case to $-8\pi G\mathcal{R}^2 P/c^2$, which leads (using the ultrarelativistic equation of state $P = \rho_R c^2/3$) to

$$\frac{d}{dt}(\rho_R\mathcal{R}^3) = -\underbrace{\frac{3P}{c^2}\dot{\mathcal{R}}\mathcal{R}^2}_{=\rho_R} \Leftrightarrow \dot{\rho}_R\mathcal{R}^3 + 3\mathcal{R}^2\dot{\rho}_R + \mathcal{R}^2\dot{\rho}_R = 0 \Leftrightarrow \frac{d}{dt}(\rho_R\mathcal{R}^4) = 0 \quad (21.8)$$

This is the equation for the **radiation-dominated** Universe where the density of the expanding Universe decreases faster than when the matter is dominated.⁴ Hence in a universe containing both

²I have changed the font for the constant below (compared to the previous version of these notes): $\mathcal{R}_0 \rightarrow R_0$, so that this const is not confused with the history function in the current epoch, $\mathcal{R}(t_0)$.

³This is dimensionally correct since $GM/c^2 \sim G\rho\mathcal{R}^3/c^2$ has the dimensionality of length (the Schwarzschild radius).

⁴Note that $\mathcal{R}^4$ means that the temperature of radiation decreases linearly with $1/\mathcal{R}$ (in agreement with the Stefan-Boltzmann law).

matter and radiation (when we should substitute $\rho = \rho_M + \rho_R$ for ρ_R in the l.h.s. of Eq. (21.8)), the density of the latter decreases faster. Then – inevitable – sometime in the past radiation was dominating.

In this scenario (and in any other reasonable one), looking back in time we inevitably see that the Universe began with a Big Bang, from $\rho = \infty$ and $\mathcal{R} = 0$, and then expanded firstly dominated by radiation, and then eventually by matter.

21.3 The history function of a closed universe

The full analytic solution for $\mathcal{R}(t)$ is straightforward. I show here how it works for simple scenarios where the curvature is positive and the cosmological term is absent (unlike in our Universe).

• **matter-dominated case** Substituting constant (21.5) into Eq. (21.1a) (multiplied by $\mathcal{R}^2$), we obtain

$$\frac{\dot{\mathcal{R}}^2}{c^2} = \frac{R_0}{\mathcal{R}} - k \Rightarrow k = \frac{R_0}{\mathcal{R}} - \frac{\dot{\mathcal{R}}^2}{c^2} = \frac{\mathcal{R}^2}{c^2} \left(\frac{c^2 R_0}{\mathcal{R}^3} - H^2 \right) = \frac{\mathcal{R}^2}{c^2} \left(\frac{8\pi G \rho}{3} - H^2 \right). \quad (21.9)$$

The formal solution of this equation is immediate:

$$ct = \int \frac{d\mathcal{R}}{\sqrt{R_0/\mathcal{R} - k}} \quad (21.10)$$

The integration is straightforward for any k , although rather involved. In particular, for a flat ($k = 0$) matter-dominated universe one easily finds the history function as

$$\mathcal{R}(t) = \left(\frac{9}{4} R_0 \right)^{1/3} (ct)^{2/3}.$$

Let us also consider the case of the closed Universe, $k = 1$ (referring the Universe of negative curvature to Problem Sheet 4). The integration in Eq. (21.10) is convenient to perform in the following parametric representation:

$$\frac{\mathcal{R}}{R_0} = \sin^2 \eta \Rightarrow \begin{cases} \left(\frac{R_0}{\mathcal{R}} - 1 \right)^{-1/2} = \tan \eta \\ d\mathcal{R} = R_0 \sin 2\eta d\eta \end{cases} \Rightarrow ct = 2R_0 \int_0^{\eta_0} \sin^2 \eta d\eta = R_0 \left(\eta_0 - \frac{1}{2} \sin 2\eta_0 \right)$$

Thus one arrives at the following parametric representation of the history function:

$$\begin{aligned} ct_0 &= R_0 \left(\eta_0 - \frac{1}{2} \sin 2\eta_0 \right) \\ \mathcal{R} &= R_0 \sin^2 \eta_0 \end{aligned} \quad (21.11)$$

One can omit the index indicating the present epoch in this final result. If one so desires, η can be expressed via $\mathcal{R}$ from the second equation and substituted then into the first one to give t as an explicit function of $\mathcal{R}$. However, $\mathcal{R}$ cannot be explicitly written as an elementary function of t . This solution confirms the qualitative picture described above: R_0 is the maximum radius of the Universe reached at $\eta = \pi/2$, i.e. at the epoch $t = \pi R_0 / 2c$.

Both R_0 and η are unmeasurable parameters. One may express them in terms of the Hubble constant, H , and the deceleration parameter, q . Denoting $\mathcal{R}' \equiv d\mathcal{R}/d\eta$, etc, one finds

$$H = \frac{\dot{\mathcal{R}}}{\mathcal{R}} = \frac{\mathcal{R}'}{t'} \frac{1}{\mathcal{R}} = \frac{c}{R_0} \frac{\cos \eta}{\sin^3 \eta}.$$

As expected, H changes sign at $\eta = \pi/2$ when the contraction starts and diverges both at the Big Bang ($\eta = 0$) and Big Crunch $\eta = \pi$. The density parameter Ω_M , and hence deceleration parameter $q = \Omega_M/2$ is found from Eq. (21.7): $\Omega_M = (1 - \mathcal{R}/R_0)^{-1} = 1/\cos^2 \eta$. In the absence of Λ , it lives up to its name and remains always positive. Hence

$$\cos \eta = \frac{1}{\sqrt{2q}} \quad \text{and} \quad R_0 = \frac{c}{H} \frac{2q}{(2q-1)^{3/2}},$$

so that $\mathcal{R}$ and t may be expressed as functions of observable H and q only.

• **radiation-dominated case.** As the $\mathcal{R}$ was smaller in the past, it is inevitable that at some earlier epoch the radiation $\propto 1/\mathcal{R}^4$ was dominating over the matter $\propto 1/\mathcal{R}^3$. In this case $\rho = B/\mathcal{R}^4$ where B is a certain constant, and the Einstein-Friedman equations reduce to

$$\dot{\mathcal{R}}^2 = \frac{B}{\mathcal{R}^2} - kc^2 \quad (21.12)$$

It's qualitative behaviour is the same as for the matter-dominated scenario, so that for the case considered here ($k = 1$) the singularity at $\mathcal{R} = 0$ – Big Bang followed at the end by Big Crunch – is unavoidable.

For the negatively curved open Universe ($k = -1$), the solution is technically similar. The result for the history function in the matter-dominated case differs from that in Eq. (21.11) by substituting $\sinh \eta$ for $\sin \eta$ there. Physically, this is totally different as the expansion goes on forever with the Universe becoming less and less dense.

However, we already know that the Universe is almost flat, and its expansion is accelerating. Let us finally consider this, more relevant case.

21.4 Accelerated expansion

As we know, in Eq. (21.2) Ω_M is no bigger than 0.3 and Ω_R is utterly negligible in the present epoch. Hence this equation is incomplete and some other contribution is required. This missing contribution is coming from “dark energy” whose physical origin is very much debated. But it can be phenomenologically allowed for by keeping the cosmological term $\Lambda g_{\mu\nu}$ in Einstein's equations (19.2). In mixed tensor components they become⁵

$$\underbrace{R^\mu_\nu - \delta^\mu_\nu R}_{\equiv G^\mu_\nu} - \delta^\mu_\nu \Lambda = \frac{8\pi G}{c^4} T^\mu_\nu. \quad (21.13)$$

This leads to adding the Λ -terms into the Einstein–Friedman equations (21.1) as follows:

$$\frac{\dot{\mathcal{R}}^2}{\mathcal{R}^2} \equiv H^2 = \frac{8\pi G \rho_M}{3} - \frac{kc^2}{\mathcal{R}^2} + \frac{\Lambda c^2}{3} \quad (21.14)$$

$$\frac{2\ddot{\mathcal{R}}}{\mathcal{R}} + \frac{\dot{\mathcal{R}}^2 + kc^2}{\mathcal{R}^2} - \Lambda c^2 = -\frac{8\pi G P(t)}{c^2} \Rightarrow \frac{\ddot{\mathcal{R}}}{\mathcal{R}} = -\frac{4\pi G}{3}(\rho_M + \rho_R) + \frac{\Lambda c^2}{3}, \quad (21.15)$$

⁵Alternatively, it can be stated that the tensor $T_{\mu\nu}$ acquires the vacuum contribution equal to $\lambda g_{\mu\nu}$. This is equivalent to adding the cosmological constant provided that $\Lambda = \kappa \lambda \equiv \frac{8\pi G}{c^4} \lambda$.

Dividing Eq. (21.14) by H^2 , we obtain the equation similar to Eq. (21.2) as before, but with the Λ -contribution and – for generality – with $\rho = \rho_M + \rho_R$ (reminder: $\rho_R c^2/3 = P$):

$$\frac{kc^2}{H^2 \mathcal{R}^2} = \frac{(\rho_M + \rho_R)}{\rho_{\text{crit}}} + \frac{\Lambda c^2}{3H^2} - 1 \equiv \Omega_M + \Omega_R + \Omega_V - 1. \quad (21.16)$$

To make notations more uniform, sometimes the energy density of vacuum, ρ_V , is introduced as $\rho_V \equiv \Lambda c^2/8\pi G$, so that the total density parameter

$$\Omega = \frac{\rho_M + \rho_R + \rho_V}{\rho_{\text{crit}}} \equiv \Omega_M + \Omega_R + \Omega_V.$$

In this scenario, instead of Eq. (21.4) we obtain for the deceleration parameter

$$q = \frac{\Omega_M + \Omega_R}{2} - \Omega_V. \quad (21.17)$$

The cosmological constant Λ is not directly measurable: the value of Ω_V comes from the fact that currently the Universe is flat, i.e.

$$\Omega = \Omega_M + \Omega_R + \Omega_V \approx 1.$$

As $\Omega_R \approx 0$ and $\Omega_M \approx 1/4$, we have $\Omega_V \approx 3/4$ so that the deceleration parameter is negative, $q \approx -0.6$. Hence the Universe expansion is currently accelerating! The nature of the cosmological constant (dark energy) is still unclear and a subject of intensive current investigations.

Solving (21.14) and (21.15) we can in principle determine the history function similar to the case of the matter-dominated closed Universe demonstrated in the previous section. Note that the first of Eqs. (21.8), where one substitute the full matter density $\rho = \rho_M + \rho_R$ for ρ_R , is not affected by Λ . Hence, qualitatively expansion is similar to that in the Friedman models without the cosmological term: in early epoch after the Big Bang it was dominated by radiation so that the full matter density was decreasing $\propto \mathcal{R}^{-4}$; in later epochs (including the current one) when density became sufficiently low the (non-relativistic) matter contribution becomes dominant, and $\rho \approx \rho_M \propto \mathcal{R}^{-3}$. The time-dependence of $\mathcal{R}$ is mostly defined by radiation in early epoch, then by matter as in Eq. (21.11) for $k = 1$ (and similarly for $k = -1$ with $\sin \eta \rightarrow \sinh \eta$). Eventually, as both $\mathcal{R}$ and ρ decrease, the history function becomes dominated by the cosmological term, which already gives the biggest contribution into Ω .

Let us briefly outline $\mathcal{R}(t)$ in the most interesting and most simple case is that of the flat Universe, $k = 0$. In the early radiation-dominated epoch, we keep only Ω_R , and in later, matter-dominated epoch, only Ω_M . In these cases Eq. (21.16) (equivalent to Eq. (21.14)) becomes, with substituting $\rho \mathcal{R}^4 = \text{const}$ for the earlier and $\rho \mathcal{R}^3 = \text{const}$ for the later epochs:

$$H^2 \equiv \frac{\dot{\mathcal{R}}^2}{\mathcal{R}^2} = \frac{8\pi G \rho}{3} \rightarrow \begin{cases} \frac{\text{const}}{\mathcal{R}^4} \Rightarrow \mathcal{R} \dot{\mathcal{R}} = \text{const} \Rightarrow \mathcal{R} \propto t^{1/2} & \text{radiation-dominated} \\ \frac{\text{const}}{\mathcal{R}^3} \Rightarrow \mathcal{R}^{1/2} \dot{\mathcal{R}} = \text{const} \Rightarrow \mathcal{R} \propto t^{2/3} & \text{matter-dominated} \end{cases} \quad (21.18)$$

It is worth noticing that only H^2 enters the Friedman equation so that we do not know the sign of the last constants from these equations alone: $\dot{\mathcal{R}}$ could be either positive or negative. But we know that the Universe is expanding - so we choose the positive sign. Then in the flat Universe the sign of $\dot{\mathcal{R}}$ could never change (as well as even in the closed but accelerated Universe, but we will not go into

any further detail). With $\mathcal{R}$ increasing, density ρ decreases (due to acceleration of the expansion) and eventually we come to the vacuum-dominated case when only Ω_V should be kept, and positive $\dot{\mathcal{R}}$ should be chosen:

$$H^2 \equiv \frac{\dot{\mathcal{R}}^2}{\mathcal{R}^2} = \frac{\Lambda c^2}{3} \quad \Rightarrow \quad \mathcal{R} \propto e^{Ht}, \quad H = c\sqrt{\frac{\Lambda}{3}}. \quad (21.19)$$

So the death of the Universe is greatly accelerated at the end when it is expanding exponentially fast.

In the last lecture, amongst other things I will briefly list the arguments that such an exponential expansion took place also at the very beginning after the Big Bang – the inflation scenario.

Lecture 22. Measuring the Universe

We have found curvature coefficient k and the history function $\mathcal{R}(t)$ in terms of cosmological observables, the Hubble constant H and the deceleration parameter q expressible in terms of the matter density Ω_M and the dark energy density Ω_Λ . We briefly discuss how distances to and receding velocities of remote galaxies are extracted by analysing astronomic observations, and what is meant by the statement that we live in an almost flat Universe.

22.1 The Hubble ‘constant’ and ‘deceleration’ parameter

The history function $\mathcal{R}(t)$ is necessarily time-dependent. Arguments against the stationary Universe: uniformly lit sky (if there is no gravitational red shift); problems with gravitation (what is the field inside a cavity?), *etc.* Anyway, although there are other ways to dealing with the paradoxes of stationary solutions, we know due to Hubble that the Universe is expanding. Also, there is no steady-state solution to Einstein’s equations (or the Poisson equation) without the cosmological term.

Let us recall the definition of the Hubble constant in terms of $\mathcal{R}$. The proper distance **at the same epoch** (a usual name for the cosmic time) is defined by measuring the interval between two spatially remote events at the same t_0 :

$$S = \int_0^{r_0} \sqrt{-ds^2} = \mathcal{R}(t_0) \int_0^{r_0} \frac{dr}{\sqrt{1 - kr^2}} = \mathcal{R}(t_0) \chi_0, \quad \chi \equiv \begin{cases} \sin^{-1} r & k = 1 \\ r & k = 0 \\ \sinh^{-1} r & k = -1 \end{cases} \quad (22.1)$$

Suppose $r = 0$ is the coordinate of our Galaxy. Thus, the recession velocity and acceleration of the galaxy at point $\chi_0 \equiv \chi(r_0)$ from the origin are

$$v = \dot{S} = \dot{\mathcal{R}}\chi_0 = \frac{\dot{\mathcal{R}}}{\mathcal{R}}S \equiv HS, \quad a = \ddot{S} = \ddot{\mathcal{R}}\chi_0 = \frac{\ddot{\mathcal{R}}\mathcal{R}}{\dot{\mathcal{R}}^2} \frac{\dot{\mathcal{R}}^2}{\mathcal{R}^2}S \equiv -qH^2S \quad (22.2)$$

where, by definition, H is the Hubble ‘constant’ and q is the deceleration parameter

$$H \equiv \frac{\dot{\mathcal{R}}}{\mathcal{R}} = \frac{\dot{S}}{S}, \quad q \equiv -\frac{\mathcal{R}\ddot{\mathcal{R}}}{\dot{\mathcal{R}}^2},$$

introduced in the previous lecture. The CP requires that in any given epoch, H is the same everywhere. Then, $\dot{S} = HS$ means that the further is a galaxy from us, the faster it recedes, and with the bigger acceleration, as we have already learned that q , Eq. (21.4), is negative.

A question: is this recession velocity limited by c ?

It was quite difficult to measure q or even its sign, although the belief in the fact that $\ddot{\mathcal{R}}$ is negative and q is positive is reflected in the name. For a while, it seemed obvious that the expansion that has started with the Big Bang could only slow down with time due to gravitation attraction between receding galaxies. However, the data accumulated during the last two decades (starting from 1998) made this name obsolete: most probably, $q \approx -0.6$ so that the Universe expansion is accelerated (see the next section for detail). Since in the absence of the **cosmological constant** in Einstein’s equations q must be positive, this became a practically undisputed argument in favour of adding Λ to Einstein’s equations. Its physical origin (hinted in the term “dark energy”) is a subject of intensive current research.

22.2 An alternative form of the FRW metric

The proper distance, χ , defined above, is convenient to rewrite the FRW metric. The new form of the metric follows immediately from $\mathcal{R}^2 d\chi^2 = \mathcal{R}^2(1 - kr^2)^{-1} dr^2$, and, inverting $\chi(r)$ in Eq. (22.1):

$$r = r(\chi) \equiv \begin{cases} \sin \chi & k = 1 \\ \chi & k = 0 \\ \sinh \chi & k = -1 \end{cases}$$

Therefore, the FRW metric takes the form

$$\boxed{ds^2 = c^2 dt^2 - \mathcal{R}^2(t) [d\chi^2 + r^2(\chi) (d\theta^2 + \sin^2 \theta d\phi^2)]}, \quad (22.3)$$

The proper distance $\chi\mathcal{R}$ would be “measured” by a chain of fundamental observers laying rulers end to end at the same cosmic instance on the spacelike geodesic from 0 to r_0 . This does not sound as a good practical recipe, although this distance can be thought of as the length of the geodesic on the rubber model. Before discussing how the distances are actually measured, let us look at this rubber model in trying to understand the nature of the expansion.

22.3 The size of the observable Universe: particle and event horizons

The expansion of the Universe could be compared to that of a rubber balloon (of spherical shape for $k = 1$, or of saddle-like shape for $k = -1$, or simply a rubber sheet for $k = 0$). Galaxies would be represented as little flakes glued to the balloon. The inflation of the balloon increases the distances between the flakes, but not their own sizes. The light propagates along null geodesics which are just great circles on the 3D sphere (for $k > 0$) embedded into a fictional 4D space (not spacetime!). A simple question is whether all the points on our rubber balloon can be causally connected. As there is no centre there, the question can be posted as whether our present position (the origin) can be reachable by light sent from any other position? In general, this is not the case: the balloon is inflated, and this inflation affects remote points stronger: we have already seen that the further a galaxy from us, the faster it recedes (this is the meaning of the Hubble ‘constant’). There might exist remote galaxies that light sent from us will never reach them. By reciprocity, it means that we will **never** see light sent from them, and thus cannot have any observational evidence whether such galaxies exist.

Hence we define the **event horizon** S_e as the most distant present event from which a signal can ever reach us (‘ever’ could mean a finite future time, $t_{\max}$, if the Universe is closed). On the other hand, there were some events which have been unobservable but will become in principle observable in the future. Thus, we define the **particle horizon** S_p as the largest comoving distance from which a light signal could have reached us if it was emitted at time $t = 0$ (at the Big Bang). Again, by reciprocity, it is the most remote point reached by a signal sent from the origin (us!) at the moment of Big Bang. Both S_p and S_e can be easily defined via the parameters of the FRW metric using the null worldlines $c dt = \mathcal{R}(t) d\chi$:

$$\begin{aligned} S_e &= c\mathcal{R}(t_0) \int_{t_0}^{\infty} \frac{dt}{\mathcal{R}(t)}, \\ S_p &= c\mathcal{R}(t_0) \int_0^{t_0} \frac{dt}{\mathcal{R}(t)} \end{aligned} \quad t_0 \text{ is the present epoch.} \quad (22.4)$$

22.4 How to measure distances

The metric information about lengths, *etc*, is naturally contained in the metric itself. In order to see it more explicitly, we shall cast the FRW metric into more convenient (for this purpose) form. However, distances between fundamental observers given by the metric could not be measured directly. We then discuss three possible ways of observational measurements of distances: by the apparent size, by the apparent luminosity, and using the cosmological red shift.

22.4.1 Cosmological red shift

Let a fundamental observer E (emitter) at the coordinate centre transmit, at epoch t_e , a pulse of light which is received by a fundamental observer R (receiver) at position χ_0 at the present epoch t_0 . We can model the light signal as two close successive bugs (representing two nodes of the light pulse) crawling with the same (proper) velocity on the rubber surface. If the surface was rigid, a distance between the bugs would remain the same. For an inflating rubber balloon, the distance between the bugs is increasing in time due to the expansion (and it would be decreasing in time for a deflating balloon). As this distance represents the wavelength, it is clear that in the expanding Universe signals experience **the cosmological red shift**. Let's stress that this is an **expansion** effect rather than a dynamical **velocity** effect. Defining, as usual, $Z = \Delta\lambda/\lambda$, one can see that

$$Z = \frac{\Delta\lambda}{\lambda_e} = \frac{\lambda_0}{\lambda_e} - 1 = \frac{\mathcal{R}(t_0) - \mathcal{R}(t_e)}{\mathcal{R}(t_e)}. \quad (22.5)$$

This can be easily obtained more formally. Due to the spherical symmetry (CP!) one can limit considerations to $d\phi = d\theta = 0$. Then – for any k – one finds from Eq. (22.3) the null worldline $c^2 dt^2 = \mathcal{R}^2 d\chi^2$, so that, on integrating we have

$$S = \mathcal{R}(t_0)\chi_{re} = c\mathcal{R}(t_0) \int_{t_e}^{t_0} \frac{dt}{\mathcal{R}(t)}. \quad (22.6)$$

Here we have also used Eq. (22.1) with χ_{re} defining the proper distance S from the receding galaxy that emitted signal to our (receiving) galaxy in the present epoch. If the duration of the pulse is Δt_e at the point of emission and Δt_0 at the point of reception, then one takes Δ of the integral above, and uses that $\Delta\chi_0 = 0$ (remember that all fundamental observers are comoving):

$$\Delta \int_{t_e}^{t_0} \frac{dt}{\mathcal{R}(t)} = \frac{\Delta t_0}{\mathcal{R}(t_0)} - \frac{\Delta t_e}{\mathcal{R}(t_e)} = 0 \implies \frac{\Delta t_0}{\Delta t_e} = \frac{\mathcal{R}(t_0)}{\mathcal{R}(t_e)}$$

Suppose that Δt_e is the time between two consecutive nodes of the light signal. Then $\lambda_e = c\Delta t_e$. Defining the cosmological red shift as the change (increase) in the wavelength at the point of reception, one arrives again at Eq. (22.5).

The redshift (22.5) defines the distance between the emitter and receiver when the history function is known. We need to invert this by defining the distance via Z . When the redshift is small, we can use Taylor's expansion of $\mathcal{R}(t_e)$ in the vicinity of t_0 – the present epoch for the receiver, *i.e.* we assume that $\delta t \equiv t_0 - t_e$ is small.¹ Note that $t_0 > t_e$: the receiver always sees the signal emitted in an earlier epoch. Hence we find

$$\mathcal{R}(t_e) = \mathcal{R}(t_0) - \dot{\mathcal{R}}(t_0)\delta t + \frac{1}{2}\ddot{\mathcal{R}}(t_0)\delta t^2 \dots = \mathcal{R}(t_0) \left[1 - H_0\delta t - \frac{1}{2}q_0 H_0^2 \delta t^2 \dots \right], \quad (22.7)$$

¹We will see that this means that the receding velocity of the observer E is nonrelativistic: $v = H_0 S \ll c$.

where H_0 and q_0 are the Hubble constant and deceleration in the present epoch. Substituting this into Eq. (22.5), we find

$$Z = H_0 \delta t + H_0^2 \left[1 + \frac{1}{2} q_0\right] \delta t^2 + \dots \quad (22.8)$$

Now we express δt via the distance S . To this end, we substitute into the integral in Eq. (22.6) the expansion (22.7) of $\mathcal{R}(t)$ with $t_e \rightarrow t$ and $\delta t = t_0 - t$:

$$\frac{1}{\mathcal{R}(t)} = \frac{1}{\mathcal{R}(t_0)} \left[1 - H_0 (t - t_0) + \left(H_0^2 - \frac{1}{2}\right) (t - t_0)^2 + \dots\right].$$

With this substitution, calculating the straightforward integral in Eq. (22.6) results in

$$S = c \delta t + \frac{1}{2} c H_0 \delta t^2 \Rightarrow c \delta t = S - \frac{H_0}{2c} S^2 + \dots \quad (22.9)$$

Next, we substitute δt into Eq. (22.8) to get the relation between the redshift and distance:

$$Z = \frac{H_0 S}{c} + \frac{1 + q_0}{2} \left(\frac{H_0 S}{c}\right)^2 + \dots \quad (22.10)$$

The small parameter of this expansion is $H_0 S/c$. In the first order we have the most straightforward relation between Z and the distance to a receding galaxy: the redshift is proportional to the distance.² Alternatively, using Eq. (22.2), we can express Z via the receding velocity $v = \dot{S}$,

$$Z = \frac{v}{c} + \frac{1 + q_0}{2} \left(\frac{v}{c}\right)^2 + \dots, \quad (22.11)$$

so that the equivalent small parameter of the expansion is v/c .

Eqs. (22.10)–(22.11) give the model-independent relations between the redshift, Hubble ‘constant’ and ‘deceleration’ parameter. To utilise Eq. (22.5) for $Z \gtrsim 1$ we need some model of $\mathcal{R}(t)$.

Consider, as an example, the simplest model of the matter-dominated flat Universe with $\Lambda = 0$. In this case $\mathcal{R}(t) \propto t^{2/3}$, Eq. (21.18). Then we find from Eq. (22.5)

$$Z = \left(\frac{t_0}{t_e}\right)^{2/3} - 1.$$

On the other hand, in this model we find from Eq. (22.6)

$$S = c t_0^{2/3} \int_{t_e}^{t_0} \frac{dt}{t^{2/3}} = 3 c t_0^{2/3} \left(t_0^{1/3} - t_e^{1/3}\right) = 3 c t_0 \left[1 - \left(\frac{t_e}{t_0}\right)^{1/3}\right] = 3 c t_0 \left[1 - \frac{1}{\sqrt{1+Z}}\right].$$

The receding velocity of galaxy E in the present epoch is found from Eq. (22.2)

$$v = \dot{S} = H_0 S \equiv \frac{\dot{\mathcal{R}}(t_0)}{\mathcal{R}(t_0)} S = \frac{2S}{3t_0} = 2c \left[1 - \frac{1}{\sqrt{1+Z}}\right]. \quad (22.12)$$

When $Z > 3$, we have $v > c$. One might think that such a remote galaxy is beyond the particle horizon but this is not the case: in this model we find from Eq. (22.4) that $S_p = 3ct_0$ so that the

²Using the current value of the Hubble constant, $H_0 = 2.2 \times 10^{-18} \text{s}^{-1}$, we find that $Z = 0.1$, up to which we can reliably use the expansion, corresponds to $S = 0.1c/H_0 \sim 5 \times 10^{16} \text{l.y.} \approx 10^{10} \text{Mps}$ – a huge distance, having in mind that galaxies are separated only by 1Mps. For $Z > 1$, the expansion definitely fails and we need to use models.

emitting galaxy is within the particle horizon for any Z . The point is that in this model the expansion is decelerated with $q = 1/2$ (as $\Omega = \Omega_M = 1$) and any photon eventually reaches us.³ Note that in this model the current age of the Universe is $t_0 = \frac{2}{3H_0} \approx 3 \times 10^{17} s \approx 10$ billion years while we know that some stars have age of 12Gyears. The current best estimate is $t_0 = 13.7 \times 10^9$ years.

Now it is clear that extracting H_0 and q_0 from observational data requires preliminary modelling of $\mathcal{R}(t)$ (or, equivalently, of Ω_M and Ω_Λ), and comparing the model predictions with observations. The redshift alone is not sufficient as one need to extract two parameters, H_0 and q_0 . So other measurements of distance are necessary. For small Z , one can measure distance from the apparent size of the object, while for large Z only the apparent luminosity is used.

22.4.2 Distance from apparent size and from luminosity

If an object of a known cross-section $d\mathcal{A}$ (e.g. a galaxy of a typical size) subtends a small solid angle $d\Theta$ (using Θ instead of usual Ω which is differently engaged in cosmology). Then in Euclidean geometry $d\mathcal{A} = S^2 d\Theta$ so that

$$S = \sqrt{d\mathcal{A}/d\Theta}.$$

This is a *naïve* formula which deliberately ignores all subtlety due to possible deviations of the space geometry from Euclidean.

We can formulate this in terms of the linear ‘physical size’ of the object, dD : $d\mathcal{A} \sim dD^2$, that should be measured at the time when the signal is emitted. Thus, $D = \int \sqrt{-ds^2} = r(\chi)\mathcal{R}(t_0) \int d\theta = r(\chi)\mathcal{R}(t_0)\delta\theta$ which gives

$$S \sim dD/d\theta = r(\chi)\mathcal{R}(t_0),$$

The past and present values of $\mathcal{R}$ are related by the formula for the cosmological red shift.

Similarly, **the distance from apparent luminosity**, b , is

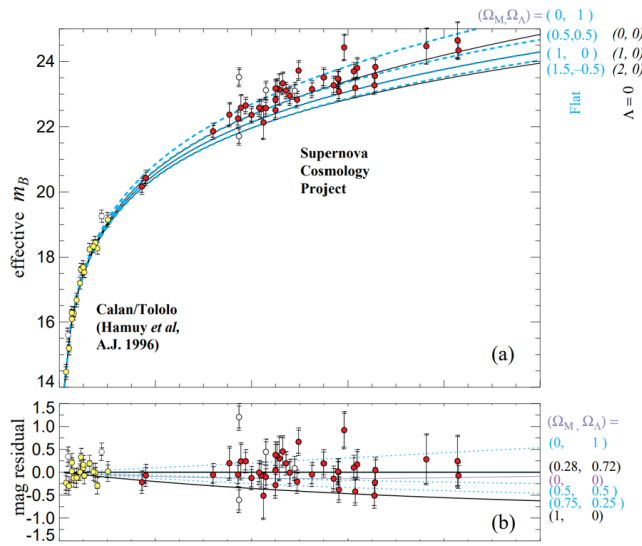
$$S = \sqrt{B/b}$$

where B is known intrinsic luminosity of an object. To make this a reliable measure, one should have "standardised candles" – objects with known intrinsic luminosity.

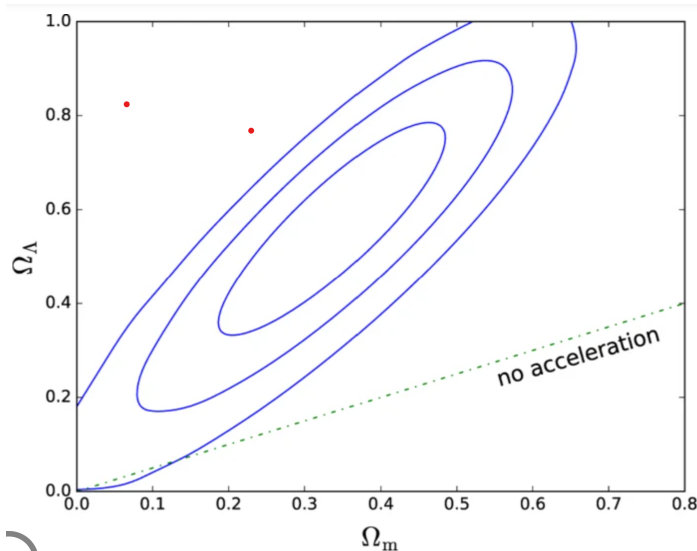
22.4.3 Observation results

The choice of standardised candles depends on the distance to stars or galaxies.

³This means that there is no event horizon in this model. Indeed, $S_e \rightarrow \infty$ in Eq. (22.4).

Contributions to Ω .

It is seen from the picture above that the most probable answer, $\Omega_\Lambda = 0.72$ and $\Omega_M = 0.28$ is statistically better than alternatives while the answer with $\Omega_\Lambda = 0$ can be certainly excluded.

Probabilities of Ω_Λ and Ω_M .

This role at cosmologically large distances is played starting from the late 1990's by a certain type of supernovae (SN Ia). By measuring the apparent distance to more and more remote (*i.e.* more and more red-shifted) SN it became possible to find with the ever increasing accuracy not only H but q (or equivalently, both Ω_M and Ω_Λ). The picture on the left is from the Supernovae Cosmology Project, S.Perlmutter et al, 1998 (similar data from the High-Z Supernova Search Team, A.G.Riess, B.P.Schmidt et al, 1998).

Statistics have substantially improved for the last two decades and is summarised in the picture on the left. It shows probabilities of different values of Ω_Λ and Ω_M with accuracy of σ , 2σ and 3σ (*i.e.* the probability being outside the smallest ellipse is less than $1/e$, the middle $1/e^2$ and that of being outside the biggest 3σ - is less than $1/e^3$.)

22.5 Horizon and flatness problems. Inflation scenario

The flatness of the Universe is quite puzzling by itself: as it is almost flat now, it should be fantastically flat at the early epoch as any deviation from that would be greatly increasing with time. How happens that it is still about 1 while $\mathcal{R}$ has increased probably by 10^{30} since the earliest, so called Planck epoch (when quantum mechanics realm so that we know nothing of the preceding time)? The approximate homogeneity and isotropy of the cosmic background radiation is also puzzling: the signals come from widely remote parts of the Universe which are totally outside of each other horizons. The radiation originates from the early epoch when – if we use the Friedman models (with or without Λ) – these parts were causally unconnected and thus could not “coordinate” their evolution, but are nevertheless observed to be at the same temperature with a high precision.

The solution to both problems (and many other in particle physics, such as the lack of symmetry

between matter and antimatter) is in the *inflation scenario* due to Guth and – independently – Linde (1972). It assumes an extremely fast exponential growth of $\mathcal{R}$ driven by vacuum (dark) energy not only at the very end, as in Eq. (21.19), but also at the very beginning. In this scenario, the Universe has been born from vacuum in a phase transition due to a quantum fluctuation and expanded dramatically (by many orders of magnitude) so that the horizon today is much larger than a naïve estimate by H^{-1} . After the exponential expansion, the Universe was still flat and huge and its further history developed according to one the Friedman models with $\Lambda \neq 0$ but within a huge horizon (no problem of the absence of causality) and with initial $\Omega = 1$. However, a consistent theory of both exponential inflation and the earliest Friedman universe requires the quantum theory of gravity which is still missing – this is left for your generation to develop!